Compact Calculus

Calculus for the practitioners

Puripat Thumbanthu
& Kunakorn Chaiyara

1E

Compact Calculus - Calculus for the practitioners

Puripat Thumbanthu, Kunakorn Chaiyara

Started: 07/28/2022

Revision: April 28, 2026

Prerequisites: *naive set theory, algebra, geometry, basic trigonometry*

Contents

Acknowledgement

Book I

Differential equations

Preface to book I

Purpose of book I

I probably made a lot of mathematicians angry by writing this book. But it misses their points. This textbook is designed for the practitioners of calculus: people that are interested in fields of sciences including biology, chemistry, and physics. Therefore, expect what you aren't expected, and don't expect what you traditionally expect.

The contents of this textbook are entirely self-contained and organized into four parts (volumes)[1]: dynamical systems and ordinary differential equations, signal analysis, real and complex analysis, and vector calculus with partial differential equations. However, the topics deviates greatly from the traditional way calculus is taught. I took a great care of how the material is arranged in better needs practitioners, which means it may deviate from a standard calculus curriculum. To maintain focus within the core chapters, the auxilaries topics typically covered in traditional courses have been moved to the appendices.

Even though the book is very self-contained, it still doesn't serve as a standalone book. Sure, the story that I'm about to tell is quite fruitful,

[1]The exact number of parts may change as more material is added during the writing process.

and you can enjoy it from the beginning to end. But, it lacks one thing: problem solving experiences. As of the revision today (April 28, 2026), there are not that many exercises in the chapters. So, you might have to find some online. I've scattered plenty of them throughout the chapters.

Layout of the book I

Layout of Part I

The first part of the book concerns differential equations: a powerful tool that can be used to describe how a system changes over time. The contents are laid out as follows:

- Chapters 1 and 2 develops the basic building blocks of calculus: derivatives and integrals, from the ground up
- Chapter 3 takes the reader through derivatives and integrals of basic functions, e.g., exponentials, and logarithms.
- Chapter 4 takes a detour from the main objective a bit, and looks at how calculus could be applied to solve various geometrical problems.
- Chapter 5 studies the interaction between calculus and various trigonometric functions. It also applies calculus to describe some physical systems, specifically oscillatory ones. From this chapter, most of the foundations of calculus is already laid down, ready to be used.
- Finally, chapter 6 uses all the knowledge developed in earlier chapters to explore many systems that exist in nature, including biological, chemical, and physical systems.

Part I

The foundation of calculus: differential equations

CHAPTER 1

Writing down nature: derivatives

Prerequisites: *graphs, functions, kinematic variables*

1.1 Invitation to calculus from a practitioner

Calculus is a mathematical language — a framework that's used to describe a system that's changing, a.k.a. a **dynamical system**. Its power comes from the ability to predict how the change affects the system over time. If you look around, much of the world is always changing; hence, the framework that calculus provides is very vast in applications. Whether it'd be the change in the weather patterns, in the movement of an object, in the evolution of biological systems, or even in your relationship, calculus can model them all.[1]

[1] That last one might be diabolical to think about for some of you.

Thinking of calculus as a mathematical framework for changing systems allows us to gain a deeper intuition about the subject, and also allows the theories to naturally develop. Still, this aspect of calculus is greatly underappreciated in introductory courses. As a remedy, I've decided to structure this whole volume around the concept of dynamical systems: systems that evolve through time, whether they are biological, chemical, or physical, in order to foster a deeper understanding for the subject.

As said in the preface, I have thrown most unrelated contents into the appendices. But as much as I'd love to do that, some purely mathematical contents and exercises are still essential for understanding and solving these dynamical systems. While I've kept those in, I tried to make them as brief as possible to not disrupt your entertainment.

Outline of the framework: The problem of dynamical systems are separated into two parts.

1. Describing the system, and
2. Predicting the system's future.

Calculus is also neatly separated into two parts which neatly align with these problems: differential calculus and integral calculus. Differential calculus uses differential equations to describe a system; those equations are then solved using integral calculus to predict the system's future. This chapter is devoted to the first part of calculus: differential calculus.

At the end of this chapter, you should develop the intuition that

1. Derivative measures the rate of change of a function w.r.t. a variable.
2. Derivatives can be thought of as the slope of a graph.

3. The universe is described in the language of differential equations.

1.1.1 The notation of differential calculus

Infinitesimals If a is a variable, then $\mathrm{d}a$ is a very small change in a, sometimes referred to as a small increment of a, or an **infinitesimal** of a. E.g., if $\mathbf{x}$ is displacement, then $\mathrm{d}\mathbf{x}$ is a very small displacement. If t is time, then $\mathrm{d}t$ is a very short time. It's very similar to the delta notation where Δx represents a difference or a change in x. Replacing the Δ with d suggests that "it is an *absolutely small* difference in x."

Infinitesimals can also be divided. E.g., the ratio between some small distance $\mathrm{d}x$ and some short time $\mathrm{d}t$ is the speed of an object during that short time: $\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t}$.[2]

1.2 Speed and instantaneous rate of change

On terminology: In this chapter, *speed* and *velocity* is used interchangeably for pedagogical purposes.

The simplest way to construct the language of change is to quantitatively analyze an object that's moving and changing position through time, e.g., a car moving, a block sliding, a ball falling, etc. One way to describe these systems is to use the **speed**, which measures how fast the object is moving through time. In other words,

> *Speed measures the change in position of an object over a time interval*

[2] This "division" works when you're working with *simple* quantity of *only one variable*.

If the object is at point $\mathbf{s}_1$ at time t_1, then moves to point $\mathbf{s}_2$ at time t_2, the speed ($\mathbf{v}$) would be

$$\mathbf{v} = \frac{\mathbf{s}_2 - \mathbf{s}_1}{t_2 - t_1}. \tag{1.1}$$

However, this equation doesn't give us the full picture, as it doesn't fully capture the behavior of an actual moving object.

Consider a ball that's been dropped from a building. Before the drop, the ball has zero speed as it isn't moving. Then, the ball gradually gets faster and faster over time until it hits the ground and does a few bounces until it stops. According to eq. (1.1), the ball might have the speed of $10\,\mathrm{m\,s^{-1}}$ in the entire journey. Does this number capture the actual behavior of the ball? No, because the speed of the ball varies over time.

Another example would be a moving car. Let's say you're driving from Thailand to Malaysia, and according to eq. (1.1), your speed is $50\,\mathrm{km\,h^{-1}}$. Does this number say anything about the congestion slowing you down? Does it say anything about whether you've been on a highway or not? Still no, as the speed of a car can be slowed down or sped up by external factors: it is also varying over time.

The quantity that eq. (1.1) spits out actually isn't the actual speed of an object; it is the **average speed** of the entire motion. It disregards the change in speed that happens along the way and averages out everything. Well, this quantity is mostly fine in daily situations. But if you want to describe a dynamical system in its entirety, it's not wise to just know the average: you want to know everything in order to accurately predict the system's future. So is there a description of motion that's more detailed and nuanced than this?

To better describe a ball that's been dropped from a building, you might calculate the speed in each phase of the ball and list it out. E.g.,[3]

- Speed before the drop: $0\,\mathrm{m\,s^{-1}}$
- Average speed from the moment of dropping to three seconds after dropping: $3\,\mathrm{m\,s^{-1}}$
- Average speed from three seconds after dropping to three seconds after landing: $10\,\mathrm{m\,s^{-1}}$
- Average speed from three seconds after landing to landing: $17\,\mathrm{m\,s^{-1}}$
- Average speed during the first bounce: $5\,\mathrm{m\,s^{-1}}$

But still, this isn't enough. What if there are sudden gusts of wind pushing the ball upwards while the ball is bouncing, suddenly changing its speed? What if the ball bounces on a little kid's head causing the average speed during the first bounce to be lower? Even though this report is more nuanced than just the average speed, it still misses some things that could happen in between.

So you subdivide the time into smaller and smaller parts, also increasing the length of the speed report. But still, any finite subdivision of time is going to miss some details in the movement for the same reason given prior.

A little sarcasm will lead you to say, if I'm going to be this picky and fussy about the speed report, why don't just subdivide the time into intervals so small — *infinitesimally small*, and report an *infinite* list of speed throughout those times. Can it be that when the time interval is so short, shorter than the variations in the velocity of the object, the average veloc-

[3] I'm just making numbers up. Don't mind me.

ity at that short time interval becomes the actual velocity of an object at that instance?

Believe it or not, calculus says that *you absolutely can*. The idea of an infinitesimally small subdivision is the core idea of differential calculus, and it gave rise to the notion of **derivatives** which a measure of rate of change at an instance.

The derivative of some quantity A with respect to (w.r.t.) another quantity B, denoted

$$\frac{\mathrm{d}A}{\mathrm{d}B} \tag{1.2}$$

is a measure of how much A changes if B changes by an *infinitesimally small* value, $\mathrm{d}B$. In the case of speed, it is a measure of how the position of an object ($\mathbf{s}$) changes if the time (t) changes or increases by an *infinitesimally small* time step $\mathrm{d}t$. Using the notation of calculus, we write that:

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{s}}{\mathrm{d}t} \tag{1.3}$$

instead. The equation above reads: "the velocity is the derivative of the distance s w.r.t. time t."

To put it in a different way, recall the notation discussed in section 1.1.1. It says that $\mathrm{d}t$ is an infinitesimally small increment of time t, and $\mathrm{d}\mathbf{s}$ is an infinitesimally small change in position. Specifically, it is the change in position in the time span $\mathrm{d}t$. The ratio between $\mathrm{d}t$ and $\mathrm{d}\mathbf{s}$ then yields the derivative: the rate of change in $\mathbf{s}$ in time.

From here on out, *derivative* and *rate of change* is used interchangeably. So, every time you see *derivative*, think *rate of change*.

1.3 An attempt to define derivatives

Mathematically, a **derivative** is a measure of a function's rate of change with respect to a variable. In the previous example, $\mathbf{s}$ is a function that's dependent on time, and its derivative w.r.t. time, $\mathbf{v}$, is how $\mathbf{s}(t)$ changes after an infinitesimally small time step $\mathrm{d}t$.

To find an explicit expression for derivative, consider two points in spacetime $(\mathbf{s}_1, t_1)$ and $(\mathbf{s}_2, t_2)$. The change in $\mathbf{x}$ is $\mathbf{x}_2 - \mathbf{x}_1$, and the change in time is $t_2 - t_1$. The derivative of position w.r.t. time is then just

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{s}}{\mathrm{d}t} = \frac{\mathbf{s}_2 - \mathbf{s}_1}{t_2 - t_1}. \tag{1.4}$$

Another way to write this would be to let $t_2 = t + \mathrm{d}t$, then $\mathbf{s}_2 = \mathbf{s}(t + \mathrm{d}t)$ and $\mathbf{s}_1 = \mathbf{s}(t)$, and we get

$$\mathbf{v} = \frac{\mathbf{s}(t + \mathrm{d}t) - \mathbf{s}(t)}{t + \mathrm{d}t - t} = \frac{\mathbf{s}(t + \mathrm{d}t) - \mathbf{s}(t)}{\mathrm{d}t} \tag{1.5}$$

We've said earlier that we want this time difference $\mathrm{d}t$ to be infinitesimally small. However, this time difference can't be zero. Because if it is zero, then we would get that

$$\mathbf{v} = \frac{\mathbf{s}(t + 0) - \mathbf{s}(t)}{0} = \frac{\mathbf{s}(t) - \mathbf{s}(t)}{0} = \frac{0}{0}, \tag{1.6}$$

which is undefined. Hence, $\mathrm{d}t$ has to be something that's small but not zero.

Calculus offers a notation to denote a variable that gets infinitesimally close to a number but isn't that number: the **limit**. If a variable a is infinitesimally close to a number b, then we write

$$\lim_{a \to b}. \tag{1.7}$$

The equation of velocity then becomes

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{s}}{\mathrm{d}t} = \lim_{\mathrm{d}t \to 0} \frac{\mathbf{s}(t + \mathrm{d}t) - \mathbf{s}(t)}{\mathrm{d}t} \tag{1.8}$$

The equation above is what we generally refer to as the *definition of derivative*. Generally, this notion can be extended to any function $f(x)$ by a substitution.

Definition 1.3.1: *Naive definition of derivative*

A derivative of a function $f(x)$ w.r.t. a variable x is the rate of change of $f(x)$ w.r.t. x, and it is written as

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} \quad \text{or} \quad \frac{\mathrm{d}}{\mathrm{d}x}f(x), \tag{1.9}$$

where

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}. \tag{1.10}$$

Equation (1.10) directly reads:

The derivative of $f(x)$ with respect to x is $\dfrac{f(x+h)-f(x)}{h}$ where h is very close to 0, but h is strictly not 0.

On notation: So far, we've been using the Leibniz's notation for derivatives, and it has the property that derivatives behave exactly like fractions, and you can cancel terms.

$$\frac{\mathrm{d}a}{\mathrm{d}b}\times\frac{\mathrm{d}b}{\mathrm{d}c} = \frac{\mathrm{d}a}{\mathrm{d}c}. \tag{1.11}$$

But, this isn't the only accepted notation. Multiple great mathematicians had come up with their own way of writing the derivative. Some are better than others in certain places. I'd introduce other notations later on if necessary.

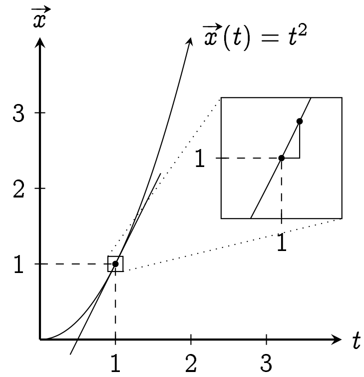


Figure 1.1 | POSITION-TIME GRAPH WHERE $\mathbf{x}(t) = t^2$.

1.4 The geometrical interpretation of the derivative

One way to interpret derivatives is by using slope. Notice that eq. (1.4) looks a lot like the slope equation for a line

$$m = \frac{y_2 - y_1}{x_2 - x_1}. \tag{1.12}$$

We've also seen that eq. (1.4) is analogous to the definition of derivative eq. (1.10). So is the derivative just the slope of a line? If it is, then what is the y-axis and the x-axis of a graph?

We can compare eq. (1.12) to eq. (1.4): the y-axis should be the position $\mathbf{x}$, and the x-axis, the time t. If we draw that out, we'll get the x-t graph, which can encode the exact trajectory of an object. An example of which is shown in fig. 1.1.

But what does this have to do with derivatives? Equation (1.12) only works for straight line! Well, here's the beauty of it. If you zoom into any points on a curve, eventually, it will look like a line. And thus, *the derivative zooms into the curve at some point, and chooses two very close points on* *the curve and calculate its slope. In which, that slope represents the rate of change of the function at that point.*

1.5 Evaluation of derivatives: method of increments

The derivative's definition can be used to directly evaluate derivatives. This is called the **method of increment**. E.g., in fig. 1.1, let's evaluate the velocity at time $t = 1\,\mathrm{s}$ where the position function, $\mathbf{x}(t) = t^2$. $t = 1\,\mathrm{s}$. We start from eq. (1.10):

$$\begin{aligned}
\mathbf{v}(1) = \frac{\mathbf{x_2} - \mathbf{x_1}}{t_2 - t_1} &= \lim_{h\to 0} \frac{\mathbf{x}(1+h) - \mathbf{x}(1)}{(1+h) - 1} \\
&= \lim_{h\to 0} \frac{(1+h)^2 - 1}{h} \\
&= \lim_{h\to 0} \frac{2h + h^2}{h} \\
&= \lim_{h\to 0} 2 + h.
\end{aligned}$$

Since h is very close to zero, we approximate $2 + h$ as 2. Imagine comparing 2 to 10^{-10}. The 10^{-10} wouldn't make a noticeable difference, and we can ignore it. Therefore, $\mathbf{v}(t = 1\,\mathrm{s}) = 2\,\mathrm{m\,s^{-1}}$.

Now, try evaluating $\mathbf{v}(t = 3\,\mathrm{s})$ for $\mathbf{x}(t) = t^3$. You should get $81\,\mathrm{m\,s^{-1}}$. As a hint, you can also ignore h^2 because if $h < 1$, then $h^2 < h$.

1.6 Higher order derivatives

In kinematics, there are a whole set of quantities that can describe an object's trajectory, e.g., the acceleration, which is defined to be the rate of change of velocity w.r.t. time:

$$\mathbf{a}(t) = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t}.$$

Order	Name
1	Velocity/Speed
2	Acceleration
3	Jerk
4	Snap/ Jounce
5	Crackle
6	Pop

Table 1.1 | HIGHER ORDER DERIVATIVES OF POSITION W.R.T. TIME

But the $\mathbf{v}$ is also the rate of change of position w.r.t. time. Thus,

$$\mathbf{a}(t) = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}\mathbf{r}(t)}{\mathrm{d}t}\right) = \frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2},$$

where $\mathbf{r}$ is any position vector.

The $\mathrm{d}^2\mathbf{r}$ and $\mathrm{d}t^2$ is just a matter of symbolic manipulation and should only be interpreted as just a shorthand. $\mathbf{a}$ is called the **second order derivative** of $\mathbf{r}$ because you've differentiated $\mathbf{r}$ twice. **Higher order derivatives** of position w.r.t. time is listed in table 1.1.

1.7 Expressing nature: basic differential equations

The dynamics of a physical system is, up to some degree, universally described by the famous Newton's second law $\mathbf{F} = m\mathbf{a}(t)$, which in derivatives form becomes:

$$\mathbf{F} = m\frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}. \tag{1.13}$$

Let's try to describe a simple system with this.

In fig. 1.2, a ball is dropped from height h. The Earth's gravity pulls the ball with force mg where m is the mass of the ball, and $\mathbf{g}$, the

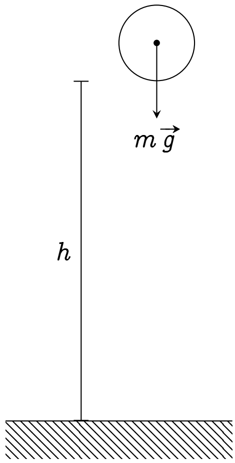


Figure 1.2 | A BALL DROPPED FROM HEIGHT h

acceleration from Earth's gravity. Newton's second law tells us that

$$\mathbf{F} = m\frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2} \tag{1.14}$$

$$m\mathbf{g} = m\frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2} \quad \text{or} \quad \mathbf{g} = \frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}, \tag{1.15}$$

which is sometimes called the **equation of motion**. This equation packs every information about this system you'd ever want. It directly reads as: "The acceleration of the ball is equals to g."

If you've stayed for this long, congratulations! You now have the power to describe every physical systems with mathematical equations using derivatives: the measure of the rate of change. But it might not be as useful as yet, just as a hammer may seem useless if used to paint, derivatives falls apart when you ask about the future of the system. E.g., how long the ball takes to reach the ground? Or, what's the position of the ball at a certain time? That's the job of the integral to solve, and we'll do so in the next chapter.

1.8 Conclusion for Chapter 1

1. The concept of approaching can be used to bypass dividing by zero.
2. Derivatives are rate of change of a function w.r.t. a variable which can be evaluated by the method of increments.
3. Derivatives can be thought as the slope of a graph, or the tangent to a curve.
4. Physical systems can be described by differential equations of different forms. One of them is the Newton's formulation stated in eq. (1.13)

Remarks 1.8.1

In section 1.2, we zoomed in on the graph to approximate the function as a line. Actually, this is quite literally the whole idea of derivatives. If we dig in further in calculus, sometimes the rate of change analogy doesn't even make sense. However, saying that the derivative tries to approximate every function as a line works in all scenario. Though, it's quite abstracted away from the world.

CHAPTER 2

Integrals and antiderivative

Prerequisites: *chapter 1, sigma summation notation*

On terminologies: Mathematically, every line, including straight lines, is a **curve**. In the standard textbook repertoire, the term "*area under the graph*" is often used to refer to integrals. But a **graph** is *a diagram consisting of a line or lines, showing how two or more sets of numbers are related to each other* (*8*), not the curve itself. Therefore, I'll refer to a curve as any line that connects two points, whether straight or not; hence, the integrals will be referred to as *the area under a curve*.

2.1 Invitation: an impossible mission

2.1.1 The mindset of integral calculus

Literally everything that involves a quantity changing is written in the language of derivatives. When derivatives are used in equations to describe a system, that equation is called *differential equation*. Its solution contains the future of the entire system in question. However, it's quite hard to solve; some, even impossible.

In physics, a **state** represents the configuration that the system *at a point in time*. To predict how the system might evolve, we need to know the initial state of a system, i.e., how the system is initially configured. If you know this initial state, and you know the rules that the system plays by (In this case, $\mathbf{F} = m\dfrac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$), it's possible to determine the state of the system at any point in time. This prediction of the system is caled the **state function**: a function that takes in some input time t and outputs the configuration of the system at that time.

Consider the example from chapter 1. The equation of motion of a ball dropped from a height h is a second-order differential equation

$$g = \frac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}, \tag{2.1}$$

with initial condition being $\mathbf{r} = h$. The state function that we ought to know is $\mathbf{r}(t)$, which takes in a time t and outputs the position of the ball at that time. You can see that this function is under the derivative sign. To solve for this function, we need to undo the derivative sign and isolate the function out. How?

It seems impossible at first, because the only information we're given is that $\mathbf{r}(t)$ must satisfy eq. (2.1), i.e., the second derivative of $\mathbf{r}(t)$ is g. But the amount of functions in mathematics are so enormous. It's like we have to search through a gigantic pool of functions that mathematics have to offer, just to find a single function $\mathbf{r}(t)$ that satisfies eq. (2.1). It'd be like finding a needle in the haystack. Even if you find one, how do you know that the function you found is the only solution?

Of course, this is a textbook, there must be a solution. If there is a will, there is a way. You might have to reverse engineer derivatives, which looks tedious at first. But you might as well find something interesting along the way.

2.1.2 Notation of integral calculus

The $\int$, a.k.a. the **integral**[1], means to sum. This integral symbol is basically sigma summation symbol, but for infinitesimals. Like the sigma notation, the integrals also have bounds: the **integral bounds**, which are written below and on top of the integral symbol to denote the start and end of summation. E.g., if we sum a lot of little time step $\mathrm{d}t$ together from t_A to t_B we get $t_B - t_A$, the total time step. Thus,

$$\int_{t_A}^{t_B} \mathrm{d}t = t_B - t_A. \tag{2.2}$$

2.2 Finding a function in the haystack

Equation (2.1) has a second order derivative, let's go slowly and undo one derivative at a time. We'll undo the derivative of the R.H.S. to get the velocity first. Then, we'll undo the derivative again to get the position.

2.2.1 Step one: the velocity function from acceleration

One good strategy for solving any kind of equation is **separation of variables**. We isolate the variable we're interested in solving, which is $\mathbf{r}$, and move everything else to the other side. Here, write $\dfrac{\mathrm{d}^2\mathbf{r}(t)}{\mathrm{d}t^2}$ as $\dfrac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t}$. Then, isolate $\mathbf{v}$ on one side and move $\mathbf{t}$ to the other.

$$\mathbf{g} = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t} \tag{2.3}$$

$$\mathrm{d}\mathbf{v}(t) = \mathbf{g}\,\mathrm{d}t. \tag{2.4}$$

[1] Which also looks like a beansprout

This equation reads

> *A small change in velocity* $\mathrm{d}\mathbf{v}$ *is product of* $\mathbf{g}$ *and a small time interval* $\mathrm{d}t$*.*

To find the total change in velocity, we sum up a lot of small changes in velocity $\mathrm{d}\mathbf{v}(t)$, which is equal to $\mathbf{g}\,\mathrm{d}t$. Because $\mathrm{d}v$ is directly proportional to $\mathrm{d}t$, the total change in $\mathbf{v}$ is simply $\mathbf{g}t$. However, *changes* doesn't say anything about the initial condition, so we add a term C to compensate. Therefore,

$$\mathbf{v}(t) = \mathbf{g}t + C. \tag{2.5}$$

To find what C is, just plug in the initial condition. If $\mathbf{v}(t = 0) = \mathbf{v}_0$, then

$$\mathbf{v}(t = 0) = v_0 = \mathbf{g} \times 0 + C \tag{2.6}$$

$$v_0 = C. \tag{2.7}$$

Thus,

$$\mathbf{v}(t) = \mathbf{g}t + \mathbf{v}_0. \tag{2.8}$$

We can also express these ideas symbolically using the integral symbol (section 2.1.2) as

$$\mathbf{g} = \frac{\mathrm{d}\mathbf{v}(t)}{\mathrm{d}t} \tag{2.9}$$

$$\mathrm{d}\mathbf{v} = \mathbf{g}\,\mathrm{d}t \tag{2.10}$$

$$\int \mathrm{d}\mathbf{v} = \int \mathbf{g}\,\mathrm{d}t \tag{2.11}$$

$$\mathbf{v} = \mathbf{g}t + \mathbf{v}_0. \tag{2.12}$$

2.2.2 Step two: the position function from velocity

Rewrite $\mathbf{v}$ as $\frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$, then do separation of variables.

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t} = \mathbf{g}t + \mathbf{v}_0 \tag{2.13}$$

$$\mathrm{d}\mathbf{r} = \mathrm{d}t(\mathbf{g}t + \mathbf{v}_0), \tag{2.14}$$

which reads,

> *A small change in position* $\mathrm{d}\mathbf{r}$ *is the product of* $\mathbf{g}t + \mathbf{v}_0$ *and a small time interval* $\mathrm{d}t$.

If we want to find the total change in position $\mathbf{r}$, we just have to sum all $\mathrm{d}\mathbf{r}$'s. This time, it's not as obvious, because there's $\mathbf{g}t + \mathbf{v}_0$, which is also changing with time. So at each point in time, the rate of change of position is different.

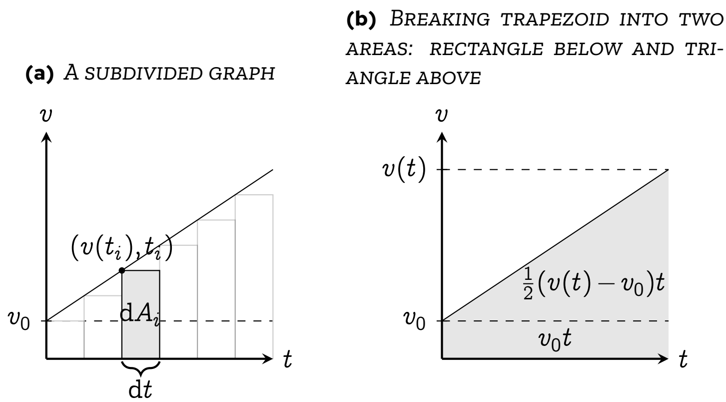


Figure 2.1 | A v-t GRAPH OF A BALL DROPPED FROM A BUILDING

A good strategy in math if you don't know what to do is to just graph the function. The graph of eq. (2.13) is shown in fig. 2.1(a). A small time interval represents a little step in the t axis. The curve shown in the graph represents $\mathbf{g}t + \mathbf{v}_0$. Therefore, $\mathrm{d}t(\mathbf{g}t + \mathbf{v}_0)$ would just represent an area of a little rectangle $\mathrm{d}A$ as shown in the figure.

The total change in position is the sum all those rectangles. When $\mathrm{d}t \to 0$, the sum of all $\mathrm{d}t(\mathbf{g}t + \mathbf{v}_0)$ approaches the area under the graph, which can be calculated geometrically as shown in fig. 2.1(b). Thus,

$$\mathbf{r} = \frac{1}{2}(\mathbf{v}(t) - \mathbf{t}_0)t + \mathbf{v}_0 t + C \tag{2.15}$$

$$= \frac{1}{2}(\mathbf{g}t - \mathbf{t}_0)t + \mathbf{v}_0 t + C \tag{2.16}$$

$$= \frac{1}{2}\mathbf{g} \times t^2 + \mathbf{v}_0 t + C \tag{2.17}$$

When $t = 0$, $\mathbf{r} = \mathbf{r}_0$. Therefore,

$$\mathbf{r}_0 = \frac{1}{2}\mathbf{g} \times 0^2 + \mathbf{v}_0 \times 0 + C$$

$$\mathbf{r}_0 = C.$$

Therefore,

$$\mathbf{r}(t) = \frac{1}{2}\mathbf{g}t^2 + \mathbf{v}_0 t + \mathbf{r}_0. \tag{2.18}$$

To model the trajectory of the ball, we set

1. $\mathbf{v}_0 = 0$ (object is dropped and starts at zero speed)
2. $\mathbf{r}_0 = 0$ (convenient initial condition placement)

thus we get,

$$\mathbf{r} = \frac{1}{2}\mathbf{g}t^2 \implies t = \sqrt{\frac{2\mathbf{r}}{\mathbf{g}}} \tag{2.19}$$

Since the ground is at $\mathbf{r} = h$, the time that the ball hits the ground is then $\sqrt{2h/\mathbf{g}}$.

2.2.3 Conclusion: area under the curve and antiderivative

The form of the differential equation that we've solved is in the form $\frac{\mathrm{d}x}{\mathrm{d}t} = f(x)$. We have to undo the derivative, and the simplest way is to do separation of variables and turn the equation into

$$\mathrm{d}x = f(x)\,\mathrm{d}t, \tag{2.20}$$

which reads,

> A small change in x, i.e., $\mathrm{d}x$ is represented by the area of a rectangle width $\mathrm{d}t$ and height $f(x)$.

And, the total change x is represented by the sum all those little rectangles, which is the area under the curve $f(x)$. Then, we add a constant C to compensate for the initial condition. In symbolic form introduced in section 2.1.2, it's just

$$\int \mathrm{d}x = \int g(x)\,\mathrm{d}t \tag{2.21}$$

$$x = \int g(x)\,\mathrm{d}t. \tag{2.22}$$

For now, we could say that integration is the reverse of derivatives. But to clearly see how this is linked for every function, we must study the fundamental theorem of calculus, which is the bridge between integration and differentiation.

Before we go there, let me clarify some terminologies. An **integral** refers to the area under the curve evaluated between two points. We say that an integral must have an **integral bound**. If we want to find the area under a function $f(x)$ from $x = a$ to $x = b$, we write it as

$$A = \int_a^b f(x)\,\mathrm{d}x. \tag{2.23}$$

This just reads

> The area A under the curve $f(x)$ from $x = a$ to $x = b$ is equal to the sum of the area of many thin stripes width $\mathrm{d}x$ height $f(x)$ that lies between $x = a$ and $x = b$.

The **antiderivative** however, refers to the function which takes in a value a, and output the integral of $f(x)$, evaluated from 0 to a. Therefore, if $A(a)$

is the antiderivative of $f(x)$, then

$$A(a) = \int_0^a f(x)\,\mathrm{d}x. \tag{2.24}$$

It also implies that if the area between a and b can be evaluated by

$$\int_a^b f(x)\,\mathrm{d}x = A(b) - A(a). \tag{2.25}$$

2.3 The fundamental theorem of calculus

The intuition of fundamental theorem of calculus states that derivatives and integrals are essentially inverse of each other. In this section, I'll clarify this fact and make it more rigorous.

Theorem 2.3.1: *The (first) fundamental theorem of calculus*

If $A(x)$ is the antiderivative of $f(x)$, then

$$\frac{\mathrm{d}A(x)}{\mathrm{d}x} = f(x). \tag{2.26}$$

If $A(x)$ is the antiderivative of $f(x)$, then

$$\int_0^x f(x)\,\mathrm{d}x = A(x). \tag{2.27}$$

The *actual* area of one of the stripes (not rectangles) width $\mathrm{d}x$ shown in fig. 2.2, it's obviously $A(x+\mathrm{d}x) - A(x)$. The riemann sum approximation approximates the area by a small rectangle area $f(x)\,\mathrm{d}x$. We can write the relation between the actual and the approximated area as

$$A(x+\mathrm{d}x) - A(x) \approx f(x)\,\mathrm{d}x. \tag{2.28}$$

To turn this into an equality, we add a correction term ε

$$A(x+\mathrm{d}x) - A(x) = f(x)\,\mathrm{d}x + \varepsilon. \tag{2.29}$$

If we let $\mathrm{d}x \to 0$, ε is negligible compared to $f(x)\,\mathrm{d}x$; therefore,

$$\lim_{\mathrm{d}x\to 0} A(x+\mathrm{d}x) - A(x) = \lim_{\mathrm{d}x\to 0} f(x)\,\mathrm{d}x. \tag{2.30}$$

Since $f(x)$ is not a variable that's controlled by the limit sign,

$$\begin{aligned} \lim_{\mathrm{d}x\to 0} A(x+\mathrm{d}x) - A(x) &= f(x) \lim_{\mathrm{d}x\to 0} \mathrm{d}x \\ \lim_{\mathrm{d}x\to 0} \frac{A(x+\mathrm{d}x) - A(x)}{\mathrm{d}x} &= f(x). \end{aligned}$$

And here, we see that the L.H.S. is just the derivative of $A(x)$ w.r.t. x, thus

$$\frac{\mathrm{d}A(x)}{\mathrm{d}x} = f(x), \tag{2.31}$$

or "*the rate of change in area is the function itself*". But the area function is given by the integral. This means

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\int f(x)\,\mathrm{d}x\right) = f(x): \tag{2.32}$$

integrals and derivatives are inverses of each other. If we rephrase theorem 2.3.1, we see that "*the integral is the cumulative effect of the function.*"

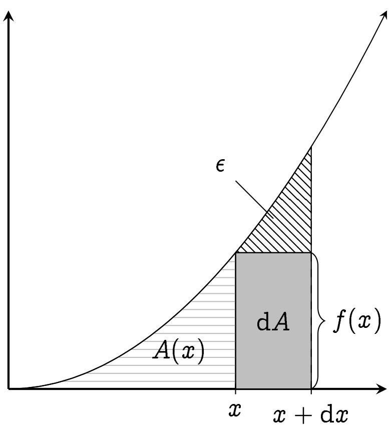


Figure 2.2 | THE GEOMETRICAL INTERPRETATION OF PART ONE OF THE FUNDAMENTAL THEOREM OF CALCULUS (THEOREM 2.3.1).

The second fundamental theorem of calculus,

Theorem 2.3.2: *The second fundamental theorem of calculus (Newton-Leibniz rule)*

If $A(x)$ is the antiderivative of $f(x)$, then the integral of $f(x)$ from $x = a$ to $x = b$ is

$$\int_a^b f(x)\,\mathrm{d}x = \int_0^b f(x)\,\mathrm{d}x - \int_0^a f(x)\,\mathrm{d}x = A(b) - A(a). \tag{2.33}$$

follows as a direct consequence of the geometrical interpretation of integrals: area under the curve.

The fundamental theorem of calculus allows us to evaluate integrals using derivatives. For example, if the derivative of x^2 is $2x$, then we also know that the integral of $2x\,\mathrm{d}x$ is just x^2, which we'll discuss how to do that in the next chapter.

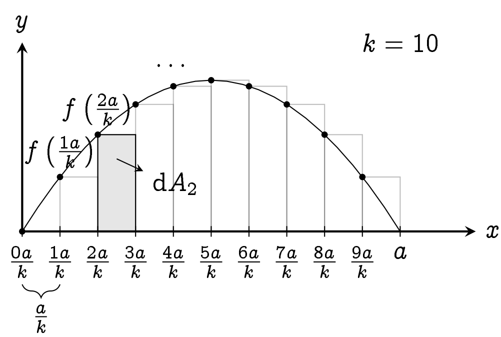


Figure 2.3 | Illustration of Riemann sum of a function $f(x)$ from 0 to a by setting $k = 10$

2.4 How to calculate an integral? Riemann sum

In section 2.2, we used geometry to find the area under a curve. However, that is not always possible, e.g., try integrating fig. 2.3 geometrically[2]. It'd be impossible. But we can still approximate its area by slicing the area under the curve into thin rectangular stripes, then summing them. The approximated area is called the **Riemann sum**. Due to its computational cost, you don't really want to use this method. However, to develop a good intuition at the integral, we should still know its symbolic form.

Let there be a function $A(s)$ that represents the actual area of a function $f(x)$ from 0 to s. The approximated area is then

$$\sum_{i=0}^{k} \mathrm{d}A_i = \sum_{i=0}^{k} \text{width} \times \text{height} \tag{2.34}$$

where k is the amount of subdivisions. From fig. 2.3, the width of each stripe is a / k, and the height of the i'th stripe is $f(ia / k)$. Therefore,

$$\sum_{i=0}^{k} \frac{a}{k} \times f\left(\frac{ia}{k}\right).$$

In fig. 2.3, we $k = 10$. The area of the second rectangle $\mathrm{d}A_2$ is

$$\mathrm{d}A_2 = \frac{a}{k} f\left(\frac{2a}{k}\right).$$

For an arbitrarily finite k, the Riemann sum is just an approximation. If you want to find the *actual* area under the curve, let $k \to \infty$. The limit as $k \to \infty$ is what we actually call the **integral**. Thus, we say

[2]This is what you'd get if you solve the simple harmonic oscillator

Definition 2.4.1: *Naive definition of integrals*

The (definite) integral, or the area under the curve of $f(x) = y$ from 0 to a, is defined as

$$\int_0^a \mathrm{d}A = \int_0^a f(x)\,\mathrm{d}x = \lim_{k\to\infty} \sum_{i=0}^{k} \frac{a}{k} \times f\left(\frac{ia}{k}\right) = A(a), \tag{2.35}$$

where $\displaystyle\int$ is the integral sign[a]. Here, 0 is the lower bound of integration, and a, the higher bound. The function $A(x)$ is called the **antiderivative**, or the **indefinite integral** of $f(x)$.

[a] Famously known for looking like a beansprout

I shall put these definitions into perspective in the next two examples. It might use a bit of series knowledge. If you don't know, you can simply search up the summation identities that I'll use in *Wikipedia* (11).

Example 2.4.2: *Riemann sum and antiderivative of x^2* Let $f(x) = x^2$, and let $A(a)$ be the antiderivative of $f(x)$, i.e., $A(a)$ is the area under the curve of $f(x)$ from 0 to a. Note that

$$\sum_{i=0}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}, \tag{2.36}$$

The Riemann sum is then

$$\begin{aligned}
\sum_{i=0}^{k} \frac{a}{k} \times f\left(\frac{ia}{k}\right) &= \sum_{i=0}^{k} \frac{a}{k} \times \left(\frac{ia}{k}\right)^2 && (2.37)\\
&= \sum_{i=0}^{k} \frac{a^3}{k^3} \times i^2 && (2.38)\\
&= \frac{a^3}{k^3} \times \frac{k(k+1)(2k+1)}{6}. && (2.39)
\end{aligned}$$

To find the antiderivative, let $k \to \infty$. Notice that the $+1$ in the parenthesis are negligible when $k \to \infty$. Therefore, we can write the antiderivative as

$$
\begin{aligned}
A(a) &= \int_0^a f(x)\,\mathrm{d}x && (2.40)\\
&= \lim_{k\to\infty} \frac{a^3}{k^3} \times \frac{k(k+1)(2k+1)}{6} && (2.41)\\
&= \lim_{k\to\infty} \frac{a^3}{k^3} \times \frac{2k^3}{6} && (2.42)\\
&= \lim_{k\to\infty} \frac{x^3}{3} = \frac{x^3}{3}. && (2.43)
\end{aligned}
$$

Example 2.4.3: *Riemann sum and antiderivative of x^3* Let $f(x) = 4x^3$, and let $A(a)$ be the antiderivative of $f(x)$, i.e., $A(a)$ is the area under the curve of $f(x)$ from 0 to a. Note that

$$
\sum_{i=0}^{k} i^3 = \left(\frac{k(k+1)}{2}\right)^2, \tag{2.44}
$$

The Riemann sum is then

$$
\begin{aligned}
\sum_{i=0}^{k} \frac{a}{k} \times f\left(\frac{ia}{k}\right) &= \sum_{i=0}^{k} \frac{a}{k} \times 4\left(\frac{ia}{k}\right)^3\\
&= 4\sum_{i=0}^{k} \left(\frac{x}{k}\right)^4 i^3\\
&= 4\left(\frac{x}{k}\right)^4 \left(\frac{k(k+1)}{2}\right)^2.
\end{aligned}
$$

To find the antiderivative, let $k \to \infty$. The $+1$ in $(k+1)$ can be ignored as $k \to \infty$. Therefore,

$$
\begin{aligned}
A(a) &= \int_0^a f(x)\,\mathrm{d}x\\
&= \lim_{k\to\infty} 4\left(\frac{x}{k}\right)^4 \left(\frac{k^2}{2}\right)^2\\
&= \lim_{k\to\infty} x^4 = x^4.
\end{aligned}
$$

2.5 Basic applications of the integrals

2.5.1 Five equations of linear motion

So far, we have developed an intuition for derivatives, integrals, and their relationship. Let's apply those concepts to derive the famous equations of linear motion:

Theorem 2.5.1: *Motion with uniformed acceleration*

$$v(t) = v_0 + at, \tag{2.45}$$
$$s = \frac{1}{2}(v_0 + v(t))t, \tag{2.46}$$
$$s = v_0(t)t + \frac{1}{2}at^2, \tag{2.47}$$
$$s = v(t)t - \frac{1}{2}at^2, \tag{2.48}$$
$$v(t)^2 = v_0^2 + 2as. \tag{2.49}$$

Here, v_0 represents the initial velocity, $v(t)$, the velocity at any time t, a, the acceleration, and s, the displacement. These equations are derived from the Newton's second law on *constant/uniformed acceleration* motion in one dimension, which we can evaluate using the geometrical interpretation of integrals developed earlier.

Because we've constrained the object to accelerate at a, the force exerted must be ma. Newton's second law $F = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2}$ then simplifies to $a = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}$. The same idea applies, we rewrite the equation in terms of v.

$$a = \frac{\mathrm{d}v}{\mathrm{d}t} \tag{2.50}$$

Which reads, *"the rate of change of v w.r.t. t is a"* or *"the slope of the v-t curve is always equals to a"*. Because a is constant, the v-t curve must be a straight line with slope a, shown in fig. 2.4.

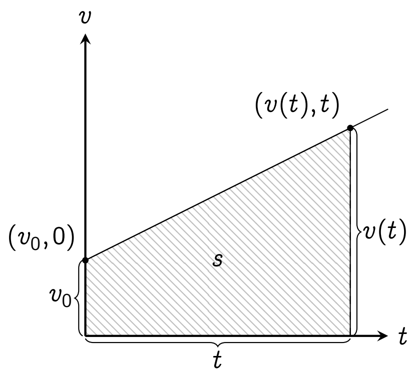


Figure 2.4 | v-t GRAPH OF AN OBJECT UNDER CONSTANT OR UNIFORMED ACCELERATION

Derivation of eq. (2.45). We take advantage of the linearness of the curve. Just pick two points on fig. 2.4, as already shown, then

$$
\begin{aligned}
m = a &= \frac{\Delta v}{\Delta t} = \frac{v(t) - v_0}{t - 0} \\
a &= \frac{v(t) - v_0}{t} \\
v(t) &= v_0 + at.
\end{aligned}
$$

□

Derivation of eq. (2.46). Use the reverse of theorem 2.3.1. We know that[3] $v = \frac{\mathrm{d}}{\mathrm{d}t} s$; therefore,

$$\int v \, \mathrm{d}t = s,$$

which reads "*The displacement s is the area under the curve of a v-t graph*". From fig. 2.4, the area under the curve is a trapezoid with side length v_0, $v(t)$, and width t. Thus,

$$s = \frac{1}{2}(v_0 + v(t))t,$$

which is just eq. (2.46): the area of a trapezoid. □

[3]Here, x is replaced with s to represent displacement in one dimension.

Derivation of eqs. (2.47) to (2.49). We can arrange eq. (2.45) into three different ways, then plug in eq. (2.46). First, $v(t) = v_0 + at$

$$\begin{aligned} s &= \frac{1}{2}(v_0 + v_0 + at)t \\ &= v_0 t + \frac{1}{2}at^2. \end{aligned}$$

Second, $v_0 = v(t) - at$

$$\begin{aligned} s &= \frac{1}{2}(v(t) - at + v(t))t \\ &= v(t)t - \frac{1}{2}at^2. \end{aligned}$$

Third, $t = \dfrac{v(t) - v_0}{a}$

$$\begin{aligned} s &= \frac{1}{2}(v(t) + v_0)\frac{(v(t) - v_0)}{a} \\ 2as &= (v(t) + v_0)(v(t) - v_0) \\ 2as &= v(t)^2 - v_0^2 \\ v(t)^2 &= 2as + v_0^2. \end{aligned}$$

□

2.5.2 The area of a circle

The perimeter of the circle is $2\pi r$. For now, we only know how to find areas of polygons, but we want to know the circle's area. So, is there any way to turn a circle into a polygon? From section 2.4, that the riemann sum can approximate areas under the curve using rectangular stripes. It'd be great if this circle can be turned into multiple rectangular stripes on a graph right?

So, we dissect a circle radially into small rings $\mathrm{d}r$ thin, as shown in fig. 2.5. Then, stretch all the rings into a very thin trapezoid-like shape. It might seem impossible at first, but considering that our ring is very thin, it's quite easy to stretch it without breaking. I encourage you to grab a piece of paper, cut a really thin ring and try it out. If you actually do it, it'll look a bit warped. However, the warpedness will go away the thinner you go.

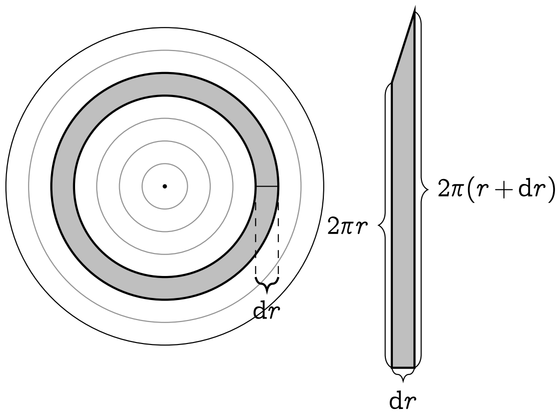


Figure 2.5 | (Left) Dissecting a circle radius R radially into rings, each ring $\mathrm{d}r$ thin. (Right) Stretching a ring into a trapezoid (Not to scale).

Since we sliced our trapezoid from a circle, for a stripe positioned at r, the inner side will be $2\pi r$ long and the outer side, $2\pi(r + dr)$. The area of the little trapezoid $\mathrm{d}A$ then becomes

$$\begin{aligned}\mathrm{d}A &= \frac{1}{2}\,\mathrm{d}r(2\pi r + 2\pi(r + dr)) \\ &= 2\pi r\,\mathrm{d}r + 2\pi\,\mathrm{d}r^2.\end{aligned}$$

As $\mathrm{d}r \to 0$, $\mathrm{d}r^2$ becomes negligible. Therefore,

$$\mathrm{d}A = 2\pi r\,\mathrm{d}r. \tag{2.51}$$

This equation says that for $\mathrm{d}r \to 0$, the trapezoid becomes a rectangle side length $\mathrm{d}r$ and $2\pi r$. Now, we have to sum it together. If we can put all these rectangles onto a graph, we can easily use the Riemann's sum to evaluate it. A natural way to do this is to put all the rectangles that we got from stretching the rings of the circle onto a graph one by one. The result would look something like fig. 2.6.

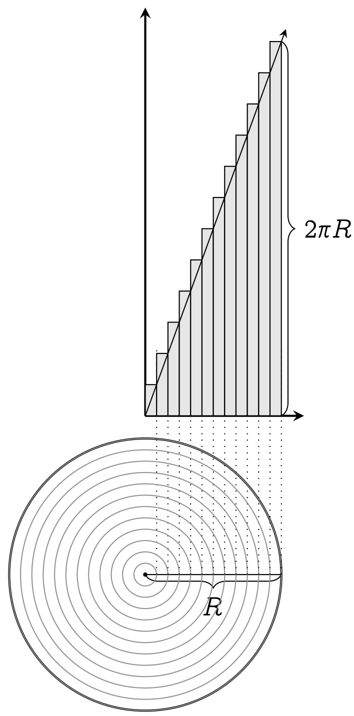


Figure 2.6 | REARRANGING ALL THE APPROXIMATED RECTANGLES ONTO A GRAPH (*NOT TO SCALE*)

For every stripe at r, its height is $2\pi r$. If you were to plot the height of all rectangles when $\mathrm{d}r \to 0$, it'll eventually look like a curve that's given by $f(r) = 2\pi r$. We're interested in the area of the circle from 0 to R. Now, it's transformed into the area under the curve of $f(r) = 2\pi r$: a triangle with base R and height $f(R) = 2\pi R$. Thus, the area of a circle becomes

$$A = \frac{1}{2}R(2\pi R) = \pi R^2.$$

And there you go, you've turned essentially turned circle into a triangle and evaluate its area from there. I'd like to end off this chapter by mentioning the spirit of mathematics. Sometimes, you can't solve the problem directly. Most of the time, you have to re-frame the problem into another more-solvable problem. Problems like these often have the most sublime connections to the foundations of mathematics. This is a common theme in most of mathematics, especailly calculus. So, be sure to keep this in mind while reading through.

2.6 Conclusion for Chapter 2

1. Riemann sum are used to approximate areas under the curve of a function by using little rectangles then summing it.

2. Integrals or anti-derivatives are functions that output the area under the graph of other functions.

3. The limit where the width of the rectangles in the Riemann sum approaches zero, the Riemann sum becomes an integral.

4. Integrals are the cumulative effect of a function.

5. Integrals and derivatives are inverses of each other, and they're related by the fundamental theorem of calculus

6. Integrals can be used in various ways by reframing questions into another simpler question.

CHAPTER 3

Basic derivatives and antiderivatives

This chapter covers the derivatives and integrals of common functions: polynomials, exponential, and logarithms; focusing on their geometrical interpretation. It prepares you for the real deal: differential equations. So, it's better if you can study these contents very thoroughly and get very familiar with doing derivatives and integrals. Most of these rules can be done mentally, and it's advised to practice these skills, so you will have an easier time solving differential equations.

Prerequisites: *binomial theorem, basic trigonometry, derivatives, and integrals*

Terminologies: The integral refers to the area under the curve. The antiderivative refers to the function $A(x')$ that outputs the area under the curve from 0 to x'.

3.1 Basic rules of differentiation and integration

The chain rule: The Leibniz' notation treats derivative as fractions. In a sense, we "can" cancel terms[1] (eq. (1.11)). We exploit this property to find derivatives of composite functions. Consider two functions $f(x)$ and $g(x)$, the derivative of $f(g(x))$ can be found by a simple substitution. First let $g(x) = u$.

$$\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = \frac{\mathrm{d}f(u)}{\mathrm{d}x}.$$

Let $1 = \frac{\mathrm{d}u}{\mathrm{d}u}$, then

$$\frac{\mathrm{d}}{\mathrm{d}x} f(u) = \frac{\mathrm{d}}{\mathrm{d}x} f(u) \times 1 = \frac{\mathrm{d}f(u)}{\mathrm{d}x} \frac{\mathrm{d}u}{\mathrm{d}u}.$$

Perform a change of denominator, then substitute back $u = g(x)$:

$$\frac{\mathrm{d}}{\mathrm{d}x} f(g(x)) = \frac{\mathrm{d}f(u)}{\mathrm{d}u} \frac{\mathrm{d}g(x)}{\mathrm{d}x}.$$

This is what we call the chain rule, or more generally

Proposition 3.1.1: *The chain rule* If y is a function of x, and u is a function of x, then $\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x}$. [a]

[a] This only works when y, u, and x follows a set of conditions that we'll discuss further in analysis.

The chain rule holds the intuition of how rate of changes relate to each other. E.g., the cheetah's speed is 10 times the bicycle's speed, which is 4 times the walking speed. The ratio between the cheetah's speed compared to walking speed would obviously be $10 \cdot 4 = 40$:

$$\frac{\mathrm{dCheetah}}{\mathrm{dWalking}} = \frac{\mathrm{dCheetah}}{\mathrm{dBicycle}} \times \frac{\mathrm{dBicycle}}{\mathrm{dWalking}}. \tag{3.1}$$

[1] The actual proof of the chain rule is very nuanced. Keep it in the back of your mind that Leibniz' notation is just a syntactic sugar, and not an actual proof for the chain rule.

Integral constant In section 2.2, each time we evaluate the antiderivative, we add an initial condition term, i.e., v_0 and r_0. So for any function $f(x)$,

$$\int f(x)\,\mathrm{d}x = A(x) + C, \tag{3.2}$$

where C is any constant. However, theorem 2.3.2 still holds for integrals with bounds.

Always add an integral constant after evaluating the antiderivative

Integral of the infinitesimal The antiderivative of the small rectangles $\mathrm{d}x$ is just the total rectangle x plus the integral constant C.

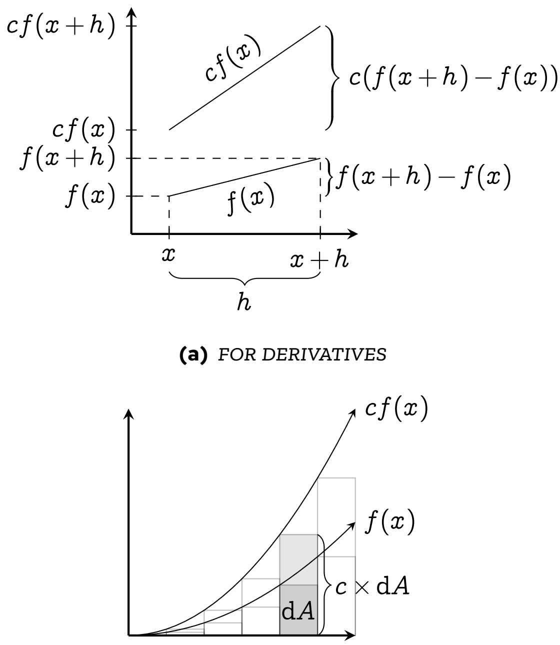
$$\int \mathrm{d}x = x + C. \tag{3.3}$$

Rules of equality If two arguments are equal, their derivatives and antiderivatives w.r.t. the same variable must also be equal.

$$\text{If } f = g\text{, then } \frac{\mathrm{d}f}{\mathrm{d}x} = \frac{\mathrm{d}g}{\mathrm{d}x}\text{, and } \int f\,\mathrm{d}x = \int g\,\mathrm{d}x + C$$

Derivative of a constant A constant doesn't change; thus, the derivative of a constant is zero.

$$\frac{\mathrm{d}}{\mathrm{d}x}(c) = 0. \tag{3.4}$$



(b) *FOR INTEGRALS*

Figure 3.1 | CONSTANT MULTIPLE RULES

3.2 Linearity of differentiation and integration

Definition 3.2.1: *Linear-ness*

A **linear** mathematical entity are defined as anything that is compatible with addition and scaling. E.g., for a function $f(x)$ to be linear, it must obey

1. Additivity: $f(x+y) = f(x) + f(y)$
2. Homogeneity of degree one: $f(ax) = af(x)$ for all constant a

The simplest linear function is a line that passes through the origin: $f(x) = ax$, in which

$$f(x+y) = a(x+y) = ax + ay = f(x) + f(y), \tag{3.5}$$

$$f(cx) = a(cx) = c(ax) = cf(x). \tag{3.6}$$

Because the line has this property, it is linear. The property of linearity however is not limited to lines - other mathematical objects can also posess this property.

Both derivatives and integrals are linear; for any function $f(x)$, $g(x)$, and constants a, b, the following rules follow.

Proposition 3.2.2: *The constant multiple rule*

$$\frac{\mathrm{d}}{\mathrm{d}x}(af(x)) = a\frac{\mathrm{d}f(x)}{\mathrm{d}x}, \qquad \text{Illustrated in fig. 3.1(a)} \tag{3.7}$$

$$\int af(x)\,\mathrm{d}x = a\int f(x)\,\mathrm{d}x. \qquad \text{Illustrated in fig. 3.1(b)} \tag{3.8}$$

The geometrical interpretation of these two rules are very simple. When a function is multiplied by a constant c, its value is increased by a factor of c everywhere. The slope must also be increased by c, and thus the function's area also.

Proposition 3.2.3: *The sum rule*

$$\frac{\mathrm{d}}{\mathrm{d}x}(f(x)+g(x)) = \frac{\mathrm{d}}{\mathrm{d}x}f(x) + \frac{\mathrm{d}}{\mathrm{d}x}g(x), \tag{3.9}$$

$$\int f(x)+g(x)\,\mathrm{d}x = \int f(x)\,\mathrm{d}x + \int g(x)\,\mathrm{d}x. \tag{3.10}$$

I.e., adding two functions increases its slope, thus also increasing its area under the graph.

3.3 Derivatives and antiderivatives of polynomials

Now that we've discussed the "trivial rules", we're ready to tackle the easiest family of functions: the **polynomials**. They're in the form

$$f(x) = a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \dots . \tag{3.11}$$

Let's focus on the antiderivative first; we'll see later how the antiderivative of polynomials can be found with just a simple substitution trick.

The form mentioned in eq. (3.11) is quite useless if we want to make progress. We can break it down by using the linearity of derivatives:

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}(a_0 + a_1 x^1 + a_2 x^2 + a_3 x^3 + \dots) \tag{3.12}$$

$$= a_1 \frac{\mathrm{d}x^1}{\mathrm{d}x} + a_2 \frac{\mathrm{d}x^2}{\mathrm{d}x} + a_3 \frac{\mathrm{d}x^3}{\mathrm{d}x} + \dots . \tag{3.13}$$

Now we're left with derivatives of *monomials* in the form of x^n. So let's do that instead.

3.3.1 Derivatives of monomials: the power rule

The method of increments allows us to quickly evaluate the derivative of x^n.

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^n) = \lim_{h \to 0} \frac{(x+h)^n - x^n}{h}. \tag{3.14}$$

By using the binomial expansion,

$$= \lim_{h \to 0} \frac{\left(\sum_{k=0}^{n} \binom{n}{k} x^{n-k} \cdot h^n\right) - x^n}{h} \tag{3.15}$$

$$= \lim_{h \to 0} \frac{\binom{n}{0} x^n h^0 + \binom{n}{1} x^{n-1} h^1 + \binom{n}{2} x^{n-2} h^2 + \dots \binom{n}{n} + x^0 h^n - x^n}{h} \tag{3.16}$$

$$= \lim_{h \to 0} \frac{x^n + n x^{n-1} h^1 + \binom{n}{2} x^{n-2} h^2 + \dots + h^n - x^n}{h} \tag{3.17}$$

$$= \lim_{h \to 0} n x^{n-1} + \binom{n}{2} x^{n-2} h + \binom{n}{3} x^{n-3} h^2 + h^{n-1} \tag{3.18}$$

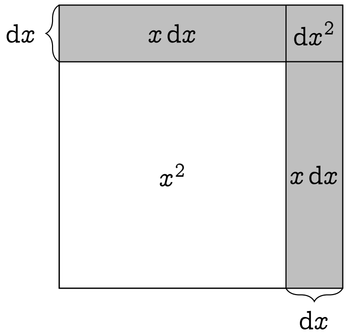


Figure 3.2 | INTERPRETATION OF $\frac{dx^2}{dx}$ (dx EXAGGERATED)

The terms with h vanishes when $h \to 0$; therefore,

$$\frac{d}{dx}(x^n) = nx^{n-1}. \tag{3.19}$$

The equation above is what we call the **power rule.** Because the binomial theorem work for all real numbers, this is true for any powers of n.

Now, what is the geometrical interpretation of the power rule? Let's try to answer this question by geometrically interpreting the simplest non-trivial monomial: x^2. This function represents the area of a square sidelength x. Its derivative then represents the ratio between the change in x, and the change of area when the sidelength is increased by dx.

Illustrated in fig. 3.2,

$$\frac{d}{dx}(x^2) = \lim_{dx \to 0} \frac{x\,dx + x\,dx + dx^2}{dx}. \tag{3.20}$$

When $dx \to 0$, the dx^2 square would just become a single point. Compared to the big $x\,dx$ on the side, it's negligible; therefore, we can safely ignore it. The equation then becomes

$$\frac{d}{dx}(x^2) = \lim_{dx \to 0} \frac{2x\,dx}{dx} = 2x,$$

which is equivalent to the result from the power rule.

Deriving $\frac{\mathrm{d}}{\mathrm{d}x}(x^3)$ geometrically shouldn't be too hard either. Take a cube sidelength x; increase its side length by $\mathrm{d}x$, and compute the ratio between the change in volume and $\mathrm{d}x$. The final answer should be $3x^2$. You could try with more higher order of n, but it'd be very hard to visualize.

Here I leave some exercises which shouldn't be too hard to do. The fourth and fifth one requires you to convert the fractions into negative exponents such as $\frac{1}{x} = x^{-1}$, then use the power rule. The sixth one requires you to convert square roots into fractional exponents. And, the final two would need you to expand the polynomial first, then use the power rule.

1. $\frac{\mathrm{d}}{\mathrm{d}x}(x^2 - 2x + 16)$ $\qquad 2x - 2$
2. $\frac{\mathrm{d}}{\mathrm{d}x}(x^3 + x^2 + x + 1)$ $\qquad 3x^2 + 2x + 1$
3. $\frac{\mathrm{d}}{\mathrm{d}x}(3x^4 + 24x^3 - 2x^2 - 32x + 88)$ $\qquad 12x^3 + 72x^2 - 4x - 32$
4. $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{x^2} + \frac{1}{x} + x\right)$ $\qquad -2x^{-3} - 1x^{-2} + 1$
5. $\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{3}{x} + x^2 + \frac{1}{4}x^{-4}\right)$ $\qquad -3x^{-2} + 2x - x^{-5}$
6. $\frac{\mathrm{d}}{\mathrm{d}x}\left(\sqrt{x} + \sqrt[3]{x}\right)$ $\qquad \frac{1}{2}x^{-1/2} + \frac{1}{3}x^{-2/3}$
7. $\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2 + 3x)^2\right)$ $\qquad 4x^3 + 12x^2 + 8x$
8. $\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2 + 1)^2(x - 2)\right)$ $\qquad 5x^4 - 8x^3 + 6x^2 - 8x + 1$

3.3.2 Antiderivatives of polynomials: the reversed power rule

For the antiderivative of x^n, we use the fundamental theorem of calculus (theorem 2.3.1) on the power rule (eq. (3.19)),

$$x^n = \int nx^{n-1}\,\mathrm{d}x.$$

Substitute n with $n+1$

$$\int (n+1)x^n\,\mathrm{d}x = x^{n+1} + C,$$

and by the linearity of integrations (eq. (3.8)), we get the **reversed power rule**:

$$\int x^n\,\mathrm{d}x = \frac{x^{n+1}}{n+1} + C. \tag{3.21}$$

Notice, this rule does not work for $n = 1$ because we can't divide by zero...*or can we???* (Discussed further in sections 3.5 and 3.7)

In just the same way, it's very important to familiarize yourself to these integrals. Therefore, here I leave some examples for practice. For the forth and fifth one, you also need to turn the fractions into a negative exponent. And the last two, expand the polynomial first.

1. $\int x^2 + 1\,\mathrm{d}x$ $\qquad \frac{x^3}{3} + x + C$
2. $\int 2x^3 + x^2 + 3\,\mathrm{d}x$ $\qquad \frac{x^4}{2} + \frac{x^3}{3} + 3x + C$
3. $\int 3x^2 - 5x + 2\,\mathrm{d}x$ $\qquad x^3 - \frac{5x^2}{2} + 2x + C$
4. $\int \sqrt{x} + \frac{1}{3}\sqrt[3]{x}\,\mathrm{d}x$ $\qquad \frac{2x^{3/2}}{3} + \frac{x^{4/3}}{4}$
5. $\int 3x^{-5/2} + \frac{x^{7/3}}{3}$ $\qquad \frac{x^{10/3}}{10} - 2x^{-3/2}$
6. $\int (x^2 + 3x + 1)^2\,\mathrm{d}x$ $\qquad \frac{x^5}{5} + \frac{3x^4}{2} + \frac{11x^3}{3} + 3x^2 + x$

7. $\displaystyle\int (x+1)(x-1)(x+2)\,\mathrm{d}x$ $\qquad \dfrac{x^4}{4} + \dfrac{2x^3}{3} - \dfrac{x^2}{2} - 2x$

3.3.3 Extending the equations of linear motion

Let's use the power rule to extend the equation of linear motions in section 2.5.1 to include a constant jerk j, i.e.,

$$\frac{\mathrm{d}a}{\mathrm{d}t} = j \tag{3.22}$$

Then, we can move the $\mathrm{d}t$ around and integrate both sides by using the reversed power rule.

$$\begin{aligned}
\int j\,\mathrm{d}t &= \int \mathrm{d}a \\
j \int \mathrm{d}t &= \int \mathrm{d}a && \text{Linearity of integrals} \\
jt + a_0 &= a && \text{Integral constant } a_0
\end{aligned}$$

Because a is the derivative of v,

$$\begin{aligned}
\frac{\mathrm{d}v}{\mathrm{d}t} &= jt + a_0 \\
\int \mathrm{d}v &= \int jt + a_0\,\mathrm{d}t \\
v &= j\int t\,\mathrm{d}t + \int a_0\,\mathrm{d}t && \text{Linearity of integrals} \\
v &= \frac{1}{2}jt^2 + a_0 t + v_0. && \text{Reversed power rule}
\end{aligned}$$

We add v_0 as an integral constant in a similar manner. Since v is the derivative of r,

$$\begin{aligned}
\frac{\mathrm{d}r}{\mathrm{d}t} &= \frac{1}{2}jt^2 + a_0 t + v_0 \\
\int \mathrm{d}r &= \int \frac{1}{2}jt^2 + a_0 t + v_0\,\mathrm{d}t \\
r &= \frac{1}{2}j\int t^2\,\mathrm{d}t + a_0\int t\,\mathrm{d}t + v_0\int \mathrm{d}t
\end{aligned}$$

$$r = \frac{1}{6}jt^3 + \frac{1}{2}a_0t^2 + v_0t + r_0.$$

And there you have it! Further extensions can be made from here: the jerk could be time-dependent. But that's probably enough to illustrate my point. If you'd like to try, extend the equation of motion to have a constant snap: $\frac{\mathrm{d}j}{\mathrm{d}t} = s$. You should get

$$r = \frac{1}{24}st^4 + \frac{1}{6}jt^3 + \frac{1}{2}a_0t^2 + v_0t + r_0.$$

3.4 Exponentials and growth

The next function that we're going to discuss is exponentials: the mathematical representation of growth and decay. As an example, let's say there's a magical drop of water that doubles its volume V every hour. I.e., for any time t,

$$V(t+1) = 2V(t). \tag{3.23}$$

If the drop starts at one unit of volume, $V(0) = 1$; thus,

$$\begin{aligned}
V(1) &= 2V(0) = 2, \\
V(2) &= 2V(1) = 2(2) = 2^2, \\
V(3) &= 2V(2) = 2(2^2) = 2^3, \\
V(4) &= 2V(3) = 2(2^3) = 2^4, \\
&\vdots
\end{aligned}$$

It's clear that the pattern is $V(t) = 2^t$: an exponential function. A natural question to ask is "*what is its rate of change?*". Let's start by plotting the function over time (fig. 3.3). As you can see, exponentials grow too quick to plot. By $V(7)$, we're already in the hundreds. So, it'd probably be better to list the values of each point on a table. Notice that on the right most

t	$V(t)$	$V(t) - V(t-1)$
0	1	
1	2	$2 - 1 = 1$
2	4	$4 - 2 = 2$
3	8	$8 - 4 = 4$
4	16	$16 - 8 = 8$
5	32	$32 - 16 = 16$
6	64	$64 - 32 = 32$
7	128	$128 - 64 = 64$
8	256	$256 - 128 = 128$

Table 3.1 | $V(t) = 2^t$ FROM 0 TO 8

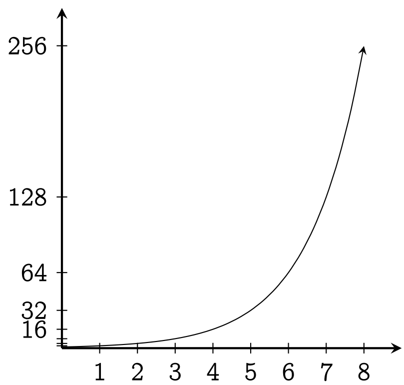


Figure 3.3 | EXPONENTIAL FUNCTION 2^x PLOTTED FROM 0 TO 8

column of table 3.1, the difference between $V(t)$ and $V(t-1)$ is exactly $V(t-1)$: the function changes as much as its past-self. So does that mean that $\frac{\mathrm{d}}{\mathrm{d}x}(2^x) = 2^x$?

Sadly not, but it's very close. See, table 3.1 only shows a *discrete* step where $\mathrm{d}x = 1$. You can write it out as

$$\frac{2^{x+1} - 2^x}{1} = 2^x \left(\frac{2-1}{1}\right) = 2^x, \tag{3.24}$$

that is why $V(t) - V(t-1) = V(t-1)$. But to evaluate the derivative of 2^x, this increment can't be one. It must be infinitesimally small. Hence, we recall the method of increments:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(2^x) &= \lim_{h\to 0}\frac{2^{x+h}-2^x}{h}\\ &= 2^x \lim_{h\to 0}\left(\frac{2^h-1}{h}\right).\end{aligned}$$

As of now, the theories of limit isn't mature enough to evaluate this. So, we resort to manually plugging in a very smaller and smaller value of h, and see where this limit goes. This term seems to not blow off to infinity, but rather stabilizes at around 0.69314 … . So,

$$\frac{\mathrm{d}}{\mathrm{d}x}(2^x) = (0.693\ldots)2^x. \tag{3.25}$$

Generally, this equation can be extended to any other bases of exponents.

$$\frac{\mathrm{d}}{\mathrm{d}x}(a^x) = \lim_{h\to 0}\frac{a^{x+h}-a^x}{h} \tag{3.26}$$

$$= a^x\left(\lim_{h\to 0}\frac{a^h-1}{h}\right) \tag{3.27}$$

I.e., *the rate of change of an exponential is itself times some constant*. For an exponential of base a, this constant is

$$\lim_{h\to 0}\frac{a^h-1}{h}. \tag{3.28}$$

In the future, we shall develop a better way to calculate this constant. For now, it's just plugging in small values of h and plotting it as shown in fig. 3.4.

Notice that between $a = 2.5$ and $a = 3$, the graph of this constant passes through one. If we can find this function, then we'd have a function that is its own derivative! So let's find this value.

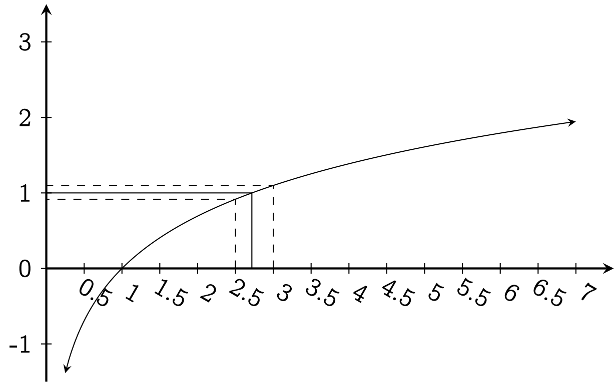


Figure 3.4 | GRAPH OF THE LEADING CONSTANT OF THE DERIVATIVE OF a

3.4.1 A function that is its own derivative: Euler's number

Let's set a goal: find the function that is its own derivative. If that function exists, it must satisfy the differential equation

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = f(x). \tag{3.29}$$

We've already seen that one possible candidate for this function is the exponential function n^x. But how do we find the number n that satisfies it?

Here is a good time where I introduce a method that's used very often to solve differential equations: **power series expansion**. Most functions have a power series expansion, written as[2]

$$f(x) = a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots . \tag{3.30}$$

E.g., $\sin(x)$ can be written as[3]

$$\sin(x) = x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots . \tag{3.31}$$

[2]Although the convergence of the series derived is quite questionable; thankfully, the power series of the exponential function converges everywhere.

[3]If you don't believe me, wait for section 5.2.1.

We can assume that the exponential function can be expanded using a power series; thus,

$$n^x \equiv a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots \,. \tag{3.32}$$

To find out each of the leading coefficient, just substitute the expansion into eq. (3.29) and see where mathematics would lead us.

A direct substitution followed by the power rule would give

$$\frac{\mathrm{d}}{\mathrm{d}x}(a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots) = a_1 + 2a_2x^1 + 3a_3x^2 + 4a_4x^3 + \dots \,. \tag{3.33}$$

By the eq. (3.29), the R.H.S must also equals n^x; therefore,

$$\begin{aligned} n^x &= a_0 + a_1x^1 + a_2x^2 + a_3x^3 + \dots \text{ and also,} \\ n^x &= a_1 + 2a_2x^1 + 3a_3x^2 + 4a_4x^3 + \dots \,. \end{aligned} \tag{3.34}$$

Since they're equal, their leading coefficient must match, and thus forming a system of equations

$$\begin{aligned} a_0 &= a_1 \\ a_1 &= 2a_2 \\ a_2 &= 3a_3 \\ a_3 &= 4a_4 \\ &\vdots \end{aligned} \tag{3.35}$$

Because n^x is an exponential, n^0 must equal one. Therefore, $x = 0$

$$\begin{aligned} 1 &= a_0 + a_1(0)^1 + a_2(0)^2 + a_3(0)^3 + \dots \\ a_0 &= 1; \end{aligned}$$

thus, by eq. (3.35), a_1 must also equal 1. Substituting $a_1 = 1$ into eq. (3.35) gives

$$a_2 = \frac{1}{2},\ a_3 = \frac{1}{2 \times 3},\ a_4 = \frac{1}{2 \times 3 \times 4}, \dots ; \tag{3.36}$$

therefore,

$$n^x = 1 + 1 + \frac{1}{2!}x + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \ldots$$

Clearly, $a_m = \dfrac{1}{m!}$. If we want to find n, just let $x = 1$.

$$n = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots = \sum_{i=0}^{\infty} \frac{1}{i!}$$

and there it is: the expression for n, which evaluates to $2.71828\ldots$.[4]

The property that the exponentiation of this number is its own derivative makes it a very important tool in mathematics. Even though it's intrinsically related to growths, it somehow manages to seek its way to places that are not even remotely related to growths: oscillations, the patterns of prime numbers, wave equations, etc. So don't be surprised if you'd see it on unrelated topics later on. Because of this wide ranging applications, this constant: $2.71828\ldots$, have a name. It's the **Euler's number**, denoted e, and[5]

$$\mathrm{e} = 1 + 1 + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \cdots = \sum_{i=0}^{\infty} \frac{1}{i!} \tag{3.37}$$

3.4.2 Another interpretation of the Euler's number: continuous bank interests

There are two types of bank interests: simple and compound. Simple interests is the thing that you don't really want: the interest is always the same and doesn't grow with your account. You can calculate it by using

$$n(t) = n_0 + tr \tag{3.38}$$

[4] Using the continuation that $0! = 1$.

[5] Not to be confused with the "Euler's constant" which is another constant written γ, and is around $0.57721\ldots$

where $n(t)$ is the total money at time t, n_0 the initial money in your bank account, and r, the interest rate.

Compound interest in the other hand calculates your interest based on how much money you have at that moment:

$$n(t+1) = n(t)r + n(t). \tag{3.39}$$

We can find the expression for $n(t)$ in a similar fashion to what we've done in eq. (3.23). You'll get

$$n(t) = (1+r)^t n_0 \tag{3.40}$$

which is an exponential function.

Let's say you deposit 100\$ into a bank and the bank is offering you two options on **compound interests** rate. 1) Take 100% interest in 1 year, 2) Take 100 / 2% twice a year, or 3) Take 100 / 356% daily. If you take option one, you'd end up with 200\$. Option two takes you to 225\$, and option three takes you to around 271.447 ... \$. You might see a theme here. If you get 100 / n% interest, n times a year, the result keeps getting higher. Is there an upper limit to this?

If we write it in terms of limits as $n \to \infty$ and use eq. (3.40), the compound interests formula,

$$x = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n n_0.$$

Where x is the total money after a year. We're interested in the upper limit, so we'll just let n_0 for now. The expression will become

$$x = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \tag{3.41}$$

Technically, we could go in and substitute a very high n, such as 1000000. But I believe you could already see that it would be a nightmare to calculate: exponentiation is not at all an easy task. However, notice that from option three earlier, the total money is 271.447 ... \$ which is suspiciously similar to e at 2.71828 ... . If eq. (3.41) equals eq. (3.35), we'd find the upper limit for this problem and solve the mystery.

We can use the binomial theorem on eq. (3.41) and get

$$\begin{aligned}
&\lim_{n\to\infty}\sum_{k=0}^{n}\binom{n}{k}1^{n-k}\frac{1}{n^k}\\
&=\lim_{n\to\infty}\binom{n}{0}\left[\frac{1}{n^0}+\binom{n}{1}\frac{1}{n^1}+\binom{n}{2}\frac{1}{n^2}+\binom{n}{3}\frac{1}{n^3}+\dots\right].
\end{aligned}$$

Substituting the definition of n choose k,

$$=1+\lim_{n\to\infty}\left[\frac{n!}{1!(n-1)!}\frac{1}{n^1}+\frac{n!}{2!(n-2)!}\frac{1}{n^2}+\frac{n!}{3!(n-3)!}\frac{1}{n^3}+\dots\right].$$

Now, we can cancel the $n!$ on the numerator to the denominator and isolate the factorials.

$$\begin{aligned}
&=1+\lim_{n\to\infty}\left[\frac{n(n-1)!}{(n-1)!}\frac{1}{1!n^1}+\frac{n(n-1)(n-2)!}{(n-2)!}\frac{1}{2!n^2}+\dots\right]\\
&=1+\frac{1}{1!}+\lim_{n\to\infty}\left[\frac{n(n-1)}{2!n^2}\right.\\
&\qquad\left.+\frac{n(n-1)(n-2)}{3!n^3}+\frac{n(n-1)(n-2)(n-3)}{4!n^4}+\dots\right]
\end{aligned}$$

Notice, as $n\to\infty$, the ratio between $n+R$ and n where R is any real numbers would be literally negligible. For every terms in our series, both the numerator and the denominator has the same polynomic degrees. Therefore, all the n's in the series cancel out and we get

$$x=1+\frac{1}{1!}+\frac{1}{2!}+\frac{1}{3!}+\dots\,. \tag{3.42}$$

This infinite sum is literally the Euler number (eq. (3.35)); thus,

$$\mathrm{e}=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n. \tag{3.43}$$

In conclusion, the upper limit that the bank can give you is e, which should make sense geometrically. Because we're gradually turn-ing a discrete interest into a continuous one, e should appear in the limit of the continuous bank interests.

The limit definition of e allows us to express the exponential function e^x in terms of limits.

$$\mathrm{e}^x = \left(\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n\right)^x \tag{3.44}$$

$$= \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^{nx} \tag{3.45}$$

If $n = u \,/\, x$, then

$$\mathrm{e}^x = \lim_{\frac{u}{x}\to\infty}\left(1+\frac{x}{u}\right)^u \tag{3.46}$$

For $\frac{u}{x} \to \infty$ to be true for all finite x, u must approach ∞; therefore,

$$\mathrm{e}^x = \lim_{u\to\infty}\left(1+\frac{x}{u}\right)^u \tag{3.47}$$

3.4.3 The antiderivative of exponential functions

It should be trivial that if e^x is the derivative of itself, so is its antiderivative

$$\int \mathrm{e}^x \,\mathrm{d}x = \mathrm{e}^x + C. \tag{3.48}$$

For other bases, we could use the fundamental theorem of calculus (theorem 2.3.1) to find its antiderivative. I.e., if

$$\frac{\mathrm{d}(n^x)}{\mathrm{d}x} = n^x \lim_{h\to 0}\frac{n^h-1}{h},$$

then

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}(n^x) &= n^x \lim_{h\to 0}\frac{n^h-1}{h} \\
n^x &= \lim_{h\to 0}\frac{n^h-1}{h}\int n^x \,\mathrm{d}x \\
\int n^x \,\mathrm{d}x &= n^x \lim_{h\to 0}\left(\frac{n^h-1}{h}\right)^{-1} + C.
\end{aligned}$$

The term in the limit sign still appears here. If we want to uncover the origin of this term, we must discuss the logarithms.

3.5 Logarithms

Logarithms are inverses of exponential, like how roots are inverses of monomials. To illustrate what I mean,

$$\text{For monomials, } \sqrt[a]{x^a} = x, \text{ but with logarithms, } \log_a(a^x) = x.$$

They have the following properties:

$$\log_a(x) + \log_a(y) = \log_a(xy), \tag{3.49}$$

$$\log_a(x) - \log_a(y) = \log_a\left(\frac{x}{y}\right), \tag{3.50}$$

$$\log_a(x) = \frac{\log_b(x)}{\log_b(a)}. \tag{3.51}$$

Since e^x is shown to be a very important function in modelling continuous growth, and is its own derivative, we give its inverse function its own name: the **natural logarithm**, written as $\ln(x)$.

What's the derivative of $\ln(x)$? By the method of increments,

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(x) = \lim_{h\to 0}\frac{\ln(x+h)-\ln(x)}{h} \tag{3.52}$$

$$= \lim_{h\to 0}\frac{\ln(\frac{x+h}{x})}{h} \tag{3.53}$$

$$= \lim_{h\to 0}\frac{\ln(1+\frac{h}{x})}{h} \tag{3.54}$$

$$= \lim_{h\to 0}\ln\left(\left(1+\frac{h}{x}\right)^{\frac{1}{h}}\right). \tag{3.55}$$

This form is very similar to the limit definition of the exponential function e^x, but with some terms swapped around. So let's try to fit this limit to the form of e^x. I shall let $h = \frac{1}{n}$. If $h \to 0$, then $n \to \infty$. Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(x) = \lim_{n\to\infty} \ln\left(\left(1+\frac{1}{nx}\right)^n\right) \tag{3.56}$$

$$= \lim_{n\to\infty} \ln\left(\left(1+\frac{1/x}{n}\right)^n\right). \tag{3.57}$$

Recall that

$$\mathrm{e}^x = \lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n. \tag{3.58}$$

Thus,

$$\lim_{n\to\infty}\left(1+\frac{1/x}{n}\right)^n = \mathrm{e}^{\frac{1}{x}}. \tag{3.59}$$

which fits the form of eq. (3.57). Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}\ln(x) = \lim_{n\to\infty}\ln(\mathrm{e}^{\frac{1}{x}}) \tag{3.60}$$

$$= \lim_{n\to\infty}\frac{1}{x} = \frac{1}{x}: \tag{3.61}$$

the derivative of the natural logarithm of x is the reciprocal of x.

This result can also be interpretted geometrically. If the exponential grows very quickly when x increases, its inverse must grow slow. The slow growing rate is directly captured by the function $\frac{1}{x}$.

Here is the perfect time that I'll give you a glimpse into calculus with many variables: multivariable calculus. I shall present another way to derive the derivative of $\ln(x)$: **implicit differentiation**.

Functions in the form $f(x) = y$ are called **explicit functions**: the relations between the two variables are written explicitly. Functions that can't be written as $f(x) = y$ are called **implicit functions**, e.g., $x^2+y^2 = r^2$. **Implicit differentiation** concerns the differentiation of implicit functions. In this case, it actually makes our life easier if we turn $\ln(x)$ into an implicit function.

If I let $\ln(x) = y$, I can raise both sides to the power of e and get

$$\mathrm{e}^{\ln(x)} = \mathrm{e}^{y}. \tag{3.62}$$

Because logarithms are inverses of exponential,

$$x = \mathrm{e}^{y}. \tag{3.63}$$

By the rule of equality, (section 3.1), we can take the derivative of both sides w.r.t. y instead of x:

$$\begin{aligned}\frac{\mathrm{d}x}{\mathrm{d}y} &= \frac{\mathrm{d}\mathrm{e}^{y}}{\mathrm{d}y} \\ \frac{\mathrm{d}x}{\mathrm{d}y} &= \mathrm{e}^{y}.\end{aligned}$$

But we're looking for the derivative of y (which is just $\ln(x)$) w.r.t. x, not the derivative of x w.r.t. y. Here's where Leibniz's notation comes into clutch: we can swap the numerator with the denominator for both sides then, substitute back $y = \ln(x)$:

$$\begin{aligned}\frac{\mathrm{d}\ln(x)}{\mathrm{d}x} &= \frac{1}{\mathrm{e}^{\ln(x)}} \\ \therefore \frac{\mathrm{d}\ln(x)}{\mathrm{d}x} &= \frac{1}{x},\end{aligned}$$

and there: the derivative of the natural logarithm is the reciprocal.

With the power of natural logarithms, we can actually go back at the derivative of n^x and finally uncover the mystery behind the proportionality term that's lingering around. Start with the manipulation of n^x.

$$n^x = \left(\mathrm{e}^{\ln(n)}\right)^x = \mathrm{e}^{x\ln(x)}. \tag{3.64}$$

With the chain rule (section 3.1), let $u = x\ln(n)$:

$$\frac{\mathrm{d}}{\mathrm{d}x}(n^x) = \frac{\mathrm{d}\mathrm{e}^{x\ln(n)}}{\mathrm{d}x}$$

$$
\begin{aligned}
&= \frac{\mathrm{d}e^u}{\mathrm{d}x} \times \frac{\mathrm{d}u}{\mathrm{d}u} \\
&= \frac{\mathrm{d}e^u}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \\
&= e^{x\ln(n)} \times \frac{\mathrm{d}x\ln(n)}{\mathrm{d}x} \\
&= n^x \ln(n).
\end{aligned}
$$

And here it is: the mystery proportionality constant. It's just a consequence of the natural logarithm. Thus, one way to define the natural log would be

$$\ln(n) = \lim_{h\to 0} \frac{n^h - 1}{h}. \tag{3.65}$$

The antiderivative of other bases exponents are then given by

$$\int n^x = \frac{1}{\ln(n)} n^x + C. \tag{3.66}$$

3.5.1 The product rule and the quotient rule

Sometimes, we have to multiply the two functions together before taking the derivatives. There are two ways to do this. To keep the spirit of visualization, I shall first introduce the geometrical way, then the analytical way.

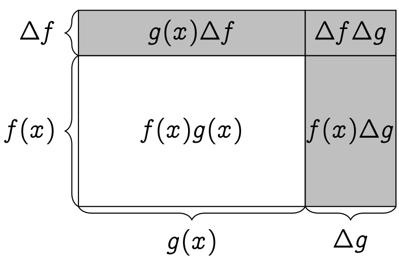


Figure 3.5 | THE GEOMETRICAL INTERPRETATION OF THE PRODUCT RULE.

The derivative of $f(x)g(x)$ w.r.t. x can be thought of a rectangle with side length that's governed by $f(x)$ and $g(x)$. As shown in fig. 3.5, the area increase on side $f(x)$ is $g(x)\,\Delta f$ and on $g(x)$, $f(x)\,\Delta g$. The $\Delta f\ \Delta g$ part is basically negligible. Therefore,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\big(f(x)g(x)\big) &= \lim_{h\to 0}\frac{g(x)\,\Delta f + f(x)\,\Delta g}{h}\\ &= \lim_{h\to 0}\frac{f(x)\big(g(x+h)-g(x)\big)+g(x)\big(f(x+h)-f(x)\big)}{h}\\ &= f(x)\lim_{h\to 0}\frac{g(x+h)-g(x)}{h}+g(x)\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}\\ &= f(x)\frac{\mathrm{d}g(x)}{\mathrm{d}x}+g(x)\frac{\mathrm{d}f(x)}{\mathrm{d}x}.\end{aligned}$$

Which is what we call the product rule. Notice the alternation between f and g in the two terms. The derivative of $f(x)$ is multiplied by $g(x)$, and the derivative of $g(x)$ is multiplied by $f(x)$. This is a direct consequence of the diagram: the change in $f(x)$ is multiplied by $g(x)$ to give the area and also the other way around. You could check this with the method of increments, and it would still be true. I encourage you to do it.

To take derivatives of quotients of functions, just plug in $1\ /\ g(x)$ instead of $g(x)$. The final form should be

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{f(x)}{g(x)}\right)=\frac{f(x)\dfrac{\mathrm{d}g(x)}{\mathrm{d}x}-g(x)\dfrac{\mathrm{d}f(x)}{\mathrm{d}x}}{f(x)^2}. \tag{3.67}$$

But how's about the product of multiple functions? We can't really geometrically interpret those anymore. So we have to turn ourselves to the analytical method.

Let's think of this through. We don't know the derivative of products, but we know the derivative of *sums*: the linearity of the derivative allows us to break derivatives of sums into sums of derivatives. So if we can turn products into sum, this problem would be so easy! Gladly, the logarithms can do exactly that. To make our life easier, we shall use the natural logarithm. Because I want to save space, let's write $a(x)b(x)c(x)\ldots$ as just

$abc\ldots$. Do know that these are all functions of x.

$$\begin{aligned}\frac{\mathrm{d}(abc\ldots)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\ln(abc\ldots)}\right)\\ &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\mathrm{e}^{\ln(a)+\ln(b)+\ln(c)+\ldots}\right).\end{aligned}$$

Now, let $\ln(a)+\ln(b)+\ln(c)+\cdots = u$ and use the chain rule,

$$\begin{aligned}&= \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^u)\cdot\frac{\mathrm{d}u}{\mathrm{d}u}\\ &= \frac{\mathrm{d}}{\mathrm{d}x}(\mathrm{e}^u)\cdot\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(a)+\ln(b)+\ln(c)+\ldots\right)\\ &= \mathrm{e}^u\left(\frac{\mathrm{d}\ln(a)}{\mathrm{d}x}+\frac{\mathrm{d}\ln(b)}{\mathrm{d}x}+\frac{\mathrm{d}\ln(c)}{\mathrm{d}x}+\ldots\right).\end{aligned}$$

Then use the chain rule again on the terms in the parenthesis

$$\begin{aligned}&= \mathrm{e}^u\left(\frac{\mathrm{d}\ln(a)}{\mathrm{d}x}\frac{\mathrm{d}a}{\mathrm{d}a}+\frac{\mathrm{d}\ln(b)}{\mathrm{d}x}\frac{\mathrm{d}b}{\mathrm{d}b}+\frac{\mathrm{d}\ln(c)}{\mathrm{d}x}\frac{\mathrm{d}c}{\mathrm{d}c}+\ldots\right)\\ &= (abc\ldots)\left(\frac{\mathrm{d}\ln(a)}{\mathrm{d}a}\frac{\mathrm{d}a}{\mathrm{d}x}+\frac{\mathrm{d}\ln(b)}{\mathrm{d}b}\frac{\mathrm{d}b}{\mathrm{d}x}+\frac{\mathrm{d}\ln(c)}{\mathrm{d}c}\frac{\mathrm{d}c}{\mathrm{d}x}+\ldots\right)\\ &= (abc\ldots)\left(\frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}x}+\frac{1}{b}\frac{\mathrm{d}b}{\mathrm{d}x}+\frac{1}{c}\frac{\mathrm{d}c}{\mathrm{d}x}+\ldots\right).\end{aligned}$$

And there we have it: the generalized product rule.

3.5.2 Alternative derivations for the power rule

The power rule can also be derived using the same technique we just used. However, we use a different property of logarithm: $\ln(x^n) = n\ln(x)$.

$$\frac{\mathrm{d}}{\mathrm{d}x}x^n = \frac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{n\ln(x)}.$$

Let $u = n\ln(x)$ then use the chain rule

$$\begin{aligned}&= \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}x}\cdot\frac{\mathrm{d}u}{\mathrm{d}u}\\ &= \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}u}\cdot\frac{\mathrm{d}u}{\mathrm{d}x}\\ &= \mathrm{e}^u\cdot\frac{\mathrm{d}}{\mathrm{d}x}\left(n\ln(x)\right)\\ &= x^n\cdot n\frac{1}{x} = nx^{n-1}.\end{aligned}$$

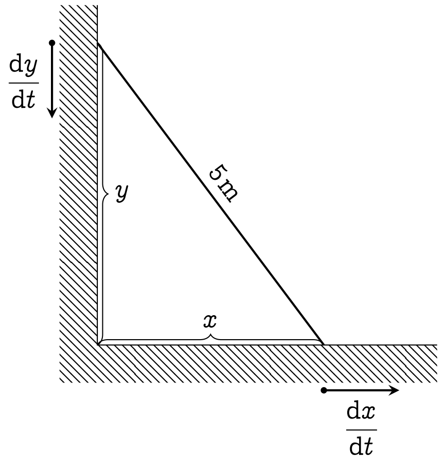


Figure 3.6 | A ladder length 5m sliding down a corner.

3.6 Implicit differentiation

Implicit differentiation also shows up in problems where two rate of changes are related to each other: the **related rates** problem. Consider a sliding ladder that's 5m long (fig. 3.6). At a certain time t, what's the sliding rate along the x-axis w.r.t. the y-axis?

The problem is asking for $\frac{dx}{dy}$. Here, the rate of sliding along the y-axis $\frac{dy}{dt}$ and along the x-axis $\frac{dx}{dt}$ is related because the ladder length is fixed. If y decreases, x must increase. The Pythagorean theorem says

$$x^2 + y^2 = 5^2. \tag{3.68}$$

We can take the derivative w.r.t. t on both side then use the chain rule

$$\begin{aligned}\frac{dx^2}{dt} + \frac{dy^2}{dt} &= \frac{d(5^2)}{dt}\\ \frac{dx^2}{dx}\frac{dx}{dt} + \frac{dy^2}{dy}\frac{dy}{dt} &= 0\\ 2x\frac{dx}{dt} + 2y\frac{dy}{dt} &= 0.\end{aligned}$$

We can take advantage of the Leibniz's notation and multiply by $\mathrm{d}t$ on both sides giving

$$2x\,\mathrm{d}x + 2y\,\mathrm{d}y = 0.$$

To find $\frac{\mathrm{d}x}{\mathrm{d}y}$, we just have to isolate the variables,

$$\frac{\mathrm{d}x}{\mathrm{d}y} = -\frac{y}{x}.$$

And this should actually make sense. Because if y is increasing by a bit, x must decrease by some amount, and that amount is $-y\,/\,x$: the higher the y, the larger the rates of sliding.

3.7 Integral of the reciprocal

By the fundamental theorem of calculus (theorem 2.3.1), the antiderivative of $\frac{1}{x}$ is $\ln(x)$. You might believe this, and it's not wrong. But I find it very disturbing and unresolved. It's like taking the result and pointing it back to the origin. It doesn't really make sense. So from this dissatisfaction, I spent a night coming up with a way to derive this using just the reversed power rule. Enjoy the transformation!

$$\begin{aligned}
\int \frac{1}{x}\,\mathrm{d}x &= \int \lim_{h\to 0}\left(\frac{1}{2}x^{-1+h} + \frac{1}{2}x^{-1-h}\right)\mathrm{d}x \\
&= \lim_{h\to 0}\int\left(\frac{1}{2}x^{-1+h} + \frac{1}{2}x^{-1-h}\right)\mathrm{d}x \\
&= \lim_{h\to 0}\int\left(\frac{1}{2}\frac{x^{-1+h+1}}{(-1+h+1)} + \frac{1}{2}\frac{x^{-1-h+1}}{(-1-h+1)}\right)\mathrm{d}x \\
&= \lim_{h\to 0}\left(\frac{1}{2}\frac{x^{h}}{h} - \frac{1}{2}\frac{x^{-h}}{-h}\right) = \lim_{h\to 0}\left(\frac{1}{2}\frac{x^{h}}{h}\frac{x^{h}}{h} - \frac{1}{2}\frac{1}{hx^{h}}\right) \\
&= \lim_{h\to 0}\left(\frac{x^{2h}-1}{2hx^{h}}\right) = \lim_{h\to 0}\left(\frac{\mathrm{e}^{2h\ln(x)}-1}{2h\ln(x)}\cdot\frac{\ln(x)}{x^{h}}\right) \\
&= \lim_{h\to 0}\left(\frac{\mathrm{e}^{2h\ln(x)}}{2h\ln(x)}\right)\cdot\lim_{h\to 0}\left(\frac{\ln(x)}{x^{h}}\right)
\end{aligned}$$

$$= \lim_{h\to 0}\left(\frac{e^{2h\ln(x)}}{2h\ln(x)}\right)\cdot\ln(x). \tag{3.69}$$

Then, we evaluate the limit at the front by letting $u = 2h\ln(x)$. When $h \to 0$, $u \to 0$ as well. Then, use the definition of e from eq. (3.41).

$$\begin{aligned}\lim_{h\to 0}\left(\frac{e^{2h\ln(x)}}{2h\ln(x)}\right) &= \lim_{u\to 0}\left(\frac{e^u - 1}{u}\right)\\ &= \lim_{u\to 0}\left(\frac{\left(\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n\right)^u}{u}\right).\end{aligned}$$

Change the limits from $n \to \infty$ into $n \to 0$. Notice, $\lim_{n\to 0}\left(1+\frac{1}{n}\right)^n = \lim_{n\to 0}(1+n)^{1/n}$. If $n \to 0$ and $u \to 0$, that means $n = u$. Substitute $n = u$ into the limit,

$$\begin{aligned}&= \lim_{u\to 0}\left(\frac{\left(\lim_{u\to 0}(1+u)^{1/u}\right)^u - 1}{u}\right)\\ &= \lim_{u\to 0}\left(\frac{1+u-1}{u}\right) = 1.\end{aligned}$$

Then, substitute this limit back into eq. (3.69), you'll see that

$$\int \frac{1}{x}\,dx = \lim_{h\to 0}\left(\frac{e^{2h\ln(x)}}{2h\ln(x)}\right)\cdot\ln(x) = \ln(x) + C,$$

completing the proof.

3.8 Amalgamation of functions

Technically, I could end the chapter now because we've explored the derivatives and antiderivatives of all non-trigonometric functions: polynomials, exponentials, and logarithms. But I want you to familiarize yourself with evaluating derivatives and antiderivatives of an amalgamation of functions first, whether it be by addition, subtraction, multiplication, division, or jamming one function inside another (composite functions). These techniques will help us solve other harder problems in later chapters. So, this section is written as a compilation of mathematical exercise, ready to be trained on.

The reason that this section is the longest in this chapter is because it contains all solution to the exercises, making it a bit exhaustive even for me to write. So, you don't have to do all the exercises. Just familiarize yourself with it, and you will be mostly fine. However, there are one place that I want you to focus: section 3.8.2, which is on substitution in antiderivatives. It is absolutely necessary, so don't skip over it. You shall see why in later chapters.

Evaluating derivatives is completely different to evaluating antiderivatives. In derivatives, there are strict rules to guide you through. Those rules are gone in antiderivatives. It is an art form; a very difficult one. But it has proven to be too useful to be discarded. So, focus on it. Without further ado, let's crack on.

3.8.1 Technique of differentiation

[6]Put simply, if there's a linear combination of functions, use the linearity of derivatives to break it down into multiple smaller derivatives

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(af(x)+bg(x)\right)=a\frac{\mathrm{d}f(x)}{\mathrm{d}x}+b\frac{\mathrm{d}g(x)}{\mathrm{d}x},$$

if there's product, then use the product rule

$$\frac{\mathrm{d}\big(f_1(x)f_2(x)\ldots f_n(x)\big)}{\mathrm{d}x}=f_1f_2\ldots f_n\left(\frac{1}{f_1}\frac{\mathrm{d}f_1}{\mathrm{d}x}+\frac{1}{f_2}\frac{\mathrm{d}f_2}{\mathrm{d}x}+\cdots+\frac{1}{f_n}\frac{\mathrm{d}f_n}{\mathrm{d}x}\right),$$

and if there are composite functions $f(g(x))$, and you know the derivative of both $f(x)$ and $g(x)$, just set $u=g(x)$ and use the chain rule.

$$\frac{\mathrm{d}f(g(x))}{\mathrm{d}x}=\frac{\mathrm{d}f(u)}{\mathrm{d}x}=\frac{\mathrm{d}f(u)}{\mathrm{d}u}\times\frac{\mathrm{d}u}{\mathrm{d}x}=\frac{\mathrm{d}f(u)}{\mathrm{d}u}\times\frac{\mathrm{d}g(x)}{\mathrm{d}x}$$

[6]The rules mentioned here are repeated without numberings for ease of reading

And, implicit derivatives can also help you simplify a whole class of problems if you're stuck. Knowing these, you're more than ready now. Let's get on with solving.

Explicit differentiation Let's start with some simple composite functions

Example 3.8.1: $\dfrac{\mathrm{d}}{\mathrm{d}x}\mathrm{e}^{2\sqrt{x}}$ This is a composite function $f(g(x))$ where $f(x) = \mathrm{e}^x$, and $g(x) = 2\sqrt{x}$. We know that the derivative of e^x is e^x (section 3.4). And by the linearity of derivatives and the power rule,

$$\frac{\mathrm{d}g(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}(2\sqrt{x}) = 2\frac{\mathrm{d}x^{1/2}}{\mathrm{d}x} = 2\left(\frac{1}{2}x^{-1/2}\right) = x^{-\frac{1}{2}} = \frac{1}{\sqrt{x}}. \tag{3.70}$$

So, setting $g(x) = 2\sqrt{x} = u$, we get

$$\frac{\mathrm{d}\mathrm{e}^{2\sqrt{x}}}{\mathrm{d}x} = \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}x} \tag{3.71}$$

$$= \frac{\mathrm{d}\mathrm{e}^u}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.72}$$

$$= \mathrm{e}^u \times \frac{\mathrm{d}}{\mathrm{d}x}(2\sqrt{x}) \tag{3.73}$$

$$= \mathrm{e}^{2\sqrt{x}}\left(\frac{1}{\sqrt{x}}\right) = \frac{\mathrm{e}^{2\sqrt{x}}}{\sqrt{x}}. \tag{3.74}$$

Now let's throw in a bit of products.

Example 3.8.2: $\dfrac{\mathrm{d}}{\mathrm{d}x}(x^3\ln(x^3))$ By the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}x}(x^3\ln(x^3)) = (x^3)\left(\frac{\mathrm{d}\ln(x^3)}{\mathrm{d}x}\right) + \ln(x^3)\left(\frac{\mathrm{d}x^3}{\mathrm{d}x}\right) \tag{3.75}$$

The power rule says that $\dfrac{\mathrm{d}}{\mathrm{d}x}x^3 = 3x^2$. We then evaluate $\dfrac{\mathrm{d}\ln(x^3)}{\mathrm{d}x}$ by using the chain rule. Let $x^3 = u$, then

$$\frac{\mathrm{d}\ln(x^3)}{\mathrm{d}x} = \frac{\mathrm{d}\ln(u)}{\mathrm{d}x} \tag{3.76}$$

$$= \frac{\mathrm{d}\ln(u)}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x}, \tag{3.77}$$

discussed in section 3.5, $\dfrac{\mathrm{d}\ln(u)}{\mathrm{d}u} = \dfrac{1}{u}$,

$$= \frac{1}{u} \times \frac{\mathrm{d}x^3}{\mathrm{d}x} \tag{3.78}$$

$$= \frac{1}{u} \times 3x^2 = \frac{1}{x^3} \times 3x^2 = \frac{3}{x}; \tag{3.79}$$

therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(x^3\ln(x^3)\right) = x^3 \times \frac{3}{x} + \ln(x^3) \times 3x^2 \tag{3.80}$$

$$= 3x^2 + 3x^2\ln(x^3) = 3x^2\left(1 + \ln(x^3)\right). \tag{3.81}$$

How's about some purely polynomial derivatives?

Example 3.8.3: $\dfrac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5(3x^3+4)^9\right)$ This derivative is quite tough to evaluate by expansion. A better way would be to use the product rule and the chain rule in combination. Firstly,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5(3x^3+4)^9\right)$$
$$= (x^2+1)^5\left(\frac{\mathrm{d}}{\mathrm{d}x}\left((3x^3+4)^9\right)\right) + (3x^3+4)^9\left(\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2+1)^5\right)\right). \tag{3.82}$$

The derivative of $(3x^3+4)^9$ can be evaluated using the chain rule by setting $u = 3x^3 + 4$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((3x^3+4)^9\right) = \frac{\mathrm{d}u^9}{\mathrm{d}x} \tag{3.83}$$

$$= \frac{\mathrm{d}u^9}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.84}$$

By the power rule, $\dfrac{\mathrm{d}u^9}{\mathrm{d}u} = 9u^8$

$$= 9u^8 \times \frac{\mathrm{d}}{\mathrm{d}x}(3x^3 + 4) \tag{3.85}$$

Using the power rule, $\dfrac{\mathrm{d}}{\mathrm{d}x}(3x^3 + 4) = 9x^2$

$$= 9(3x^3 + 4)^8 \times 9x^2 = 81x^2(3x^3 + 4)^8. \tag{3.86}$$

Similarly, use the chain rule to evaluate $(x^2 + 1)^5$ and set $u = x^2 + 1$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2 + 1)^5\right) = \frac{\mathrm{d}u^5}{\mathrm{d}x} \tag{3.87}$$

$$= \frac{\mathrm{d}u^5}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.88}$$

By the power rule, $\dfrac{\mathrm{d}u^5}{\mathrm{d}u} = 5u^4$

$$= 5u^4 \times \frac{\mathrm{d}(x^2 + 1)}{\mathrm{d}x} \tag{3.89}$$

By the power rule, $\dfrac{\mathrm{d}(x^2 + 1)}{\mathrm{d}x} = 2x$

$$= 5(x^2 + 1)^4 \times 2x = 10x(x^2 + 1)^4. \tag{3.90}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left((x^2 + 1)^5(3x^3 + 4)^9\right) = (x^2 + 1)^5 \times 81x^3(3x^3 + 4)^8 \tag{3.91}$$
$$+ (3x^3 + 4)^9 \times 10x(x^2 + 1)^4$$

$$= 81x^3(x^2 + 1)^5(3x^3 + 4)^8 \tag{3.92}$$
$$+ 10x(x^2 + 1)^4(3x^3 + 4)^9.$$

There are certain cases of differentiation where using algebraical rules is better than using the chain rule with the product rule. Here is such example

Example 3.8.4: $\dfrac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\sqrt{\dfrac{x+1}{x-1}}\right)\right)$ Breaking this derivative with the chain rule will be a nightmare. As said, let's use the algebraical properties of logarithm to help us with this. Namely, $\ln\left(\frac{a}{b}\right) = \ln(a) - \ln(b)$, and $\ln(a^b) = b\ln(a)$.

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\sqrt{\frac{x+1}{x-1}}\right)\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln(\sqrt{x+1}) - \ln(\sqrt{x-1})\right) \tag{3.93}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left((x+1)^{1/2}\right) - \ln\left((x-1)^{1/2}\right)\right) \tag{3.94}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{2}\ln(x+1) - \frac{1}{2}\ln(x-1)\right) \tag{3.95}$$

By the linearity of derivative,

$$= \frac{1}{2}\left(\frac{\mathrm{d}\ln(x+1)}{\mathrm{d}x} - \frac{\mathrm{d}\ln(x-1)}{\mathrm{d}x}\right). \tag{3.96}$$

These smaller derivatives are in the form $\dfrac{\mathrm{d}\ln(u)}{\mathrm{d}x}$ where in the first one, $u = x+1$, and in the second one, $u = x-1$. By the chain rule and the derivative of logarithms (section 3.5),

$$\frac{\mathrm{d}\ln(u)}{\mathrm{d}x} = \frac{\mathrm{d}\ln(u)}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{u} \times \frac{\mathrm{d}u}{\mathrm{d}x}. \tag{3.97}$$

Since in both cases, $\dfrac{\mathrm{d}u}{\mathrm{d}x} = 1$,

$$\frac{\mathrm{d}\ln(x+1)}{\mathrm{d}x} = \frac{1}{x+1} \quad \text{and} \quad \frac{\mathrm{d}\ln(x-1)}{\mathrm{d}x} = \frac{1}{x-1}; \tag{3.98}$$

thus,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\sqrt{\frac{x+1}{x-1}}\right)\right) = \frac{1}{2}\left(\frac{1}{x+1} - \frac{1}{x-1}\right) = -\frac{1}{x^2-1}. \tag{3.99}$$

Implicit differentiation Implicit differentiation can help us with evaluating the derivative of functions in the form $f(x)^{g(x)}$. Let $y = f(x)^{g(x)}$, then using the algebraic properties of logarithms,

$$\ln(y) = \ln\left(f(x)^{g(x)}\right) \tag{3.100}$$

$$\ln(y) = g(x)\ln\big(f(x)\big) \tag{3.101}$$

$$\frac{\mathrm{d}\ln(y)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\big(g(x)\ln\big(f(x)\big)\big). \tag{3.102}$$

Using the chain rule on the L.H.S. and the product rule on the R.H.S.,

$$\frac{\mathrm{d}\ln(y)}{\mathrm{d}y} \times \frac{\mathrm{d}y}{\mathrm{d}x} = g(x)\frac{\mathrm{d}\ln\big(f(x)\big)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x} \tag{3.103}$$

The derivative of the $\ln(x)$ is $\frac{1}{x}$ (section 3.5), and use the chain rule on the R.H.S.

$$\frac{1}{y} \times \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{g(x)}{f(x)}\frac{\mathrm{d}f(x)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x} \tag{3.104}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{g(x)}{f(x)}\frac{\mathrm{d}f(x)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x}\right)y \tag{3.105}$$

Since $y = f(x)^{g(x)}$,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \left(\frac{g(x)}{f(x)}\frac{\mathrm{d}f(x)}{\mathrm{d}x} + \ln\big(f(x)\big)\frac{\mathrm{d}g(x)}{\mathrm{d}x}\right)f(x)^{g(x)} \tag{3.106}$$

Frankly, it can also be done explicitly by rewriting $f(x)^{g(x)}$ as $\mathrm{e}^{g(x)\ln(f(x))}$ and using the chain rule, but it's just neater this way. Let's try some examples. First, with the infamous x^x.

Example 3.8.5: $\frac{\mathrm{d}x^x}{\mathrm{d}x}$ This function x^x is in the form $f(x)^{g(x)}$ where $f(x) = x$, and $g(x)$ is also equal to x. Trivially, $\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}g(x)}{\mathrm{d}x} = 1$, so we can use eq. (3.106) right away and get

$$\frac{\mathrm{d}x^x}{\mathrm{d}x} = \left(\frac{x}{x}(1) + \ln(x)(1)\right)x^x = x^x\left(\ln(x) + 1\right). \tag{3.107}$$

Implicit differentiation can also be used if you are too lazy to deal with roots, e.g.,

Example 3.8.6: $\dfrac{\mathrm{d}\sqrt{2\ln(x)^2-1}}{\mathrm{d}x}$ It is very possible to go through this and use the chain rule and the power rule. But who wants to deal with derivative of roots? They are confusing. Well, we can actually use implicit differentiation to turn the square root into an integer power. Set $y=\sqrt{2\ln(x)^2+1}$, then

$$y=\sqrt{2\ln(x)^2+1} \tag{3.108}$$

$$y^2=2\ln(x)^2+1. \tag{3.109}$$

Take the derivative on both sides w.r.t. x. Use the linearity of derivative to break down the R.H.S., and use the chain rule to take care of the rest

$$\frac{\mathrm{d}y^2}{\mathrm{d}x}=2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x}+\cancelto{0}{\frac{\mathrm{d}}{\mathrm{d}x}(1)} \tag{3.110}$$

$$\frac{\mathrm{d}y^2}{\mathrm{d}y}\times\frac{\mathrm{d}y}{\mathrm{d}x}=2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} \tag{3.111}$$

$$2y\times\frac{\mathrm{d}y}{\mathrm{d}x}=2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} \tag{3.112}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{1}{y}2\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x} \tag{3.113}$$

Use the chain rule to evaluate $\dfrac{\mathrm{d}\ln(x)^2}{\mathrm{d}x}$. Set $u=\ln(x)$

$$\frac{\mathrm{d}\ln(x)^2}{\mathrm{d}x}=\frac{\mathrm{d}u^2}{\mathrm{d}x} \tag{3.114}$$

$$=\frac{\mathrm{d}u^2}{\mathrm{d}u}\times\frac{\mathrm{d}u}{\mathrm{d}x} \tag{3.115}$$

$$=2u\times\frac{\mathrm{d}\ln(x)}{\mathrm{d}x} \tag{3.116}$$

$$=\frac{2\ln(x)}{x} \tag{3.117}$$

Therefore,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{y}2\left(\frac{2\ln(x)}{x}\right) \tag{3.118}$$

Substitute in $y = \sqrt{2\ln(x)^2+1}$, then

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{4\ln(x)}{x\sqrt{2\ln(x)^2+1}}. \tag{3.119}$$

Or maybe, you just have a very wonky equation that you can't even rearrange in terms of $y = f(x)$, e.g.,

Example 3.8.7: *Find* $\frac{\mathrm{d}y}{\mathrm{d}x}$ *given that* $x^y = y^x$ e start from taking the logarithm of both sides

$$\ln(x^y) = \ln(y^x) \tag{3.120}$$

$$y\ln x = x\ln y \tag{3.121}$$

Then, take the derivative w.r.t. x by using the product rule

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\ln x) = \frac{\mathrm{d}}{\mathrm{d}x}(x\ln y) \tag{3.122}$$

$$y\frac{\mathrm{d}\ln x}{\mathrm{d}x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = x\frac{\mathrm{d}\ln y}{\mathrm{d}x} + \ln y\cancelto{1}{\frac{\mathrm{d}x}{\mathrm{d}x}} \tag{3.123}$$

First, the derivative of $\ln(x)$ is $\frac{1}{x}$, and we can use the chain rule to take care of the rest

$$\frac{y}{x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = x\frac{\mathrm{d}\ln y}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x} + \ln y \tag{3.124}$$

$$\frac{y}{x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{x}{y}\frac{\mathrm{d}y}{\mathrm{d}x} + \ln y \tag{3.125}$$

$$\frac{x}{y}\frac{\mathrm{d}y}{\mathrm{d}x} + \ln x\frac{\mathrm{d}y}{\mathrm{d}x} = \ln y - \frac{y}{x} \tag{3.126}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x}(x - \ln x) = \ln y - \frac{y}{x} \tag{3.127}$$

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\ln y - \frac{y}{x}}{x - \ln x}. \tag{3.128}$$

3.8.2 Technique of integration: substitution

Before we finish this chapter off, I'd want to take a look at a very useful technique for dealing with complicated integrals: substitution.

Substitution is a technique to simplify algebraic expressions. I believe you've already seen a multitude of them. Basically, we substitute some expression of x with u, then use everything in our power to change every x to u, including the differential element: from $\mathrm{d}x$ to $\mathrm{d}u$.

u-substitution works when the integrals are of the form

$$\int f\left(g(x)\right)\frac{\mathrm{d}g(x)}{\mathrm{d}x}\,\mathrm{d}x. \tag{3.129}$$

In this notation, it's easy to see that the $\mathrm{d}x$ will cancel, leaving us with

$$\int f\left(g(x)\right)\mathrm{d}g(x). \tag{3.130}$$

If $g(x) = u$, then our integral simplifies to

$$\int f(u)\,\mathrm{d}u. \tag{3.131}$$

Let's apply this knowledge to solve some integrals. Starting with the simplest: integrand with polynomials.

Example 3.8.8: $\int x(x^2+1)^{10}\,\mathrm{d}x$ If $f(x) = x^{10}$, and $u = x^2+1$, then

$$\int x(x^2+1)^{10}\,\mathrm{d}x = \int u^{10}x\,\mathrm{d}x. \tag{3.132}$$

Because $x = \frac{1}{2}\frac{\mathrm{d}(x^2+1)}{\mathrm{d}x}$,

$$\int u^{10} x \,\mathrm{d}x = \int u^{10} \frac{1}{2}\frac{\mathrm{d}(x^2+1)}{\mathrm{d}x}\,\mathrm{d}x = \frac{1}{2}\int u^{10}\,\mathrm{d}(x^2+1) \tag{3.133}$$

Since $u = x^2 + 1$, our integral simplifies to

$$\frac{1}{2}\int u^{10}\,\mathrm{d}u. \tag{3.134}$$

Using the reversed power rule (eq. (3.19)), we get

$$\frac{1}{2}\frac{u^{11}}{11} = \frac{u^{11}}{22} = \frac{(x^2+1)^{11}}{22} + C. \tag{3.135}$$

Example 3.8.9: $\int x^2 \mathrm{e}^{4x^3+4}\,\mathrm{d}x$ The integrand is a composite function: $f(u) = \mathrm{e}^u$ and $u = 4x^3 + 4$. In which, $\frac{\mathrm{d}u}{\mathrm{d}x} = 12x^2$; therefore,

$$\begin{aligned}
\int x^2 \mathrm{e}^{4x^3+4}\,\mathrm{d}x &= \frac{1}{12}\int \mathrm{e}^u 12x^2\,\mathrm{d}x && (3.136)\\
&= \frac{1}{8}\int \mathrm{e}^u \frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x && (3.137)\\
&= \frac{1}{8}\int \mathrm{e}^u\,\mathrm{d}u = \frac{1}{8}\mathrm{e}^u && (3.138)\\
&= \frac{1}{12}\mathrm{e}^{4x^3+4} + C. && (3.139)
\end{aligned}$$

Notice any patterns? If the integral has a polynomial u with degree n nested inside another function, there must be a term outside with the degree $n-1$ to substitute with. In ex. 3.8.8, $u = x^2 + 1$ and there's an x multiplied outside waiting to be substituted. In ex. 3.8.10, $u = 4x^3 + 4$, and there's an x^2 outside. This pattern is a consequence of the power rule, and is pretty common. So, here are some practice problems with answers and hints on the right.[7] Try not to look at it much.

1. $\displaystyle\int 3x^2\sqrt{x^3+5}\,\mathrm{d}x$ — $\dfrac{2}{3}(x^3+5)^{\frac{3}{2}}+C$, $u=x^3+5$
2. $\displaystyle\int \sqrt{4+3x}\,\mathrm{d}x$ — $\dfrac{2}{9}(3x+4)^{\frac{3}{2}}+C$, $u=4+3x$
3. $\displaystyle\int \frac{24x}{(4x^2+4)^2}\,\mathrm{d}x$ — $\dfrac{3}{4x^2+4}+C$, $u=4x^2+4$

Now let's move on from polynomials to other functions that are nested inside another function. For the substitution to work, there must be the function's derivative multiplied, waiting to be substituted. A classic example is:

Example 3.8.10: $\displaystyle\int \frac{\ln(x)}{x}\,\mathrm{d}x$ This integral is not quite a composite function, but rather just a product between a function and its derivative. If $u=\ln(x)$, then $\dfrac{\mathrm{d}u}{\mathrm{d}x}=\dfrac{1}{x}$; therefore,

$$\int \frac{\ln(x)}{x}\,\mathrm{d}x = \int u\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x \tag{3.140}$$

$$= \int u\,\mathrm{d}u \tag{3.141}$$

$$= \frac{u^2}{2} = \frac{\ln(x)^2}{2} + C. \tag{3.142}$$

Example 3.8.11: $\displaystyle\int x\mathrm{e}^{x^2}(\mathrm{e}^{x^2}+3)\,\mathrm{d}x$ Let $\mathrm{e}^{x^2}+3=u$, then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{\mathrm{d}\mathrm{e}^{x^2}+3}{\mathrm{d}x} \tag{3.143}$$

$$= \frac{\mathrm{d}\mathrm{e}^{x^2}}{\mathrm{d}x} \tag{3.144}$$

[7] Practice problems taken from (2).

Let $x^2 = v$, then by the chain rule,

$$= \frac{\mathrm{d}e^v}{\mathrm{d}x} \cdot \frac{\mathrm{d}v}{\mathrm{d}v} \tag{3.145}$$

$$= \frac{\mathrm{d}e^v}{\mathrm{d}v} \cdot \frac{\mathrm{d}x^2}{\mathrm{d}x} \tag{3.146}$$

$$= 2xe^v = 2xe^{x^2}. \tag{3.147}$$

Thus,

$$\int xe^{x^2}(e^{x^2}+3)\,\mathrm{d}x = \frac{1}{2}\int u\left(2xe^{x^2}\right)\mathrm{d}x \tag{3.148}$$

$$= \frac{1}{2}\int u\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x = \frac{1}{2}\int u\,\mathrm{d}u \tag{3.149}$$

$$= \frac{1}{2}\frac{u^2}{2} = \frac{(e^{x^2}+3)^2}{4} + C. \tag{3.150}$$

Sometimes, they need some *creative* algebraic manipulations, and sometimes even multiple substitution. E.g.,[8]

Example 3.8.12: $\int \frac{1}{x(x^5+1)}\,\mathrm{d}x$ If we want to use power rule substitution, there's only x^5 but no x^4 to substitute with. So, we need to get a bit creative.

First, factor the x^5 out of the inner parenthesis and get

$$\int \frac{1}{x^6\left(1+\frac{1}{x^5}\right)}\,\mathrm{d}x. \tag{3.151}$$

Now, we can let $u = 1 + \frac{1}{x^5}$. By the power rule, $\frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{5}{x^6}$. Our integral then becomes

$$-5\int \frac{1}{u}\left(-\frac{5}{x^6}\right)\mathrm{d}x = -\frac{1}{5}\int \frac{1}{u}\frac{\mathrm{d}u}{\mathrm{d}x}\,\mathrm{d}x \tag{3.152}$$

[8]Taken from MIT Integration Bee 2019, I_9 (1)

$$= -\frac{1}{5}\int \frac{1}{u}\,\mathrm{d}u = -5\ln(x) \tag{3.153}$$

$$= -\frac{1}{5}\ln\left(1+\frac{1}{x^5}\right)+C \tag{3.154}$$

In certain problems, there aren't any outside terms to substitute with, so you have to create them. E.g.,

Example 3.8.13: $\int (1+\sqrt{x})^4\,\mathrm{d}x$ We want to get rid of the term inside the parenthesis. So, let $u = 1+\sqrt{x}$, and by the power rule, $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2\sqrt{x}}$. However, there is no $\frac{1}{\sqrt{x}}$ outside to substitute with. Let's be adventurous and substitute u in anyway and see what happens.

$$\int (1+\sqrt{x})^4\,\mathrm{d}x = \int u^4\,\mathrm{d}x \tag{3.155}$$

If $\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2\sqrt{x}}$, then $\mathrm{d}x = 2\sqrt{x}\,\mathrm{d}u$,

$$= \int u^4(2\sqrt{x})\,\mathrm{d}u. \tag{3.156}$$

We're almost there, now we just have to replace the leftover x with u. Since $u = 1+\sqrt{x}$, $\sqrt{x} = u-1$; thus,

$$\int u^4(2\sqrt{x})\,\mathrm{d}u = 2\int u^4(u-1)\,\mathrm{d}u \tag{3.157}$$

$$= 2\int u^5 - u^4\,\mathrm{d}u = 2\left(\frac{u^6}{6}-\frac{u^5}{5}\right) \tag{3.158}$$

$$= 2\left(\frac{(1+\sqrt{x})^6}{6}-\frac{(1+\sqrt{x})^5}{5}\right). \tag{3.159}$$

These problems definitely take a while to get used to. So I highly encourage you to practice integrating using a problem set ((1), (2)). The unfortunate thing is that most problem sets include trigonometry. So, you'd have to pick the one without them to practice with. Or, you can just wait until we discuss trigonometry and then practice from there.

3.9 Formula for Chapter 3

3.9.1 Formula for derivatives of functions

1. $f(x) = g(x) \implies \dfrac{\mathrm{d}f(x)}{\mathrm{d}x} = \dfrac{\mathrm{d}g(x)}{\mathrm{d}x}$ (Rules of Equality)
2. For $c \in \mathbb{R}$, $\dfrac{\mathrm{d}(c)}{\mathrm{d}x} = 0$ (Derivative of a constant)
3. $\dfrac{\mathrm{d}x}{\mathrm{d}y} = \dfrac{\mathrm{d}x}{\mathrm{d}u} \times \dfrac{\mathrm{d}u}{\mathrm{d}y}$ (Chain rule)
4. $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(af(x) + bg(x)\big) = a\dfrac{\mathrm{d}f(x)}{\mathrm{d}x} + b\dfrac{\mathrm{d}g(x)}{\mathrm{d}x}$ (Linearity of differentiation)
5. $\dfrac{\mathrm{d}}{\mathrm{d}x}(ax^n) = anx^{n-1}$ (Power rule)
6. $\dfrac{\mathrm{d}}{\mathrm{d}x}(n^x) = n^x \ln(n)$, $\dfrac{\mathrm{d}(\mathrm{e}^x)}{\mathrm{d}x} = \mathrm{e}^x$ (Derivative of exponentials)
7. $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(\ln(x)\big) = \dfrac{1}{x}$ (Derivative of natural logarithms)
8. $\dfrac{\mathrm{d}}{\mathrm{d}x}\big(f_1(x)f_2(x)\big) = f_1(x)\dfrac{\mathrm{d}f_2}{\mathrm{d}x} + f_2(x)\dfrac{\mathrm{d}f_1}{\mathrm{d}x}$ (Product rule for two functions)
9. $\dfrac{\mathrm{d}}{\mathrm{d}x}(f_1 f_2 \dots f_n) = f_1 f_2 \dots f_n \left(\dfrac{1}{f_1}\dfrac{\mathrm{d}f_1}{\mathrm{d}x} + \dfrac{1}{f_2}\dfrac{\mathrm{d}f_2}{\mathrm{d}x} + \cdots + \dfrac{1}{f_n}\dfrac{\mathrm{d}f_n}{\mathrm{d}x}\right)$ (Generalized product rule)

3.9.2 Formula for antiderivatives of functions

1. $f(x) = g(x) \implies \displaystyle\int f(x)\,\mathrm{d}x = \int g(x)\,\mathrm{d}x$ (Rules of Equality)

2. $\int af(x) + bg(x)\,\mathrm{d}x = a\int f(x)\,\mathrm{d}x + b\int g(x)\,\mathrm{d}x$ (Linearity of integration)

3. $n \neq -1$, $\int ax^n\,\mathrm{d}x = a\dfrac{x^{n+1}}{n+1} + C$ (Reversed power rule)

4. $\int n^x\,\mathrm{d}x = \dfrac{1}{\ln(n)}n^x + C$, $\int \mathrm{e}^x\,\mathrm{d}x = \mathrm{e}^x + C$ (Antiderative of exponentials)

5. $\int \dfrac{1}{x}\,\mathrm{d}x = \ln(x) + C$ (Antiderivative of the reciprocal)

6. $\int f(g(x))\dfrac{\mathrm{d}g}{\mathrm{d}x}\,\mathrm{d}x = \int f(u)\,\mathrm{d}u; u = g(x)$ (u-Substitution)

3.9.3 Definition for various functions and constants

1. $\mathrm{e}^x = \lim\limits_{k\to 0}\sum\limits_{i\to 0}^{k}\dfrac{1}{i!} = 1 + \dfrac{x^1}{1!} + \dfrac{x^2}{2!} + \dfrac{x^3}{3!} + \cdots = \lim\limits_{n\to\infty}\left(1 + \dfrac{x}{n}\right)^n$

2. $\mathrm{e} = \sum\limits_{i=0}^{\infty}\dfrac{1}{i!} = 1 + \dfrac{1}{1!} + \dfrac{1}{2!} + \dfrac{1}{3!} + \cdots = \lim\limits_{n\to\infty}\left(1 + \dfrac{1}{n}\right)^n$

3. $\ln(x) = \lim\limits_{n\to 0}\left(\dfrac{x^h - 1}{h}\right)$

CHAPTER 4

A study of graphs with calculus

4.1 Invitation: sketching mountain ranges

If I told you to *sketch* out x^2, it'd be easy, right? Just a simple parabola would do. But if I say: plot $x^4 - 2x^3 + x^2 - x + 1$? It isn't so easy now is it. You could substitute various x and plot it into the graph directly. But I'm asking for a rough sketch, not a plot. I just want the general shape and structure of the function: where are the highest and lowest points, the x-intercept, and the y-intercept. We're going to learn how to do exactly that in this chapter.

Imagine a function as a vast mountain range. When sketching this landscape, we're interested in identifying its peaks, valleys, and plateaus. A peak represents the sharp summit of a mountain, where the land falls away in all directions; a valley is a sunken area, nestled lower than the surrounding terrain; and a plateau is a level stretch, a temporary calm before the ascent resumes toward higher peaks or descends into deeper valleys. Standing at a peak, the land around you dips downward. In contrast, in a valley, everything rises above you. On a plateau, the ground is flat under your feet, but when gazed into the distance, one side raises higher, but the other reveals the path down. These important features are all characterized by a flat ground: a place with zero slope. If the mountain range represents function, then all of these points have zero first derivative.

Still, the information about zero first derivative is not sufficient to determine whether the point is a peak, a valley, or a plateau. What helps classify these points is the sign of the second derivative.

The second derivative suggests the concavity of a graph at that point. If the first derivative gives you the slope, the second derivative will give you the rate of change of that slope. If the second derivative is positive, it means that the function is increasing around that point; thus, that point must be a valley. If it's negative, then everything is decreasing around; thus, it's a peak. If it's zero, it's simply a plateau.

Just like a mountain, a function can have many points that have zero derivatives. Functions with more than one peaks exists. The first derivative does not provide us with any information on which one of those peaks are the highest. It only tells you where the peak is. In order to find which one is the highest, we'd just have to directly find the function's value at those points, then compare it.

In mathematics terminology, the points that have zero first derivative is called a **stationary point**: the value of the surrounding function is stationary. The valleys are called, **local minima**; the peaks, **local maxima**, and the plateau, **inflection point**. The word *local* suggests that locally, these are either the maximum or minimum point, but it might not be globally. Like how you can stand on your house and say that "this is the highest point of my house," but you're still not higher than Mount Everest. We call the global maximum or minimum point, the **absolute maximum** and **absolute minimum**.

4.2 Minima and maxima

This section focuses on finding the minima and maxima of functions and sketching it. These methods are applicable for all kinds of function, but we'll only study the important ones: quadratics and cubics.

4.2.1 Shape of quadratics

Let's start off with quadratics: a function with only one peak or valley. For any given quadratic

$$f(x) = ax^2 + bx + c, \tag{4.1}$$

Its derivative can be found using the power rule (eq. (3.19)).

$$\frac{\mathrm{d}}{\mathrm{d}x} f(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(ax^2 + bx + c\right) \tag{4.2}$$

$$= 2ax + b. \tag{4.3}$$

The peak or valley of $f(x)$ is at $x = x_0$ where $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\Big|_{x=x_0} = 0$. The notation $\frac{\mathrm{d}}{\mathrm{d}x} f(x)\Big|_{x=x_0}$ tells you that this derivative needs to be evaluated at $x = x_0$. For a quadratic, this means

$$2ax_0 + b = 0 \tag{4.4}$$

$$x_0 = -\frac{b}{2a}, \tag{4.5}$$

which matches with the usual result from traditional algebra. The y value of these points can be found by plugging in $x = -\frac{b}{2a}$ into $f(x)$.

$$f\left(-\frac{b}{2a}\right) = a\left(-\frac{b}{2a}\right)^2 + b\left(-\frac{b}{2a}\right) + c \tag{4.6}$$

$$= \frac{b^2}{4a} - \frac{b^2}{2a} + c \tag{4.7}$$

$$= c - \frac{3b^2}{4a} \tag{4.8}$$

Other important points include the y and x intercept. The y-intercept can easily be found by setting $x = 0$:

$$y_0 = a(0)^2 + b(0) + cy_0 = c. \tag{4.9}$$

Sadly, calculus does not offer any tools to find the exact root of an equation; you still have to use the quadratic equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}. \tag{4.10}$$

Classically, the direction that the quadratic The second derivative of a quadratic is determined by the sign of a. The same result can be derived from the second derivative.

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2} f(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right) \tag{4.11}$$

$$= \frac{\mathrm{d}}{\mathrm{d}x}(2ax + b) \tag{4.12}$$

$$= 2a. \tag{4.13}$$

Since we only care about the sign, the two in $2a$ does not matter. This result tell us that if a is positive, it's a right side up parabola, and the stationary point is a local minimum. If it isn't, then it's upside-down and the stationary point is a local maxima.

A parabola doesn't have any inflection point because it only occurs when the second derivative is zero. However, the only way that $\frac{\mathrm{d}^2}{\mathrm{d}x^2} f(x) = 0$ is for a to be zero. If it happens, then $f(x)$ isn't a parabola anymore, but rather a straight line.

Example 4.2.1: *Sketch the graph of* $f(x) = -2x^2 + x + 6$ Because a is negative, the parabola is upside-down. The vertex is at

$$x = -\frac{b}{2a} = -\frac{1}{2(-2)} = \frac{1}{4}, \tag{4.14}$$

$$y = -2\left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right) + 6 = 6.125. \tag{4.15}$$

The y-intercept is trivially 6. The x intercept are the roots of this polynomial, given by

$$x = \frac{-(1) \pm \sqrt{(1)^2 - 4(-2)(6)}}{2(-2)} = 2, -\frac{3}{2}. \tag{4.16}$$

Factorization would also yield the same result.

$$\begin{aligned} -2x^2 + x + 6 &= -(2x^2 - x - 6) \qquad (4.17)\\ &= -(x-2)(2x+3) \to x = 2, -\frac{3}{2} \qquad (4.18) \end{aligned}$$

Therefore, the sketched graph would be

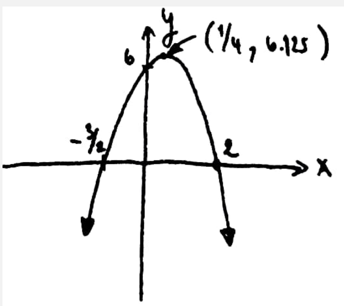


Figure 4.1 | SKETCH OF THE FUNCTION $-2x^2 + x + 6$.

We can also determine the amount of roots by just looking at the y-value of the vertex point, given by eq. (4.8), and the direction of the parabola. But it's more computationally expensive than the determinant method which says,

> A parabola has no roots if $b^2 - 4ac < 0$, one root if $b^2 - 4ac = 0$, and two roots if $b^2 - 4ac > 0$.

This is a direct result of the quadratic formula: $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$ only have real solution if the terms in the root, $b^2 - 4ac$, is positive.

4.2.2 Shape of cubics

Cubics are third order polynomials

$$f(x) = ax^3 + bx^2 + cx + d. \tag{4.19}$$

It's one of the many families of function that can contain more than one stationary point. The x coordinate of the stationary points can be found by using the same method as earlier.

$$\begin{aligned}
\frac{\mathrm{d}f(x)}{\mathrm{d}x} &= \frac{\mathrm{d}}{\mathrm{d}x} ax^3 + bx^2 + cx + d && (4.20)\\
&= 3ax^2 + 2bx + c, && (4.21)
\end{aligned}$$

in which

$$\begin{aligned}
\left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_0} &= 0 = 3ax_0^2 + 2bx_0 + c && (4.22)\\
x_0 &= \frac{-2b \pm \sqrt{(2b)^2 - 4(3a)(c)}}{2(3a)} && (4.23)\\
&= \frac{-2b \pm 2\sqrt{b^2 - 3ac}}{6a} && (4.24)\\
&= \frac{-b \pm \sqrt{b^2 - 3ac}}{3a}. && (4.25)
\end{aligned}$$

By the terms under the root, a cubic has no stationary point if $\frac{b^2-3ac}{3a} < 0$, one if $\frac{b^2-3ac}{3a} = 0$, and two if $\frac{b^2-3ac}{3a} > 0$.

The function $f(x)$ intersects the y axis when $x = 0$:

$$f(0) = a(0)^3 + b(0)^2 + c(0) + d = d; \tag{4.26}$$

thus, the y intercept is at $(0, d)$. For the x-intercept, again, there is no tool in calculus that can be used to find the exact root. So, you'd just have to either factor the function, or use the very lengthy cubic formula.

$$x = \sqrt[3]{q + \sqrt{q^2 + (r - p^2)^3}} + \sqrt[3]{q - \sqrt{q^2 + (r - p^2)^3}} + p \tag{4.27}$$

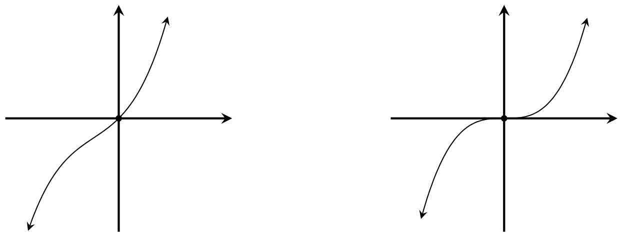
where

$$p = -\frac{b}{3a}, \quad q = p^3 + \frac{bc - 3ad}{6a^2} \quad \text{and} \quad r = \frac{c}{3a}. \tag{4.28}$$

Another topic that I'd like to discuss is the amount of root of cubics. Normally, we would use factorization to determine the amount. From the shape of the cubic, one side shoots of to positive infinity, and the other, negative infinity. And by continuity, it must intersect the x axis somewhere in between and create a root. Therefore, all cubics must have at least one root over the reals. But what if we want to know whether it has a second or third roots or not? We can just look at the amount and the position of the stationary point, given by eq. (4.25).

A key feature of stationary points is that they indicate where a function changes direction. If a function is increasing and then begins to decrease, there must be a stationary point marking this transition. So, if a function lacks a stationary point, and one end tends toward positive infinity while the other falls to negative infinity, we can conclude the function is always either increasing or decreasing. Once the function intersects the x-axis to form a root, it cannot turn back to create another root.

A cubic with zero stationary points ($\frac{b^2-3ac}{3a} < 0$) are such kind of function. One side goes to infinity, and the other, negative infinity. Since



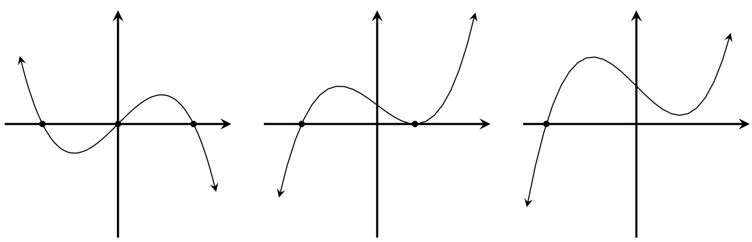
(a) $x^3 + x^2 + x$: NO VERTEX

(b) x^3: ONE VERTEX

Figure 4.2 | CUBICS WITH ZERO AND ONE VERTEX

it has no stationary point, it must only have one root. One such example is the function $f(x) = x^3 + x^2 + x$, plotted in fig. 4.2(a).

Similarly, cubics with only one stationary point ($\frac{b^2-3ac}{3a} = 0$) only have one root. Illustrated in fig. 4.2(b), the stationary point must be an inflection point, as it cannot represent a local maximum or minimum. This is due to the behavior of the function, which extends toward both positive and negative infinity. If the stationary point were a local maximum or minimum, it would turn into a parabola, and thus needs another point to reverse its direction again to be a cubic. Hence, cubic functions with only one stationary point have just one root.



(a) $x^3 + x$: 3 ROOTS **(b)** $x^3 - \frac{3}{4}x + \frac{1}{4}$: 2 ROOTS **(c)** $x^3 - x$: 1 ROOT

Figure 4.3 | TWO VERTICES CUBICS WITH VARYING AMOUNT OF ROOTS

For cubics with two vertices, one of those must be a local maximum, the other, a local minimum. The amount of roots can be directly determined by the position of these points. A cubic will have three roots if the vertices aren't on the same side of the x axis, it must have three roots, illustrated in fig. 4.3(a). If one of the vertex is at $y = 0$, then it has two roots; one of them being the vertex itself, illustrated in fig. 4.3(b). If both vertices are on the same side, then it must have only one root; illustrated in fig. 4.3(c). It is left as an exercise for the reader to find the position of vertices of the cubics given in fig. 4.3

4.2.3 Optimization problems

The method of finding local maxima, minima, and inflection points are very useful for sketching graphs. Are there any real life applications to this? The answer is in optimization.

Optimization is about finding the best solution for a problem. What's the best time to harvest the crop? What's the best placement for an air conditioner? What's the best way to make profit?, etc. These problems are all about finding the maximum values of some variables. That's exactly what our tools have been able to do: finding the maxima and minima of some functions. Thus, I'd like to discuss some applications of maxima and minima on some physical problems.

Classic maximum area problem. Given a rope length L, what's the maximum rectangular area that this rope can enclose?

To solve this problem, we start by constructing our rectangle. Drawn in fig. 4.4, the perimeter of a rectangle is given by $2(a + b) = L$.

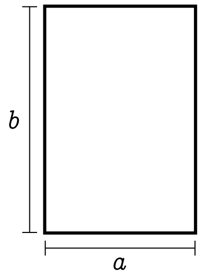


Figure 4.4 | A RECTANGLE FOR OPTIMIZATION

Without loss of generality, let side a has length l. Thus,

$$2(l+b) = L \tag{4.29}$$

$$b = \frac{L}{2} - l \tag{4.30}$$

The area of the rectangle, which is a function of l, $A(l)$ is then

$$A(l) = a \times b = l\left(\frac{L}{2} - l\right) = -l^2 + l\frac{L}{2}. \tag{4.31}$$

Since this function is a parabola, its stationary point must be the local maxima, and must also be the global maxima. This point is where our function is the highest. It can be found by using the derivative w.r.t. l.

$$\frac{\mathrm{d}}{\mathrm{d}l}A(l) = -2l + \frac{L}{2} \tag{4.32}$$

Using the same argument as earlier, we want to find the point l_0 where $\frac{\mathrm{d}}{\mathrm{d}l}A(l)\Big|_{l=l_0} = 0$:

$$0 = -2l_0 + \frac{L}{2} \tag{4.33}$$

$$l_0 = \frac{L}{4}; \tag{4.34}$$

thus, the maximum $A(l)$ is

$$A\left(\frac{L}{4}\right) = -\left(\frac{L}{4}\right)^2 + \left(\frac{L}{4}\right)\frac{L}{2} = \frac{L^2}{16} \tag{4.35}$$

Funnily enough, the maximum area that a perimeter length L can increase is also a parabola $\frac{1}{16}L^2$.

Modified maximum area problem. Given a fence length L. What's the maximum area that I can enclose, given that one of the sides doesn't need any fence?

If our rectangle has sides a and b, and one side doesn't need a fence, without loss of generality, the perimeter is $a + 2b = L$. If $a = l$, then $b = \frac{L-l}{2}$. The area of the rectangle then becomes

$$A(l) = a \times b = l\left(\frac{L-l}{2}\right) = -\frac{1}{2}l^2 + \frac{L}{2}l. \tag{4.36}$$

Using the same argument,

$$\frac{\mathrm{d}}{\mathrm{d}l}A(l) = -l + \frac{L}{2} \tag{4.37}$$

$$\frac{\mathrm{d}}{\mathrm{d}l}A(l)\bigg|_{l=l_0} = 0 = -l_0 + \frac{L}{2} \tag{4.38}$$

$$l_0 = \frac{L}{2}, \tag{4.39}$$

and

$$A(l_0) = -\frac{1}{2}\left(\frac{L}{2}\right)^2 + \frac{L}{2}\left(\frac{L}{2}\right) = \frac{3}{8}L^2. \tag{4.40}$$

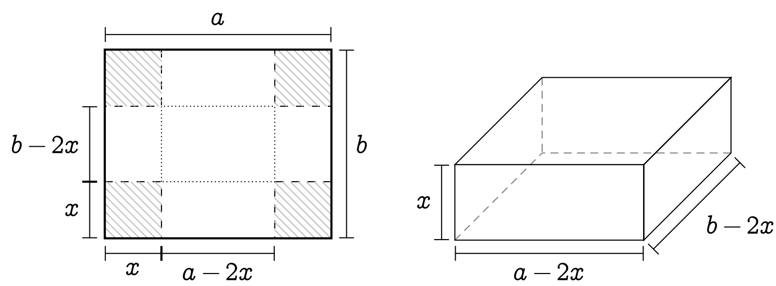
Volume of a box. Let's say you have a paper size $a \times b$, shown in fig. 4.5(a). If you want to fold it up into a box without a closing top by cutting a square size x on each corner, shown in fig. 4.5(b), find the maximum volume that the box can enclose.

Shown in fig. 4.5(a), if we cut out a square side length x from all corners, then the side length of the final box will be $a - 2x$, $b - 2x$, and x. The volume $V(x)$, is then

$$V(x) = x(a-2x)(b-2x) = 4x^3 + (-2a-2b)x^2 + (ab)x \tag{4.41}$$

This function is a cubic, and we can use eq. (4.25) to find the x coordinate of the vertex.

$$x_0 = \frac{-(-2a-2b) \pm \sqrt{(-2a-2b)^2 - 3(4)(ab)}}{3(4)} \tag{4.42}$$



(a) *Unfolded. Gray part is cut out.* **(b)** *Folded*

Figure 4.5 | Box volume optimization

$$= \frac{2a + 2b \pm \sqrt{4a^2 + 4b^2 - 4ab}}{12} \tag{4.43}$$

$$= \frac{a + b \pm \sqrt{a^2 + b^2 - ab}}{6} \tag{4.44}$$

If $a^2 + b^2 - ab = 0$, there's only one stationary point at $x_0 = \frac{a+b}{6}$, which is an inflection point. Otherwise, it must have two stationary points. Since the leading coefficient of $V(x)$ is positive, $V(x)$ must approach negative infinity when x approaches infinity. Meaning that, the highest local maximum is on the right vertex; thus, we select the negative root:

$$x_0 = \frac{a + b - \sqrt{a^2 + b^2 - ab}}{6}. \tag{4.45}$$

The full solution is left out due to arithmetical tediousness. It'd be better if you just plug in the numbers.

4.3 Tangent to a curve

In this section, I'll show you how to find a line that's tangent to a curve. But to do it the traditional way would be a bit boring, as it's quite dry and actually doesn't help us with graph sketching at all. So, I shall introduce it via the Newton-Raphson's root finding algorithm; as it's an algorithm that directly takes advantage of the line tangent to a function, and it shows up in many places. So much so that it exists inside the core of the fast square root algorithm which is implemented in every computer.

4.3.1 Newton-Raphson method

Polynomials appears everywhere. Most of the time, you're required to find its root. But it might come as a surprise that it is *impossible* to find a general formula for finding the root of polynomials with degrees higher than four.[1] So most of the time, we resort to root approximation methods.

The root approximation method which is relevant to calculus is the Newton-Raphson's root finding algorithm. I'd try to explain this algorithm has much as I can, but I think it's better if you see it.

The Newton-Raphson algorithm takes advantage of slopes descent. Let's say you want to approximate the root of $f(x)$. You put down a random guess, say x_1. If that point is a root, $f(x_1) = 0$. If it's not, the root is either above ($f(x_1) < 0$) the point or under ($f(x_1) > 0$). What you do next is that you draw a line tangent to the function $f(x)$ at $x = x_1$ and let it cross the x-axis, shown in fig. 4.6. Intuitively, the tangent line will intersect the x-axis at x_2 where x_2 should hopefully be closer to the root. Our new guess will then be $(x_2, f(x_2))$. We then repeat this method until the approximation are satisfactory.

Here, I shall use the Newton-Raphson root finding algorithm to illustrate how calculus can be applied to geometry.

[1] This is a consequence of the Galois theory, which is a profound result from abstract algebra. You'd usually learn in graduate mathematics, but if you want to just scratch the surface, the video by *Math Visualized*, on *Galois Theory Explained Simply* (6), does explain it in brief. Still, I had to watch it a couple of times before understanding it.

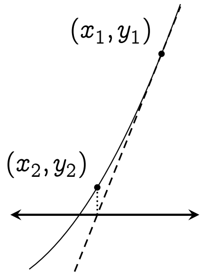


Figure 4.6 | NEWTON-RAPHSON'S METHOD ILLUSTRATED

The first part of the Newton-Raphson method is to be able to find the tangent to a curve. Let's first start with a line equation

$$y = mx + c. \tag{4.46}$$

If my first guess is on the point $(x_1, f(x_1))$, the line that passes through that point and tangent to $f(x)$ is

$$y = \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1} x + c, \tag{4.47}$$

where $m = \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1}$. And c is the intersection with the x axis.

But we don't have to actually find c. For a line tangent to a curve, we just want to move the line from the origin to the tangent point $(x_1, f(x_1))$. To move the line to the right by x_1 units, we substitute x with $x - x_1$, and to shift it up by y_1, substitute y with $y - y_1$. Thus, the line equation turns into

$$y - y_1 = (x - x_1) \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1} \tag{4.48}$$

$$y - f(x_1) = (x - x_1) \left.\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right|_{x=x_1}. \tag{4.49}$$

For simplicity of notation, I shall let $\mathrm{D}_x f(x_1)$ represent $\frac{\mathrm{d}f(x)}{\mathrm{d}x}$:

$$y - f(x_1) = (x - x_1)\, \mathrm{D}_x f(x_1). \tag{4.50}$$

Then, Newton-Raphson method tells us that the intersection between the tangent line and the x axis is what our next guess, x_2, should be. It can be easily evaluated by setting $y = 0$.

$$0 - f(x_1) = (x_2 - x_1)\,\mathrm{D}_x f(x_1) \tag{4.51}$$

$$x_2 = x_1 - \frac{f(x_1)}{\mathrm{D}_x f(x_1)} \tag{4.52}$$

Generally, we can repeat this forever. Thus,

$$x_{n+1} = x_n - \frac{f(x_n)}{\mathrm{D}_x f(x_n)} \tag{4.53}$$

a recurrence relation for finding roots of functions. Notice that we don't even need to know the y value for this algorithm to work. And that's the beauty of it.

Let's use this root finding method to approximate $\sqrt{2}$, which is the root of the polynomial $x^2 - 2 = 0$. Since the derivative of $x^2 - 2$ is $2x$, Newton-Raphson method (eq. (4.53)) gives

$$x_{n+1} = x_n - \frac{x_n^2 - 2}{2x_n} \tag{4.54}$$

$$= \frac{x_n^2 + 2}{2x_n}. \tag{4.55}$$

Since $\sqrt{2} \approx 1.414213\ldots$, let's use $x_1 = 1$ as our first guess.

$$x_2 = \frac{x_1^2 + 2}{2x_1} \tag{4.56}$$

$$= \frac{1^2 + 2}{2(1)} = 1.5, \tag{4.57}$$

which is closer to 1.41 than our first guess. Next,

$$x_3 = \frac{x_2^2 + 2}{2x_2} \tag{4.58}$$

$$= \frac{1.5^2 + 2}{2(1.5)} = 1.4166\ldots, \tag{4.59}$$

which is off by just a 0.2% error. By the forth iteration, we get $1.414215\ldots$, which is off by a mere 0.0002%. And we can stop here.

4.3.2 Numerical version of the Newton-Raphson method

For a function that you can't find a derivative, or you're just too lazy to find the derivative explicitly, you can turn the Newton-Raphson method into a numerical algorithm via the definition of derivative. From eq. (4.53),

$$x_{n+1} = x_n - \frac{f(x_n)}{\lim_{h\to 0}\frac{f(x_n+h)-f(x_n)}{h}} \tag{4.60}$$

$$= x_n - \lim_{h\to 0}\frac{f(x_n)h}{f(x_n+h)-f(x_n)}, \tag{4.61}$$

where h can be set to any arbitrarily small values, e.g., 0.01. If you have some experiences with coding, this can be easily implemented by using a `for` loop. Here's a simple python snippet that I've written to do just that. Just put in the function in `def f(x):` block, put the desired iteration in `iterationCount`, your first guess in `firstGuess`, and the value of h in `derivativeStep`. If you have some experiences, I highly encourage you to tinker around with this algorithm, and you might find something beautiful.

```
import matplotlib.pyplot as plt
import math

def f(x):  # Insert function here
    return x**3 - 2 * x + 1

iterationCount = 10  # Amount of iteration
firstGuess = -2  # First guess
derivativeStep = 0.01  # For calculating the method of
↪  increments
```

```python
approximationList = [float(firstGuess)]  # List of approximations
for i in range(iterationCount):
    diff = (
        f(approximationList[i] + derivativeStep) - f(approximationList[i])
    ) / derivativeStep  # Finding the derivative at a certain point
    nextGuess = (
        approximationList[i] - f(approximationList[i]) / diff
    )  # Newton-Raphson's method
    approximationList.append(nextGuess)  # Add next guess to the list of approximations

print(approximationList)  # Print out the list of approximations
fig, axes = plt.subplots()  # Initiate plot
axes.plot(list(range(iterationCount + 1)), approximationList)  # Plotting
axes.set_xlabel("Iteration")  # Setting the x-axis label
axes.set_ylabel("Guess")  # Setting the y-axis label
plt.show()  # Showing the plot
```

4.4 Normal lines

Alongside the tangent line, there is another line that we're interested in: the normal. A normal line is a line that's perpendicular to the tangent. By analytical geometry, two lines are perpendicular *iff* the product of their slope is -1. Let the slope of the tangent be m_T and the normal be m_N, then

$$m_T \times m_N = -1 \tag{4.62}$$

$$m_N = -\frac{1}{m_T}. \tag{4.63}$$

Since the slope of the tangent at a point $(x_0, f(x_0))$ is $\mathrm{D}_x f(x_0)$,

$$m_N = -\frac{1}{\mathrm{D}_x f(x_0)}. \tag{4.64}$$

The normal line equation can then be formed by translating the line with slope m_N to the point $(x_0, f(x_0))$:

$$y - f(x_0) = -\frac{x - x_0}{\mathrm{D}_x f(x_0)}. \tag{4.65}$$

4.5 Arc length of functions

Before we wrap off this chapter, here's a fun geometrical problem to think about. Given a function $f(x)$, what's the length of the line drawn by the function from a to b? Illustrated in fig. 4.7, the total arc length s can be subdivided into smaller arc lengths $\mathrm{d}s$ that we can sum up. How? The answer is: integrals.

The total arc length s of the function $f(x)$ from a to b is just

$$\int_a^b \mathrm{d}s \tag{4.66}$$

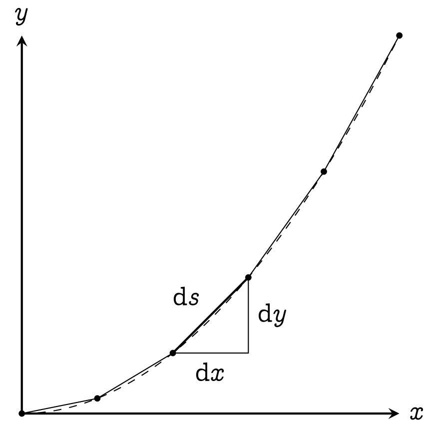


Figure 4.7 | ARC LENGTH CALCULATION

By the Pythagorean theorem $\mathrm{d}s = \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2}$. Can we plug this directly into the integral? Of course, we can:

$$s = \int_{x=a}^{x=b} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2}. \tag{4.67}$$

Factorizing $\mathrm{d}x$ out of the square root would yield an integral

$$\int_{x=a}^{x=b} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = \int_{x=a}^{x=b} \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x \tag{4.68}$$

$$= \int_{x=a}^{x=b} \sqrt{1 + \left(\frac{\mathrm{d}f(x)}{\mathrm{d}x}\right)^2}\,\mathrm{d}x. \tag{4.69}$$

Simply plug in the derivative of $f(x)$ and evaluate the integral, and you would get the arc length.

CHAPTER 5

Calculus, trigonometry, and oscillations

Prerequisites: *basic trigonometry.*

5.1 Invitation: oscillation and trigonometry

Trigonometry is already pretty useful on its own, as it connects the angles of a triangle to its side. But as you study further, trigonometric functions appears in many unexpected places, especially in oscillatory systems. Whether it's a motion of a spring, the synchronized blinks of fireflies, or even dynamics of love; they're somehow all modeled using trigonometric functions. Sure, the usual sine and cosine is already oscillating up and down with a definite pattern. But there are so many oscillatory functions in this world! Why these two? Is it baked into the nature of oscillation? You'll find out in this chapter.

Before moving on, I'd like to point out the whole goal of the rest of the chapters in this volume. After this chapter, the foundation of calculus are in a sense, complete. From here on out, we shall be focusing on various differential equations that describe nature; starting with this chapter. But honestly, I have no idea how I could introduce you to oscillations without studying how trigonometric functions interact with calculus first. So, in the first few sections, it's just going to be mathematics which, I tried to keep it as short but yet as complete as possible.

5.1.1 Newton's fluxion notation

Here's the notation of derivative that will be used throughout the chapter, called the **Newton's fluxion notation** or, **dot notation**. *This notation is used only when the derivative is taken w.r.t. time*. It places a dot over the variables, e.g., the first derivative of position r w.r.t. time is $\dot{r}$.

Higher derivatives are written with more dots, e.g., the second derivative of position r w.r.t. is $\ddot{r}$. The third derivative is $\dddot{r}$, forth derivative, $\ddddot{r}$ and so on.

5.2 Derivative of trigonometric functions

5.2.1 Two fundamental functions: sine and cosine

The derivative of sine is cosine. We'll prove it later, but the geometrical interpretation is that the derivative of sine oscillates along off-sync with the function itself. When $x = 0$, there's a 45° rising slope. At $x = \frac{\pi}{2}$, the slope flattens down to zero. Later at $x = \pi$, the slope is going 45° down, etc. If we plot the slope of the sine function at each point (shown as dots in fig. 5.1), it resembles a cosine curve.

To prove this, let's start with the definition of the derivative given in eq. (1.10),

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \lim_{h\to 0}\frac{\sin(x+h)-\sin(x)}{h}. \tag{5.1}$$

Using the sum of sines ($\sin(x+h) = \sin(x)\cos(h) + \cos(x)\sin(h)$),

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \lim_{h\to 0}\frac{\sin(x)\cos(h)+\cos(x)\sin(h)-\sin(x)}{h} \tag{5.2}$$

When $h \to 0$, $\cos(h) \to 1$

$$= \lim_{h\to 0}\frac{\sin(x)+\cos(x)\sin(h)-\sin(x)}{h} \tag{5.3}$$

$$= \cos(x)\lim_{h\to 0}\frac{\sin(h)}{h}. \tag{5.4}$$

For small angles, $\sin(h) \approx h$, therefore

$$= \cos(x)\lim_{h\to 0}\frac{h}{h} = \cos(x). \tag{5.5}$$

In conclusion, the derivative of sine is just the cosine just as we predicted with the graph.

Similarly, the derivative of cosine can also be found using the method of increments and using the sum of cosines formula $\cos(x+h) = \cos(x)\cos(h) - \sin(x)\sin(h)$

$$\frac{\mathrm{d}}{\mathrm{d}x}(\cos(x)) = \lim_{h\to 0}\frac{\cos(x+h)-\cos(x)}{h} \tag{5.6}$$

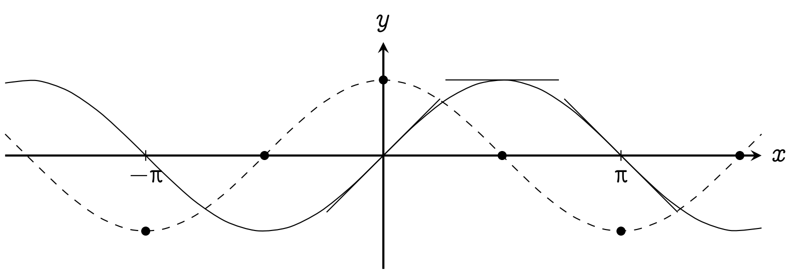


Figure 5.1 | The relation between sine (in black) and cosine (in dashed)

$$= \lim_{h \to 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h} \tag{5.7}$$

When $h \to 0$, $\cos(h) = 1$

$$= \lim_{h \to 0} \frac{\cos(x) - \cos(x) - \sin(x)\sin(h)}{h} \tag{5.8}$$

$$= \lim_{h \to 0} \frac{-\sin(x)\sin(h)}{h} \tag{5.9}$$

For small angles of h, $\sin(h) \approx h$

$$= \lim_{h \to 0} \left(-\sin(x)\right) = -\sin(x) \tag{5.10}$$

I.e., the derivative of cosine is the negative of sine. From those two alone, we can form a set of formulas for differentiating sines and cosines:

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\left(\sin(x)\right) &= \cos(x) \\
\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\sin(x)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\cos(x)\right) = -\sin(x) \\
\frac{\mathrm{d}^3}{\mathrm{d}x^3}\left(\sin(x)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(-\sin(x)\right) = -\cos(x) \\
\frac{\mathrm{d}^4}{\mathrm{d}x^4}\left(\sin(x)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(-\cos(x)\right) = \sin(x)
\end{aligned} \tag{5.11}$$

We can see that the derivative of these two functions cycles in four: $\sin(x), \cos(x), -\sin(x), -\cos(x)$. From these alone, we can use the rules developed in chapter 3, e.g., the chain rule, power rule, etc., in order to find the derivative of other trigonometric functions.

5.2.2 Derivative of other trigonometric functions

Note that you can continue reading the next section straightaway. This section serves as a reference. But for those who are still interested, I'm glad to have you here. This is one of the places that uses the chain rule, product rule, and power rule a lot. It's a good exercise. Without further ado, let's start with the cosecant ($\csc(x) = \frac{1}{\sin(x)}$).

$$\frac{\mathrm{d}}{\mathrm{d}x}\csc(x) = \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\sin(x)} \tag{5.12}$$

$$\begin{aligned}
&= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{u}\cdot\frac{\mathrm{d}u}{\mathrm{d}u} && \text{Chain rule: } \sin(x) = u && (5.13)\\
&= \frac{\mathrm{d}}{\mathrm{d}u}u^{-1}\cdot\frac{\mathrm{d}u}{\mathrm{d}x} && && (5.14)\\
&= -1u^{-2}\cdot\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) && \text{Power rule, } u = \sin(x) && (5.15)\\
&= -\csc(x)\cot(x) && \cot(x) = \frac{\cos(x)}{\sin(x)}. && (5.16)
\end{aligned}$$

And the secant function ($\sec(x) = \frac{1}{\cos(x)}$),

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\sec(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{\cos(x)}\right) && && (5.17)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{1}{u}\right)\cdot\frac{\mathrm{d}u}{\mathrm{d}u} && \text{Chain rule: } \cos(x) = u && (5.18)\\
&= \frac{\mathrm{d}}{\mathrm{d}u}u^{-1}\cdot\frac{\mathrm{d}u}{\mathrm{d}x} && && (5.19)\\
&= -1u^{-2}\cdot\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) && \text{Power rule, } u = \cos(x) && (5.20)\\
&= -\frac{1}{\cos^2(x)}(-\sin(x)) && \frac{\mathrm{d}}{\mathrm{d}x}\cos(x) = -\sin(x) && (5.21)\\
&= \sec(x)\tan(x). && && (5.22)
\end{aligned}$$

For the tangent function ($\tan(x) = \frac{\sin(x)}{\cos(x)}$), we use the product rule and the results from earlier

$$\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\tan(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{\sin(x)}{\cos(x)} && (5.23)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\sin(x)\sec(x) && (5.24)\\
&= \sin(x)\frac{\mathrm{d}}{\mathrm{d}x}\sec(x) + \sec(x)\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) && (5.25)\\
&= \sin(x)\sec(x)\tan(x) + \sec(x)\cos(x) && (5.26)\\
&= \sin(x)\frac{1}{\cos(x)}\frac{\sin(x)}{\cos(x)} + \frac{1}{\cos(x)}\cos(x) && (5.27)\\
&= \left(\frac{\sin(x)}{\cos(x)}\right)^2 + 1 = \tan^2(x) + 1 = \sec^2(x) && (5.28)
\end{aligned}$$

The Pythagorean identity is used in the last line to convert $1 + \tan^2(x)$ to

$\sec^2(x)$. Of course, we cannot miss its reciprocal function: $\cot(x) = \frac{1}{\tan(x)}$.

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}x}\cot(x) &= \frac{\mathrm{d}}{\mathrm{d}x}\frac{1}{\tan(x)} && (5.29)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\frac{\cos(x)}{\sin(x)} && (5.30)\\
&= \frac{\mathrm{d}}{\mathrm{d}x}\cos(x)\csc(x) && (5.31)\\
&= \cos(x)\frac{\mathrm{d}}{\mathrm{d}x}\csc(x) + \csc(x)\frac{\mathrm{d}}{\mathrm{d}x}\cos(x) && (5.32)\\
&= \cos(x)\big(-\csc(x)\cot(x)\big) + \csc(x)\big(-\sin(x)\big) && (5.33)\\
&= -\cos(x)\frac{1}{\sin(x)}\frac{\cos(x)}{\sin(x)} - \frac{1}{\sin(x)}\sin(x) && (5.34)\\
&= -\big(\cot^2(x) + 1\big) = -\csc^2(x), && (5.35)
\end{aligned}
$$

where we've again used the Pythagorean identity $\cot^2(x) + 1 = -\csc^2(x)$.

Altogether, I've summarized all the derivatives of trigonometric functions into table 5.1.

$f(x)$	$\frac{\mathrm{d}}{\mathrm{d}x}f(x)$
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$\tan(x)$	$\sec^2(x)$
$\csc(x)$	$-\csc(x)\cot(x)$
$\sec(x)$	$\sec(x)\tan(x)$
$\cot(x)$	$-\csc^2(x)$

Table 5.1 | THE DERIVATIVES OF TRIGONOMETRIC FUNCTIONS

5.3 Power series of trigonometric functions

The approximation method that I'd be discussing are commonly taught to high schoolers with the name **small angle approximation**, i.e., for $\theta \to 0$,

$$\sin(\theta) \approx \tan(\theta) \approx \theta \quad \text{and} \quad \cos(\theta) \approx 1 - \frac{\theta^2}{2} \tag{5.36}$$

These approximations are very useful for calculating limits, e.g.,[1]

$$\lim_{\theta \to 0} \frac{\sin(n\theta)}{\theta} = \lim_{\theta \to 0} \frac{n\theta}{\theta} = n. \tag{5.37}$$

Here, I want to focus on where these approximations really comes from: the power series development of trigonometric functions.

5.3.1 Power series of sine

Assume that the sine function can be written as a power series

$$\sin(x) = s_0 + s_1 x^1 + s_2 x^2 + s_3 x^3 + \ldots \tag{5.38}$$

Because $\sin(0) = 0$,

$$\sin(0) = s_0 + s_1(0)^1 + s_2(0)^2 + s_3(0)^3 + \ldots \tag{5.39}$$

$$s_0 = 0; \tag{5.40}$$

Because sine is an odd function, $\sin(-x) = -\sin(x)$; thus,

$$\begin{aligned} s_1(-x)^1 &+ s_2(-x)^2 \\ &+ s_3(-x)^3 + s_4(-x)^4 + \ldots = -s_1x^1 - s_2x^2 \\ &\qquad - s_3x^3 - s_4x^4 + \ldots \end{aligned} \tag{5.41}$$

$$\begin{aligned} -s_1x^1 &+ s_2x^2 \\ &- s_3x^3 + s_4x^4 - \ldots = -s_1x^1 - s_2x^2 - \\ &\qquad s_3x^3 - s_4x^4 - \ldots \end{aligned} \tag{5.42}$$

[1] Normally these limits are evaluated using the squeeze theorem. However, I don't want the mathematical intricacies to disrupt the flow of our journey right now. The full theorem will be discussed later at the end of the book.

In the L.H.S., the sign alternates between negative and positive. In order for the L.H.S. to match the R.H.S., all the positive terms must vanish, leaving only the matching negative terms; therefore,

$$\sin(x) = s_1x^1 + s_3x^3 + s_5x^5 + \ldots . \tag{5.43}$$

This is what we mean by "sine is an odd function": there are only odd powers in its power series.

To find s_1, take the derivative of sine

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \cos(x) = s_1 + 3s_3x^2 + 5s_5x^4 + \ldots , \tag{5.44}$$

then substitute $x = 0$

$$\cos(0) = 1 = s_1 + 3s_3(0)^2 + 5s_5(0)^4 + \ldots \tag{5.45}$$

$$1 = s_1. \tag{5.46}$$

We can then take the derivative of sine again to get a recurrence relation.

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\sin(x) = 3\cdot 2s_3x + 5\cdot 4s_5x^3 + 7\cdot 6s_7x^5 + \ldots \tag{5.47}$$

$$-\sin(x) = \frac{3!}{1!}s_3x + \frac{5!}{3!}s_5x^3 + \frac{7!}{5!}s_7x^5 + \ldots \tag{5.48}$$

$$-s_1x - s_3x^3 - s_5x_5 - \ldots = \frac{3!}{1!}s_3x + \frac{5!}{3!}s_5x^3 + \frac{7!}{5!}s_7x^5 + \ldots \tag{5.49}$$

By matching terms with the same order,

$$\begin{aligned} -s_1 &= \frac{3!}{1!}s_3x, \\ -s_3 &= \frac{5!}{3!}s_5x, \\ -s_5 &= \frac{7!}{5!}s_7x. \\ &\vdots \end{aligned} \tag{5.50}$$

From $s_1 = 1$, we get that $s_3 = -\frac{1}{3!}$, $s_5 = -\frac{1}{5!}$, $s_7 = -\frac{1}{7!}$ etc. The negative sign alternates every term. We can then summarize the whole thing as

$$\sin(x) = \sum_{n=0}^{\infty}\frac{(-1)^{2n+1}}{(2n+1)!}x^{2n+1} \tag{5.51}$$

$$= x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \dots, \tag{5.52}$$

which is the polynomial expansion of the sine function.

5.3.2 Power series of cosine

The power series of cosine can be developed using the same method as discussed earlier. But notice that the derivative of sine is cosine. Hence, we can just take the derivative of the sine's power series once to get the cosine's power series.

$$\frac{\mathrm{d}}{\mathrm{d}x}\sin(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7\right) \tag{5.53}$$

$$\cos(x) = 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 + \cdots = \sum_{n=0}^{\infty}\frac{(-1)^{2n}}{(2n)!}x^{2n} \tag{5.54}$$

5.3.3 Power series of other trigonometric functions

The same method could be used to find the power series expansion of secant, cosecant, tangent, etc. However, their power series doesn't actually represent the function because the other functions are all ratios of sine and cosines. To see what I mean, consider the behavior of the tangent function.

The tangent function is defined as $\tan\theta = \frac{\sin\theta}{\cos\theta}$. From this definition, $\tan\theta$ approaches infinity whenever $\cos\theta$ approaches zero, which is at $n\pi$ where $n \in \mathbb{Z}$[2]. As seen, there are infinitely many places in which the function approaches infinity. However, polynomials only have two infinities: at $x \to \infty$, and $x \to -\infty$; therefore, any polynomial expansion can't possibly represent the tangent function. The same logic also works for other trigonometric functions.

[2] $\mathbb{Z}$ is the set of integers

5.3.4 Approximation of trigonometric functions

One way a function can be approximated is by truncating its power series. To see why this is viable, consider the limit

$$\lim_{x\to 0} \sin(x) = \lim_{x\to 0}\left(x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \ldots\right) \tag{5.55}$$

When $x \to 0$, all the higher order degrees term vanishes; thus,

$$\lim_{x\to 0} \sin(x) = x. \tag{5.56}$$

It tells us that when $x \approx 0$, the sine function behaves like a linear function. Illustrated in fig. 5.2,

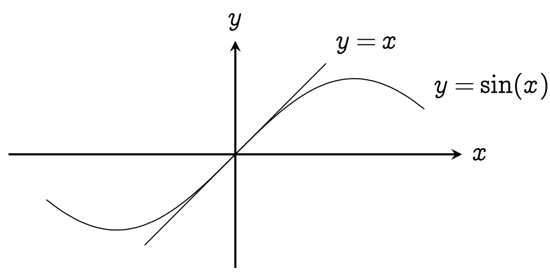


Figure 5.2 | COMPARISON OF $y = \sin(x)$ AND $y = x$ NEAR $x = 0$

When x gets larger, the higher order terms in the power series contributes more and more, so we need to include them. An excellent approximation for the sine function when $x \in [-\pi, \pi]$ is a truncation at the fifth term:

$$\sin(x) \approx x - \frac{1}{3!}x^3 + \frac{1}{5!}x^5 - \frac{1}{7!}x^7 + \frac{1}{9!}x^9, \tag{5.57}$$

plotted in fig. 5.3. It is extremely accurate in that range. For cosine,

$$\lim_{x\to 0} \cos(x) = \lim_{x\to 0}\left(1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots\right) \approx 1 - \frac{x^2}{2!} \approx 1. \tag{5.58}$$

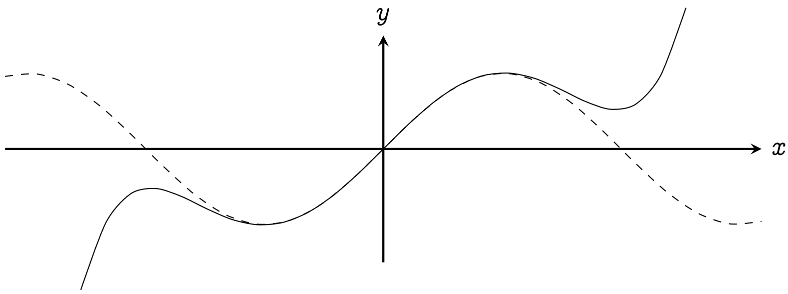


Figure 5.3 | An approximation of sine by truncating its power series at the fifth term.

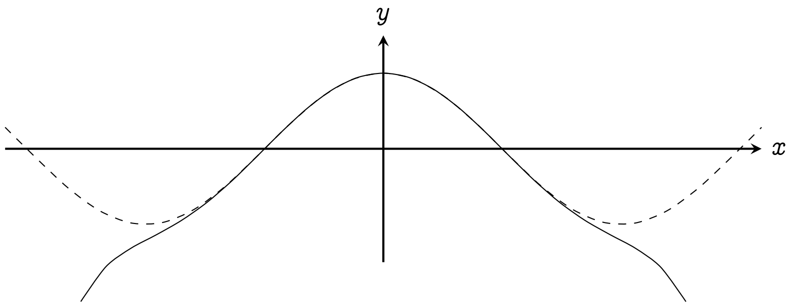


Figure 5.4 | An approximation of cosine by truncating its power series at the fourth term

An acceptable approximation around zero is 1 or $1 - \frac{x^2}{2}$. Similarly, an excellent approximation when $x \in [-\pi, \pi]$ is also a truncation at the fourth term:

$$\cos(x) \approx 1 - \frac{1}{2!}x^2 + \frac{1}{4!}x^4 - \frac{1}{6!}x^6 \tag{5.59}$$

plotted in fig. 5.4

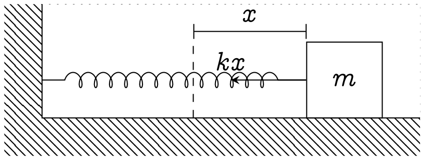


Figure 5.5 | SETUP OF THE HARMONIC OSCILLATOR

5.4 The harmonic oscillator

Finally, we're ready to tackle our first oscillatory system: the harmonic oscillator. Illustrated in fig. 5.5, consider a mass m connected to a spring. The point that the spring is neither stretched nor compressed is called the **equilibrium position**, drawn as the dashed line. The force that the spring acts on the mass depends on the displacement from the equilibrium position x:

$$F = -kx, \tag{5.60}$$

where k is the stiffness of the spring. Higher k represents a stiffer spring, and lower k represents a looser spring. Newton's second law reads

$$-kx = m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} \tag{5.61}$$

$$-\frac{k}{m}x = \frac{\mathrm{d}^2x}{\mathrm{d}t^2}. \tag{5.62}$$

We're interested in the solution of eq. (5.62) given an initial condition that at time $t = 0$, the mass is distance A away from the equilibrium position, and it's released with zero initial velocity, i.e.,

$$x(t=0) = A \quad \text{and} \quad \dot{x}(t=0) = 0. \tag{5.63}$$

5.4.1 Your first trigonometric substitution

The equation that we got, eq. (5.62), is an inseparable second order differential equation. It is extremely hard to work with. But by exploiting the chain rule, we can reduce it into a first order differential equation.

$$\frac{\mathrm{d}^2x}{\mathrm{d}t^2}=\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)=\frac{\mathrm{d}\dot{x}}{\mathrm{d}t}=\frac{\mathrm{d}\dot{x}}{\mathrm{d}x}\frac{\mathrm{d}x}{\mathrm{d}t}=\dot{x}\frac{\mathrm{d}\dot{x}}{\mathrm{d}x}. \tag{5.64}$$

Equation (5.62) then becomes

$$-\frac{k}{m}x=\dot{x}\frac{\mathrm{d}\dot{x}}{\mathrm{d}x} \tag{5.65}$$

$$-\frac{k}{m}x\,\mathrm{d}x=\dot{x}\,\mathrm{d}\dot{x} \tag{5.66}$$

$$-\frac{k}{m}\int x\,\mathrm{d}x=\int\dot{x}\,\mathrm{d}\dot{x} \tag{5.67}$$

$$-\frac{k}{m}\frac{x^2}{2}=\frac{\dot{x}^2}{2}+C \tag{5.68}$$

$$-\frac{k}{m}x^2=\dot{x}^2+C. \tag{5.69}$$

To find C, substitute in the initial condition eq. (5.63).

$$-\frac{k}{m}(A)^2=(0)^2+C \tag{5.70}$$

$$C=-\frac{k}{m}A^2. \tag{5.71}$$

Thus,

$$-\frac{k}{m}x^2=\dot{x}^2-\frac{k}{m}A^2 \tag{5.72}$$

$$\dot{x}^2=\frac{k}{m}\left(A^2-x^2\right) \tag{5.73}$$

$$\dot{x}=\sqrt{\frac{k}{m}}\sqrt{A^2-x^2}. \tag{5.74}$$

Because I'm too lazy to write square roots, let's use $\omega=\sqrt{k\,/\,m}$; thus,

$$\dot{x}=\omega\sqrt{A^2-x^2} \tag{5.75}$$

$$\frac{\mathrm{d}x}{\mathrm{d}t}=\omega\sqrt{A^2-x^2} \tag{5.76}$$

$$\omega\,\mathrm{d}t = \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x \tag{5.77}$$

$$\omega \int \mathrm{d}t = \int \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x. \tag{5.78}$$

The L.H.S. directly evaluates to ωt. But on the R.H.S., power rule substitution wouldn't work because there's a polynomial with degree two the root, but no terms with lower degree outside. If $u = A^2 - x^2$, we'd be going in loops. So how do we evaluate this?

The answer is, you can't do anything about it if we're going to use the methods so far. It's time for a new tool: **trigonometric substitution**.

We want to take advantage of the Pythagorean identity $1 - \sin^2\theta = \cos^2\theta$ to bring $A^2 - x^2$ out of the root. Substitute in $x = A\sin\theta$ and see what happens.

$$\int \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{A^2 - A^2\sin^2\theta}}\,\mathrm{d}x \tag{5.79}$$

$$= \frac{1}{A}\int \frac{1}{\sqrt{1 - \sin^2\theta}}\,\mathrm{d}x \tag{5.80}$$

$$= \frac{1}{A}\int \frac{1}{\sqrt{\cos^2\theta}}\,\mathrm{d}x = \frac{1}{A}\int \frac{1}{\cos\theta}\,\mathrm{d}x \tag{5.81}$$

Taking care of the $\mathrm{d}x$; if $x = A\sin\theta$, then

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = A\frac{\mathrm{d}\sin\theta}{\mathrm{d}\theta} \tag{5.82}$$

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = A\cos\theta \tag{5.83}$$

$$\mathrm{d}x = A\cos\theta\,\mathrm{d}\theta. \tag{5.84}$$

Substituting back into the integral gives

$$\frac{1}{A}\int \frac{1}{\cos\theta}\big(A\cos\theta\,\mathrm{d}\theta\big) = \int \mathrm{d}\theta = \theta + C. \tag{5.85}$$

Because $x = A\sin\theta$, $\theta = \arcsin\big(x\,/\,A\big)$

$$\int \frac{1}{\sqrt{A^2 - x^2}}\,\mathrm{d}x = \arcsin\left(\frac{x}{A}\right) + C. \tag{5.86}$$

Thus, eq. (5.78) becomes

$$\omega t = \arcsin\left(\frac{x}{A}\right) + C. \tag{5.87}$$

Then, substitute in the initial condition (eq. (5.63)) to find C:

$$\omega(0) = \arcsin\left(\frac{A}{A}\right) + C \tag{5.88}$$

$$C = \frac{\pi}{2}. \tag{5.89}$$

Thus,

$$\omega t = \arcsin\left(\frac{x}{A}\right) + \frac{\pi}{2} \tag{5.90}$$

$$\arcsin\left(\frac{x}{A}\right) = \omega t + \frac{\pi}{2} \tag{5.91}$$

$$x = A\sin\left(\omega t + \frac{\pi}{2}\right). \tag{5.92}$$

Since $\sin(\theta + \frac{\pi}{2}) = \cos(\theta)$,

$$x(t) = A\cos(\omega t) \tag{5.93}$$

which is the solution of the differential equation. It represents where the mass is relative to the equilibrium point over time. Taking the derivative of this once,

$$\dot{x}(t) = A\frac{\mathrm{d}\cos(\omega t)}{\mathrm{d}t} \tag{5.94}$$

Let $u = \omega t$. By the chain rule,

$$= A\frac{\mathrm{d}\cos(u)}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}t} \tag{5.95}$$

$$= -A\sin(u)\frac{\mathrm{d}}{\mathrm{d}t}(\omega t) \tag{5.96}$$

$$= -\omega A\sin(\omega t), \tag{5.97}$$

we get the velocity as a function w.r.t. time. Similarly, taking the derivative of velocity again gives the acceleration

$$\ddot{x} = \frac{\mathrm{d}\dot{x}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(-\omega A\sin(\omega t)\right) = -\omega^2 A\sin(\omega t). \tag{5.98}$$

5.4.2 The power series method

Perhaps a faster method to solve eq. (5.62) is to just assume that the solution can be expanded in terms of power series

$$x(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3 + \dots \,. \tag{5.99}$$

Consequently,

$$\dot{x}(t) = a_1 + 2a_2 t + 3a_3 t^2 + 4a^4 t_3 + \dots \tag{5.100}$$

$$\ddot{x}(t) = 2a_2 + 3 \cdot 2a_3 t + 4 \cdot 3a_4 t^2 + 5 \cdot 4a_5 t_3 + \dots \tag{5.101}$$

The initial condition given in eq. (5.63) ($x(t = 0) = A$, $\dot{x}(t = 0) = 0$) can then be used to find a_0 and a_1.

$$x(t = 0) = a_0 + a_1(0) + a_2(0)^2 + a_3(0)^3 + \dots \tag{5.102}$$

$$a_0 = A \tag{5.103}$$

and from eq. (5.100),

$$\dot{x}(t = 0) = a_1 + 2a_2(0) + 3a_3(0)^2 + 4a_4(0)^3 + \dots \tag{5.104}$$

$$a_1 = 0 \tag{5.105}$$

The rest of the terms can be found by using the differential equation itself. First, rearrange the equation into

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \omega^2 x = 0. \tag{5.106}$$

Then, substitute in eqs. (5.99) and (5.101)

$$0 = \frac{\mathrm{d}^2 x}{\mathrm{d}t^2} + \omega^2 x \tag{5.107}$$

$$0 = \left(2a_2 + 3 \cdot 2a_3 t + 4 \cdot 3a_4 t^2 + 5 \cdot 4a_5 t^3 + 6 \cdot 5a_6 t^4 + 6 \cdot 7a_7 t^5 + \dots\right)$$
$$+ \left(\omega^2 A + a_2 \omega^2 t^2 + a_3 \omega^2 t^3 + a_4 \omega^2 t^4 + a_5 \omega^2 t^5 \dots\right)$$

$$
\begin{aligned}
0 = &\left(2a_2 + \omega^2 A\right) \\
&+ (3 \cdot 2a_3)t \\
&+ (4 \cdot 3a_4 + \omega^2 a_2)t^2 \\
&+ (5 \cdot 4a_5 + \omega^2 a_3)t^3 \\
&+ (6 \cdot 5a_6 + \omega^2 a_4)t^4 \\
&+ (7 \cdot 6a_7 + \omega^2 a_5)t^5 + \ldots
\end{aligned}
\tag{5.108}
$$

For the R.H.S. to be 0, all terms must vanish; thus, we get an infinite series of equations to be evaluated

$$
\begin{aligned}
0 &= \left(2a_2 + \omega^2 A\right) \\
0 &= (3 \cdot 2a_3) \\
0 &= (4 \cdot 3a_4 + \omega^2 a_2) \\
0 &= (5 \cdot 4a_5 + \omega^2 a_3) \\
0 &= (6 \cdot 5a_6 + \omega^2 a_4) \\
0 &= (7 \cdot 6a_7 + \omega^2 a_5) \\
&\vdots
\end{aligned}
\tag{5.109}
$$

Starting from the second one: $0 = 3 \cdot 2a_3$ means that $a_3 = 0$. This eliminates every term that includes a_3. The fourth equation says $5 \cdot 4a_5 + \omega^2 a_3 = 0$, thus $a_5 = 0$. This pattern continues, eliminating every odd terms in the series. We're then left with

$$
\begin{aligned}
0 &= 2a_2 + \omega^2 A \\
0 &= 4 \cdot 3a_4 + \omega^2 a_2 \\
0 &= 6 \cdot 5a_6 + \omega^2 a_4 \\
0 &= 8 \cdot 7a_8 + \omega^2 a_6 \\
&\vdots
\end{aligned}
\tag{5.110}
$$

Starting from the first one, we get $a_2 = -\frac{\omega^2}{2}A$. The second equation gives $a_4 = \frac{\omega^4}{4!}A$. The third gives $a_6 = \frac{\omega^6}{6!}A$. This pattern of $a_{2n} = \frac{\omega^{2n}}{2n}A$ continues on forever. The power series expansion for $x(t)$ gives

$$x(t) = a_0 + \cancelto{0}{a_1} + a_2 t^2 + \cancelto{0}{a_3} t^3 + a_4 t^4 + \cancelto{0}{a_5} t^5 + a_6 t^6 + \dots \tag{5.111}$$

$$= A + A\frac{\omega^2}{2!}t^2 + A\frac{\omega^4}{4!}t^4 + A\frac{\omega^6}{6!}t^6 + \dots \tag{5.112}$$

$$= A\left(\frac{(\omega t)^2}{2!} + \frac{(\omega t)^4}{4!} + \frac{(\omega t)^6}{6!} + \dots\right) \tag{5.113}$$

Looks familiar yet? *This*, the result that we got is exactly the power series expansion for cosine (eq. (5.54)); thus,

$$x(t) = A\cos(\omega t). \tag{5.114}$$

5.4.3 Physical aspects of the harmonic oscillator

Now that we have solved the harmonic oscillator, let's bring our attention to the qualitative description of the system.

The mass is released with zero velocity at $x = A$. As the spring tries to restore its equilibrium, it accelerates the mass towards the equilibrium position. The further the mass is from the equilibrium, the stronger the spring pulls. Initially, the spring pulls very hard on the mass, causing it to accumulate much velocity. But as it approaches the equilibrium position, the spring's force diminishes and there is barely any force left to stop the mass, so it overshoots the other side, compressing the spring. This oscillatory cycle continues on forever. Intuitively, we can predict that the mass will have the highest velocity and lowest acceleration as it directly passes through the equilibrium; and, the lowest velocity and the highest acceleration when the spring is fully stretched or compressed. But does the mathematics agree with this?

In fig. 5.6, the position function (eq. (5.93)) is plotted in thick lines, and the velocity function (eq. (5.97)) are plotted in dashed lines.

The furthest point that a mass could be from a spring is at $x = \pm A$. This is called the oscillation **amplitude**. Mathematically, this represents the local maxima and minima of the function, which can be obtained by taking the first derivative and setting it to zero (section 4.2). The derivative of the position w.r.t. time is just the velocity (eq. (5.97)), and it reaches zero whenever $\sin(\omega t)$ is zero

$$\sin(\omega t) = 0 \tag{5.115}$$

$$\omega t = \arcsin(0) + 2n\pi \qquad n \in \mathbb{Z} \tag{5.116}$$

$$t = \frac{(2n+1)\pi}{\omega}. \tag{5.117}$$

Another way to derive this fact is to just notice that the range of the cosine function only extends from -1 to 1; therefore, the highest that $A\cos(\omega t)$ can go is just A. So, another equation that we can form is

$$\cos(\omega t) = 1 \tag{5.118}$$

$$\omega t = \arccos(1) + 2n\pi \qquad n \in \mathbb{Z} \tag{5.119}$$

$$t = \frac{(2n+1)\pi}{\omega}, \tag{5.120}$$

which has the same solution as $\sin(\omega t) = 0$.

The time that the mass goes through the equilibrium position can be found by just setting $A\cos(\omega t)$ to zero

$$A\cos(\omega t) = 0 \tag{5.121}$$

$$\omega t = \arccos(0) + 2n\pi \qquad n \in \mathbb{Z} \tag{5.122}$$

$$t = \frac{2n\pi}{\omega} \tag{5.123}$$

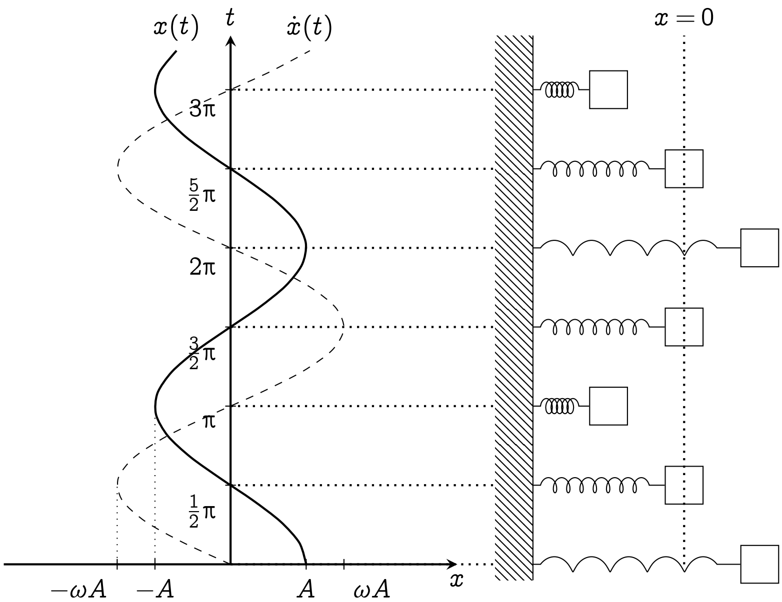


Figure 5.6 | THE TRAJECTORY OF A MASS IN THE HARMONIC OSCILLATOR SYSTEM. (LEFT) THE POSITION IS PLOTTED IN THICK LINES, AND THE VELOCITY, IN DOTTED LINES. (RIGHT) DEPICTION OF AN OSCILLATING MASS, ACCORDING TO THE SOLUTION OF THE DIFFERENTIAL EQUATION.

Since the velocity function (eq. (5.97)) is a sine function, it must oscillate π radians off-phase relative to the position function. Therefore,

1. Whenever the mass is furthest away from the equilibrium, the velocity is zero

2. Whenever the mass is directly passing through the equilibrium, the velocity reaches its maximum at $\pm\omega A$

The first fact can be easily verified by setting the velocity function, $-\omega A\sin(\omega t)$

equals to zero

$$-\omega A\sin(\omega t) = 0 \tag{5.124}$$
$$\omega t = \arcsin(0) + 2n\pi \tag{5.125}$$
$$t = \frac{2n\pi + 1}{\omega}, \tag{5.126}$$

which directly matches with the point of furthest position (eq. (5.117)). The second one can be verified by taking the derivative of velocity, which is just the acceleration, found in eq. (5.98). Setting that equals to zero yields

$$-\omega^2 A\cos(\omega t) = 0 \tag{5.127}$$
$$\omega t = \arccos(0) + 2n\pi \tag{5.128}$$
$$t = \frac{2n\pi}{\omega}, \tag{5.129}$$

which matches our result from eq. (5.123).

The acceleration function is again, a cosine function that's in phase with the position. Therefore, the local minima and maxima from the position function must match. We can quickly conclude that the acceleration is at its highest at $t = \frac{2n\pi}{\omega}$, and is zero at $\frac{(2n+1)\pi}{\omega}$.

You would think that we have completed the physical aspects of the harmonic oscillator. But there is something I left over: the ω term. Since the beginning, I have assigned $\omega = \sqrt{\frac{k}{m}}$ as a notational trick for convenience. Is there any real physical meaning behind ω?

If you don't know much about units, it's fine if you skip these paragraphs straightaway. But one clue that suggests the physical meaning of ω is its unit. Since $\omega = \sqrt{\frac{k}{m}}$, we have to analyze the units of its components separately. k is a spring stiffness constant which is got from

$$F = -kx \quad \text{or} \quad k = -\frac{F}{x}. \tag{5.130}$$

F is the unit of force (Newton N), which is just $\mathrm{kg}\frac{\mathrm{m}}{\mathrm{s}^2}$), and x, the units of distance (Meters m); therefore, k is the unit of

$$\frac{\mathrm{kg}\frac{\mathrm{m}}{\mathrm{s}^2}}{\mathrm{m}} = \frac{\mathrm{kg}}{\mathrm{s}^2}. \tag{5.131}$$

And since m has the unit of kilogram (kg), ω which is $\sqrt{\frac{k}{m}}$ has the unit of

$$\sqrt{\frac{\frac{\mathrm{kg}}{\mathrm{s}^2}}{\mathrm{kg}}} = \sqrt{\frac{1}{\mathrm{s}^2}} = \frac{1}{\mathrm{s}} \tag{5.132}$$

Funnily enough, this is the unit of Hertz, ($\frac{1}{\mathrm{s}} = \mathrm{Hz}$) which describes frequencies. But it cannot possibly be just a unit of frequency because the solution of the differential equation says so.

The solution of our equation is $x(t) = A\cos(\omega t)$. If ω is really a unit of Hertz, then ωt would have no units, which contradicts that the cosine function either accepts radians (rad), or degrees (deg). But these are units that are dimensionless, meaning that we can just multiply it into ω; therefore, ω must have the unit of $\mathrm{rad\,s^{-1}}$, which is the unit of angular frequencies.

The conclusion that ω is the angular frequencies can also be directly deducted from the solution of the equation. From both of eq. (5.117) and eq. (5.123), the time between each maximum and minima point is divided by ω. If ω is very high, then the time between the maximum and minimum is decreased. If it's low, then the time is increased. Therefore, ω is directly associated to the natural frequency of the system. The more ω is, the more frequent the oscillator oscillates.

Another way ω can be interpreted is by looking at the expression of ω itself: $\sqrt{\frac{k}{m}}$. The spring stiffness constant k is on the top, so it means the stiffer the spring, the higher omega is. A stiffer spring would allow the spring to pull the mass faster; thus, increasing the frequencies. The mass

m is on the bottom: the larger the mass, the lower the omega. That just means a more massive block is harder for the spring to move; therefore, the mass will travel slower, decreasing the frequencies.

5.4.4 Ansatz method, and solution for other differential equations

There's actually a method that I've been hiding all along. In reality, the differential equation that describes the harmonic oscillator is "*too easy*" to be integrated by hand. Why integrate when you can just *guess* the solution?

The **ansatz method**[3] is a guessing method that's commonly taught in MIT (Massachusetts Institute of Technology). Let's go back to the harmonic oscillator differential equation

$$\frac{\mathrm{d}^2 x}{\mathrm{d}t^2} = -\omega^2 x. \tag{5.133}$$

It's asking

> What function, when differentiated twice is the negative of itself times a certain constant?

This should already ring a bell that it's obviously either the cosine or sine function. Which one is it?

To deal with this, notice that the differential equation is linear. That is, if $f(t)$ and $g(t)$ are solutions, then $Af(t) + Bg(t)$ where A and B are constants is a solution also. The proof is as follows. First, $f(t)$ and $g(t)$ are solutions mean

$$\frac{\mathrm{d}^2 f(t)}{\mathrm{d}t^2} + \omega^2 f(t) = 0 \quad \text{and} \quad \frac{\mathrm{d}^2 g(t)}{\mathrm{d}t^2} + \omega^2 g(t) = 0. \tag{5.134}$$

[3] Borrowed from the German noun, *ansetzen*, which means to estimate.

Thus,

$$\frac{\mathrm{d}^2\big(Af(t)+Bg(t)\big)}{\mathrm{d}t^2}+\omega^2\big(Af(t)+Bg(t)\big)=0 \tag{5.135}$$

$$A\left(\frac{\mathrm{d}^2f(t)}{\mathrm{d}t^2}+\omega^2f(t)\right)+B\left(\frac{\mathrm{d}^2g(t)}{\mathrm{d}t^2}+\omega^2g(t)\right)=0 \tag{5.136}$$

$$0+0=0, \tag{5.137}$$

which is true. Since both sines and cosines represents solution of the differential equation, the actual solution could contain both of them, so we shall guess that

$$x(t)=A\cos(\omega t)+B\sin(\omega t) \tag{5.138}$$

Indeed, it does satisfy the differential equation. Using the chain rule twice,

$$\frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2}=\frac{\mathrm{d}^2}{\mathrm{d}t^2}\big(A\cos(\omega t)+B\sin(\omega t)\big) \tag{5.139}$$

$$=-\omega^2A\cos(\omega t)-\omega^2B\sin(\omega t) \tag{5.140}$$

$$=-\omega^2\big(A\cos(\omega t)+B\sin(\omega t)\big) \tag{5.141}$$

$$\frac{\mathrm{d}^2x(t)}{\mathrm{d}t^2}=-\omega^2x(t). \tag{5.142}$$

Similarly, this is just a guess. To actually find what A and B is, we have to plug in the initial condition.

Here is where it gets quite interesting: by the simplicity of this method, we are not restricted to the initial condition that $x(t=0)=A$ and $v(t=0)=0$. So, let's try solving for the case where $x(t=0)=X$ and $v(t=0)=V$.

Frankly, there's nothing restricting us in the first place, but I decided not to show it using the direct integration method because the process is too tedious. I prefer this method to analyze other conditions. So, let's start solving!

Let's use the ansatz $x(t) = A\cos(\omega t) + B\sin(\omega t)$. Now, let's substitute our initial condition into $x(t)$. At $t = 0$, $x(t = 0) = X$

$$x(0) = A\cos\big(\omega(0)\big) + B\sin\big(\omega(0)\big) \tag{5.143}$$

$$X = A\cancelto{1}{\cos(0)} + B\cancelto{0}{\sin(0)} \tag{5.144}$$

$$X = A \tag{5.145}$$

Thus, $x(t) = X\cos(\omega t) + B\sin(\omega t)$. To find B, we just have to differentiate this expression once and substitute in $v(t = 0) = V$

$$v(t = 0) = -\omega X\sin\big(\omega(0)\big) + \omega B\cos\big(\omega(0)\big) \tag{5.146}$$

$$\frac{V}{\omega} = -\cancelto{0}{X\sin(0)} + B\cancelto{1}{\cos(0)} \tag{5.147}$$

$$\frac{V}{\omega} = B \tag{5.148}$$

Therefore,

$$x(t) = X\cos(\omega t) + \frac{V}{\omega}\sin(\omega t). \tag{5.149}$$

In this form, it's quite hard to see what is the amplitude and the point of maximum and minimum velocity is. To make it easier for us to interpret, we can combine the cosine and sine term using the **harmonic addition theorem**

$$a\cos\theta + b\sin\theta = c\cos(\theta + \phi) \tag{5.150}$$

where

$$c = \operatorname{sgn}(a)\sqrt{a^2 + b^2} \quad \text{and} \quad \phi = \arctan\left(-\frac{b}{a}\right). \tag{5.151}$$

Equation (5.149) then becomes

$$\sqrt{X^2 + \frac{V^2}{\omega^2}}\cos\left(\omega t + \arctan\left(-\frac{V}{\omega X}\right)\right) \tag{5.152}$$

In this form, it's easy to see that the amplitude: the point that the mass is furthest away from the spring, is $x = \sqrt{X^2 + \frac{V^2}{\omega^2}}$. But what is that $\arctan\left(-\frac{V}{\omega X}\right)$ doing there? What does it represent? We shall discuss this in detail in section 6.2

5.5 Trigonometric substitutions

As seen earlier, trigonometric substitution can be used in evaluating integrals which has terms of $\sqrt{A^2 - x^2}$ by substituting $x = \cos\theta$, and taking advantage of the Pythagorean identity. But it doesn't stop at just $\sin^2\theta + \cos^2\theta = 1$, we can use other Pythagorean identities to evaluate integrals which contains $\sqrt{A^2 + x^2}$ and $\sqrt{x^2 - A^2}$ also. We shall explore integral of these forms and solve some integrals in this section.

Honestly, when I was writing this section, I feel bored. But that picture when I first saw these integrals four years ago, caught me:

> *"How in the world is it possible? A simple substitution could do such wonderful things! It just destroyed this integrals involving square roots and turned it into a much more managable form. Amazing!"*

The magic is in the interplay between two seemingly unrelated things: square roots and trigonometric functions. So, I try to find examples that would hopefully let you appreciate these things more. I hope you find the excitement in it, and let's crack on.

Integrals containing $\sqrt{A^2 - x^2}$

Let the integral be described by

$$\int f\left(\sqrt{A^2 - x^2}, x\right) \mathrm{d}x. \tag{5.153}$$

where f is a function of $\sqrt{A^2 - x^2}$ and x. As said earlier, substitute $x = a \sin\theta$; thus,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(A\sin\theta) \tag{5.154}$$

$$\mathrm{d}x = A\cos(\theta)\,\mathrm{d}\theta. \tag{5.155}$$

Using $1 - \cos^2\theta = \sin^2\theta$, our integral becomes

$$A\int f\left(\sqrt{A^2 - A^2\cos^2\theta}, A\sin\theta\right)\cos\theta\,\mathrm{d}\theta$$

$$= A\int f\left(A\sqrt{1 - \cos^2\theta}, A\sin\theta\right)\cos\theta\,\mathrm{d}\theta \tag{5.156}$$

$$= A\int f\left(A\sqrt{\sin^2\theta}, A\sin\theta\right)\cos\theta\,\mathrm{d}\theta \tag{5.157}$$

$$= A\int f\left(A\sin\theta, \sin\theta\right)\cos\theta\,\mathrm{d}\theta \tag{5.158}$$

Now let's see some magic: two magic tricks with circles.

The next two examples evaluates the perimeter and area of a circle by using trigonometric substitution. The equation of a circle radius R centered at the origin is given by $x^2 + y^2 = R^2$, which is in an implicit form. To integrate this, we have to turn it into an explicit form

$$y = f(x) = \sqrt{R^2 - x^2}. \tag{5.159}$$

Since $f(x)$ contains a square root, the function can only output a positive value; therefore, it only represents half a circle. After calculating the perimeter and area covered by $f(x)$, we have to double that to get the actual perimeter and area of a circle.

Example 5.5.1: *Perimeter of a circle* This is an excuse for me to revisit the arc length formula in section 4.5. The perimeter of a circle is essentially double of the arc length of $\sqrt{R^2 - x^2}$ from $-R$ to R which from eq. (4.69), is given by

$$\int_{-R}^{R} \sqrt{1 + \left(\frac{\mathrm{d}\sqrt{R^2 - x^2}}{\mathrm{d}x} \right)^2} \, \mathrm{d}x. \tag{5.160}$$

The derivative inside can be evaluated using the chain rule:

$$\begin{aligned}
\frac{\mathrm{d}\sqrt{R^2 - x^2}}{\mathrm{d}x} &= \frac{\mathrm{d}\sqrt{u}}{\mathrm{d}u} \times \frac{\mathrm{d}u}{\mathrm{d}x} && u = R^2 - x^2 && (5.161) \\
&= \frac{\mathrm{d}u^{\frac{1}{2}}}{\mathrm{d}u} \times \frac{\mathrm{d}R^2 - x^2}{\mathrm{d}x} && && (5.162) \\
&= \frac{1}{2\sqrt{u}}(-2x) && \text{Power rule} && (5.163) \\
&= -\frac{x}{\sqrt{R^2 - x^2}} && u = R^2 - x^2 && (5.164)
\end{aligned}$$

Our integral then becomes

$$\begin{aligned}
\int_{-R}^{R} \sqrt{1 + \left(-\frac{x}{\sqrt{R^2 - x^2}} \right)^2} \, \mathrm{d}x &= \int_{-R}^{R} \sqrt{1 + \frac{x^2}{R^2 - x^2}} \, \mathrm{d}x && (5.165) \\
&= \int_{-R}^{R} \sqrt{\frac{R^2 - x^2 + x^2}{R^2 - x^2}} \, \mathrm{d}x && (5.166) \\
&= R \int_{-R}^{R} \frac{1}{\sqrt{R^2 - x^2}} \, \mathrm{d}x. && (5.167)
\end{aligned}$$

This form can be easily evaluated by substituting $x = R\sin\theta, \mathrm{d}x = R\cos\theta \, \mathrm{d}x$. The bounds are then changed by $\theta = \arcsin\left(\frac{x}{R}\right)$ to

$$\begin{aligned}
x = -R \to \theta = \arcsin\left(\frac{-R}{R}\right) &= -\frac{\pi}{2}, \\
x = R \to \theta = \arcsin\left(\frac{R}{R}\right) &= \frac{\pi}{2}.
\end{aligned} \tag{5.168}$$

Our integral then becomes

$$R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{\sqrt{R^2 - R^2 \sin^2\theta}} R\cos\theta \, \mathrm{d}\theta$$

$$= R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{R\sqrt{1-\sin^2\theta}} R\cos\theta \, \mathrm{d}\theta \tag{5.169}$$

$$= R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\theta}{\cos\theta} \, \mathrm{d}\theta = R \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \mathrm{d}\theta \tag{5.170}$$

$$= R(\theta|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = R\left(\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right) = R\pi; \tag{5.171}$$

therefore, the perimeter of a whole circle radius R is $2\pi R$.

Example 5.5.2: *Area of a circle* The area of a circle radius R is double the integral of $\sqrt{R^2 - x^2}$:

$$\int_{-R}^{R} \sqrt{R^2 - x^2} \, \mathrm{d}x. \tag{5.172}$$

Perform a trigonometric substitution similar to the example above: $x = R\sin\theta$, $\mathrm{d}x = R\cos\theta \, \mathrm{d}\theta$, $x = -R \to \theta = -\frac{\pi}{2}$, and $x = R \to \theta = \frac{\pi}{2}$. Our integral then becomes

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{R^2 - R^2\sin^2\theta} R\cos\theta \, \mathrm{d}\theta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} R^2\sqrt{1-\sin^2\theta}\cos\theta \, \mathrm{d}\theta$$
$$= R^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^2\theta \, \mathrm{d}\theta. \tag{5.173}$$

To evaluate this, we have to use the trigonometric identity $\cos(2\theta) = 2\cos^2\theta - 1$. Rearranging the identity gives $\cos^2\theta = \frac{\cos(2\theta)+1}{2}$; therefore, our integral becomes

$$R^2 \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos(2\theta)+1}{2} \, \mathrm{d}\theta = \frac{R^2}{2}\left[\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos(2\theta) \, \mathrm{d}\theta + \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \mathrm{d}\theta\right]. \tag{5.174}$$

The second integral simply evaluates to $\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)$, which is just π. The first integral requires another substitution.

Integrals containing $\sqrt{x^2 - A^2}$

Let the integral be described by

$$\int f\left(\sqrt{x^2 - A^2}, x\right) \mathrm{d}x. \tag{5.175}$$

We want to take advantage of $\sec^2\theta - 1 = \tan^2\theta$, so substitute $x = A\sec\theta$. From table 5.1,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(A\sec\theta) \tag{5.176}$$

$$\mathrm{d}x = A\sec\theta\tan\theta\,\mathrm{d}\theta. \tag{5.177}$$

Our integral then becomes

$$\int f\left(\sqrt{A^2\sec^2\theta - A^2}, A\sec\theta\right) A\sec\theta\tan\theta\,\mathrm{d}\theta$$

$$= A\int f\left(A\sqrt{1 - \sec^2\theta}, A\sec\theta\right)\sec\theta\tan\theta\,\mathrm{d}\theta \tag{5.178}$$

$$= A\int f\left(A\sqrt{\tan^2\theta}, A\sec\theta\right)\sec\theta\tan\theta\,\mathrm{d}\theta \tag{5.179}$$

$$= A\int f\left(A\tan\theta, A\sec\theta\right)\sec\theta\tan\theta\,\mathrm{d}\theta. \tag{5.180}$$

Integrals containing $\sqrt{A^2 + x^2}$

Let the integral be described by

$$\int f\left(\sqrt{A^2 + x^2}, x\right) \mathrm{d}x. \tag{5.181}$$

We want to take advantage of $1 + \tan^2\theta = \sec^2\theta$, so substitute $x = A\tan\theta$. From table 5.1,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(A\tan\theta) \tag{5.182}$$

$$\mathrm{d}x = A\sec^2\theta. \tag{5.183}$$

Our integral then becomes

$$\int f\left(\sqrt{A^2 + A^2\tan^2\theta}, A\tan\theta\right) A\sec^2\theta\,\mathrm{d}\theta$$

$$= \int f\left(A\sqrt{1+\tan^2\theta}, A\tan\theta\right)\sec^2\theta\,\mathrm{d}\theta \tag{5.184}$$

$$= \int f\left(A\sqrt{\sec^2\theta}, A\tan\theta\right)\sec^2\theta\,\mathrm{d}\theta \tag{5.185}$$

$$= \int f\left(A\sec\theta, A\tan\theta\right)\sec^2\theta\,\mathrm{d}\theta. \tag{5.186}$$

CHAPTER 6

Merging calculus and dynamical systems

With the merge of calculus and trigonometry, all cards are on the deck; finally, unlocking the door to analyze many interesting dynamical systems scattered all over the world.[1]

Since this is an exploratory chapter, I can't really provide a main *theme* for the chapter. It's just chapter that explores *differential equations*, if you could even call that a theme. So, I've decided to structure the chapter into examples and theories. A section will start with an example of a dynamical system. Then, a description of the system using differential equations. Then, a solution and an analysis on its physical and mathematical properties.

The chapter is planned out as follows:

1. Prelude: more advanced techniques of integration

[1]These dynamical systems are partially my excuses to teach various technique of integrations. So, be prepared for it.

2. Solution sketching
3. Exponential growth and decay (Separated into two sections)
4. Logistic growths
5. Movement with drag forces
6. Realization of gravitational systems
7. Alternating currents

6.1 Prelude: fancier techniques of integration

6.1.1 Rational quadratic integrals

In this chapter, you'd see many integrals of the form

$$\int \frac{Ax + B}{Cx^2 + Dx + E}\,\mathrm{d}x \tag{6.1}$$

There are multiple ways to integrate this. Each one have different styles. Here, I shall introduce you to all but first, let's do some preliminary simplification.

It'd be great if the numerator is just a simple constant. Well, it's indeed possible to break this integral into two smaller integrals using the linearity property:

$$\begin{aligned} &\int \frac{Ax + B}{Cx^2 + Dx + E}\,\mathrm{d}x \\ &= \int \frac{Ax}{Cx^2 + Dx + E}\,\mathrm{d}x + B\int \frac{1}{Cx^2 + Dx + E}\,\mathrm{d}x \end{aligned} \tag{6.2}$$

The second integral already has a constant in the denominator. What about the first?

A simple change of variable would do the trick. If we let u be the denominator, then $\mathrm{d}u = (2Cx + D)\,\mathrm{d}x$. We're just missing a constant D, so let's add it in and subtract it out:

$$\int \frac{Ax}{Cx^2+Dx+E}\,\mathrm{d}x = \frac{A}{2C}\int \frac{2Cx+D-D}{Cx^2+Dx+E}\,\mathrm{d}x$$

$$= \frac{A}{2C}\int \frac{2Cx+D}{Cx^2+Dx+E} - \frac{AD}{2C}\frac{1}{Cx^2+Dx+E}\,\mathrm{d}x \tag{6.3}$$

$$= \frac{A}{2C}\int \frac{1}{u}\,\mathrm{d}u - \frac{AD}{2C}\frac{1}{Cx^2+Dx+E}\,\mathrm{d}x \tag{6.4}$$

$$= \frac{A}{2C}\ln(u) - \frac{AD}{2C}\frac{1}{Cx^2+Dx+E}\,\mathrm{d}x \tag{6.5}$$

$$= \frac{A}{2C}\ln(Cx^2+Dx+E) - \frac{AD}{2C}\frac{1}{Cx^2+Dx+E}\,\mathrm{d}x \tag{6.6}$$

We can combine this result with eq. (6.2) in order to get

$$\begin{aligned}&\int \frac{Ax+B}{Cx^2+Dx+E}\,\mathrm{d}x\\ &= \frac{A}{2C}\ln(Cx^2+Dx+E) - \left(\frac{AD}{2C}+B\right)\int \frac{1}{Cx^2+Dx+E}\,\mathrm{d}x\end{aligned} \tag{6.7}$$

As seen, we're left with integrals of the form

$$\int \frac{1}{Cx^2+Dx+E}\,\mathrm{d}x, \tag{6.8}$$

which is way easier to deal with.

Firstly, let's pull out the leading constant from the denominator:

$$\int \frac{1}{Cx^2+Dx+E}\,\mathrm{d}x = \frac{1}{C}\int \frac{1}{x^2+\frac{D}{C}x+\frac{E}{C}}\,\mathrm{d}x \tag{6.9}$$

Notice that the form of this integral is suspiciously similar to the arctangent integral:

$$\int \frac{1}{x^2+a^2}\,\mathrm{d}x = \frac{1}{a}\arctan\left(\frac{x}{a}\right). \tag{6.10}$$

The arctangent integral is just missing the term $\frac{D}{C}x$. If we can remove this term from our integral in question, then it would be solvable using the arctangent integral. Let's do that.

Every quadratics can be written into the form $a(x-h)^2+k$. This can be easily achieved by completing the square.

$$x^2+\frac{D}{C}x+\frac{E}{C} = (x)^2+2(x)\left(\frac{D}{2C}\right)+\left(\frac{D}{2C}\right)^2-\left(\frac{D}{2C}\right)^2+\frac{E}{C} \tag{6.11}$$

$$= \left(x + \frac{D}{2C}\right)^2 + \left(\frac{E}{C} - \frac{D^2}{4C^2}\right) \tag{6.12}$$

$$= \left(x + \frac{D}{2C}\right)^2 + \left(\frac{4CE - D^2}{4C^2}\right) \tag{6.13}$$

If we let $u = x + \frac{D}{2C}$, then $\mathrm{d}u = \mathrm{d}x$, and

$$\frac{1}{C}\int \frac{1}{x^2 + \frac{D}{C}x + \frac{E}{C}}\,\mathrm{d}x$$

$$= \frac{1}{C}\int \frac{1}{\left(x + \frac{D}{2C}\right)^2 + \left(\frac{4CE-D^2}{4C^2}\right)}\,\mathrm{d}x \tag{6.14}$$

$$= \frac{1}{C}\int \frac{1}{u^2 + \left(\frac{4CE-D^2}{4C^2}\right)}\,\mathrm{d}u \tag{6.15}$$

If $\frac{4CE-D^2}{4C^2}$ is positive, then we can use the arctangent integral to get

$$\frac{1}{C}\sqrt{\frac{4C^2}{4CE - D^2}}\,\arctan(u\frac{4C^2}{4CE - D^2}) \tag{6.16}$$

$$= \frac{2}{\sqrt{4CE - D^2}}\arctan\left(\frac{4C^2(x + \frac{D}{2C})}{4CE - D^2}\right) \tag{6.17}$$

$$= \frac{2}{\sqrt{4CE - D^2}}\arctan\left(\frac{4C^2x + 4CD}{4CE - D^2}\right) \tag{6.18}$$

6.2 Phase space

CHAPTER 7

Basic techniques of integration

Abstract

This chapter explores various techniques you can use to integrate stuffs. You still have to keep in mind that even though this chapter is very abstracted away, *integrals still have its geometric meaning: area under the graph.* This geometric interpretation is going to somewhat appear quite often throughout the chapter, so be aware of it.

Also, some integrals are definite: they have clear boundaries of integrations. You also have to keep track of what variable is bounded by that bound, otherwise you might end up evaluating incorrectly.

As an educator myself, I usually don't teach this topic but leave it as an exercise. Techniques of integrations come naturally with much experiences. Therefore, I list a table of integrations at the end of the chapter for the physicists out there.

7.1 Change of variables

Change of variables, or integration by substitution is a method to convert an unknown integrals into a more familiar one. Consider $\int x\mathrm{e}^(x^2)\,\mathrm{d}x$. This integral might seem daunting at first but, notice that the derivative of x^2 is exactly x. I shall introduce a new variable u where $u = x^2$. If we take the derivative w.r.t. x on both sides,

$$\frac{\mathrm{d}u}{\mathrm{d}x} = 2x$$

$$\mathrm{d}u = 2x\,\mathrm{d}x.$$

7.2 Integrals of inverse functions

7.3 Integrals of symmetric functions

7.3.1 Even and odd functions

7.3.2 Functions with axial and spherical symmetry

7.4 Integration by part

7.5 Recurrence relations and reduction formulas

7.6 Working with complex numbers and exponentials

7.7 Trigonometric substitution

7.8 Rational functions

7.8.1 Partial fraction decomposition

7.8.2 Quadratic under radicals

7.8.3 Ostrogradski method

7.9 Feynman's trick for integration

7.10 Polar integration

7.11 Gaussian integral

7.12 Cauchy's formula for repeated integrations

7.13 Numerical integration

CHAPTER 8

Advanced techniques of integration

Abstract

This is the chapter that explores more advanced techniques of integrations. You could go on with your life skipping this chapter and it'd still be fine. However, for the curious, there are some very interesting maths in here, so be sure to check this chapter out before going on to the next part of the book.

8.1 Rational functions

8.1.1 Quadratics denominator

An integral in the form

$$\int \frac{1}{ax^2 + bx + c} \, \mathrm{d}x \tag{8.1}$$

can be evaluated by completing the squares:

$$ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + \left(c - \frac{b^2}{4a}\right) \tag{8.2}$$

8.1.2 Denominator quadratics under radicals

8.2 Mathematical interlude: trigonometric substitution

The trigonometric substitution technique earlier works for all the Pythagorean identity. The most used ones being

$$1 - \cos^2\theta = \sin^2\theta, \quad 1 + \tan^2\theta = \sec^2\theta \quad \text{and} \quad 1 - \sec^2\theta = \tan^2\theta.$$

They're used to take terms outside the square root, whether it'd be $\sqrt{a^2 - x^2}$, $\sqrt{a^2 + x^2}$, or $\sqrt{x^2 - a^2}$. Now let's go through these one by one.

Integrals with $\sqrt{a^2 - x^2}$: Let $x = a\sin\theta$; therefore, $\theta = \arcsin\left(\frac{x}{a}\right)$, and

$$\sqrt{a^2 - x^2} = a\sqrt{1 - \sin^2\theta} = a\sqrt{\cos^2\theta} = a\cos\theta. \tag{8.3}$$

The differential element $\mathrm{d}x$ transforms into

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(a\sin\theta) \tag{8.4}$$

$$\mathrm{d}x = a\cos\theta\,\mathrm{d}\theta. \tag{8.5}$$

Example 8.2.1: $\int \frac{\sqrt{4-x^2}}{x^2}\,\mathrm{d}x$ This integral has a square root of the form $\sqrt{a^2 - x^2}$ where $a = 2$; thus, substitute in $x = 2\sin\theta$. Then,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}(2\sin\theta) \tag{8.6}$$

$$\mathrm{d}x = 2\cos\theta\,\mathrm{d}\theta \tag{8.7}$$

The integral then becomes

$$\int \frac{\sqrt{4-4\cos^2\theta}}{4\sin^2\theta} 2\cos\theta \,\mathrm{d}\theta = \int \frac{2\sqrt{1-\cos^2\theta}}{4\sin^2\theta} 2\cos\theta \,\mathrm{d}\theta \tag{8.8}$$

$$= \int \frac{2\sin^2\theta}{4\sin^2\theta} 2\cos\theta \,\mathrm{d}\theta \tag{8.9}$$

$$= \int \cos\theta \,\mathrm{d}\theta = \sin\theta \tag{8.10}$$

Since $x = 2\sin\theta$, $\sin\theta = \frac{x}{2}$, our integral evaluates to $\frac{x}{2} + C$.

Integrals with $\sqrt{a^2+x^2}$: Let $x = a\tan\theta$; therefore, $\theta = \arctan\left(\frac{x}{a}\right)$, and

$$\sqrt{a^2+x^2} = a\sqrt{1+\tan^2\theta} = a\sqrt{\sec^2\theta} = a\sec\theta. \tag{8.11}$$

By **??**, the differential element $\mathrm{d}x$ transforms into

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = \frac{\mathrm{d}}{\mathrm{d}\theta}\left(a\sec\theta\right) \tag{8.12}$$

$$\mathrm{d}x = a\sec\theta\tan\theta. \tag{8.13}$$

8.2.1 Feynman's trick for integration

Formerly known as Leibniz's rule but popularized by Richard Feynman, is a very powerful integral technique that allows you to differentiate the function in the integral sign. Sadly, it only works for definite integrals. However, it's still a very nice trick to have just in case.

The idea is: integrals and derivatives can be swapped in

8.3 Integrals of other trigonometric functions

Part II

Differential equations: qualitative aspects

Part III

Signal analysis

Book II

Real analysis

Preface to Book II

In contrast to the first book, this book would make a lot of mathematicians happy, as this part of the book is a bit out of touch with the others. I named this whole series "Compact Calculus - Calculus for the practitioners," because I envision it to be a gentle and compact introduction to calculus that is used by most practitioners. One could argue that real analysis isn't for the practitioners, as it is considered to be one of the hardest pure mathematics course that an undergraduate can take. It's dedicated to rigorously analyzing the behaviors of the reals, redefining many of the hand-wavy definitions that we've used, and proving even the most trivial properties. Some proofs are so abstract that most have trouble wrapping their heads around them. Why did I decide to write about real analysis?

After reading many analysis book[1], I've developed a deeper appreciation for the nature of mathematics. It's remarkable that from just a few axioms, mathematics has managed to prosper into the vast interconnecting webs of theories that we know today. So, I'd like to express the beautifulness of mathematics, and write them down on a book. I tried to make the book as comprehensive as possible, so those who are interested can also see the beauty in the foundations of the mathematics.

[1]Even though I don't even have the point of doing so; I do physics, not mathematics.

Part IV

Fundamentals of analysis: logic, number systems and arithmetrization

CHAPTER 9

Mathematical logic

9.1 Prelude: mathematical analysis and rigorousness

You are about to begin your second exposure to serious mathematics: mathematical analysis. It's sole purpose is to return to the foundations we've covered and rebuild everything from scratch. But why?

Prior, I've emphasized on geometrical methods: things that can be *drawn out* and *seen*. E.g., if a function is an increasing function, then its derivative must be positive. The derivative represents the slope, and the slope of an increasing function should be positive. Try draw an increasing function with a negative slope, you'd quickly find it's virtually impossible! But virtually impossible for us to draw doesn't mean that it can't exist as a weird counterexamples. Another such example might be how the integral is the reverse of derivative due to the geometry of cumulative changes, as explained in chapter 2. But as you'd see, it's very possible to draw a function that has an integral but

The flaw of geometry is: it's impossible to draw *all* possible configurations. In chapter 2, we've seen that integration is the "reverse" of derivatives. So it's natural to assume that if a function can be integrated, its integral must have a derivative which yields back the same function:

$$\frac{\mathrm{d}}{\mathrm{d}x}\int_0^x f(t)\,\mathrm{d}t = f(x) \tag{9.1}$$

However, this is not the case. An example would be the **Thomae's function** shown in fig. 9.1, defined as

$$T(x) = \begin{cases} \frac{1}{b} & \text{if } x = \frac{a}{b} \in \mathbb{Q}; a, b \in \mathbb{Z} \\ 0 & \text{if } x \in \mathbb{R}\backslash\mathbb{Q} \end{cases}. \tag{9.2}$$

This function is also called the **popcorn function**. It is a function with infinitely thin strips, so naturally, its Riemann integral must be zero everywhere, and in fact it is. However, the derivative of zero is just zero. It does not return the Thomae's function.

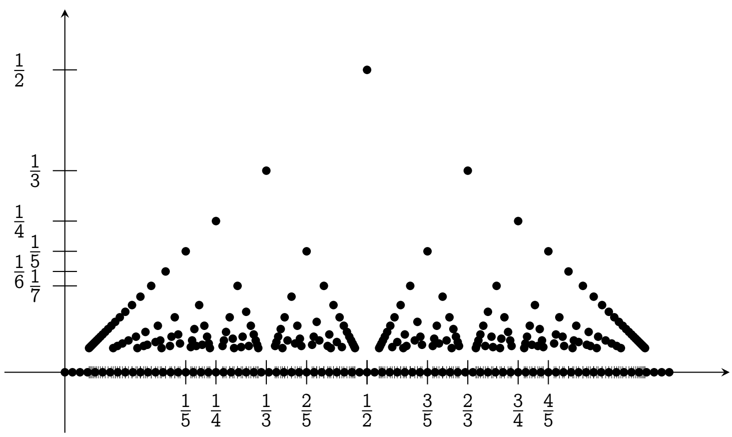


Figure 9.1 | THOMAE'S FUNCTION, DRAWN BY USER121799 (9)

Another example is in chapter 4: a function has an extrema, i.e., lowest or highest point in a neighborhood, if its derivative is zero. And that makes sense until we realize that the function that we've been drawing are the nice ones. The ones that follows this property. There are many other functions which don't have this property. The easiest to understand would be a linear function with its domain restricted to some closed intervals.

Let $f(x) = mx$ where m is a positive real number, and consider the restriction of f to some closed interval $[\![a, b]\!]$. Notice that $f(b)$ is the global maxima, and $f(a)$ is the global minima, but the derivative of f as those point aren't zero: it's m.

Algebra also has its own flaws. If handled carelessly, it's possible to arrive at a contradiction. E.g.,

Example 9.1.1: *One equals two* Let $a = b$, then

$$a^2 = ab \tag{9.3}$$

$$a^2 - b^2 = ab - b^2 \tag{9.4}$$

$$(a - b)(a + b) = b(a - b) \tag{9.5}$$

$$a + b = b \tag{9.6}$$

$$b + b = b \tag{9.7}$$

$$2b = b \tag{9.8}$$

$$2 = 1 \tag{9.9}$$

If you have quick eyes, you would've noticed that in eq. (9.6), we've divided by $a - b$, which is zero because $a = b$, causing the whole contradiction.

Albeit, the example prior is quite obvious. So, here's an examples that's much subtler.

Example 9.1.2: $\pi = 0$

$$\int_0^\pi \cos^2 x \,\mathrm{d}x = \int_0^\pi \frac{1+\cos 2x}{2}\,\mathrm{d}x \tag{9.10}$$

$$= \int_0^\pi \frac{1}{2}\,\mathrm{d}x + \int_0^\pi \frac{\cos 2x}{2}\,\mathrm{d}x \tag{9.11}$$

$$= \frac{\pi}{2} + \left(\frac{1}{4}\sin 2x\right|_0^\pi \tag{9.12}$$

$$= \frac{\pi}{2} + 0 = \frac{\pi}{2} \tag{9.13}$$

But then,

$$\int_0^\pi \cos^2 x \,\mathrm{d}x = \int_0^\pi \cos x \cos x \,\mathrm{d}x \tag{9.14}$$

$$= \int_0^\pi \cos x \sqrt{1-\sin^2 x}\,\mathrm{d}x \tag{9.15}$$

Substitute $u = \sin x$, then $\mathrm{d}u = \cos x\,\mathrm{d}x$. If $x = 0$, then $u = 0$. If $x = \pi$, then $u = 0$, so the integral turns into

$$= \int_0^0 \sqrt{1-u^2}\,\mathrm{d}u = 0 \tag{9.16}$$

In conclusion, the same integral evaluates to both $\frac{\pi}{2}$ and 0, implying that both are equal; hence, $\pi = 0$.

The mistake is that change of variables only work when u is related to x by a bijective, or a one-to-one function. The sine function is *not* bijective on the domain of integration: $\sin 0 = \sin \pi = 0$. Hence, the substitution does not give the correct answer.

Historically, mathematical analysis was born from mistakes of these flaws in geometry and algebra that culminated into a pile of unusable results: Taylor series that don't converge, Fourier series that don't approximate, errors in swapping integral bounds, errors in substitution, etc. So, mathematicians try found a way to *unite* the whole subject and provide a proper rigorous mathematical grounding: a rulebook that when followed, no contradictions can occur. It took them more than twenty years to develop, and now, it's your turn to learn it. Starting with the most fundamental part: logic.

I'd like to admit that logic in its entirety is very rich and fruitful; *it would be a disgrace* to claim that the entire field is contained in this one puny chapter. It's not. I'd say that this chapter serves as an overview for the subject and is *hopefully enough logic for you to work on analysis*. With that out of the way, let's begin.

9.2 The building block of logical sentences

Logic is the study of concrete, provable statements: statements that are either true or false. It leaves no room for maybes or mights. These statements which can either be true or false are called **propositions**, e.g.,

- You are a virgin. (You are either a virgin or not a virgin)
- Your father is getting milk. (Your father is either getting milk or not)
- A proposition is a statement that is either true or false.
- I have your IP address.

It's also wise to assign symbols to these statements to avoid writing long sentences. E.g., the proposition "You are a virgin," can be represented with p, "Your father is getting milk," with q, etc. In symbols, the $:=$ sign, a.k.a., the **walrus sign** is used to say "*is declared to be.*" E.g.,

$$A := \text{"Plants are not green"} \tag{9.17}$$

reads "A is declared to be the proposition 'Plants are not green.'"

Note that a proposition need not carry a truth value in this immediate moment. It must just follow the rule that when it is eventually evaluated, it can only be either true or false. In this spirit, there are some statements that don't look like a proposition but actually are propositions.

- There might be an asteroid hitting your house tomorrow. (The truthfulness right now is still not determined as it is not tomorrow. But it is still a proposition as the only two allowed cases are still true or false. When tomorrow comes, it either hits or it doesn't. There is no in between, as otherwise it'd be a Schrödinger's asteroid)
- Tomorrow, there might be a baby whose name is "U vuvwevwevwe Onyetenyevwe Ugwemuhwem Osas," (More accurately, their parents
named them that keysmash, perhaps after the child's first squeals. The truthfulness is still not determined, but either a baby is born tomorrow with this name or no baby with this name at all.)
- Your romantic partner might be cheating on you, assuming you have a partner. (Your partner is either cheating or not cheating. They can't cheat and not cheat at the same time. Quantum love still doesn't exists... yet...)

The truthfulness of some statements depends on external factors, but they're still either true or false, so they qualify to be propositions. Statements that looks *obviously true* or *obviously false* might not always that way because said external factors. E.g.,

- The earth is flat (Obviously false, but there might be another universe out there that the laws of physics aren't the same as ours, causing the earth to be flat)

- Salt tastes salty (Obviously true, but there might be an animal who perceives salt as umami or sweet)
- You can write with a pencil (Obviously true, but there might be some universes where pencils are used to erase and erasers are used to write)

These statements are called **contingent statements**, and appears everywhere in mathematics. E.g., the truthfulness of

$$1 + 1 = 2 \tag{9.18}$$

is only determinable when 1, 2, $=$, and $+$ is defined to be the standard system of arithmetic. There are other worlds where $1 + 1 = 0$[1], or $1 + 1 = 6$. The truth value of these statements depend solely on the propositions that comes before it. The statement $1 + 1 = 2$ depends on the proposition that governs how equality, numbers, and addition works. And you can ask further: why does equality work this way, and you will find deeper propositions about it. However, you can't always keep asking why something is true. Because eventually, you would reach the bottom of everything: the **axioms**.

Axioms are propositions which are deemed to be true. E.g., we can always draw a straight line between two points. It looks so simple and so rudimentary that there's no point to find a deeper reason to justify it. We just give up and say that *if it's so rudimentary and so trivial that it should've been true, it must be true*. Even though these axioms are arbitrary, they are the starting point of which the mathematical landscape sprouts.

[1]Define 0 and 1 usually on the natural numbers, then define $+$ as being addition modulo two.

There might be a world where it's impossible to draw a line between two points, but in this normal world, you can do it. Heck, even logic itself needs to have axioms. Here are three core logical axioms:

Axiom 9.2.1: *Axioms of logic*

1. **Law of identity:** A proposition is identical to itself.
2. **Law of non-contradiction:** A proposition cannot be both true and false at the same time and in the same sense.
3. **Law of excluded middle:** A proposition can either be true or false. There is no third option.

Even with the axioms, there are some statements that we don't know how to prove yet — they are still contingent. Math is riddled with said statements, e.g., the Riemann's hypothesis, or the Collatz' conjecture. They're either true or not true, but their truth value is still unknown as humanity can't prove them yet.

If a statement doesn't have any truthfulness, then they aren't propositions. They might be questions or commands. E.g.,

- Do your homework! (Propositions aren't commands, as commands doesn't have a truth value assigned to it.)
- When did the dinosaurs die? (Propositions aren't questions.)
- Does quantum love exists? (Propositions can't be yes or no questions either. A proposition asserts, not ask.)

These statements have an answer, but aren't provable in the logical sense. Hence, they are not the subject of our concerns

Going back to propositions, if a proposition is true, it carries the truth values of true, and is equivalent to $\top$. If it isn't, then it carries the truth value of false, and is equivalent to $\perp$. The equivalence symbol, $\equiv$ is used to denote relations between truth value of propositions. E.g., if

$$p \coloneqq \text{"Your father is getting milk,"} \tag{9.19}$$

and p carries the truth value of true ($\top$), we write

$$p \equiv \top, \tag{9.20}$$

which reads, p is *equivalent to* ($\equiv$) true ($\top$). If proposition

$$q \coloneqq \text{"You are a virgin"} \tag{9.21}$$

carries the truth value of false ($\perp$), we write

$$q \equiv \perp. \tag{9.22}$$

The three horizontal line symbol ($\equiv$), called the **equivalence sign**, is not the same as the equality sign ($=$). Equality is the absolute state of being equal: two sides are *identical in every way*. But the equivalence symbol weaker. Two wildly different statements can be equivalent. They just have to *carry* the same truth value. E.g.,

$$(1 + 1 = 2) \equiv (10 \times 2 = 20) \tag{9.23}$$

$$\text{The sun rises in the west} \equiv \text{Hydrogen has twenty electrons} \tag{9.24}$$

The two sides are all true, hence they must be equivalent. However, they are not the same statement.

Soon, we shall learn about **logical inference**: a method to conclude that something is true using other true statements that has already been proven or stated. E.g., from the law of excluded middle, we can deduct that

$$p \coloneqq \text{"The proposition } A \text{ is both true and false"} \tag{9.25}$$

is definitely false from the law of excluded middle which is axiomatized (assumed) to be true. In other words, the law of excluded middle *entails* the falsehood of p.

One can also do logical inference with multiple statements. If these statements:

1. Superman wears a cape,
2. Capes are only worn by heroes,
3. All heroes that wear capes can fly,

are assumed to be true, then it's possible to infer that superman can fly. From items 1 and 2, superman is a hero. And from item 3, superman can fly, as he is a hero. The inferred conclusion is then "Superman can fly." In the logical form, we can say that these statement *entails* that "Superman can fly." If these statements are true, it's impossible for superman to not be able to fly.

As you could see, logical inference is all about **entailment**. If whenever $p_1, p_2, \ldots$ are all true, q is impossible to be false, then we say that $p_1, p_2, \ldots$ *entails* q, and denote this relation by

$$p_1, p_2, \ldots \models q. \tag{9.26}$$

Propositions $p_1, p_2, \ldots$ are called the **antecedent**, q, the **consequence**, and $\models$, the entailment symbol. However, note that the converse

$$q \models p_1, p_2, \ldots \tag{9.27}$$

might not hold.

To make this more concrete, let

- $p :=$ "Your dad drove outside,"

- $q :=$ "Your dad went out to get milk," and

then if

$$p \vDash q \tag{9.28}$$

this means that whenever your dad drive outside, you can confirm that he is definitely getting milk. However, it doesn't mean that if he's getting milk, he has to drive outside. He might've walked, or flown, or swam to get milk, etc. Entailment doesn't always work in the reverse.

Similarly, entailment also doesn't guarantee that the antecedent is true. It just guarantees that in *every* universe that the antecedent is true, the consequence must also be true. Your dad doesn't have to drive outside for this entailment to hold. The entailment just says that in *every* universe that he drives outside, he is going to get milk.

Soon, you'll also see that there are **rules of inference**, which aids logical inference. But as of now, our palette is still limited: we know what a proposition is, how letters can represent them, $\top$, $\bot$, and the equivalence between propositions. It is still not possible to chain propositions together to create chain of reasonings that'd make use of rules of inference. Hence, we shall see how propositions are linked together using logical connectives first.

9.3 Chaining propositions: logical connectives and truth tables

9.3.1 Negation

The simplest logical connectives is the **negation**.

Definition 9.3.1: *Negation*

The **negation** of a proposition p is a proposition denoted $(\neg p)$ which has the truth value opposite to p.

The symbol $\neg$ is called the **negation symbol**.

If proposition p represents "Your father is getting milk," the negation of p, $(\neg p)$ then represents "Your father is **not** getting milk." If the assertion that p is true, i.e., your father is getting milk is true, then the assertion that $\neg p$ is true, i.e., your father is **not** getting milk must be false.

The negation can also be used on true and false symbol:

$$\top \equiv \neg\bot \quad \text{and} \quad \bot \equiv \neg\top. \tag{9.29}$$

The equivalence relation $(\equiv)$ still holds when we apply negation to both sides, i.e., if

$$\begin{aligned} \text{You can stand on Jupiter} &\equiv \bot \\ \neg(\text{You can stand on Jupiter}) &\equiv \neg\bot \\ \text{You can } \textbf{not} \text{ stand on Jupiter} &\equiv \top. \end{aligned}$$

It is left as an exercise to the reader to test these propositions by flying to Jupiter.

It's useful to summarize the relation between a proposition and another by using a **truth table** as shown in table 9.1.

p	$\neg p$
$\top$	$\bot$
$\bot$	$\top$

Table 9.1 | CHARACTERISTIC TRUTH TABLE OF NEGATION

9.3.2 Logical conjunction, logical disjunction

Definition 9.3.2: *Logical conjunction*

The **conjunction** of propositions p and q is a proposition denoted by $(p \wedge q)$. This proposition carries the truth value of true if and only if both p and q are true.

The symbol $\wedge$ is called the **logical conjunction**, **conjunction**, or **AND** symbol.

E.g., let

1. $p :=$ "Your father is getting milk."
2. $q :=$"Your father is getting butter."

Then, $(p \wedge q)$ represents the proposition "Your father is getting milk *and* butter." If $(p \wedge q)$ has been observed or proven to be true, then you can infer that he's getting both. In contrary, if $(p \wedge q)$ is false, we can say that either

1. "Your father is not getting anything,"
2. "Your father is getting only the milk, or"
3. "Your father is getting the butter."

Definition 9.3.3: *Logical disjunction*

The **disjunction** of propositions p and q is a proposition denoted by $(p \vee q)$. This proposition carries the truth value of true if either one of the propositions are true.

The symbol $\vee$ is called the **logical disjunction**, **disjunction**, or the **OR** symbol.

Let's use the example prior. If $(p \vee q)$ has been observed or proven to be true, then there are three possible cases

1. "Your father is getting just the milk,"
2. "Your father is getting just the butter, or"
3. "Your father is getting both the milk and the butter."

If $(p \vee q)$ has been observed to be false, then there's only one possible case: "Your father is coming home empty-handed."

The characteristic truth table of both conjunction and disjunction operations is shown in table 9.2

p	q	$(p \wedge q)$	$(p \vee q)$
$\top$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\bot$	$\top$
$\bot$	$\bot$	$\bot$	$\bot$

Table 9.2 | TRUTH TABLE FOR $(p \wedge q)$ AND $(p \vee q)$

Notice that swappin the propositions of the conjunction and disjunction does not change the outcome.

$$(p \vee q) \equiv (q \vee p), \quad \text{and} \quad (p \wedge q) \equiv (q \wedge p). \tag{9.30}$$

The connectives that swapping the input position does not change the output is said to have the **commutative property**.

9.3.3 Implications: connectives of entailment

One important thing about entailment is that the sentence $p \models q$ is not a proposition. It does not carry any truth value. Rather, it is an assertion that forces q to be true whenever p is true. Hence, $p \models q$ does not carry any truth value and is not useful for chaining propositions together.

To capture the behavior of entailment into the language of logic, we need implication.

Definition 9.3.4: *Material implication*

The **material implication** of propositions p and q is a proposition denoted $(p \implies q)$ which is true if and only if p entails q.

In $(p \implies q)$, p is called the antecedent and q is called the consequent.

The material implication is also known as implication, or conditional. The symbol $\implies$ is called the **implication** or **conditional** symbol.

To arrive at the characteristic truth table for the implication, we can go through these evaluations and assign truth values to them one by one to get the desirable behavior of implication.

Since the only way that entailment would not hold is for the antecedent to be false and the consequent to be true, the only time that $(p \implies q)$ should be false is if p is false, but q is true. Hence, we write the characteristic truth table shown in table 9.3.

p	q	$(p \implies q)$
$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\top$
$\bot$	$\bot$	$\top$

Table 9.3 | CHARACTERISTIC TRUTH TABLE OF IMPLICATION

To illustrate the usage, consider the following propositions:

- $p :=$ It rained — The antecedent
- $q :=$ The ground is wet — The consequence

There are then four evaluations possible.

1. It rained, and the ground is wet ($p \equiv \top$, $q \equiv \top$)
2. It rained, but the ground isn't wet ($p \equiv \top$, $q \equiv \bot$)
3. It didn't rain, and the ground is wet ($p \equiv \bot$, $q \equiv \top$)
4. It didn't rain, and the ground isn't wet ($p \equiv \bot$, $q \equiv \bot$)

By our model of reasoning, if it rained, then the ground should be wet. If it didn't rain, then the ground could either be dry or wet. I might've spilled some water on the ground, or a dog might've urinated on the ground, causing the ground to be wet without any rain. These reasoning makes sense, and these are reflected in the assertion of the material conditional: if $p \models q$, then either of these cases are possible:

- $p \equiv \top, q \equiv \top$,
- $p \equiv \bot, q \equiv \top$,
- $p \equiv \bot, q \equiv \bot$.

In all these cases listed here, the material implication should evaluate to true. But,

- $p \equiv \top, q \equiv \bot$

is not possible if $p \models q$. Hence, the material implication evaluates to false.

In contrast with the logical conjunction and disjunction, the implication is *not* commutative.

$$(p \implies q) \quad \text{is not the same as} \quad (q \implies p). \tag{9.31}$$

as you can easily check from the characteristic truth table.

Assuming that $(p \implies q)$ is the same thing as $(q \implies p)$ is a logical fallacy that's named **affirming the consequent** or **fallacy of the converse**. For example, if p represents "I am a human" and q represents "I am a mammal." Since a human is a type of mammal, $p \models q$; thus, $(p \implies q)$ which reads "I am a human, hence I am also a mammal," is true. But then, does $q \models p$? In this world that we live in, being mammal doesn't mean being a human. A dog is also a mammal, a cat is also a mammal, etc. Hence, there are certain cases where q is true but p is false. Hence q does not entail p but p entails q.

This fallacy might sound trivial, but it will creep through the page and cause your proof to be invalid if you're not careful. A classic one would be the square root argument. If $x = 3$, then $x^2 = 9$. But $x^2 = 9$ does *not* mean $x = 3$ because $x = -3$ would also suffice. Another advanced one would be the limits and boundedness of functions. The existence of the limit $\lim_{x \to a} f(x)$ entails the boundedness of $f(x)$ is bounded in some neighborhood of a that doesn't contain a itself. It sounds plausible for the converse to be true: if $f(x)$ is bounded near a, then $\lim_{x \to a} f(x)$ should exists when in reality, there are many functions which doesn't have this property. E.g.,

$$f(x) = \sin\left(\frac{1}{x}\right) \tag{9.32}$$

This is also called the **topologist's sine curve**. The function is bounded on its entire domain. However, the limit $\lim_{x \to 0} f(x)$ does not exist because the graph is forever oscillating as shown in fig. 9.2.

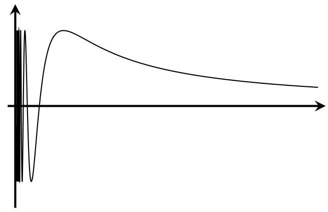


Figure 9.2 | TOPOLOGIST SINE CURVE
$f(x) = \sin\left(\frac{1}{x}\right)$

9.3.4 Biconditional: connectives of equivalence

Definition 9.3.5: *Logical biconditional*

The **logical biconditional** of propositions p and q is a proposition denoted $(p \iff q)$ in which

$$(p \iff q) := ((p \implies q) \wedge (q \implies p)). \tag{9.33}$$

By the rules of logical conjunction, $(p \iff q)$ is true when both $(p \implies q)$ and $(q \implies p)$ are true. I.e., the proposition that's formed via the biconditional is true only when the entailment goes both ways. E.g., consider these propositions:

1. $p :=$ "You can see the stars."
2. $q :=$ "It is currently nighttime."

Obviously,

$$p \models q \quad \text{and} \quad q \models p : \tag{9.34}$$

if it's nighttime you must see the stars[2], and if you can see the stars it must be nighttime. Since the entailment goes both ways, the proposition:

[2]Unless you are in a very bright city then that's just depressing.

p	q	$(p \iff q)$
$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$
$\bot$	$\top$	$\bot$
$\bot$	$\bot$	$\top$

Table 9.4 | CHARACTERISTIC TRUTH TABLE OF $p \iff q$

- $(p \iff q) =$ "You can see the stars because it is night, and you can tell that it is night by seeing that there are stars"

must be true. In a similar manner,

- $((\neg q) \iff (\neg p)) =$ "You can't see the stars because it is day, and you can tell that it's day by seeing that there are no stars."

must also be true aswell.

The characteristic truth table of the biconditional is given in table 9.4

The biconditional is related to the equivalence symbol like how material implication is related to entailment. If the proposition $(p \iff q)$ is true, then $p \equiv q$ and vice versa.

Since the phrase "if and only if," is quite long, it's common to abbreviate the biconditional to ***iff*** with a double "f" and in italics. This is to differentiate the biconditional from the implication.

9.3.5 A well-formed sentence

As I've hinted in the definitions prior, a proposition can be formed by using logical connectives on already known propositions to create a new proposition. This is captured in the following definition:

Definition 9.3.6: *A well-formed sentence*

If p and q are propositions, then nothing else is a proposition except for the following

- $(\neg p)$
- $(p \wedge q)$
- $(p \vee q)$
- $(p \implies q)$
- $(p \iff q)$

This definition prevents gibberish from being a proposition that we would have to consider. E.g.,

$$\neg \wedge \vee p \vee q \implies p \quad \text{or} \quad ABBC \wedge q \iff \implies \neg X \tag{9.35}$$

would not be a proposition because they don't obey the rule of a well-formed propositions.

Let's start with a simple relationship between three propositions:

1. p represents "Your mom calls your father."
2. q represents "Your father is out to get milk."
3. r represents "Your father will never come back."

We could create a new proposition that says

$$\big(p \implies (q \wedge r)\big), \tag{9.36}$$

which reads:

> If your mother calls your father, then he's going to be out to get milk and never come back.

p	q	r	$(q \wedge r)$	$(p \implies (q \wedge r))$
$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\top$	$\bot$	$\bot$	$\bot$
$\top$	$\bot$	$\top$	$\bot$	$\bot$
$\top$	$\bot$	$\bot$	$\bot$	$\bot$
$\bot$	$\top$	$\top$	$\top$	$\top$
$\bot$	$\top$	$\bot$	$\bot$	$\top$
$\bot$	$\bot$	$\top$	$\bot$	$\top$
$\bot$	$\bot$	$\bot$	$\bot$	$\top$

Table 9.5 | TRUTH TABLE OF THE PROPOSITION $(p \implies (q \wedge r))$

There are eight possible situations for eq. (9.36) which can be written as a truth table shown in table 9.5.

If the proposition $(p \implies (q \wedge r))$ is true, then $p \models (q \wedge r)$ and these situations are allowed to happen:

1. Your mom did call your father. He went out to get milk, and isn't going to come back.
2. Your mom did not call your father, but he still went out to get milk, and isn't going to come back.
3. Your mom did not call your father, but he went out to get milk, and is coming back.
4. Your mom did not call your father, and he didn't go out to get milk, but isn't going to come back.
5. Your mom did not call your father, he didn't go out to get milk, and is going to come back.

If the proposition is false, then these three cases are then allowed to happen as well.

1. Your mom did call your father. He went out and get milk, but he came back.
2. Your mom did call your father. He didn't go out to get milk, but he's never coming back.
3. Your mom did call your father, but he didn't go out to get milk, and he's never coming back.

We can always extend this method of writing a truth table for more propositions, but that's not in our interests right now. If you want to try it out, then there are exercises are in the end of the chapter.

When the propositions gets really long, we drop some of the parenthesis according to the following rules in order to minimize clutter.

Notation 9.3.7: *Rules for dropping parenthesis in propositions*

1. Drop the outer-most parenthesis
2. If a negation is infront of a single-letter proposition, then drop the parenthesis that covers it. E.g.,

$$(\neg p) \wedge q = \neg p \wedge q, \tag{9.37}$$

$$(A \implies (\neg B)) \wedge (\neg C) = (A \implies \neg B) \wedge \neg C. \tag{9.38}$$

However, note that if the negation is infront of a parenthesis, it could not be dropped. E.g.,

$$\neg(A \vee B) \neq \neg A \vee B. \tag{9.39}$$

3. If propositions are chained together by the same commutative logical connectives, drop the parenthesis. E.g.,

$$p \wedge (((q \wedge r) \wedge (s \wedge t)) \wedge u) = p \wedge q \wedge r \wedge s \wedge t, \tag{9.40}$$

$$(p \iff q) \iff ((r \iff s) \iff t) \\ = p \iff q \iff r \iff s \iff t, \tag{9.41}$$

$$(p \vee q) \vee (r \vee (s \vee t)) = p \vee q \vee r \vee s \vee t \tag{9.42}$$

But note that if the connectives are *not* the same, you can't drop the parenthesis. E.g.,

$$p \vee (q \wedge r) \neq p \vee q \wedge r. \tag{9.43}$$

9.4 Logical structure. Tautologies and contradiction

With these logical connectives, there deem to be some propositions that are always true no matter how its evaluated or false no matter how it is evaluated: its truth value depends on the logical structure of how the propositions are chained together, not how its evaluated. E.g.,

$$p \vee \neg p \tag{9.44}$$

is always true. If p is true, then $\neg p$ is false. If p is false, then $\neg p$ is true. In all evaluations of p and $\neg p$, either one of these must be true. Hence, their disjunction is **always true** all the time. We call these statements: a **tautology**. Since $\top$ is always true, it is also commonly referred to as a tautology, and every tautology is equivalent to $\top$.

In the other hand, there are also statements that are always false. E.g.,

$$p \wedge \neg p \tag{9.45}$$

These are called **contradictions**. Similarly, $\bot$ is also called a contradiction.

9.5 Useful logical identities

All the interactions between tautologies and contradiction via logical operators are already given in tables 9.1, 9.3 and 9.4; thus, they won't be mentioned here. Now, let p be a proposition. Then, the following seven propositions hold:

Proposition 9.5.1 $p \lor \top \equiv \top$.

The logical disjunction only requires one proposition to be true. If there's one tautology, the whole statement is a tautology.

Proposition 9.5.2 $p \lor \bot \equiv p$.

If one proposition is a contradiction, the truth value of the statement is dependent on p: p being a tautology makes the statement true, and p being a contradiction will make the statement false.

Proposition 9.5.3 $p \land \top \equiv p$.

The logical conjunction requires both propositions to be true. If one is already a tautology, the truth value of the statement is dependent on the other value.

Proposition 9.5.4 $p \land \bot \equiv \bot$.

It only takes one false proposition for the logical conjunction to output a contradiction.

Proposition 9.5.5 $p \implies \top \equiv \top$.

If p is true, then the reasoning is valid. If p is false, the consequence could be from anything else, and the reasoning would still be sensible. Therefore, $p \implies \top$ is a tautology.

Proposition 9.5.6 $p \to \bot \equiv \neg p$.

If p is false, then $\bot \to \bot$ is true. If p is true, then $\top \to \bot$ is false.

Proposition 9.5.7 $p \iff \top \equiv p$, and $p \iff \bot \equiv \neg p$.

The statement could only be true if both match; therefore, the truth value of the statement is completely dependent on p.

9.6 Rules of inferences and logical structure

Before we go any further, note that there are going to be a lot of weird names coming up. You don't have to remember them. Most of them are trivial rules that you'll be able to make sense of. Just think through them logically, and you'll be fine.

9.6.1 Modus ponens

Let's start with a simple example called **modus ponens**. It says that, if p is true and $p \to q$ is true, q must also be true. I.e.,

$$(p \wedge (p \implies q)) \vDash q \tag{9.46}$$

To prove this, we translate the rule into mathematical form

$$(p \wedge (p \implies q)) \implies q, \tag{9.47}$$

Then, we can either construct the truth table (table 9.6), or we could prove it by contradiction

p	q	$(p \to q)$	$p \land (p \to q)$	$(p \land (p \to q)) \to q$
$\top$	$\top$	$\top$	$\top$	$\top$
$\top$	$\bot$	$\bot$	$\bot$	$\top$
$\bot$	$\top$	$\top$	$\bot$	$\top$
$\bot$	$\bot$	$\top$	$\bot$	$\top$

Table 9.6 | TRUTH TABLE OF MODUS PONENS

It might be a hard concept to wrap your head around at first, but stay with me here. **Proof by contradiction** is a form of proof where we assume that the statement is false, then show that it leads to a logical fallacy; therefore, the statement cannot be false: it must be a tautology. I shall illustrate the concept by proving the modus ponens:

1. Assume that $(p \land (p \to q)) \to q$ is false.
2. Because the statement is in two parts that are joined by a logical condition, the reason $p \land (p \to q)$ must be true, and the consequence q must be false in order for the statement to be false.
3. Since $q \equiv \bot$, $(p \land (p \to q))$ becomes $(p \land (p \to \bot))$.
4. $p \to \bot$ is $\neg p$; thus, $(p \land (p \to \bot))$ simplifies to $p \land \neg p$
5. Because $p \land \neg p$ is joined by a logical disjunction, and one of them is always false, the reason as a whole is false.
6. But from item 2, the reason must be true, which contradicts with item 5, thus creating a logical fallacy.
7. This logical fallacy can only be resolved if the original statement is true in the first place. Therefore, the modus ponens is a tautology.

We then write an inference table shown in table 9.7, where every sentence above the black line is proven to be a true statement, and we put the conclusion under the black line.

$$\begin{array}{rl} & p \\ & p \to q \\ \hline \therefore & q. \end{array}$$

Table 9.7 | INFERENCE TABLE FOR MODUS PONENS

9.6.2 Modus tollens

The inference table for modus tollens is shown in table 9.8. The table reads "If $p \to q$ is a reasoning that is proven to be sensible, and q didn't happen, then p never happened as well."

$$\begin{array}{rl} & p \to q \\ & \neg q \\ \hline \therefore & \neg p. \end{array}$$

Table 9.8 | INFERENCE TABLE FOR MODUS TOLLENS

To prove this, we turn the inference table into $((p \to q) \land \neg q) \to \neg p$ then use proof by contradiction:

1. Assume that $((p \to q) \land \neg q) \to \neg p$ is false.
2. Because the statement is in two parts that are joined by a logical condition, the reason $((p \to q) \land \neg q)$ must be true and the consequence $\neg p$ must be false in order for the statement to be false.
3. If $\neg p$ is false, then p is true.
4. Because the reason is joined by a logical conjunction, both $p \to q$ and $\neg q$ must be true in order for the reason to be true.
5. If $\neg q$ is true, then q is false.
6. Substitute q is false and p is false from item 3 into the reason. $(p \to q) \land \neg q$ becomes $(\bot \to \bot) \land \neg\top \equiv \bot$: the reason is false.

7. Item 6 contradicts with item 2 which says that the reason should be true, thus creating a logical fallacy.
8. This logical fallacy can only be resolved if the original statement is true in the first place. Therefore, the modus tollens is a tautology.

9.6.3 Biconditional introduction and elimination

$$\begin{array}{rl} & p \to q \\ \hline \therefore & p \iff q, \end{array}$$

Table 9.9 | INFERENCE TABLE FOR THE BICONDITIONAL INTRODUCTION

The inference table for the biconditional introduction is shown in table 9.9 which is literally the definition of the biconditional (**??**). The biconditional elimination goes the other way: if you know that $p \iff q$ is true, you can infer that both $p \to q$ and $q \to p$ are true.

9.6.4 Conjunction introduction and elimination

The inference table for the conjunction introduction is shown in table 9.10 which reads: "If p is true and q is true, then $p \vee q$ must also be true." This statement is trivial because $\vee$ needs both proposition to be true, and both p and q are already true. The conjunction elimination goes the

$$\begin{array}{rl} & p \\ & q \\ \hline \therefore & p \wedge q, \end{array}$$

Table 9.10 | INFERENCE TABLE FOR THE CONJUNCTION INTRODUCTION other way around: if $p \wedge q$ is known to be true, then we can infer that p is true, and q is true.

9.6.5 Disjunction introduction aka addition

$$\begin{array}{rl} & p \\ \hline \therefore & p \vee q\,, \end{array}$$

Table 9.11 | Inference table for the disjunction introduction

The inference table for the disjunction introduction is shown in table 9.11 which reads, if p is true, then p or q is always true. This should also be trivial: the disjunction outputs true when one of the statements is true, which p is already proven to be true.

9.6.6 Disjunction elimination

The inference table for the disjunction elimination is shown in table 9.12. It is a complex one, so I'm going to give a word example. Let there be these propositions:

1. p representing "Your father is getting milk."
2. q representing "Your father is not coming back."
3. r representing "Your father is getting butter."

Thus,

$$\begin{array}{rl} & p \to q \\ & r \to q \\ \hline \therefore & q\,. \end{array}$$

Table 9.12 | Inference table for the disjunction elimination

1. $p \rightarrow q$ means "If your father is getting milk, he's not coming back."
2. $r \rightarrow q$ means "If your father is getting butter, he's not coming back."
3. $p \vee r$ means "Your father is getting either milk or butter."

The conclusion q represents "Your father is not coming back".

To prove this rule mathematically, we turn the inference table into $((p \rightarrow q) \wedge (p \rightarrow r) \wedge (p \vee r)) \rightarrow q$, then proceed with proof by contradiction:

1. Assume that $((p \rightarrow q) \wedge (p \rightarrow r) \wedge (p \vee r)) \rightarrow q$ is false.
2. Because the whole statement is in two parts that are joined by a logical condition, the reason $((p \rightarrow q) \wedge (r \rightarrow q) \wedge (p \vee r))$ must be true, and the consequence q must be false in order for the statement to be false.
3. Substitute $q \equiv \bot$ into the reason and get $((p \rightarrow \bot) \wedge (r \rightarrow \bot) \wedge (p \vee r))$ which simplifies to $\neg p \wedge \neg r \wedge (p \vee r)$.
4. From item 2, the reason must be true. Since the reason is in three parts which are all joined logical conjunction, we can use conjunction elimination to conclude that $\neg p$, $\neg r$, and $p \vee r$ are all true.
5. Since both $\neg p$ and $\neg r$ are true, both p and r are false.
6. $p \vee r$ evaluates to $\bot$, which contradicts with item 5 and creates a logical fallacy.
7. This logical fallacy can only be resolved if the original statement is true in the first place. Therefore, the rule of disjunctive elimination is a tautology.

$$\begin{array}{rl} & p \lor q \\ \hline \therefore & q. \end{array}$$

Table 9.13 | INFERENCE TABLE FOR THE DISJUNCTIVE SYLLOGISM

9.6.7 Disjunctive syllogism

The inference table is shown in table 9.13. This one is also a hard one, so let me give a word example. Let

1. p — Your father is getting milk,
2. q — Your father is getting butter.

Therefore,

1. $p \lor q$ means "Your father is getting either the milk or the butter,"
2. $\neg p$ means "Your father is not getting the milk,"

which concludes in q, which means "Your father is getting the butter." To prove this, we turn the rule of disjunctive syllogism into

$$((p \lor q) \land \neg p) \equiv q, \tag{9.48}$$

then proceed with proof by contradiction: assume that the disjunctive syllogism is false. Then,

1. Because the statement is in two parts that are joined by a logical condition, in order for the statement to be false, the reason $((p \rightarrow q) \land \neg p)$ must be true, and the consequence q must be false.
2. Substitute $q \equiv \perp$ into the reason and get $(p \rightarrow \perp) \land \neg p$ which simplifies to $\perp \land \neg p$.

$$\begin{array}{rl} & p \to q \\ \hline \therefore & p \to r. \end{array}$$

Table 9.14 | INFERENCE TABLE FOR THE HYPOTHETICAL SYLLOGISM

3. Since the logical conjunction between a contradiction and anything is false, it contradicts with item 1 which says that the reason should be true; creating a logical fallacy.
4. This logical fallacy can only be resolved if the original statement is true in the first place. Therefore, the rule of disjunctive syllogism is a tautology.

9.6.8 Hypothetical syllogism

The inference table for table 9.14 is shown in table 9.14. You should probably know the drill by now. Assume that

$$((p \to q) \wedge (q \to r)) \to (p \to r), \tag{9.49}$$

is false. Then,

1. Because the statement is in two parts that are joined by a logical condition, the reason $(p \to q) \wedge (q \to r)$ must be true, and the consequence $p \to r$ must be false in order for the statement to be false.
2. The consequence, $p \to r$ is joined by a logical condition, and is false. Therefore, the reason to this consequence p must be true, and the consequence for this consequence, r, must be false.
3. Substitute $p \equiv \top$ and $r \equiv \bot$ into the reason and get $(\top \to q) \wedge (q \to \bot)$.
4. The first parentheses evaluate to q, and the second evaluate to $\neg q$. Thus, the statement $(\top \to q) \wedge (q \to \bot)$ is false.

$$\begin{array}{rl} & p \rightarrow q \\ & r \rightarrow s \\ \hline \therefore & q \vee s. \end{array}$$

Table 9.15 | INFERENCE TABLE FOR THE CONSTRUCTIVE DILEMMA

$$\begin{array}{rl} & p \rightarrow q \\ & r \rightarrow s \\ \hline \therefore & \neg p \vee \neg r. \end{array}$$

Table 9.16 | INFERENCE TABLE FOR THE DESTRUCTIVE DILEMMA

5. Item 4 and item 2 contradicts and creates a logical fallacy. This fallacy can only be resolved if the original statement is true. Therefore, the rule of hypothetical syllogism is a tautology.

9.6.9 Constructive and destructive dilemma

Both the **constructive dilemma** (table 9.15) and **destructive dilemma** (table 9.16) are quite complex, so I'll give a word example. For the constructive dilemma, let

1. $p \rightarrow q$ represents "If your father goes out, he will get milk."
2. $r \rightarrow s$ represents "If your mother goes out, she will go and find your father."
3. $p \vee r$ represents "Either your father goes out, or your mother goes out."
4. Therefore, $q \vee s$ represents "Either your father gets milk, or your mother will go find your father."

To prove this, we turn the inference table into

$$((p \to q) \land (r \to s) \land (p \lor r)) \to (q \lor s), \tag{9.50}$$

then proceed with proof by contradiction.

1. Assume that $((p \to q) \land (r \to s) \land (p \lor r)) \to (q \lor s)$ is false.
2. Because the statement is in two parts that are joined by a logical condition, the reason $(p \to q) \land (r \to s) \land (p \lor r)$ must be true, and the consequence $q \lor s$ must be false in order for the statement to be false.
3. The consequence $q \lor s$ is false and joined by a logical disjunction. Thus, both q and s are false.
4. Substitute $q \equiv s \equiv \bot$ into the reason and get $(p \to \bot) \land (r \to \bot) \land (p \lor r)$.
5. The reason evaluates to $\neg p \land \neg r \land (p \lor r)$, which by conjunction elimination, $\neg p \equiv \neg r \equiv (p \lor r) \equiv \top$.
6. $\neg p \equiv \neg r \equiv \top$ means $p \equiv r \equiv \bot$.
7. $p \equiv r \equiv \bot$ means $p \lor r \equiv \bot$ which contradicts with item 5, creating a logical fallacy.
8. This fallacy can only be resolved if the original statement is true. Therefore, the rule of constructive dilemma is a tautology.

The **destructive dilemma** is just the constructive dilemma but with "Either your father doesn't get milk, or your mother doesn't go find your father". It is left as an exercise to the reader to prove that the rule of destructive dilemma is always true.

9.7 Conclusion for the second half of chapter 9

1. Rules of inference are used to determine what is true using other true statements.
2. Each of the rules of inference has to be a tautology, and we prove that by either constructing a truth table, or writing down a proof by contradiction.
3. Proof by contradiction is a form of proof that establishes that a proposition is a tautology by showing that when we assume that the proposition is false, it leads to a logical fallacy.

From now on, I shall write most mathematical statements in terms of logical symbols in order to get rid of ambiguity in the natural language.

CHAPTER 10

Set theory

CHAPTER 11

The foundations: natural numbers

Prerequisites: *logic, propositions, truthfulness and falsehood, logical operations, sets, functions*

11.1 Invitation: foundations of analysis

A standard course on analysis usually progresses like this.

- Glaze over the properties of real numbers
- Sequences, limit of sequences, limit of functions
- Continuity of functions and derivatives
- The Riemann integrals.

Notice that the construction of number systems aren't in there. Why did I choose to differ?

I believe that the name of the subject, *analysis*, should reflect the contents of the subject: to *analyze* mathematics. Hence, we should go back to its root: the natural numbers, and build everything up from there. Although this method is more tedious than the standard course, it paces the course so that the reader can build up mathematical maturity. At the end, you'd be able to construct a mathematical system of your own, and hopefully, be able to appreciate the wonders that nature gave us through the humble numbers.

Since this book is *not* about basic mathematics, we wouldn't go over how to count, how the number line works, how objects are represented as naturals, etc. They should be trivial for the voyagers of mathematical analysis. Hence, I shall direct all attention to the axiomatic construction of natural numbers. This chapter is planned out as follows:

1. Axiomatic construction of the naturals
2. The mathematical induction theorem
3. Addition, multiplication, and their properties
4. Order and the well-ordering principle

Most of the things stated here are trivial, but is crucial for maturing the skills of writing a good proof, which would be necessary for later chapters. Without the intention of developing this skill, this chapter is merely just a brain-decoration: practically useless to read. Hence, *please* try to prove or give reasons to each statement first before looking at the solution in order to get the most out of this chapter.

11.2 Axiomatic formulation of natural numbers

The big question here is: *what does it mean to be a number*? Let's say there's an alien species coming to us. How would you explain the concept of a number to them? You might point to your finger and say "This is the number one." The alien then thinks: "Ah, that finger is the number one". Clearly, this is incorrect. There's also other objects which can represent the number one. So you might point to your head and say that "I have only one head!" The alien then thinks: "Oh, the number one could be both your finger or your head." Again, the alien still misses the essence of what a number truly means.

The challenge arises because numbers are abstract concepts that aren't tied to any physical objects. There is no object in this world that can represent the concept of *number one* because a number is a *concept*, not an object. So, it's practically impossible to tell the alien what a number is using real world objects. Instead, one must rely on *abstractions*: using purely mathematical concepts to tell the alien what a number is and how it behaves. Here, we shall only allow ourselves to use the following:

1. Logic
2. Sets and their operations
3. Equality
4. Functions and mappings

We first start by the notation of natural numbers

Notation 11.2.1: *The set of natural numbers*

$\mathbb{N}$ is the set that contains all natural numbers, also sometimes called *the naturals*.

To construct the natural numbers, we must define what is in and isn't $\mathbb{N}$ by specifying what it means for an object to *be* a natural number.

There are three standard ways to do this: Peano's axiom, set-theoretical method, and the Church numerals. This chapter uses both the

Peano's axiom and the set-theoretical method. But we would not discuss the Church numerals, as it is a topic of computation theory, not analysis.

Conceptually, the **natural number system** is a collection of random symbols in a set T; each symbol in T is connected by a relation that I shall informally call the *next to* relation. E.g., 1 is *next to* 0, 2 is *next to* 1, etc. If all the elements in T form *one single ordered line* when they are arranged according to this relation, then "the set T equipped with that relation" is a system of natural numbers. To honor the natural-ness of T, we also rename T to $\mathbb{N}$.[1]

In the usual natural number system that we know and love, this relation acts like an arrow that points 0 to 1, 1 to 2, 2 to 3, and so on. When you draw this relation out, you get fig. 11.1: the **natural number line**.

However, this number line isn't tied to any kind of symbols. One could replace the Hindus-Arabic symbols with the Greek numerals, Thai numerals, Babylonian numerals, and still get a fully functioning natural number line. The only thing that matters for "a set equipped with a relation" to be considered *the natural number system*, is for every element to form *one single ordered line*.

This detachment from symbols makes manually listing every possible sets and relations that can form the natural number system impossible, because the amount of symbols is infinite. Thus, a rule set is needed for ruling out what qualifies as a natural number system, and

[1] That last part was a joke.

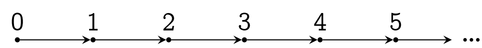


Figure 11.1 | THE NATURAL NUMBER LINE

what doesn't.

Peano's defined the system of naturals to be:

> *Any set that is equipped with a relation, such that the elements with the relation forms* one single ordered line.

This statement is readable by humans, but is not readable for mathematics. If you feed this for mathematics to read, it will say: what is a relation? what does equipped means? what does a line mean? what does it mean for a line to be ordered?, etc. Hence, Peano broke the statement above into four mathematical viable rules listed here:

Rule 1) There is a well-defined start of the line. I.e., there is an element that has no other elements preceding it. Peano chose this element to be zero.

Rule 2) There is always a next one along the line.

Rule 3) There are no branches and loops.

Rule 4) There can only be one line, and no more.

These four rules will soon be converted into the five Peano axioms.

Firstly, these rules do not care about the symbols in the set; hence, all the following lines can form a natural number system:

$$0, 1, 2, 3, 4, \ldots$$

$$2, 4, 6, 8, 10, \ldots$$

$$\mathrm{A}, \mathrm{B}, \mathrm{C}, \mathrm{D}, \mathrm{E}, \ldots$$

$$=, +, <, !, -, \ldots$$

$$\mathrm{I}, \mathrm{II}, \mathrm{III}, \mathrm{IV}, \mathrm{V}, \ldots,$$

Secondly, these rules can also be used to tell what doesn't form the natural number system. E.g., all sets that are equipped with a relation presented in fig. 11.2 can't be the naturals because they violate the following:

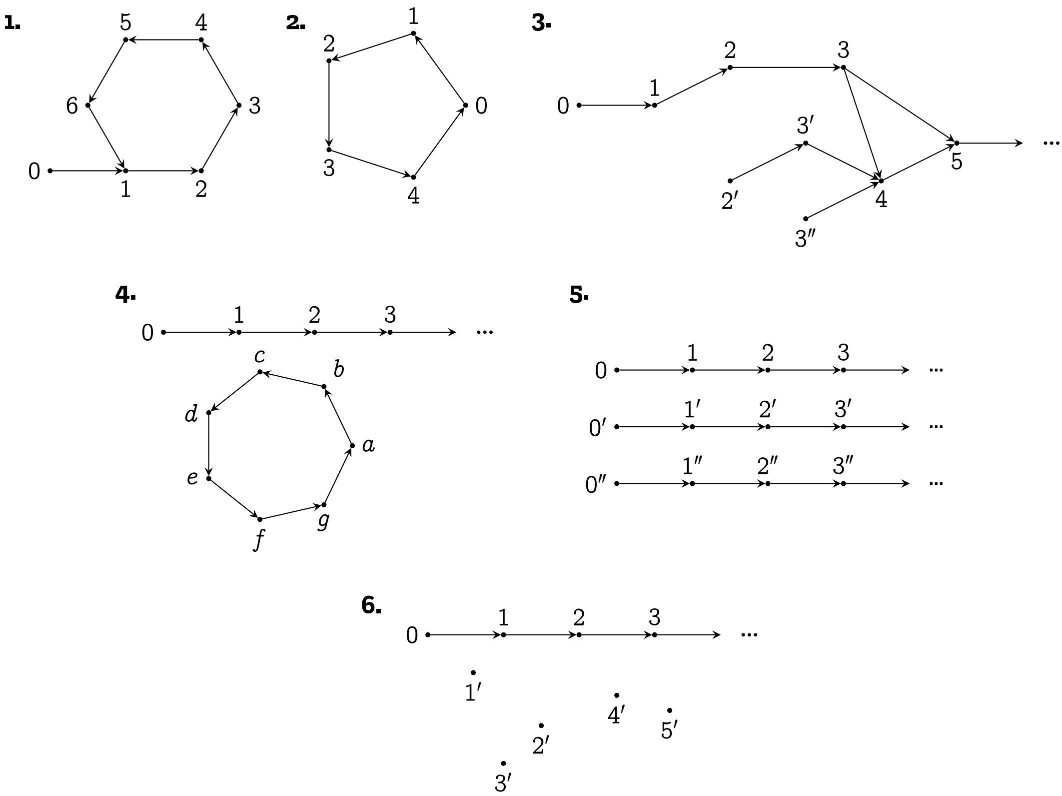


Figure 11.2 | Examples of number systems that are not the naturals

Ex 1. There are loops; hence, it can't be the naturals. (Rule 1)

Ex 2. There are loops; hence, it can't be the naturals. Also, there is no number which has no elements preceding it. Every number has a preceding element; hence, this can't be the naturals. (Rule 1)

Ex 3. There are branches; hence, it can't be the naturals. (Rule 3)

Ex 4. Even though there is an ordered chain of numbers, there are a loop in the set which is disconnected from the chain. Hence, it can't be the naturals. (Rule 3 and Rule 4)

Ex 5. There are multiple ordered chains. Hence, it can't be the naturals. (Rule 4)

Ex 6. There are elements ($1', \ldots, 5'$) that has no naturals next to it (Rule 2)

With these intuitions in mind, it should be easy to see how the mathematical forms of Peano's axiom arose.

Rule 1 is enforced by the first two axioms. The first one postulates the existence of zero, which is the start of the chain.

Axiom 11.2.2: *Existence of zero as a natural number*

The number 0 is a natural number, i.e., $0 \in \mathbb{N}$

The second axiom postulates that zero has no preceding numbers, i.e., there is no number in which the next number is zero. In Peano's universe, the *next to* relation is described using the **successor function** S. If m is next to n, then

$$m = S(n), \tag{11.1}$$

and m is called the **successor** of n. In the natural number system that we know and love, $1 = S(0)$, $2 = S(1)$, and so on. The second axiom then translates to

Axiom 11.2.3: *Zero is not the successor of any numbers*

$$\forall(n \in \mathbb{N}) : S(n) \neq 0.$$

Remarks 11.2.4: *On the beginning of the natural numbers*

A looser yet perfectly valid way to express axioms 11.2.2 and 11.2.3 would be: there exists an element in the naturals that is *not* the successor of any other numbers. Peano just decided to use zero. But generally, this element could be anything. One, two, a cat, you, etc.

To simplify the rest of the discussion, I shall refer to the number system that is formed by the Hindus-Arabic numerals under the following relations:

$$\begin{aligned} S(0) &:= 1, \\ S(S(0)) = S(1) &:= 2, \\ S(S(S(0))) = S(2) &:= 3, \\ S(S(S(S(0)))) = S(3) &:= 4, \\ &\vdots \end{aligned} \tag{11.2}$$

as the **ideal natural number system**, or shortly, **the naturals**

The rest of the rules tell you how each element are connected by the successor function: they dictate the behavior of the successor function.

Rule 2 requires all naturals to have a successor. Otherwise, there would be some element in the number line that is left floating and disconnected from the line of naturals as in the sixth example of fig. 11.2. This is done through the third axiom:

Axiom 11.2.5: *The successor function is closed in the naturals*

$$\forall(n \in \mathbb{N}) : S(n) \in \mathbb{N}$$

This axiom contains more than the disconnection of number lines. It also contains the requirement that the successor of a natural number is still is in the set. Without this, S could take a number from the ideal natural number system and point it to π, e, or even a sheep!

Rule 3 says that the natural number system must have no branch. This is enforced in two ways:

- The natural numbers must not branch outwards, and
- There must be no other branch that connects to the main branch.

Under the condition that the successor is a *function*, there is no branching outwards because a *function* can't be one-to-many; therefore, things like

$$S(0) = 1, \quad \text{and} \quad S(0) = 1', \tag{11.3}$$

can't happen. This condition alone however is not enough to stop a branch that originated elsewhere to conjoin with the main branch.

E.g., let T be a set that includes the ideal naturals, and let $1'$ and $2'$ be an element that aren't equal to any of the naturals such that

$$1', 2' \in T, \quad S(1') = 2' \quad \text{and} \quad S(2') = 3. \tag{11.4}$$

The set T is illustrated in fig. 11.3. Notice that the condition S must be a

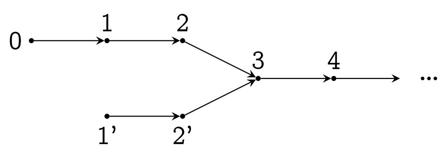


Figure 11.3 | Pathological example of naturals with unrelated branches joining to the main branch

function still allows this situation to happen. So, we need to impose injectivity of S to prevent this. Let's see how.

Axiom 11.2.6: *The successor function is an injection*

$$\forall(n, m \in \mathbb{N}) : \big(S(n) = S(m)\big) \implies (n = m).$$

From fig. 11.3,

$$S(2') = 3 \quad \text{and} \quad S(2) = 3, \tag{11.5}$$

which means

$$S(2') = S(2) \tag{11.6}$$

If S is injective, then this implies

$$2' = 2, \tag{11.7}$$

which contradicts our assumption that $2' \neq 2$. In order to resolve this contradiction, $2'$ must not be a part of the natural number system in the first place. This would leave $1'$ to still be a part of the natural number system. But since $1'$ has no successor now, axiom 11.2.5 says that $1'$ can't be a part of the natural number system as well. Hence, we're left with the ideal natural number system as desired.

Let's now consider a more illustrative pathological example. Consider a set T which contains the ideal naturals and two other elements a and b which is not in the ideal naturals. These two extra elements have the relation such that

$$S(a) = b, \quad \text{and} \quad S(b) = a. \tag{11.8}$$

It should be easy to convince yourself that all the axioms stated earlier (axioms 11.2.3, 11.2.5 and 11.2.6) do permit T to qualify as the naturals. But this shouldn't happen! The naturals should only form *one singular chain* and no more loops!

Another such example is to construct a set R which has the ideal naturals and another chain of numbers such that

$$S(0') = 1', S(1') = 2', S(2') = 3', \dots . \tag{11.9}$$

Similarly, all axioms stated earlier do permit R to be the naturals. But this shouldn't happen because the naturals can't have more than one chain.

Now, I shall state the induction axiom and illustrate how it eliminates all these other pathological number systems.

Axiom 11.2.7: *Induction axiom*

Consider any set T. If all subsets $A \subseteq T$ that has the property that $0 \in A$ and $\forall (n \in A) : S(n) \in A$ also has the property that $A = T$, then T is the naturals.[a] I.e.,

$$\begin{aligned} \big[\forall (A \subseteq T) : [(0 \in A) \wedge (\forall (n \in A) : S(n) \in A)] \implies (A = T)\big] \\ \implies (T \cong \mathbb{N}) \end{aligned}$$

where $\cong$ roughly means, having the same properties. So $T \cong \mathbb{N}$ means, T have the same properties as $\mathbb{N}$

[a] The converse of this statement is proven in corollary 11.3.11

Remarks 11.2.8: *On the beginning element of induction*

In a looser sense, A doesn't have to include zero, but it just has to include an element that is not a successor of any other elements.

Let's now go back to the set T. Let $A = \{0, 1, 2, \dots\}$ $(a, b \notin A)$. Since $0 \in A$ and all elements in A have a following element: $S(0) =$

$1, S(1) = 2, \dots$ but $A \neq T$ because $a, b \notin A$, T can't form the natural number system by the induction axiom.

For the set R, let $A = \{0, 1, 2, \dots\}$. A has zero and all elements in A have a following element. However, $A \neq R$ because $1', 2', \cdots \notin A$, A can't form the natural number system by the induction axiom.

The consequence of this axiom is: if T is a set with multiple lines, then you can choose A to be a set that contains just one of the lines in T. Because there are other lines in T, $A \neq T$; hence, T can't form the natural number system. The exact proof of the concept is in the well-ordering principle, which will be discussed soon. But for now, this should convince you that the induction axiom is capable of getting rid of systems with multiple lines or loops.

Now that the fundamental behaviors of the natural number system has been established. It's time to construct the arithmetical systems from said axioms.

11.3 Mathematical induction

The first step of constructing the system of arithmetic is to cast the induction axiom into a more practical form: mathematical induction.

Theorem 11.3.1: *Mathematical induction*

Let there be a proposition $P(n)$ where the truth value of this proposition depends on the natural number n. If $P(0)$ is true, and $P(n)$ is true implies $P(S(n))$, then $P(n)$ is true for all natural numbers. I.e.,

$$\big[P(0) \wedge \big[\forall (n \in \mathbb{N}) : P(n) \implies P(S(n))\big]\big] \implies \big[\forall (m \in \mathbb{N}) : P(m)\big].$$

The statement $P(0)$ is called the **base case of induction**, and the step of proving that $P(n) \implies P(S(n))$ is called the **induction step**. Since we have to first hypothesize that $P(n)$ is true, then use that to prove that $P(S(n))$ must also be true, the assumption that $P(n)$ is true is called the **inductive hypothesis**.

Mathematical induction is sometimes thought as a chain of dominoes. Each $P(n)$ is like a domino that's waiting to be knocked down. If you can somehow show that each domino is arranged in a way such that if one fall down, the next one would also fall, i.e., $P(n)$ implies $P(S(n))$, and that the first one has already fallen, i.e., $P(0)$ is true, then you can conclude that all dominoes must have also fallen, i.e., $P(n)$ is true for all natural numbers.

Before we prove that this is equivalent to the established induction axiom, we shall go through how we could use this in proofs first. For the ease of pedagogy, let's *informally* assume that all the laws of arithmetics on the naturals have been constructed. We shall also *informally* assume that $S(n) = n + 1$.

11.3.1 Basics of proof by induction

Proposition 11.3.2 $\forall(n \in \mathbb{N}) : 0 + 1 + 2 + \cdots + n = \frac{1}{2}n(n+1)$

Proof. Let the proposition $P(n)$ be true *iff*

$$0 + 1 + 2 + \cdots + n = \frac{1}{2}n(n+1) \tag{11.10}$$

is also true.

To begin the induction, first check that the base case ($P(0)$) is true. The truth value of $P(0)$ is equivalent to the truthfulness of the sentence

$$0 = \frac{1}{2}(0)\big((0)+1\big). \tag{11.11}$$

Since the R.H.S. also evaluates to zero, the sentence is true; hence, $P(0)$ is also true, proving the base case of induction.

Now, let's hypothesize that $P(n)$ is true, i.e.,

$$0+1+2+\cdots+n = \frac{1}{2}n(n+1) \tag{11.12}$$

is true. This is our induction hypothesis. From this, we show that $P(S(n))$ must also be true.

The truth value of $P(S(n))$ is equivalent to the truthfulness of the sentence

$$0+1+2+\cdots+n+S(n) = \frac{1}{2}S(n)(S(n)+1) \tag{11.13}$$

Starting from the L.H.S.,

$$\begin{aligned}
&0+1+2+\cdots+n+S(n) \\
&= (0+1+2+\cdots+n)+S(n) \\
&= \frac{1}{2}n(n+1)+S(n) && \text{Induction hypothesis} \\
&= \frac{1}{2}n(n+1)+(n+1) && \text{Assumption that } S(n)=n+1 \\
&= (n+1)\left[\frac{1}{2}n+1\right] \\
&= (n+1)\left[\frac{n+2}{2}\right] \\
&= \frac{1}{2}(n+1)\big[(n+1)+1\big] \\
&= \frac{1}{2}S(n)\big(S(n)+1\big). && \text{Assumption that } S(n)=n+1
\end{aligned}$$

I.e., the sentence is true; hence, $P(S(n))$ must also be true.

In conclusion, the base case is true and $P(S(n))$ follows from $P(n)$. By the mathematical induction theorem (theorem 11.3.1), $P(n)$ must be true for all naturals. Hence,

$$\forall(n \in \mathbb{N}) : 0 + 1 + 2 + \cdots + n = \frac{1}{2}n(n+1) \qquad \square$$

You might need to reread the previous proof multiple times, and that's normal. Induction does take a while to get used to. Hence, I've prepared another example.

Proposition 11.3.3 $\forall(n \in \mathbb{N}) : 2^0 + 2^1 + \cdots + 2^n = 2^{n+1} - 1$

Proof. Let the proposition $P(n)$ be true *iff*

$$2^0 + 2^1 + \cdots + 2^n = 2^{n+1} - 1 \tag{11.14}$$

is also true.

The base case $P(0)$ is obviously true because the L.H.S. of the mathematical sentence associated to $P(0)$ evaluates to $2^0 = 1$, and the R.H.S. also evaluates to $2^{0+1} - 1 = 1$.

Now, let's assume that the sentence associated to $P(n)$ is true, i.e., eq. (11.14) is true. This assumption is the inductive hypothesis. Let's now prove that the sentence associated to $P(S(n))$ is also true, i.e.,

$$2^0 + 2^1 + \cdots + 2^n + 2^{S(n)} = 2^{S(n)+1} - 1. \tag{11.15}$$

Starting from the L.H.S.,

$$\begin{aligned}
&2^0 + 2^1 + \cdots + 2^n + 2^{S(n)} \\
&= (2^0 + 2^1 + \cdots + 2^n) + 2^{S(n)} \\
&= (2^{n+1} - 1) + 2^{S(n)} && \text{Inductive hypothesis} \\
&= 2^{n+1} + 2^{S(n)} - 1 \\
&= 2^{n+1} + 2^{n+1} - 1 && \text{Assumption that } S(n) = n + 1
\end{aligned}$$

$$= 2^{n+2} - 1$$

$$= 2^{(n+1)+1} - 1$$

$$= 2^{S(n)+1} - 1, \qquad \text{Assumption that } S(n) = n + 1$$

i.e., the mathematical sentence is true; hence, $P(S(n))$ must also be true.

In conclusion, the base case is true and $P(S(n))$ follows from $P(n)$. By the mathematical induction theorem (theorem 11.3.1), $P(n)$ must be true for all naturals as well. Hence,

$$\forall(n \in \mathbb{N})2^0 + 2^1 + \cdots + 2^n = 2^{n+1} - 1 \qquad \square$$

On the next example, I shall shorten the proof a bit to save on some spaces and some typing time.

Proposition 11.3.4 $\forall(n \in \mathbb{N}) : 2^n \geq n$

Proof. Let $P(n)$ be a proposition associated with

$$2^n \geq n$$

Base case: $P(0)$ is associated with

$$2^0 \geq 0 \tag{11.16}$$

The L.H.S. is 2^0, which evaluates to 1. The R.H.S. is 0. Because $1 \geq 0$ is true, $P(0)$ is also true, proving the base case.

Inductive step: Assume that $P(n)$ is true, i.e.,

$$2^n \geq n$$

We shall then prove that $P(S(n))$ follows, i.e., $2^{S(n)} \geq S(n)$. Starting from the L.H.S.,

$$2^{S(n)} = 2^{n+1} \qquad \text{Assumption that } S(n) = n + 1$$

$$= 2^n \times 2$$
$$\geq 2n \qquad \text{Inductive hypothesis}$$

To go forward, we shall prove that $2n \geq S(n)$ for all naturals in order to conclude that $2^{S(n)} \geq S(n)$

1. For $n = 0$, $2n \geq S(n)$ is false because $2 \times 0 = 0$, $S(0) = 0 + 1 = 1$, and $0 \not\geq 1$. However, this doesn't matter because the original statement that we're trying to prove is $P(S(0))$, which states that $2^{S(0)} \geq S(0)$. The L.H.S. evaluates to 1, the R.H.S. evaluates to 1 as well, and $1 \geq 1$. Hence, $P(S(0))$ is true, and we move on to other cases.
2. For $n \geq 1$, we get that $2n = n + n \geq n + 1 = S(n)$. Hence, $2n \geq S(n)$

Since the base case and the inductive step is true, the theorem of mathematical induction assures that $P(n)$ is true for all naturals, closing the induction. □

Remarks 11.3.5

The ability to split the natural numbers into three parts is called the trichotomy of order. I.e., for all natural numbers n, and a fixed natural number m, either

$$n > m, \quad n = m, \quad \text{or} \quad n < m.$$

Or stated in the other way,

$$\{n \in \mathbb{N} : n > m\} \cup \{n \in \mathbb{N} : n < m\} \cup \{m\} = \mathbb{N}$$

Even though this property is trivial, it has its own proof, and it opens the door to so much more. This property will be discussed in detail when the notion of order has been constructed.

11.3.2 Induction that doesn't start at zero

Sometimes, there are statements that isn't true for naturals up to m, but true for all naturals that is greater than m. Such examples are

$$2^n < n!.$$

As you should verify, this statement is true for all naturals except $1, 2, 3$. In order to tackle these proofs, I shall remind you of a very important concept in mathematical logic:

> *From falsehood,* ***anything*** *follows,*

i.e., *false implies anything* is true. This quirk of the implication symbol is abused to pave ways for induction on said statements.

There are two ways of proving a statement that is true for all naturals that is greater than a certain natural number. The first way is to use the said vacuous truth, which I think is richer in theory, and the second way is to use a repeated application of the successor function, which I honestly think is just brute force.

Proposition 11.3.6 $\forall(n \in \mathbb{N}; n \geq 4) : 2^n < n!$

Let $P(n)$ be a proposition that's true if

$$2^n < n!$$

is true.

To prove that $P(n)$ is true for all naturals number greater than 3 using the second method, create a new proposition $Q(n)$ that's associated with the sentence

$$2^{S(S(S(S(n))))} < S(S(S(S(n))))!.$$

$Q(n)$ is true if $P(n+4)$ is true. Since $S(S(S(S(0)))) = 4$, and $2^4 = 16 < 4! = 24$, the base case of $Q(n)$ is true. If it is possible to show that $Q(n) \implies Q(S(n))$ and $Q(0)$ is true, then $Q(n)$ is true for all naturals and $P(n)$ is true for all naturals greater than four.

This is a valid proof, but in my humble opinion, it is by no means fruitful. I see it as hacky trick. So, I've provided the method of vacuous truth, which is perhaps a bit less hacky.

Proof. In order to prove this, the proposition has to be cast to

$$\forall(n \in \mathbb{N}) : (n \geq 4) \implies (2^n < n!)$$

The base case is vacuously true because if $n = 0$, then $n \geq 4$ is false. From falseness, anything follows. Hence, the implication that $(0 \geq 4) \implies (2^n < n!)$. The sentence is also vacuously true for $n = 1, 2, 3$.

Let's now move on to the inductive step to show that *for all* natural numbers, $P(n) \implies P(S(n))$. Let's split the natural numbers into three parts. Less than four, equal to four, and more than four.

For naturals less than four, the implication is vacuously true.

At four, the implication is *not* vacuously true but is still true because $2^4 = 16 < 4! = 24$.

For naturals more than four, let's assume that $P(n)$ is true, i.e., $2^n < n!$, then prove that $P(S(n))$ is also true, i.e., $2^{S(n)} < n!$

$$\begin{aligned} 2^{S(n)} &= 2^{n+1} \\ &= 2 \times 2^n \\ &< 2 \times n! && \text{Induction hypothesis} \end{aligned}$$

Since $n > 4$, $S(n) > 2$.

$$< S(n) \times n!$$

$$= S(n)!$$

In conclusion, $2^{S(n)} < S(n)!$ for follows $2^n < n!$ for all natural numbers, closing the induction. □

11.3.3 Induction axiom, inductive sets, and mathematical induction

In this subsection, I shall finally give a proof that mathematical induction is a consequence of the induction axiom. Before that, I'd like to introduce the notion of **inductive set**.

Definition 11.3.7: *Inductive sets*

A set A is **inductive** under the successor function S *iff* $0 \in A$ and $(n \in A) \implies (S(n) \in A)$. I.e.,

$$(0 \in A) \wedge \left[\forall (n \in A) : S(n) \in A\right]$$

This notion is introduced to simplify the induction axiom to

Corollary 11.3.8: *Induction axiom using inductive set* Let T be a set. If all inductive subsets $A \subseteq T$ under the successor function S is equal to T, then T along with S form a system of natural numbers.

The ideal natural number system forms an inductive set; however, it is not the only inductive set. The fourth and fifth pathological examples from fig. 11.2 and the pathological example from fig. 11.3 also forms an inductive set. Hence, the natural number system is just a special case of an inductive set. What sets the natural number system apart is that it is the *smallest inductive set*.

To prove this, we need to develop some operations on inductive sets

> **Lemma 11.3.9:** *The intersection of inductive sets is inductive* If A and B are inductive under S, then $A \cap B$ is also inductive under S.

Proof. By the property of inductive sets, both $0 \in A$ and $0 \in B$. Hence, $0 \in A \cap B$ as well.

To prove that $A \cap B$ is inductive, we shall do some elements chasing. Let $x \in A \cap B$. By the definition of the intersection, $x \in A \wedge x \in B$. Since both sets are inductive, $S(x) \in A \wedge S(x) \in B$. By the definition of the intersection again, $S(x) \in A \cap B$.

In conclusion, we've proven that $0 \in A$ and $\forall (x \in A \cup B) : S(x) \in A \cap B$; hence, $A \cap B$ must be inductive under S. $\square$

The next thing to prove is that if a set T equipped with a successor function S forms a natural number system, i.e., all the inductive subsets $A \subseteq T$ is equal to T, then it T is the smallest inductive set.

> **Lemma 11.3.10:** *The natural number system is the smallest inductive set* If I is an inductive set under the successor S and T satisfies the induction axiom (axiom 11.2.7) under S, then $T \subseteq I$.

Proof. Let T be a set that satisfies the induction axiom, and let I be *any* inductive set.

Consider the set $A = I \cap T$. By the definition of intersection, $A \subseteq T$. Since both I and T are inductive, by lem. 11.3.9 A is also inductive.

Because T satisfies the induction axiom, $A = T$ because A is an inductive subset of T. By the construction of A, $A = I \cap T$ as well; hence,

$$A = T = I \cap T \tag{11.17}$$

By the definition of the intersection, T must be contained in I. Since this is true for all inductive sets I, T is contained in every inductive set. Hence, it must be the smallest inductive set. □

Before proving the mathematical induction theorem, I'd like to remind you about the converse of the induction axiom is also true:

Corollary 11.3.11: *Converse of the induction axiom*

$$(T \cong \mathbb{N})$$
$$\implies [\forall (A \subseteq T) : [(0 \in A) \wedge \forall (n \in A) : S(n) \in A] \implies (A = T)]$$

Proof. Since $T \cong \mathbb{N}$, it's possible to find a subset $A \subseteq T$ such that $0 \in A$, and for all $n \in T$, $S(n) \in T$. Hence, A is inductive.

By lem. 11.3.10, T is the smallest inductive set because $T \cong \mathbb{N}$ but since $A \subseteq T$ and A is inductive as well and $A \subseteq T$, $A = T$. □

Here, we can finally prove the mathematical induction theorem.

Proof of theorem 11.3.1. Let A be a set that contains every natural number n that makes $P(n)$ true, i.e.,

$$A = \{n \in \mathbb{N} : P(n)\} \tag{11.18}$$

This also has a pretty little consequence that $A \subseteq \mathbb{N}$

Assume that $P(0)$ is true and the implication $P(n) \implies P(S(n))$ is also true, then by the construction of the set A,

$$0 \in A \quad \text{and} \quad (n \in A) \implies \big(S(n) \in A\big). \tag{11.19}$$

This means that A is inductive.

Since the naturals are an inductive set that follows the induction axiom, by lem. 11.3.10, $\mathbb{N} \subseteq A$. But since $A \subseteq \mathbb{N}$ as well, A must equal $\mathbb{N}$. Hence, if both the base case and the inductive step has been proven for $P(n)$, A must be equal to $\mathbb{N}$ and by the construction of A, $P(n)$ must be true for all natural numbers. $\square$

11.4 Set theoretical description of the naturals*

This section can be skipped without any loss of contents flow.

The Peano's axiom, even though fruitful, doesn't mention what the natural numbers actually are. It is an **implicit definition**, i.e., anything that follows axioms 11.2.2, 11.2.3 and 11.2.5 to 11.2.7 are the naturals. It's like saying that if it looks like a duck, swims like a duck, and quacks like a duck, then it *is* a duck.

Even though this abstract definition works, it might lack the intuitive tangibility one would expect for such a simple thing like the naturals. Thus in this section, I shall demonstrate the construction of the naturals using sets using the von-Neumann's method. [2]

The von-Neumann's method replaces zero with an empty set:

[2] If you're skeptical, you might ask: how can you be so sure that this is only the naturals? The solution to this question is greatly beyond the scope of this book, but for the curious I shall reference a brilliant answer from Mathematics Stack Exchange (10).

Definition 11.4.1: *Von-Neumann's zero*

$$0 := \emptyset = \{\}.$$

It also replaces the successor function with

Definition 11.4.2: *Von-Neumann's successor*

The successor of a natural number n is defined as

$$S(n) := n \cup \{n\}.$$

Referring back to the ideal naturals in eq. (11.2), they can all be cast into sets

$$\begin{aligned}
S(0) &:= 1 = 0 \cup \{0\} = \emptyset \cup \{0\} &&= \{0\} \\
S(1) &:= 2 = 1 \cup \{1\} = \{0\} \cup \{1\} &&= \{0, 1\} \\
S(2) &:= 3 = 2 \cup \{2\} = \{0, 1\} \cup \{2\} &&= \{0, 1, 2\} \\
S(3) &:= 4 = 3 \cup \{3\} = \{0, 1, 2\} \cup \{3\} &&= \{0, 1, 2, 3\} \\
&\vdots
\end{aligned} \tag{11.20}$$

The proof that this construction obeys the first four Peano's axioms is quite routine; hence, I shall leave it as an exercise for the reader. However, proving that this construction obeys the induction axiom is a bit tricky, and is a subject of set theory which I shall not cover in this text. For the curious, you can look up the proof in Goldrei's book, *Classic set theory* (3).

The reason that I've brought up this construction is because maximums and minimums are easier to construct here.

Definition 11.4.3: *Maximum and minimum*

Let a, b be von-Neumann natural numbers, define

$$\max\{a, b\} = a \cup b, \quad \text{and} \quad \min\{a, b\} = a \cap b \tag{11.21}$$

It's easy to see why this holds. E.g.,

$$\max\{2,4\} = \{0,1\} \cup \{0,1,2,3\} \tag{11.22}$$
$$= \{0,1,2,3\} \tag{11.23}$$
$$= 4 \tag{11.24}$$

and,

$$\min\{2,4\} = \{0,1\} \cap \{0,1,2,3\} \tag{11.25}$$
$$= \{0,1\} \tag{11.26}$$
$$= 2. \tag{11.27}$$

11.5 Addition on the naturals

Now that the naturals are constructed, we then construct a system of arithmetic from the ground up. Starting with addition.

Definition 11.5.1: *Addition on the naturals*

$$\forall(n, m \in \mathbb{N}) : n + S(m) = S(n + m). \tag{11.28}$$

Definition 11.5.2: *Zero and addition*

$$\forall(n \in \mathbb{N}) : n + 0 = n. \tag{11.29}$$

There has been nothing said about commutativity or associativity of addition yet; therefore, the order in which the successor function or zero is placed is very strict. Zero is defined solely as the **right identity element** of addition; therefore, we can't blindly conclude that $0 + n = n$. Def. 11.5.1 does not say about the successor function being placed on n instead of m either; therefore, statements like $S(n) + m = S(n + m)$ which seems trivial, still have to be rigorously proven. But first, let's see an example of how def. 11.5.1 and 11.5.2 captures the behavior of addition that we know.

Example 11.5.3: $2+2=4$ Given this chain of naturals

$$\begin{aligned} S(0) &\equiv 1, \\ S(S(0)) = S(1) &\equiv 2, \\ S(S(S(0))) = S(2) &\equiv 3, \\ S(S(S(S(0)))) = S(3) &\equiv 4, \\ &\vdots \end{aligned}$$

we have

$$\begin{aligned} 2+2 &= 2+S(1) && (11.30) \\ &= S(2+1). && \text{Def. 11.5.1} \quad (11.31) \end{aligned}$$

and,

$$\begin{aligned} S(2+1) &= S(2+S(0)) && (11.32) \\ &= S(S(2+0)) && \text{Def. 11.5.1} \quad (11.33) \\ &= S(S(2)). && \text{Def. 11.5.2} \quad (11.34) \end{aligned}$$

Since $2 = S(S(0))$,

$$S(S(2)) = S(S(S(S(0)))) = 4; \tag{11.35}$$

therefore, $2+2=4$.

Since we're building this from the ground up, it's going to be a very long process. So, let's set a roadmap. For all natural numbers n, m and k,

1. $0 + n = n$ (Lem. 11.5.4)
2. $S(n) + m = S(n + m)$ (Lem. 11.5.5)
3. $n + m = m + n$ (Lem. 11.5.6)
4. $(n + m) + k = n + (m + k)$ (Lem. 11.5.7)
5. $(m + n = k + n) \implies (m = k)$ (Lem. 11.5.8)
6. $n + 1 = S(n)$ (Corollary 11.5.9)

It's also worth knowing that there are no rules of arithmetic yet. The only thing available for us are just the five Peano's axiom, mathematical induction, and the rules of equality. The most powerful tool we have right now is the mathematical induction theorem. It's littered *everywhere*. Hence, both this section and the next section (section 11.6) shall serve as a very great way to practice induction.

Lemma 11.5.4: *Zero and addition from the left*

$$\forall(n \in \mathbb{N}) : 0 + n = n.$$

Proof. Let $P(n)$ be a statement that is true if $0 + n = n$.

Base case: Show that $P(0)$ is true. By def. 11.5.2,

$$0 + 0 = 0 + 0 = 0. \tag{11.36}$$

Inductive step: As an inductive hypothesis, assume that $P(n)$ is true, i.e., $0 + n = n$. Use this to prove that $P(S(n))$ is true, i.e., $0 + S(n) = S(n)$.

$$\begin{aligned} 0 + S(n) &= S(0 + n) && \text{Def. 11.5.1} && (11.37)\\ &= S(n), && \text{Inductive hypothesis} && (11.38) \end{aligned}$$

completing the induction ☐

Before proving that addition is commutative, we must first prove that def. 11.5.1 works from both sides:

Lemma 11.5.5: *Addition from the left*

$$\forall(n, m \in \mathbb{N}) : S(n) + m = S(n + m).$$

Proof. We induct on m while keeping n fixed; thus, let $P(m)$ be a statement that's true when $S(n) + m = S(n + m)$.

Base case: Show that $P(0)$ is true.

$$S(n) + 0 = S(n) \qquad \text{Def. 11.5.1} \tag{11.39}$$

$$= S(n + 0). \qquad \text{Def. 11.5.2} \tag{11.40}$$

Inductive step: Assume inductively that $P(m)$ is true, i.e., $S(n) + m = S(n+m)$. Now, prove that $P(S(m))$ is true, i.e., $S(n)+S(m) = S(n+S(m))$. Starting with the L.H.S.,

$$S(n) + S(m) = S(S(n) + m) \qquad \text{Def. 11.5.1} \tag{11.41}$$

$$= S(S(n + m)) \qquad \text{Inductive hypothesis} \tag{11.42}$$

$$= S(n + S(m)), \qquad \text{Def. 11.5.1} \tag{11.43}$$

completing the induction. □

We're now ready to prove that addition is **commutative**, i.e., changing the order in which the numbers are positioned does not affect the answer.

Lemma 11.5.6: *Commutativity of addition*

$$\forall(n, m \in \mathbb{N}) : n + m = m + n.$$

Proof. We induct on m while keeping n fixed; thus, let $P(m)$ be a statement that's true when $n+m=m+n$.

Base case: Show that $P(0)$ is true, i.e., $n+0 = 0+n$. By def. 11.5.2 and lem. 11.5.4,

$$n+0=0+n=n. \tag{11.44}$$

Inductive step: As an inductive hypothesis, assume that $P(m)$ is true, i.e., $n+m=m+n$. Use this to prove that $P(S(m))$ is true, i.e., $n+S(m)=S(m)+n$.

$$\begin{aligned} n+S(m) &= S(n+m) && \text{Def. 11.5.1} && (11.45)\\ &= S(m+n) && \text{Inductive hypothesis} && (11.46)\\ &= S(m)+n, && \text{Lem. 11.5.5} && (11.47)\end{aligned}$$

completing the induction. □

The next thing is that addition is **associative**, i.e., changing the order of groupings does not affect the result.

Lemma 11.5.7: *Associativity of addition on the naturals*

$$\forall(n,m,k\in\mathbb{N}):(n+m)+k=n+(m+k).$$

Proof. We induct on k while keeping both n and m fixed; thus, let $P(k)$ be a statement that's true if $(n+m)+k=n+(m+k)$.

Base case: Show that $P(0)$ is true, i.e., $(n+m)+0=n+(m+0)$.

$$\begin{aligned} (n+m)+0 &= n+m && \text{Def. 11.5.2} && (11.48)\\ &= n+(m+0). && \text{Def. 11.5.2} && (11.49)\end{aligned}$$

Inductive step: As an inductive hypothesis, assume that $P(k)$ is true, i.e., $(n+m)+k = n+(m+k)$. Use this to prove that $P(S(k))$ is true, i.e., $(n+m)+S(k) = n+(m+S(k))$.

$$\begin{aligned}
(n+m)+S(k) &= S((n+m)+k) && \text{Def. 11.5.1} && (11.50)\\
&= S(n+(m+k)) && \text{Inductive hypothesis} && (11.51)\\
&= n+S(m+k) && \text{Def. 11.5.1} && (11.52)\\
&= n+(m+S(k)), && \text{Def. 11.5.1} && (11.53)
\end{aligned}$$

completing the induction. □

Finally, we give ourselves the ability to cancel two sides of the equation.

Lemma 11.5.8: *Cancellation law of the addition on the naturals*

$$\forall(n,m,k \in \mathbb{N}) : (m+n = k+n) \implies (m=k).$$

Proof. We induct on n while keeping both m and k fixed; thus, let $P(n)$ be a statement that's true if $(m+n = k+n) \implies (m=k)$.

Base case: Show that $P(0)$ is true, i.e., $(m+0 = k+0) \implies (m=k)$.

$$\begin{aligned}
m+0 &= k+0 && && (11.54)\\
m &= k. && \text{Lem. 11.5.4} && (11.55)
\end{aligned}$$

Inductive step: As an inductive hypothesis, assume that $P(n)$ is true, i.e., $(m+n = k+n) \implies (m=k)$. Use this to prove that $P(S(n))$ is true, i.e. $(m+S(n) = k+S(n)) \implies (m=k)$.

$$\begin{aligned}
m+S(n) &= k+S(n) && && (11.56)\\
S(m+n) &= S(k+n) && \text{Lem. 11.5.5} && (11.57)
\end{aligned}$$

$$m + n = k + n \qquad \text{Axiom 11.2.6} \tag{11.58}$$

$$m = k. \qquad \text{Inductive hypothesis} \tag{11.59}$$

which completes the induction. □

To ease the usability of this system, we shall define the notion of *adding one*. Intuitively, the concept of *adding one* is analogous to the successor function. It is then left for us to establish the connection formally. Note that induction isn't used to prove the following corollary.

Corollary 11.5.9: *The successor function is analogous to addition of one*

$$S(0) \equiv 1, \quad \forall (n \in \mathbb{N}) : S(n) = n + 1.$$

Proof.

$$\forall (n \in \mathbb{N}) : S(n) = S(n + 0) \qquad \text{Lem. 11.5.4} \tag{11.60}$$

$$= n + S(0) \qquad \text{Lem. 11.5.5} \tag{11.61}$$

$$= n + 1. \qquad \text{Definition of number 1}$$

□

Now that's we've proven that $n + 1 = S(n)$, it's very tempting to deprecate the successor function in favor of the plus sign. But when this is done in induction proofs, it leads to clarity issues. Therefore, we shall keep it the successor notation in induction proofs, but use the plus one notation elsewhere in general arithmetic.

11.6 Multiplication

Similar to how addition is defined, it's also possible to define multiplication based on just two definitions. Informally, the first definition captures the fact that multiplication distributes over addition.

Definition 11.6.1: *Multiplication on the naturals*

$$\forall(n, m \in \mathbb{N}) : S(n) \times m = (n \times m) + m.$$

And, the second states that there exists an **absorbing element**, i.e., an element that when combined with any other element in the set yields the absorbing element itself.

Definition 11.6.2: *Absorbing element of multiplication*

$$\forall(n \in \mathbb{N}) : 0 \times n = 0.$$

We shall then prove the following: for all natural numbers n, m and k,

1. $n \times 0 = 0$ (Lem. 11.6.3)
2. $n \times S(m) = (n \times m) + n$ (Lem. 11.6.4)
3. $n \times m = m \times n$ (Lem. 11.6.5)
4. $n \times (m + k) = (n \times m) + (n \times k)$ (Lem. 11.6.6)
5. $(n \times m) \times k = n \times (m \times k)$ (Lem. 11.6.7)
6. $n \times 1 = n$ (Lem. 11.6.8)

In favor of conciseness, I shall quickly go through these and not spend much time on the induction process. Also, since addition is already built, I shall no longer reference the lemmas in this chapter. Let addition work normally as expected.

Lemma 11.6.3: *Zero is the absorbing element of multiplication*

$$\forall(n \in \mathbb{N}) : n \times 0 = 0 \times n = 0$$

Proof. Let $P(n)$ be true whenever $n \times 0 = 0$ is true.

Base case: This is similar to the proof of lem. 11.5.4

$$0 \times 0 = 0 \times 0 = 0. \tag{11.62}$$

Inductive step: Assume that $n \times 0 = 0$; prove that $S(n) \times 0 = 0$

$$\begin{aligned} S(n) \times 0 &= (n \times 0) + 0 && && (11.63)\\ &= n \times 0 && \text{Inductive hypothesis} && (11.64)\\ &= 0, && && (11.65) \end{aligned}$$

completing the induction. □

Lemma 11.6.4: *The definition of multiplication works from both sides*

$$\forall (n, m \in \mathbb{N}) : n \times S(m) = (n \times m) + n$$

Proof. We induct on n while keeping m fixed.

Base case:

$$\begin{aligned} 0 \times S(m) &= 0 && \text{Lem. 11.6.3} && (11.66)\\ &= 0 + 0 && && (11.67)\\ &= (0 \times m) + 0. && \text{Lem. 11.6.3} && (11.68) \end{aligned}$$

Inductive step: Assume that $n \times S(m) = (n \times m) + n$; prove that $S(n) \times S(m) = (S(n) \times m) + S(n)$.

$$\begin{aligned} S(n) \times S(m) &= (n \times S(m)) + S(m) && \text{Def. 11.6.1} && (11.69)\\ &= ((n \times m) + n) + S(m) && \text{Inductive hypothesis} && (11.70)\\ &= (n \times m) + n + S(m) && && (11.71)\\ &= (n \times m) + S(n) + m && && (11.72) \end{aligned}$$

$$= ((n \times m) + m) + S(n) \tag{11.73}$$

$$= (S(n) \times m) + S(n), \qquad \text{Def. 11.6.1} \tag{11.74}$$

completing the induction □

Lemma 11.6.5: *Commutativity of multiplication*

$$\forall (n, m \in \mathbb{N}) : n \times m = m \times n$$

Proof. We induct on n while keeping m fixed.

Base case: By def. 11.6.1 and lem. 11.6.4,

$$0 \times m = 0 = m \times 0. \tag{11.75}$$

Inductive step: Assume that $n \times m = m \times n$; prove that $S(n) \times m = m \times S(n)$.

$$S(n) \times m = (n \times m) + m \qquad \text{Def. 11.6.1} \tag{11.76}$$

$$= (m \times n) + m \qquad \text{Inductive hypothesis} \tag{11.77}$$

$$= m \times S(n), \qquad \text{Lem. 11.6.4} \tag{11.78}$$

completing the induction. □

Lemma 11.6.6: *Distributivity of multiplication over addition*

$$\forall (n, m, k \in \mathbb{N}) : n \times (m + k) = (n \times m) + (n \times k)$$

Proof. We induct on k while keeping n and m fixed.

Base case:

$$n \times (m + 0) = n \times m \tag{11.79}$$

$$= n \times m + 0 \tag{11.80}$$

$$= (n \times m) + (n \times 0). \qquad \text{Def. 11.6.2} \qquad (11.81)$$

Inductive step: Assume that $n \times (m + k) = (n \times m) + (n \times k)$; prove that $n \times (m + S(k)) = (n \times m) + (n \times S(k))$

$$\begin{aligned} n \times (m + S(k)) &= n \times S(m + k) && (11.82) \\ &= \big(n \times (m + k)\big) + n && \text{Lem. 11.6.4} \quad (11.83) \\ &= (n \times m) + (n \times k) + n && \text{Inductive hypothesis} \quad (11.84) \\ &= (n \times m) + \big(n \times S(k)\big), && \text{Lem. 11.6.4} \quad (11.85) \end{aligned}$$

completing the induction. □

Lemma 11.6.7: *Associativity of multiplication*

$$\forall (n, m, k \in \mathbb{N}) : (n \times m) \times k = n \times (m \times k).$$

Proof. We induct on k while keeping n and m fixed.

Base case:

$$\begin{aligned} (n \times m) \times 0 &= 0 && \text{Def. 11.6.2} \quad (11.86) \\ &= n \times 0 && (11.87) \\ &= n \times (m \times 0). && \text{Def. 11.6.2} \quad (11.88) \end{aligned}$$

Inductive step: Assume that $(n \times m) \times k = n \times (m \times k)$; prove that $(n \times m) \times S(k) = n \times (m \times S(k))$.

$$\begin{aligned} (n \times m) \times S(k) &= (n \times m) \times k + (n \times m) && \text{Def. 11.6.1} \quad (11.89) \\ &= n \times (m \times k) + (n \times m) && \text{Inductive hypothesis} \quad (11.90) \\ &= n \times \big((m \times k) + m\big) && \text{Lem. 11.6.6} \quad (11.91) \\ &= n \times (m \times S(k)), && \text{Lem. 11.6.4} \quad (11.92) \end{aligned}$$

completing the induction. □

Structurally, it would make more sense if I just introduce the cancellation law for multiplication straightaway. However, the cancellation law is only true for natural numbers that aren't zero. As we've seen earlier, the concept of vacuous truth is not sufficient. It requires trichotomy to break the natural number line into three parts: less than zero (empty set), equal to zero, and more than zero. Since trichotomy hasn't been developed yet, I shall postpone the proof of the cancellation law for multiplication until we've fully discussed these concepts in section 11.8. For those who want to skip through and read it straightaway, it's at lem. 11.8.6.

Next, we prove that one is the identity of multiplication

Lemma 11.6.8: *Identity of multiplication*

$$\forall(n \in \mathbb{N}) : n \times 1 = n.$$

Proof. **Base case:** $0 \times 1 = 0$

Inductive step: Assume that $n \times 1 = n$, prove that $S(n) \times 1 = S(n)$.

$$S(n) \times 1 = (n \times 1) + 1 \qquad \text{Def. 11.6.1} \tag{11.93}$$

$$= n + 1 \qquad \text{Induction hypothesis} \tag{11.94}$$

$$= S(n), \qquad \text{Corollary 11.5.9} \tag{11.95}$$

completing the induction. □

11.7 Positivity of natural numbers

This section concerns the property of **positive numbers** and the ordering of naturals.

Definition 11.7.1: *Positive natural numbers*

$$(n \text{ is a positive natural}) \iff \left[(n \neq 0 \in \mathbb{N})\right].$$

To be fair, we wouldn't be using this definition much because we could just say $n \neq 0$, and it's more compact notationally.

The definition of positive natural numbers split the naturals into two: zero and positive.

Corollary 11.7.2: *Dichotomy of natural numbers* For all natural numbers n, either n is zero, or n is positive.

Proof. If n is zero, then by def. 11.7.1, it isn't positive. If it is positive, then by def. 11.7.1, it isn't zero. □

The next corollary is a simple deduction of axiom 11.2.3 which says that there are no naturals in which its successor is zero.

Corollary 11.7.3: *Positive numbers are a successor of other numbers*

$$\forall(n \neq 0 \in \mathbb{N})\exists(m \in \mathbb{N}) : S(m) = n$$

For the rest of this section, we shall develop how positivity interacts with the arithmetical operations that we've defined. Firstly, addition.

Lemma 11.7.4: *The sum of a positive number and a natural is positive*

$$\forall(n \neq 0 \in \mathbb{N})\forall(m \in \mathbb{N}) : n + m \neq 0.$$

Note that n is restricted to be positive, but m isn't.

Proof. From corollary 11.7.3,

$$(n \neq 0 \in \mathbb{N}) \implies \left[\exists(k \in \mathbb{N}) : n = S(k)\right]; \tag{11.96}$$

therefore,

$$n + m = S(k) + m \tag{11.97}$$

$$= S(k + m). \tag{11.98}$$

From axiom 11.2.3, $S(k + m)$ can't be zero, meaning that $n + m$ isn't zero either; hence, $n + m$ is positive. □

The converse statement is also true

Corollary 11.7.5

$$\forall(n, m \in \mathbb{N}) : (n + m = 0) \implies \big((n = 0) \wedge (m = 0)\big) \tag{11.99}$$

This is the first time that a proof by contradiction is used. To prove a statement by contradiction, we begin by assuming that its negation is true. Next, we show that the assumption that the negation is true leads to a contradiction, implying that the negation cannot possibly hold; therefore, the original statement must be true. Since the corollary above is a three part statement, let's declare some propositions to ease the proof.

Proof. Let p represents $n + m = 0$, q represents $n = 0$, and r represents $m = 0$. The original statement can then be written as $p \implies (q \wedge r)$. The negation of this statement is then

$$\neg\big(p \implies (q \wedge r)\big) \tag{11.100}$$

$$\equiv \neg\big(\neg p \vee (q \wedge r)\big) \qquad \text{Definition of conditional}$$

$$\equiv p \wedge \neg(q \wedge r) \qquad \text{de Morgan's law} \tag{11.101}$$

$$\equiv p \wedge (\neg q \vee \neg r) \qquad \text{de Morgan's law} \tag{11.102}$$

$$\equiv (n + m = 0) \wedge \big((n \neq 0) \vee (m \neq 0)\big) \tag{11.103}$$

For this statement to be true, both $n + m = 0$ and $(n \neq 0) \vee (m \neq 0)$ must hold at the same time. Let's prove that this can't be the case.

Assume that both of the statements are true. For $(n \neq 0) \vee (m \neq 0)$ to be true, either $n \neq 0$ or $m \neq 0$. Let's assume first that $n \neq 0$ but $m = 0$. If $n \neq 0$, then it is a successor of some naturals a. Hence,

$$n + m = S(a) + 0 \tag{11.104}$$

$$= S(a) \tag{11.105}$$

Because zero isn't the successor of any natural numbers (axiom 11.2.3), $S(a)$ can't be zero, contradicting the assumption that $n + m = 0$.

If both $n \neq 0$ and $m \neq 0$, then $n = S(a)$ and $m = S(b)$ for some naturals a and b; then,

$$n + m = S(a) + S(b) \tag{11.106}$$

$$= S(a + S(b)). \tag{11.107}$$

The same argument from earlier applies.

As seen, assuming that the opposite is true always lead to a contradiction. The only way for the statement to be true is for the original statement to be true in the first place, closing the proof. $\square$

We next deal with the interaction between positivity and multiplication.

Lemma 11.7.6: *The product of two positive numbers is positive*

$$\forall(n, m \in \mathbb{N}) : ((n \neq 0) \wedge (m \neq 0)) \implies (n \times m \neq 0).$$

Proof. Let p represents $n \neq 0$, q represents $m \neq 0$, and $n \times m \neq 0$. The original statement can be written as $(p \wedge q) \implies r$. The negation of this statement is then

$$\neg((p \wedge q) \implies r)$$

$$\equiv \neg(\neg(p \wedge q) \vee r) \qquad \text{Def. of conditional} \tag{11.108}$$

$$\equiv p \wedge q \wedge \neg r \qquad \text{de Morgan's law} \tag{11.109}$$

Since the negation of *for all* ($\forall$) is *there exists* ($\exists$), the negated statement can be written as

$$\exists(n, m \in \mathbb{N}) : (n \neq 0) \wedge (m \neq 0) \wedge (n \times m = 0) \tag{11.110}$$

Since both n and m isn't zero, they are positive. □

Conversely,

Corollary 11.7.7 $\forall(n, m \in \mathbb{N}) : n \times m = 0 \implies ((n = 0) \vee (m = 0))$

Proof. Use de Morgan's law on the lemma above. □

11.8 Ordering of the natural numbers

The ultimate goal of this section is to show that the natural numbers have the well-ordering property, i.e., every non-empty subset of the naturals have the smallest element. The statement of well-ordering property is actually equivalent to the induction axiom, and one can be used to prove the other. To do that, we first have to discuss the concept of order.

Definition 11.8.1: *Orders on the naturals*

Let $n, m \in \mathbb{N}$. The number n is **strictly greater than** m and m is **strictly less than** n if there exists a positive number a such that $n = m + a$. This relation is denoted as $n > m$ and $m < n$ respectively.

$$\forall(n, m \in \mathbb{N}) : \\ \big[(n > m) \iff (m < n)\big] \iff \big[\exists(a \neq 0 \in \mathbb{N}) : n = m + a\big].$$

If a isn't restricted to be positive, we say that n is **greater than or equal to** m and m is **less than or equal to** n, denoted as $n \geq m$ and $m \leq n$ respectively.

$$\forall(n, m \in \mathbb{N}) : \big[(n \geq m) \iff (m \leq n)\big] \iff \big[\exists(a \in \mathbb{N}) : n = m + a\big].$$

In text, we usually write *greater than* and *less than* instead of the *strictly greater than* and *strictly less than* in favor of concision.

There's also another commonly accepted definition of non-strict ordering that's equivalent to def. 11.8.1, i.e.,

$$\forall(n, m \in \mathbb{N}) : \big[(n > m) \iff (m < n)\big] \\ \iff \big[\big(\exists(a \in \mathbb{N}) : n = m + a\big) \wedge (n \neq m)\big].$$

It's trivial to prove that this definition and the one of def. 11.8.1 is actually equivalent, and is left as a side-exercise for the reader.

We then begin to prove the basic properties of order. Since these proofs are pretty short, I've contained all of them in one proposition.

Proposition 11.8.2: *Basic properties of order on the naturals* For all natural numbers n, m and k,

1. $n \geq n$ (Reflexivity)
2. $\big[(n \geq m) \wedge (m \geq k)\big] \implies (n \geq k)$ (Transitivity)
3. $\big[(n \geq m) \wedge (m \geq n)\big] \implies (n = m)$ (Symmetric)
4. $(n \geq m) \iff (n + k \geq m + k)$
5. $(n > m) \iff (n \geq S(m))$

Proof of item 1 of prop. 11.8.2. From def. 11.8.1, we must show that there exists a natural a which for all naturals n, $n = n + a$. Obviously, $a = 0$, proving that $n \geq n$ for all n. $\square$

Proof of item 2 of prop. 11.8.2. From def. 11.8.1,

$$n \geq m \iff \big[\exists(a \in \mathbb{N}) : n = m + a\big], \tag{11.111}$$

$$m \geq k \iff \big[\exists(b \in \mathbb{N}) : m = k + b\big]. \tag{11.112}$$

Substituting $m = k + b$ into $n = m + a$ yields

$$n = k + (a + b). \tag{11.113}$$

Let $(a + b) = c$, we get $n = k + c$; thus by def. 11.8.1, $n \geq k$. $\square$

Proof of item 3 of prop. 11.8.2. From def. 11.8.1,

$$n \geq m \iff \big[\exists(a \in \mathbb{N}) : n = m + a\big], \tag{11.114}$$

$$m \geq n \iff \big[\exists(b \in \mathbb{N}) : m = n + b\big]. \tag{11.115}$$

Combining $n = m + a$ and $m = n + b$ yields

$$n + m = m + n + a + b \tag{11.116}$$

$$0 = a + b. \qquad \text{Lem. 11.5.8} \tag{11.117}$$

By corollary 11.7.5, $a = 0$ and $b = 0$ meaning that $n = m + 0$ and $m = n + 0$; thus, $n = m$. $\square$

Proof of item 4 of prop. 11.8.2. Since this is a biconditional, we must prove both the forwards and backwards direction, i.e.,

$$\forall(n, m, k \in \mathbb{N}) : \big[(n \geq m) \implies (n + k \geq m + k)\big] \wedge \\ \big[(n + k \geq m + k) \implies (n \geq m)\big]. \tag{11.118}$$

Either ways, we can start with def. 11.8.1

$$(n \geq m) \iff \big[\exists(a \in \mathbb{N}) : n = m + a\big]. \tag{11.119}$$

To prove the forward direction, we can add $k = k$ to $n = m + a$ to get

$$n + k = m + k + a. \tag{11.120}$$

By def. 11.8.1, $n + k \geq m + k$. And for the backwards direction, just use the cancellation law for addition on the naturals (lem. 11.5.8). □

Proof of item 5 of prop. 11.8.2. Similarly, this is also a biconditional. Let's start with the forward direction, i.e.,

$$\forall(n, m \in \mathbb{N}) : (n > m) \implies \big(S(n) \geq m\big). \tag{11.121}$$

By def. 11.8.1,

$$(n > m) \iff \big[\exists(a \in \mathbb{N}) : (a \neq 0) \wedge (n = m + a)\big], \tag{11.122}$$

where a is positive. Because positive numbers are a successor of other naturals (corollary 11.7.3), there exists a natural b in which $a = S(b)$; therefore,

$$n = m + S(b) \tag{11.123}$$

$$= S(m) + b. \tag{11.124}$$

Since b isn't constrained to be positive, $n \geq S(m)$. To prove the backwards direction, i.e.,

$$\forall(n, m \in \mathbb{N}) : \big(n \geq S(m)\big) \implies (n > m), \tag{11.125}$$

we first notice that $n \geq S(m)$ means there exists some natural a in which $n = S(m)+a$. Rewriting this gives $n = m+S(a)$. By axiom 11.2.3, $S(a) \neq 0$, meaning $S(a)$ is positive. Thus, $n > m$ by def. 11.8.1. □

Finally, we've arrived at the trichotomy property of the naturals.

Lemma 11.8.3: *Order trichotomy* For all natural numbers n and m, only one of these statements are true: $n = m$, $n > m$, or $n < m$.

Trichotomy means a *division into three categories*. If something is trichotomous, then that thing can be in only one of the three categories at a time. Trichotomous statements also appear in the construction of the integers, rationals and reals, where the numbers are distinctly categorized into negatives, zero, and positive.

As seen earlier in section 11.3.2, trichotomy is useful for proofs which requires different methods depending on whether the number is less than, more than, or equal to some other number m. On the naturals, order trichotomy is also used to develop a stronger form of mathematical induction called the *strong induction*, and also used to prove the cancellation law for multiplication.

The logical form of a trichotomous statement is given as follows. Let there be an object x, and three disjoint sets A, B, and C. Let p represents $x \in A$, q represents $x \in B$, and r represents $x \in C$. If x is trichotomous, then

$$(p \wedge \neg q \wedge \neg r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge r). \tag{11.126}$$

An equivalent form of this statement is

$$(p \vee q \vee r) \wedge \neg((p \wedge q) \vee (q \wedge r) \vee (p \wedge r)) \tag{11.127}$$

which is easier to both interpret and prove. The first part $(p \vee q \vee r)$ directly translates to "*at least one of the statement is true.*" The second part $(p \wedge q) \vee (q \wedge r) \vee (p \wedge r)$ translates to "*at least one of the statement is true.*" Together, they ensure that one of the statements are guaranteed to be true. With this simplified logical form, we can then prove the order trichotomy of the naturals.

Proof. Let there be two naturals n and m. Firstly, we prove that no two statements can hold simultaneously.

1. Proving that $n < m$ and $n = m$ can't be true simultaneously: If $n < m$, then by def. 11.8.1, $n \neq m$
2. Proving that $n > m$ and $n = m$ can't be true simultaneously: If $n > m$, then by def. 11.8.1, $n \neq m$.
3. Proving that $n < m$ and $n > m$ can't be true simultaneously: For the sake of contradiction, assume that $n < m$ and $n > m$. Then by prop. 11.8.2, $n = m$ which by the two statement that's proven earlier, is a contradiction.

And secondly, we prove that at least one of the statement is true. We induct on n while keeping m fixed. Let $P(n)$ be the statement $(n < m) \vee (n > m) \vee (n = m)$.

Base case: When $n = 0$, $n \leq m$ for all naturals m; thus, either $n = m$ or $n < m$.

Inductive step: Assume that $P(n)$ is true, check that $P(S(n))$ is also true.

1. If $n < m$, then by prop. 11.8.2, $S(n) \leq m$, i.e., either $S(n) < m$ or $S(n) = m$. Therefore, $P(S(n))$ is true because there is at least one true statement.

2. If $n = m$, then by prop. 11.8.2, $S(n) > m$: $P(S(n))$ is true.

3. If $n > m$, then by prop. 11.8.2, $S(n) > m$: $P(S(n))$ is true.

Therefore, $P(n)$ must hold true for all natural numbers, i.e., either $n = m$, $n < m$, or $n > m$. Combined with the earlier proof that no two statements can hold simultaneously, we've shown that the order trichotomy is true for all natural numbers n and m. □

The order trichotomy allows us to define two other operations: maximum and minimum.

Definition 11.8.4: *Maximum and minimum*

For all natural numbers n, m,

$$\max\{n,m\} = \begin{cases} n & n > m \\ m & n \leq m \end{cases}, \quad \text{and} \quad \min\{n,m\} = \begin{cases} n & n \leq m \\ m & n > m \end{cases}$$

The maximum and minimum is well-defined because it uses order: a well-defined property. However, it wouldn't be of much use right now. Still, I shall list down some properties of these operations for completeness' sake.

Corollary 11.8.5: *Properties of maximum and minimum* For all natural numbers n, m and c,

1. $\max\{n,m\} = \max\{n,m\}$ (Commutativity)
2. $\max\{n,m\} + c = \max\{n+c, m+c\}$

Proof. W.I.P. □

Finally, we can complete the arithmetical systems on the naturals by proving the cancellation law for multiplication

Lemma 11.8.6: *Cancellation law for multiplication*

$$\forall(n, m \in \mathbb{N})\forall(c \neq 0 \in \mathbb{N}) : (n \times c = m \times c) \implies (n = m).$$

Proof. Using proof by contradiction, we assume that the negation of the statement is true, i.e., let

$$\exists(n, m \in \mathbb{N})\exists(c \neq 0 \in \mathbb{N}) : (n \times c \neq m \times c) \land (n = m) \tag{11.128}$$

We assume that $n \times c \neq m \times c$. By the trichotomy of order, either $n \times c < m \times c$, $n \times c = m \times c$, or $n \times c > m \times c$. □

11.9 What if zero isn't in the naturals?

It is highly debated whether zero is in naturals or not. The debate is so fiery, that some even propose that the naturals start at $\frac{1}{2}$ instead of 0 as a compromise. Figure 11.4 shows my impression of the debate in greater detail.

As the figure suggests, I'm not into politics. Therefore, I shall settle both side by also providing a template for how the naturals *can* be constructed without zero.

Axiom 11.9.1: *Peano's axiom without zero*

The first three axioms stay the same, but we modify the last two.

1. $1 \in \mathbb{N}$
2. $\forall(n \in \mathbb{N}) : S(n) \in \mathbb{N}$
3. $\forall(n, m \in \mathbb{N}) : (S(n) = S(m)) \implies (n = m)$

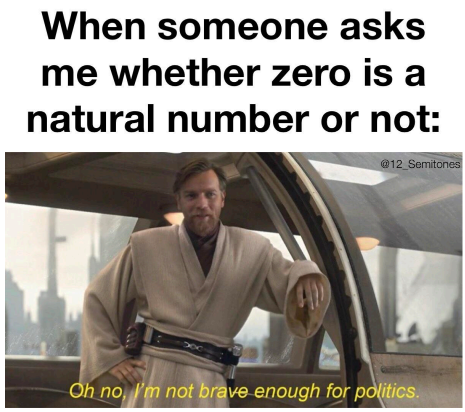


Figure 11.4 | Is zero in $\mathbb{N}$? I don't know!

4. $\forall(n \in \mathbb{N}) : S(n) \neq 1$
5. $\forall T : \left((1 \in T) \wedge \left[\forall(n \in T) : S(n) \in T\right]\right) \implies T \cong \mathbb{N}$

Even though it's possible to exclude zero, I personally think that the inclusion of zero allows us to develop a richer theory of arithmetic, as it functions as the identity of addition and the absorbing element of multiplication. If zero isn't included, we'd have to define addition and multiplication by using one instead:

Definition 11.9.2: *Addition and multiplication without zero*

For all $n, m \in \mathbb{N}$:

1. $n + 1 = S(n)$
2. $n + S(m) = S(n + m)$
3. $n \times 1 = n$
4. $S(n) \times m = (n \times m) + m$

Since I don't want to start over and prove everything again, I shall leave the construction of naturals without zero to the reader. Or if you're impatient, check out Edmund Landau's book on *Foundations of analysis* (4) and C. H. C. Little's book on *The number systems of analysis* (5) for an in-depth treatment of naturals without zero.

The next step is to extend the natural numbers to the integers.

CHAPTER 12

The integers

Notation 12.0.1: *The integers*

The integers will be marked with a dagger. E.g., 2 is a natural number, but $2^\ddagger$ is an integer.

12.1 An informal introduction to the integers

In this section, I shall not refrain myself from using the subtraction sign, as it better illustrates what we're going to be doing in this chapter.

Defining the integers is like defining subtraction and negatives. In a sense, the integers is an extension of the naturals, because as you would see, the system of the naturals are already sufficient to define the integers.

One can define an integer to be a difference between two naturals:

$$a^\ddagger = n - m. \tag{12.1}$$

The integer $a^\ddagger$ is negative if $n < m$, positive if $n > m$, and zero if $n = m$. This means, the integer is just an ordered pair (n, m), which formally represents $n - m$. But notice that this definition alone wouldn't suffice, as there are many pairs of naturals that can represent one integer. E.g., $(5, 3)$ and $(4, 2)$, and $(10, 8)$ are all different pairs of naturals, but they all represent the integer $2^\ddagger$. So, we need some way to tell whether two pair of naturals represent the same integer or not.

Consider two integers (n, m) and (k, l). These two integers are equivalent *iff*

$$n - m = k - l. \tag{12.2}$$

From the assumed knowledge of subtraction, we can move the terms around to get

$$n + l = k + m. \tag{12.3}$$

Thus, we could say that two integers (n, m) and (k, l) are equivalent *iff*

$$n + l = k + m. \tag{12.4}$$

This equivalency is denoted symbolically as

$$(n, m) \sim (k + m) \tag{12.5}$$

Addition and multiplication of integers can also be defined via the assumed knowledge of arithmetic. I.e., if $a^\ddagger = (n, m)$ and $b^\ddagger = (k, l)$, then

$$a^\ddagger + b^\ddagger = (n, m) + (k, l) \tag{12.6}$$

$$= (n - m) + (k - l) \tag{12.7}$$

$$= (n + k) - (m + l) \tag{12.8}$$

$$= (n + k, m + l), \tag{12.9}$$

and

$$a^\ddagger b^\ddagger = (n,m) \times (k,l) \tag{12.10}$$
$$= (n-m)(k-l) \tag{12.11}$$
$$= nk - nl - mk + ml \tag{12.12}$$
$$= (nk+ml) - (nl+mk) \tag{12.13}$$
$$= (nk+ml, nl+mk). \tag{12.14}$$

Negative integers can also be defined by reversing the pair. If $(n,m) = n - m = a^\ddagger$, then $(m,n) = m - n = -a^\ddagger$. Then, subtraction is simply defined by $a^\ddagger - b^\ddagger = a^\ddagger + (-b^\ddagger)$

12.2 Construction of the integers

Definition 12.2.1: *Integers*

An integer is an ordered pair of naturals. If $a^\ddagger$ is an integer, then $\exists(n,m \in \mathbb{N}) : a^\ddagger = (n,m)$.

Then, we construct the set of all integers

Definition 12.2.2: *Set of all integers ($\mathbb{Z}$)*

The set $\mathbb{Z}$ contains all integers, i.e.,

$$\mathbb{Z} = \{(n,m) : n,m \in \mathbb{N}\} \tag{12.15}$$

The equivalence relation is then stated as follows:

Definition 12.2.3: *Equivalence relation for integers*

Let $a^\ddagger = (n,m)$ and $b^\ddagger = (k,l)$ be integers. If

$$n + l = m + k, \tag{12.16}$$

these two integers are equivalent, and we denote the equivalency by

$$a^\ddagger \sim b^\ddagger, \quad \text{or} \quad (n, m) \sim (k, l) \tag{12.17}$$

Because we want to later define arithmetic on the integers, we must verify that the equivalence relation for integers also satisfy the three equality axioms, i.e.,

1. Reflexivity $(\forall (a^\ddagger \in \mathbb{Z}) : a^\ddagger \sim a^\ddagger)$
2. Symmetry $(\forall (a^\ddagger, b^\ddagger \in \mathbb{Z}) : a^\ddagger \sim b^\ddagger \iff b^\ddagger \sim a^\ddagger)$
3. Transitivity $(\forall (a^\ddagger, b^\ddagger, c^\ddagger \in \mathbb{Z} : (a^\ddagger \sim b^\ddagger) \wedge (b^\ddagger \sim c^\ddagger) \implies a^\ddagger \sim c^\ddagger)$

Once these three are verified, we can actually replace the equivalence sign $(\sim)$ with the equality sign $(=)$.

Corollary 12.2.4: *Equivalence of integers is reflexive*

$$\forall (a^\ddagger \in \mathbb{Z}) : a^\ddagger \sim a^\ddagger. \tag{12.18}$$

Proof. Let $\exists (n, m \in \mathbb{N})$ where $(n, m) = a^\ddagger$. Since $n + m = n + m$, $(n, m) \sim (n, m)$; thus, $a^\ddagger \sim a^\ddagger$. □

Corollary 12.2.5: *Equivalence of integers is symmetric*

$$\forall (a^\ddagger, b^\ddagger \in \mathbb{Z}) : a^\ddagger \sim b^\ddagger \iff b^\ddagger \sim a^\ddagger \tag{12.19}$$

Proof. Since the backwards direction can be obtained by switching a and b, we shall only prove the forward direction.

Let $\exists (n, m, k, l \in \mathbb{N})$ where $(n, m) = a^\ddagger$ and $(k, l) = b^\ddagger$. Since $a^\ddagger \sim b^\ddagger$, $(n, m) \sim (k, l)$; thus, $n + l = m + k$. By the symmetry of equality,

$m+k=n+l$. Because addition is commutative, $k+m=n+l$. Therefore, we get $(k,l)\sim(n,m)$, i.e., $b^\ddagger\sim a^\ddagger$. □

Corollary 12.2.6: *Equivalence of integers is transitive*

$$\forall(a^\ddagger,b^\ddagger,c^\ddagger\in\mathbb{Z}:(a^\ddagger\sim b^\ddagger)\wedge(b^\ddagger\sim c^\ddagger)\implies a^\ddagger\sim c^\ddagger \tag{12.20}$$

Proof. Let $\exists(n,m,k,l,p,q\in\mathbb{N})$ where $(n,m)=a^\ddagger$, $(k,l)=b^\ddagger$, and $(p,q)=c^\ddagger$.

$$a^\ddagger\sim b^\ddagger \iff (n,m)\sim(k,l)\iff n+l=m+k, \tag{12.21}$$

$$b^\ddagger\sim c^\ddagger \iff (k,l)\sim(p,q)\iff k+p=l+q. \tag{12.22}$$

Combining these two equations together we get

$$n+l+k+p=m+k+l+q. \tag{12.23}$$

By the cancellation law on addition of naturals,

$$n+p=m+q, \tag{12.24}$$

which is equivalent to saying $(n,m)\sim(p,q)$, i.e., $a^\ddagger\sim c^\ddagger$. □

By corollaries 12.2.4 to 12.2.6, we can now use the equal sign instead of the equivalence sign to represent the equality of integers. But as said earlier, there are still one thing that is still not verified: the substitution axiom, i.e., if $f(n)$ be an arithmetical function of the integers, then

$$\big(f(n)=m\big)\wedge\big(n=n'\big)\implies f(n')=m. \tag{12.25}$$

12.3 Arithmetic on the integers

Definition 12.3.1: *Addition of integers*

Let there be two integers $a^\ddagger$ and $b^\ddagger$ where $\exists(n, m, k, l \in \mathbb{N})$ such that $(n, m) = a^\ddagger$ and $(k, l) = b^\ddagger$. The sum of $a^\ddagger$ and $b^\ddagger$, denoted $a^\ddagger + b^\ddagger$, is defined as:

$$a^\ddagger + b^\ddagger = (n + k, m + l). \tag{12.26}$$

Definition 12.3.2: *Product of integers*

Let there be two integers $a^\ddagger$ and $b^\ddagger$ where $\exists(n, m, k, l \in \mathbb{N})$ where $(n, m) = a^\ddagger$ and $(k, l) = b^\ddagger$. Then, the product of $a^\ddagger$ and $b^\ddagger$, $a^\ddagger + b^\ddagger$ is defined as:

$$a^\ddagger \times b^\ddagger = a^\ddagger b^\ddagger = (nk + ml, nl + mk). \tag{12.27}$$

Here, we prove that both addition and multiplication of integers obey the substitution axiom

Corollary 12.3.3: *Addition and multiplication of integers obey the substitution axiom* Let n, m, n', m', k, l be natural numbers. If $(n, m) = (n', m')$, then

$$(n, m) + (k, l) = (n', m') + (k, l), \tag{12.28}$$

$$(n, m) \times (k, l) = (n', m') \times (k, l) \tag{12.29}$$

Proof. The relation $(n, m) = (n', m')$ means $n + m' = m + n'$.

Firstly, we prove that addition obeys the substitution axiom. Evaluate both sides separately

$$(n, m) + (k, l) = (n + k, m + l) \tag{12.30}$$

$$(n', m') + (k, l) = (n' + k, m' + l) \tag{12.31}$$

Which, for an equivalence test, we check whether $(n+k, m+l)$ is equal to $(n'+k, m'+l)$ or not.

$$(n + k, m + l) \stackrel{?}{=} (n' + k, m' + l) \tag{12.32}$$

$$n + k + m' + l \stackrel{?}{=} m + l + n' + k. \tag{12.33}$$

Using cancellation of addition on natural numbers, we get

$$n + m' \stackrel{?}{=} m + n'. \tag{12.34}$$

From our assumption that $(n, m) = (n', m')$, $n + m'$ does indeed equal $m + n'$. Therefore, our statement is true: addition obeys the substitution axiom.

For multiplication,

$$(n, m) \times (k, l) = (nk + ml, nl + mk) \tag{12.35}$$

$$(n', m') \times (k, l) = (n'k + m'l, n'l + m'k). \tag{12.36}$$

Then, we check whether $(nk+ml, nl+mk)$ is equal to $(n'k+m'l, n'l+m'k)$ or not.

$$(nk + ml, nl + mk) \stackrel{?}{=} (n'k + m'l, n'l + m'k) \tag{12.37}$$

$$nk + ml + n'l + m'k \stackrel{?}{=} nl + mk + n'k + m'l \tag{12.38}$$

$$k(n + m') + l(m + n') \stackrel{?}{=} l(n + m') + k(m + n'). \tag{12.39}$$

Since $n + m' = m + n'$, the two sides must be equal. Therefore, our statement is true: multiplication obeys the substitution axiom. □

CHAPTER 13

The rationals

CHAPTER 14

Real numbers and sequences

14.1 Invitation: why study sequences?

I'm sure that you're already familiar with **sequences**. $1, 2, 3, 4, 5, \ldots$ is a sequence of number that increases by one; $2, 4, 8, 16, 32, \ldots$ is a sequence of numbers which are doubling, etc. If the sequence is increasing by d each time, it's an arithmetic sequence. If it's multiplied by r each time, it's a geometric sequence. It's a concept so basic that it's already infused itself in high school algebra. So, why bother analyzing it here?

The answer is two-fold. Firstly, sequences of rationals are used to construct the reals. Secondly, sequences are used to define limit of functions, which are the building blocks of derivatives. We shall answer the problem of constructing the reals in this chapter, and the limit of functions in later chapters.

The real numbers are constructed differently than all number systems prior. We can't just introduce another operation and throw in some equivalence classes. You might say that we can just construct rational exponentiation to get the reals, or even logarithms. Indeed, there are infinitely many real numbers that's obtainable by rational exponentiation and logarithms; however, there are still infinitely many more numbers which cannot be constructed by these operations. Those numbers are called **non-algebraic numbers**, aka **transcendental numbers**. Our construction of reals must be able to capture all these non-algebraic numbers. How? The answer is through the analysis of sequences.

Let's define what a sequence is

Definition 14.1.1: *Sequence*

A **sequence** a_n of a set $\mathbb{F}$ is a mapping a from $\mathbb{N}$ to $\mathbb{F}$.

Notation 14.1.2

A sequence of a set $\mathbb{F}$ is written as $(a_n)_{n=i}^{f}$ where the starting point $i \in \mathbb{N}$ and the ending point is $f \in \mathbb{N}$. If $f = \infty$, then $(a_n)_{n=i}^{\infty}$ is an infinite sequence which starts at i.

For example, a finite sequence of rationals, $(a_n)_{n=2}^{7}$ defined as $a_n = \frac{1}{n}$ forms the set

$$\left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \frac{1}{7}\right\}.$$

A finite sequence of rationals, $(b_n)_{n=5}^{7}$, defined as $b_n = n^2$ forms the set

$$\{25, 36, 49, 64, 81, 100\}.$$

An infinite sequence of rationals $(c_n)_{n=3}^{\infty}$ defined as $c_n = n + 2$ starting at $n = 3$ is the set

$$\{5, 6, 7, 8, 9, 10, ...\}.$$

Cauchy constructed the real numbers by saying that the real numbers are the *limit* of these *converging rational sequences*. To illustrate what *limits* and *converging sequences* are, let's look at some examples. Consider the sequence

$$3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \dots .$$

This sequence, although filled with rational numbers, are *gradually approaching* the real number π, which is not rational. Consider another sequence

$$1, 1.4, 1.41, 1.4142, 1.41421, \dots . \tag{14.1}$$

This sequence is a sequence of rationals that *gradually approaches* $\sqrt{2}$. These numbers here, π and $\sqrt{2}$, though not in the set of rationals, are the number that these sequence *gradually approaches*. In math lingo, if a sequence *gradually approaches* some number L, then that sequence is a **convergent sequence** and L is said to be the **limit** of that sequence, i.e., the sequence converges to L. Here, you can clearly see how the reals can be possibly defined to be the limit of said converging sequence. However, we quickly run into a problem.

As you'll see later, the problem with convergence is that one has to know the limit beforehand before showing that it actually *converges*. E.g., we have to know what the exact value of π in order to show that the sequence

$$3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \dots$$

is actually a convergence sequence in the first place. I.e., we want to construct the reals by using the limit of a rational sequence, but we need to know what the reals are in order to show that the sequence actually converges. This is circular reasoning, and is undesirable in mathematics. To break out of this reasoning loop, we must show that a sequence of rational converges without knowing its limit in the first place.

Cauchy's constructed the reals by showing that all convergent sequences are Cauchy sequence. Informally,

> *A Cauchy sequence is a sequence whose elements become arbitrarily close to each other as the sequence progresses.*

After which, the algebraic operations on the reals will then be defined as the operation on each term of the sequence. E.g., if the real x is defined as the limit of the sequence $(a_n)_{n=i}^{\infty}$ and the real y as the limit of the sequence $(b_n)_{n=i}^{\infty}$. Their sum can then be represented as the limit of $(a_n + b_n)_{n=i}^{\infty}$, their product as the limit of $(a_n b_n)_{n=i}^{\infty}$, and their ratio as the limit of $\left(\frac{a_n}{b_n}\right)_{n=i}^{\infty}$, etc.

From now on, I'd like to focus on sequence of rationals because we want to construct the reals from the rationals. So, every sequence from now on is assumed to be a rational sequence. When the reals ($\mathbb{R}$) are finally constructed, one can replace all $\mathbb{Q}$ with $\mathbb{R}$.

14.2 Cauchy sequences

Definition 14.2.1: *Cauchy sequences*

If a rational sequence $(a_n)_{n=m}^{\infty}$ is Cauchy, then

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N > m \in \mathbb{N})\forall(j, k \geq N \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon. \tag{14.2}$$

I understand that this definition looks very unfriendly, so let's break it down and see how each element come together to capture concept of a sequence whose elements get closer together as the sequence progresses. In my humble opinion, the approach taken by Terence Tao in his *Analysis I* book (7) is one of the most intuitive; therefore, I shall follow his teachings and deconstruct the definition of Cauchy sequences into two bite-sized ones: ε-steadiness and eventual ε-steadiness.

The first definition defines what it means for *two elements to be close to each other.*

Definition 14.2.2: *ε-closeness*

Two numbers a and b are ε-close if $|a-b| \leq \varepsilon$

This definition measures the distance on the number line. If the distance between a and b is less than ε, we say that a and b are ε-close.

Definition 14.2.3: *ε-steadiness*

A rational sequence $(a_n)_{n=i}^{f}$ is ε-steady if for all j, k between i and f, a_j and a_k are ε-close.

One interpretation of a sequence that's ε-steady is that the range of the sequence ($\max\left((a_n)_{n=i}^{f}\right) - \min\left((a_n)_{n=i}^{f}\right)$) is restricted to be less than or equal to ε:

$$\max\left((a_n)_{n=i}^{f}\right) - \min\left((a_n)_{n=i}^{f}\right) \leq \varepsilon. \tag{14.3}$$

Because of this, the distance between each term in an ε-steady sequence is always less than or equal to ε.

This definition also works for infinite sequence. However, the concept of maxima and minima is currently a bit loose. Sometimes, the maxima and minima doesn't even exist, so it'd be wiser for now to think of an ε-steady sequence as a sequence that the distance between each and every term is less than ε. Now, let's look at some examples and non-examples.

Example 14.2.4: *ε-steady sequences*

1. The sequence $(a_n)_{n=0}^{\infty}$: $3, 3.1, 3.14, 3.141, 3.1415, 3.14159, \dots$, is 0.1-steady but not 0.01-steady, because

$$|a_0 - a_1| = |3 - 3.1| = 0.1, \quad \text{which is more than} \quad 0.01.$$

2. The sequence $(b_n)_{n=0}^{\infty}$: $-1, 0, 1, 0, -1, 0, \dots$ is 2-steady, but not 1-steady because

$$|b_0 - b_2| = |-1 - 1| = 2, \quad \text{which is more than} \quad 1.$$

3. The sequence $(c_n)_{n=4}^{9}$: $2, 3, 1, 5, 4, 6$ is 10-steady, but not 4-steady because

$$|c_6 - c_9| = |6 - 1| = 5, \quad \text{which is more than} \quad 4.$$

Before moving on, I'd like to discuss the image of ε-steady sequence. Consider a sequence $(a_n)_{n=0}^{\infty}$ shown in fig. 14.1. Let's draw a rectangle which infinite extends along the x-axis but has the height of ε. The sequence is ε-steady *iff* you can find a position of this rectangle such that every term in the sequence lies in that rectangle. Here, $(a_n)_{n=0}^{\infty}$ is ε-steady because there's a rectangle configuration in which every point of the sequence lies in the rectangle shown in the right most figure. The left most and middle figure doesn't count because there are some points, drawn in red, which is outside the rectangle.

The concept of a sequence being ε-steady is quite simple, but it doesn't capture the limiting behavior of a sequence. Since we want to construct the reals from the *limit* of a *converging sequence*, we shouldn't really

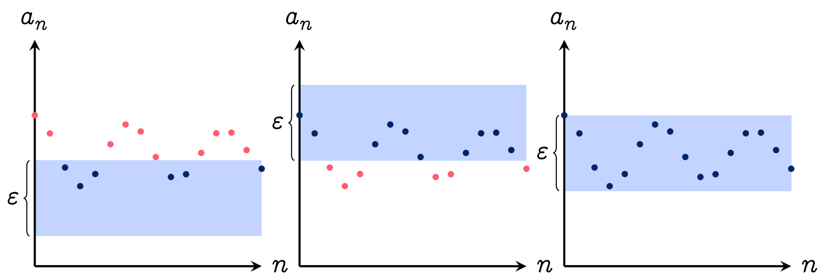


Figure 14.1 | EXAMPLES OF RECTANGLE CONFIGURATIONS TO CHECK THAT A SEQUENCE IS ε-STEADY.

care about the initial terms. Therefore, we improve on the definition of ε-steadiness to eventual ε-steadiness: a definition that captures what it means for *the elements of a sequence to get close together as the sequence progresses*.

Definition 14.2.5: *Eventual ε-steadiness*

A rational sequence $(a_n)_{n=i}^{f}$ is eventually ε-steady if there exists some natural $N \geq i$ in which the sequence $(a_n)_{n=N}^{f}$ is ε-steady.

The intuition for this is that, if it's possible to find some number N in which the removal of the first N terms of the sequence makes the sequence ε-steady, then the sequence is eventually ε-steady. Let's look at some examples and non-examples.

Example 14.2.6: *Eventually ε-steady sequences*

1. The sequence $(a_n)_{n=1}^{\infty}$: $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$, is not $\frac{1}{2}$-steady because

$$|a_0 - a_2| = \left|1 - \frac{1}{3}\right| = \frac{2}{3}, \quad \text{which is more than} \quad \frac{1}{2};$$

however, the sequence is eventually $\frac{1}{2}$-steady because the removal of the first term yields a sequence that is $\frac{1}{2}$-steady:

$$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots,$$

2. The sequence $(b_n)_{n=0}^{\infty}$: $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$, is not $\frac{1}{4}$-steady because

$$|a_2 - a_4| = \left|\frac{1}{2} - \frac{1}{8}\right| = \frac{3}{8}, \quad \text{which is more than} \quad \frac{1}{4};$$

however, the sequence is eventually $\frac{1}{4}$-steady because the removal of the first two terms yields a sequence that is $\frac{1}{4}$-steady:

$$\frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots.$$

3. The sequence $(c_n)_{n=0}^{\infty}$: $69, 2, 2, 2, 2, \dots$ is not 3-steady because for all $j > 0$,

$$|a_0 - a_j| = |69 - 2| = 67, \quad \text{which is more than} \quad 3;$$

however, the sequence is eventually 3-steady because the removal of the first term yields a sequence that is 3-steady:

$$2, 2, 2, 2, 2, \dots.$$

And now, we're ready to define what Cauchy sequences are in simple terms:

Corollary 14.2.7: *Cauchy sequences* A Cauchy sequence is an infinite sequence which is eventually ε-steady for all $\varepsilon > 0 \in \mathbb{Q}$. I.e., for a sequence $(a_n)_{n=m}^{\infty}$ to be Cauchy,

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N > m \in \mathbb{N})\forall(j, k \geq N \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon$$

A natural question to ask from here would be, *how to prove that a sequence is Cauchy?* It is quite hard, but the core concept is to show that for every ε, there exists a corresponding N which makes the sequence $\left(a_n\right)_{n=N}^{\infty}$, ε-steady. This would be one of the first proof that must have a preparation stage before the actual proof. We do some arithmetic first to decide what N might be in terms of ε, and use that to show that for all $j, k \geq N$, the sequence is ε-steady.

Example 14.2.8: *Show that $\left(a_n\right)_{n=0}^{\infty}$ is Cauchy* We prepare the proof by first fixing ε and find N in terms of ε. For all $n \geq N$, we demand that

$$|a_j - a_k| \leq \varepsilon \tag{14.4}$$
$$\left|\frac{1}{j} - \frac{1}{k}\right| \leq \varepsilon \tag{14.5}$$
$$\left|\frac{1}{j}\right| + \left|\frac{1}{k}\right| \leq \varepsilon. \qquad \text{Triangle inequality} \tag{14.6}$$

If we demand both $\left|\frac{1}{j}\right|$ and $\left|\frac{1}{k}\right|$ to be less than $\frac{\varepsilon}{2}$, then we would get $|a_j - a_k| \leq \varepsilon$ as desired; thus, let

$$\left|\frac{1}{j}\right|, \left|\frac{1}{k}\right| \leq \frac{\varepsilon}{2} \tag{14.7}$$

Since $j, k \geq N$, we have $\frac{1}{j}, \frac{1}{k} \leq \frac{1}{N}$. Combining this with the equation above gives

$$\frac{1}{N} \leq \frac{\varepsilon}{2} \tag{14.8}$$
$$N \geq \frac{2}{\varepsilon}. \tag{14.9}$$

Here, we've shown that $N \geq \frac{2}{\varepsilon}$ causes $|a_j - a_k|$ where $j, k \geq N$ to be less than ε for every ε; thus, the sequence is Cauchy. Now onto the formal proof.

Proof.

$$|a_j - a_k| = \left|\frac{1}{j} - \frac{1}{k}\right| \tag{14.10}$$

$$\left|\frac{1}{j} - \frac{1}{k}\right| \leq \left|\frac{1}{j}\right| + \left|\frac{1}{k}\right| \qquad \text{Triangle's inequality} \tag{14.11}$$

$$\left|\frac{1}{j}\right| + \left|\frac{1}{k}\right| \leq \frac{1}{N} + \frac{1}{N} \tag{14.12}$$

Choose $N \geq \frac{2}{\varepsilon}$, then

$$\frac{1}{N} + \frac{1}{N} \geq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \tag{14.13}$$

$$\frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \tag{14.14}$$

Therefore, $N \geq \frac{2}{\varepsilon}$ causes $|a_j - a_k|$ to be less than ε for all $j, k \geq N$.

Thus, the sequence is Cauchy. □

Before constructing the reals, I'd like to discuss the notion of boundedness which is later used to prove that multiplication on the reals is well-defined

Definition 14.2.9: *Boundedness of sequences*

A sequence $(a_n)_{n=i}^{f}$ is bounded by a rational $M > 0$ if for all $n \in \mathbb{N}$ that's between i and f, $|a_n| \leq M$.

Corollary 14.2.10: *Finite sequences are bounded* Every finite sequence $(a_n)_{n=i}^{f}$ is bounded

Proof. We shall use strong induction to prove this.

Base case: Trivially, the sequence a_1 is bounded by $M = |a_1|$.

Inductive step: Assume that the sequence $(a_n)_{n=i}^{N}$ where $i < N < f$ is bounded by some rational M. Then, the sequence $(a_n)_{n=i}^{S(N)}$ must also be bounded by the rational $M + |a_{S(N)}|$; therefore, any finite sequence is bounded, closing the induction. □

The next thing is to establish a connection between Cauchy sequences and the concept of boundedness.

Proposition 14.2.11 Every Cauchy sequence is bounded

Proof. Since the sequence is Cauchy, for every ε, there exists a natural N such that for all $j, k \geq N$, $|a_j - a_k| \leq \varepsilon$. The sequence can then be split into two parts:

$$a_1, a_2, \dots, a_{N-1}, \quad \text{and} \quad a_N, a_{N+1}, a_{N+2}, \dots .$$

By corollary 14.2.10, the first part of the sequence is bounded by some rational M. Since the distance between every term in the second part is less than ε. If the smallest term in the second part of the sequence is K, then the entire part must be bounded by $K + \varepsilon$; therefore, the whole sequence is bounded by $M + K + \varepsilon$. □

14.3 Construction of the reals

Now we are ready to construct the reals. Similar to how we've constructed other objects, we do not care what the object actually is, but rather, how it interacts with others. Therefore, let's define a function that takes in a Cauchy sequence and spits out a real number:

Definition 14.3.1: *Real numbers*

A real number is of the form $\mathfrak{R}(a_n)$ where $(a_n)_{n=m}^{\infty}$ is a Cauchy sequence.

Similar to how integers are defined, there are many Cauchy sequences which represent the same real number. I.e.,

$$1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \dots, \quad \text{and} \quad 1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots, \tag{14.15}$$ [1]

both of which seems to approach the number 0. Therefore, we must define the concept of equivalence sequences:

Definition 14.3.2: *Equivalent sequences*

Two sequences, $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are equivalent *iff* for all rational ε, there exists a natural $N \geq \max(m_a, m_b)$ in which for all $n \geq N$, a_n and b_n are ε-close, i.e.,

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N \geq \max(m_a, m_b) \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : \\ |a_n - b_n| \leq \varepsilon. \tag{14.16}$$

$(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ represent the same number, i.e., $\mathfrak{R}(a_n) = \mathfrak{R}(b_n)$ *iff* they are both Cauchy and equivalent

Notation 14.3.3: *Equivalent sequences*

If $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are equivalent, we write

$$(a_n)_{n=m_a}^{\infty} \sim (b_n)_{n=m_b}^{\infty}.$$

The first thing to notice is that the notion of equivalent (def. 14.3.2) doesn't care about the starting point of the sequences. We analyze the equivalency for $n \geq \max(m_a, m_b)$ only, because we only care about the behavior of the sequence as it *progresses*. Similar to how we've defined Cauchy sequences, we can also break def. 14.3.2 into two parts: ε-close,

[1]It is left as an exercise for the reader to prove that these two sequences are Cauchy.

and eventual ε-close. I shall go over them very briefly as it is analogous to how Cauchy sequences (def. 14.2.1) are defined.

Two sequences $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are ε-close *iff*

$$|a_n - b_n| \leq \varepsilon \tag{14.17}$$

These sequences are eventually ε-close if there exists a number $N > \max(m_a, m_b)$ in which $(a_n)_{n=N}^{\infty}$ and $(b_n)_{n=N}^{\infty}$ are ε-close. Finally, these sequences are equivalence if it is eventually ε-close for all rational ε.

The second thing to notice is that def. 14.3.2 doesn't care about whether the sequences are Cauchy or not. E.g.,

$$1, 2, 3, 4, 5, 6, \ldots, \quad \text{and} \quad 3, 3, 3, 4, 5, 6, \ldots, \tag{14.18}$$

are equivalent but not Cauchy.[2] But if two sequences are equivalent and one of them is Cauchy, we can guarantee that the other one must also be Cauchy.

Corollary 14.3.4 Let $(a_n)_{n=m_a}^{\infty} \sim (b_n)_{n=m_b}^{\infty}$. If one of the sequence is Cauchy, then the other must also be. If one of the sequence isn't Cauchy, then the other is also not Cauchy.

Proof. By def. 14.3.2, $(a_n)_{n=m_a}^{\infty} \sim (b_n)_{n=m_b}^{\infty}$ means

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists\big(N \geq \max(m_a, m_b) \in \mathbb{N}\big)\forall(n \geq N \in \mathbb{N}) : |a_n - b_n| \leq \varepsilon. \tag{14.19}$$

Let's assume that $(a_n)_{n=m_a}^{\infty}$ is Cauchy, i.e.,

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists\big(N \geq \max(m_a, m_b) \in \mathbb{N}\big)\forall(j, k \geq N \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon. \tag{14.20}$$

[2] It is left as an exercise for the reader to prove that these two sequence aren't Cauchy, but are still equivalent.

We then want to prove that $(b_n)_{n=m_b}^{\infty}$ is also Cauchy, i.e.,

$$\forall(\delta > 0 \in \mathbb{Q})\exists\big(N \geq \max(m_a, m_b) \in \mathbb{N}\big)\forall(j, k \geq N \in \mathbb{N}) : |b_j - b_k| \leq \delta.$$

This may seem a bit outrageous, but notice that

$$|b_j - b_k| = |b_j - a_j + a_j - a_k + a_k - b_k|. \tag{14.21}$$

By the triangle's inequality, we get

$$|b_j - a_j + a_j - a_k + a_k - b_k| \leq |b_j - a_j| + |a_j - a_k| + |a_k - b_k|. \tag{14.22}$$

From eq. (14.20) and section 14.3,

$$|b_j - a_j| + |a_j - a_k| + |a_k - b_k| \leq 3\varepsilon, \tag{14.23}$$

or $|b_j - b_k| \leq 3\varepsilon$. If $\delta = 3\varepsilon$, then $|b_j - b_k| \leq \delta$ as desired. □

14.4 Addition and multiplication on the reals

As said earlier, we want to define arithmetic on the reals as operations on Cauchy sequences. Since the definition of division has much nuance, we shall focus on addition, subtraction, and multiplication for now. Similar to how addition and multiplication on lower number systems, we shall define the operations, then prove that they are well-defined.

Definition 14.4.1: *Addition on the reals*

Let there be two Cauchy sequences $(a_n)_{n=m_a}^{\infty}$ which represents the real $x = \Re(a_n)$, and $(b_n)_{n=m_b}^{\infty}$ which represents the real $y = \Re(b_n)$. The sum of x and y, denoted $x + y$, is given by the real number $\Re(a_n + b_n)$ represented by the sequence $(a_n + b_n)_{n=\max(m_a, m_b)}^{\infty}$, i.e.,

$$x + y = \Re(a_n) + \Re(b_n) = \Re(a_n + b_n). \tag{14.24}$$

To actually prove that addition on the reals is well-defined, we have to make sure that the sum of two Cauchy sequences are Cauchy first.

> **Lemma 14.4.2:** *The sum of two Cauchy sequences are Cauchy* Let there be two Cauchy sequences $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$. Then, $(a_n + b_n)_{n=\max(m_a,m_b)}^{\infty}$ is also a Cauchy sequence.

Proof. $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are Cauchy means that for all rational ε,

$$\exists(A > m_a \in \mathbb{N})\forall(j,k \geq A \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon, \tag{14.25}$$

$$\exists(B > m_b \in \mathbb{N})\forall(j,k \geq A \in \mathbb{N}) : |b_j - b_k| \leq \varepsilon. \tag{14.26}$$

Let $C = \max(A, B)$; therefore,

$$\forall(j,k \geq C \in \mathbb{N}) : \big(|a_j - a_k| \leq \varepsilon\big) \wedge \big(|b_j - b_k| \leq \varepsilon\big). \tag{14.27}$$

We want to show that $(a_n + b_n)_{n=\max(m_a,m_b)}^{\infty}$ is Cauchy, i.e.,

$$\forall(\delta > 0 \in \mathbb{Q})\exists\big(N > \max(m_a, m_b)\big)\forall(j,k \geq C \in \mathbb{N}) : \\ \Big|(a_j + b_j) - (a_k + b_k)\Big| \leq \delta \tag{14.28}$$

From eq. (14.27), we get that for all $j, k > C$,

$$|a_j - a_k| + |b_j - b_k| \leq 2\varepsilon \tag{14.29}$$

By the triangle's inequality, $|a_j - a_k + b_j - b_k| \leq |a_j - a_k| + |b_j - b_k|$; thus,

$$\Big|a_j - a_k + b_j - b_k\Big| \leq 2\varepsilon \tag{14.30}$$

$$\Big|(a_j + b_j) - (a_k + b_k)\Big| \leq 2\varepsilon. \tag{14.31}$$

Let $\delta = 2\varepsilon$, and we get $\Big|(a_j + b_j) - (a_k + b_k)\Big| \leq \delta$ as desired. □

The direct consequence of lem. 14.4.2 is that the product of two Cauchy sequence is also bounded because Cauchy sequences are bounded. Then, we prove that addition on the reals is well-defined.

Proposition 14.4.3: *Addition on the reals is well-defined* Let there be two Cauchy sequences in which $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$, and another Cauchy sequence $(b_n)_{n=m_b}^{\infty}$. If $x = \mathfrak{R}(a_n)$, $x' = \mathfrak{R}(a'_n)$, and $y = \mathfrak{R}(b_n)$ then

$$x + y = x' + y \tag{14.32}$$

One other way to write this proposition is that sums of equivalent Cauchy sequences are equivalent.

Proof. Let $m = \max(m_a, m_{a'}, m_b)$, then $(a_n)_{n=m}^{\infty} \sim (a'_n)_{n=m}^{\infty}$ can be written as

$$\exists(N \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : |a_n - a'_n| \leq \varepsilon. \tag{14.33}$$

We want to show that $x + y = x' + y$, i.e., $(a_n + b_n)_{n=m}^{\infty} \sim (a'_n + b_n)_{n=m}^{\infty}$. By lem. 14.4.2, these two sequences must be Cauchy, so the final result is guaranteed to be a real number. This proof uses the same trick as the one before: adding and subtracting the same term. From eq. (14.33),

$$|a_n - a'_n| \leq \varepsilon \tag{14.34}$$

$$|a_n - b_n + b_n - a'_n| \leq \varepsilon \tag{14.35}$$

$$|(a_n - b_n) - (a'_n + b_n)| \leq \varepsilon, \tag{14.36}$$

meaning that $(a_n + b_n)_{n=m}^{\infty} \sim (a'_n + b'_n)_{n=m}^{\infty}$; thus,

$$\mathfrak{R}(a_n + b_n) = \mathfrak{R}(a'_n + b_n). \tag{14.37}$$

By def. 14.4.1,

$$\mathfrak{R}(a_n) + \mathfrak{R}(b_n) = \mathfrak{R}(a'_n) + \mathfrak{R}(b_n) \tag{14.38}$$

$$x + y = x' + y$$

□

Because I do not want to write separate proof for addition and subtraction, I shall embed the concept of negative numbers in the reals and define $x - y = x + (-y)$. This way, we only have to introduce the negation of the real numbers, and prove that it is well-defined instead of defining subtraction as its own operation, which would be more cumbersome.

Definition 14.4.4: *Negation of a real number*

Given a Cauchy sequence $(a_n)_{n=m}^{\infty}$ in which $x = \mathfrak{R}(a_n)$, we define

$$-x = \mathfrak{R}(-a_n). \tag{14.39}$$

Proposition 14.4.5: *Negation of a real is well-defined* If two Cauchy sequences $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$, and $x = \mathfrak{R}(a_n)$ and $x' = \mathfrak{R}(a'_n)$. Then,

$$-x = -x' \tag{14.40}$$

Another way to say this is that the negation of two equivalent Cauchy sequences are equivalent.

Proof. $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$ means for all rational ε,

$$(N \geq \max(m_a, m_b) \in \mathbb{N}) \forall (n \geq N \in \mathbb{N}) : |a_n - a'_n| \leq \varepsilon. \tag{14.41}$$

From the property of absolute values, $|a_n - a'_n| = |(-a'_n) - (-a_n)|$; therefore, $(-a_n)_{n=m_a}^{\infty} \sim (-a'_n)_{n=m_{a'}}^{\infty}$ and $x = -x'$ as desired. □

Subtraction is then defined as addition of a negated real number.

Definition 14.4.6: *Subtraction on the reals*

If $x = \mathfrak{R}(a_n)$ and $y = \mathfrak{R}(b_n)$, then the difference between x and y is

$$x - y = x + (-y) = \mathfrak{R}(a_n) + \mathfrak{R}(-b_n).$$

Since both negation and addition is well-defined, and negation is written using addition, subtraction must also be well-defined on the reals. We then move on to define multiplication.

Definition 14.4.7: *Multiplication on the reals*

Let there be two Cauchy sequences, $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$, in which $x = \mathfrak{R}(a_n)$ and $y = \mathfrak{R}(b_n)$. The product of x and y, denoted xy, is given by the real number $\mathfrak{R}(a_n b_n)$ represented by the sequence $(a_n b_n)_{n=\max(m_a, m_b)}^{\infty}$, i.e.,

$$xy = \mathfrak{R}(a_n)\mathfrak{R}(b_n) = \mathfrak{R}(a_n b_n). \tag{14.42}$$

Similarly, we must also prove that the product of two Cauchy sequences are Cauchy:

Lemma 14.4.8: *The product of two Cauchy sequences are Cauchy* Let there be two Cauchy sequences $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$. Then, $(a_n b_n)_{n=\max(m_a, m_b)}^{\infty}$ is also a Cauchy sequence.

Proof. $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ are Cauchy means that for all rational ε,

$$\exists(A > m_a \in \mathbb{N})\forall(j, k \geq A \in \mathbb{N}) : |a_j - a_k| \leq \varepsilon, \tag{14.43}$$

$$\exists(B > m_b \in \mathbb{N})\forall(j, k \geq A \in \mathbb{N}) : |b_j - b_k| \leq \varepsilon. \tag{14.44}$$

Let $C = \max(A, B)$; therefore,

$$\forall(j, k \geq C \in \mathbb{N}) : \left(|a_j - a_k| \leq \varepsilon\right) \wedge \left(|b_j - b_k| \leq \varepsilon\right). \tag{14.45}$$

We want to show that $(a_n b_n)_{n=\max(m_a, m_b)}^{\infty}$ is Cauchy, i.e.,

$$\forall(\delta > 0)\exists(N \geq C \in \mathbb{N})\forall(j, k \geq N \in \mathbb{N}) : |a_j b_j - a_k b_k| \leq \varepsilon. \tag{14.46}$$

Notice that

$$|a_j b_j - a_k b_k| = |a_j b_j - a_k b_j + a_k b_j - a_k b_k| \tag{14.47}$$

$$= |b_j(a_j - a_k) + a_k(b_j - b_k)| \tag{14.48}$$

$$|b_j(a_j - a_k) + a_k(b_j - b_k)| \leq |b_j||a_j - a_k| + |a_k||b_j - b_k| \tag{14.49}$$

From eq. (14.45),

$$|b_j||a_j - a_k| + |a_k||b_j - b_k| \leq \varepsilon\big(|b_j| + |a_k|\big). \tag{14.50}$$

It is quite tempting to let $\delta = \varepsilon\big(|b_j| + |a_k|\big)$, but it'd be better if the R.H.S. is dependent on the property of the sequence as a whole and not restricted to individual terms. Here is where boundedness comes in. From prop. 14.2.11, every Cauchy sequences are bounded. So, let's make U and V be the bound of $(a_n)_{n=m_a}^{\infty}$ and $(b_n)_{n=m_b}^{\infty}$ respectively, i.e.,

$$\forall (n \geq C \in \mathbb{N}) : \big(|a_n| \leq U\big) \wedge \big(|b_n| \leq V\big); \tag{14.51}$$

thus, $\varepsilon\big(|b_j| + |a_k|\big) \leq \varepsilon(U + V)$. If $\delta = \varepsilon(U + V)$, we get $|a_j b_j - a_k b_k| \leq \delta$ as desired. □

The direct consequence of lem. 14.4.8 is, the product of two Cauchy sequence is also bounded. Then, we prove that multiplication on the reals is well-defined.

Proposition 14.4.9: *Multiplication on the reals is well-defined* Let there be two Cauchy sequences in which $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$, and another Cauchy sequence $(b_n)_{n=m_b}^{\infty}$. If $x = \mathfrak{R}(a_n)$, $x' = \mathfrak{R}(a'_n)$, and $y = \mathfrak{R}(b_n)$ then

$$xy = x'y \tag{14.52}$$

One other way to write this proposition is that the product of equivalent Cauchy sequences are equivalent.

Proof. Let $C = \max(m_a, m_{a'}, m_b)$, then $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$ means that for all rational ε,

$$\exists(N \geq M \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : |a_n - a'_n| \leq \varepsilon. \tag{14.53}$$

Because all the sequences are Cauchy,

$$\begin{aligned}\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N \geq C)\forall(j, k \geq N \in \mathbb{N}) :& \\ \left(\left|b_j - b_k\right| \leq \varepsilon.\right) \wedge \left(\left|a_j - a_k\right| \leq \varepsilon\right) \wedge \left(\left|a'_j - a'_k\right| \leq \varepsilon\right).&\end{aligned} \tag{14.54}$$

We want to show that $(a_n b_n)_{n=C}^{\infty} \sim (a'_n b_n)_{n=C}^{\infty}$, i.e.,

$$\forall(\delta > 0 \in \mathbb{Q})\exists(M \geq C)\forall(m \geq M \in \mathbb{N}) : |a_m b_m - a'_m b_m| \leq \delta. \tag{14.55}$$

First,

$$|a_m b_m - a'_m b_m| = |b_m||a_m - a'_m| \tag{14.56}$$

- From $(a_n)_{n=m_a}^{\infty} \sim (a'_n)_{n=m_{a'}}^{\infty}$, $|a_m - a'_m| \leq \delta$.
- Because $(b_n)_{m=C}^{\infty}$ is Cauchy, it is bounded by some rational B; thus, $\forall(m \leq C) : |b_m| \leq B$.

Combining these together we get

$$|b_m||a_m - a'_m| \leq B\varepsilon. \tag{14.57}$$

If $\delta = B\varepsilon$, then $|a_m b_m - a'_m b_m| \leq \delta$ as desired. □

14.5 Reciprocation and division on the reals

Since division is just multiplication by the reciprocal, we shall only focus on reciprocation for now.

The reciprocal of $(a_n)_{n=m}^{\infty}$ is defined *iff* ,

1. The sequence $(a_n)_{n=m}^{\infty}$ is *not* equivalent to the zero sequence $(0)_{n=m}^{\infty}$: any sequences which are equivalent to the zero sequence represents zero, and the reciprocal of zero is not defined.

2. None of the terms in the sequence are zero. If one of the term in the sequence is zero, say $a_j = 0$, then $\frac{1}{a_j}$ is ∞. Since ∞ isn't a rational number, the sequence $\left(\frac{1}{a_n}\right)_{n=m}^{\infty}$ is not a rational sequence, thus by def. 14.3.1, it is not a real number.

These two restrictions forces us to define what sequence is allowed to be reciprocated. If a sequence doesn't follow these criteria, then its reciprocation is either not a Cauchy sequence of rationals; thus, can't be used to represent real numbers. Let's see some examples.

Example 14.5.1: *Sequences which cannot be reciprocated*

1. Consider the sequence $(a_n)_{n=m_a}^{\infty}$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots \tag{14.58}$$

The reciprocation of this sequence, $\left(\frac{1}{a_n}\right)_{n=m_a}^{\infty}$ is the sequence

$$1, 2, 3, 4, 5, \dots, \tag{14.59}$$

which is obviously not Cauchy. This is because the original sequence is actually equivalent to the zero sequence $(0)_{n=m}^{\infty}$.

2. Consider the sequence $(b_n)_{n=m_b}^{\infty}$:

$$0, 0.9, 0.99, 0.999, 0.9999, \dots \tag{14.60}$$

The reciprocation of this sequence, $\left(\frac{1}{b_n}\right)_{n=m_b}^{\infty}$ is the sequence

$$\infty, 1.11111\dots, 1.01010\dots, 1.00100\dots, 1.00010\dots, \dots. \tag{14.61}$$

Even though this looks like a Cauchy sequence that's somehow approaching the number one, the sequence itself isn't Cauchy because ∞ is not a number. This is because there are 0 in the original sequence.

This problem leads us to define a type of sequence that can be used for reciprocation: *sequences bounded away from zero*.

Definition 14.5.2: *Sequences bounded away from zero*

A sequence $(a_n)_{n=m}^{\infty}$ is bounded away from zero *iff* there exists some rational c in which for all natural $n \geq m$, $|a_n| \geq c$, i.e.,

$$\exists(c > 0 \in \mathbb{Q})\forall(n \geq m \in \mathbb{N}) : |a_n| \geq c. \tag{14.62}$$

This definition is actually a very strong definition which requires the sequence to be strictly non-zero and can't get arbitrarily close to zero either. An immediate consequence is that if a sequence is bounded away from zero, it is not equivalent to the zero sequence. This is because the sequence isn't c-close to zero. A natural question to ask is why we say that there exists some rational $c > 0$ in which $|a_n| \geq c$ instead of just saying $|a_n| > 0$?

Example 14.5.3: *Sequences bounded away from zero vs. Non-zero sequences* Consider the sequence $(a_n)_{n=m}^{\infty}$:

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots . \tag{14.63}$$

Although each term of the sequence is non-zero, i.e., $|a_n| > 0$ for all $n > m \in \mathbb{N}$, it is not bounded away from zero. You can find a term that's ε-close to zero for every ε. By def. 14.5.2, there doesn't exist any rational c in which all $|a_n| \geq c$ because for all c, there is always a term, $a_{\frac{1}{c}+1}$ that's smaller than c. E.g., if $c = \frac{1}{10}$, then $a_{\frac{1}{\frac{1}{10}}+1} = a_{11} = \frac{1}{11}$ is smaller than $\frac{1}{10}$.

Now we can finally define reciprocation.

Definition 14.5.4: *Reciprocation on the reals*

The reciprocal of $x = \mathfrak{R}(a_n)$ where $(a_n)_{n=m}^{\infty}$ is

$$\frac{1}{x} = \mathfrak{R}\left(\frac{1}{a_n}\right) \tag{14.64}$$

where $\frac{1}{a_n}$ is the rational reciprocation of a_n.

For this definition to be valid, the reciprocation on the reals must give a real number, i.e., $\left(\frac{1}{a_n}\right)_{n=m}^{\infty}$ is Cauchy, and also that it is well-defined. First, we prove that reciprocation is closed in the reals

Proposition 14.5.5: *Reciprocation is closed on the reals* If $(a_n)_{n=m}^{\infty}$ is a Cauchy sequence that's bounded away from zero, then $\left(\frac{1}{a_n}\right)_{n=m}^{\infty}$ is also a Cauchy sequence that's bounded away from zero.

Proof. $(a_n)_{n=m}^{\infty}$ is Cauchy means:

$$\forall(\varepsilon > 0 \in \mathbb{Q})\exists(N \in \mathbb{N})\forall(i, j > N \in \mathbb{N}) : |a_i - a_j| < \varepsilon \tag{14.65}$$

□

14.6 Ordering the reals

14.7 Exponentiation

14.8 The supremum and least upper bound

14.9 Boundedness of sequences

14.10 Convergence

You may need to grab a cup of tea and think of what this definition actually means. It's important for you to understand it fully. Let it sink in for a while. Well, I'd tell you that this definition is quite a humongous one. So, it'd be better if I break it down into two more digestible chunks: ε-boundedness and eventual ε-boundedness, in order to ease the understanding.[3]

Definition 14.10.1: *(L, ε)-boundedness*

A sequence of rationals $(a_n)_{n=i}^{f}$ is (L, ε)-bounded if for all terms in the sequence,

$$|a_n - L| \leq \varepsilon.$$

Expanding the absolute value yields

$$-\varepsilon \leq a_n - L \leq \varepsilon \tag{14.66}$$

$$L - \varepsilon \leq a_n \leq L + \varepsilon, \tag{14.67}$$

i.e., all a_n lies inside $L - \varepsilon$ and $L + \varepsilon$. Let's consider a sequence of numbers $(a_n)_{n=0}^{\infty}$ illustrated.

[3]These two definitions are not standard in the literature.

Example 14.10.2: *(L, ε)-bounded sequences*

1. The sequence $-1, 1, -1, 1, -1, \dots$ is $(0, 1)$-bounded, but not $(1, 1)$-bounded because -1 doesn't lie between $1 - 1 = 0$ and $1 + 1 = 2$.

2. The sequence $0.9, 0.99, 0.999, 0.9999, 0.99999, \dots$ is $(0.2, 1)$-bounded, but not $(0.001, 1)$-bounded because 0.9 and 0.99 doesn't lie between $1 + 0.01 = 1.01$ and $1 - 0.01 = 0.99$.

3. The sequence $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots$ is $(0.5, 0.5)$-bounded, but not $(0.5, 0.25)$-bounded because every term that isn't $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$, doesn't lie between $0.5 + 0.25 = 0.75$ and $0.5 - 0.25 = 0.25$.

Definition 14.10.3: *Eventual (L, ε)-boundedness*

A sequence of rationals $(a_n)_{n=i}^{f}$ is eventually (L, ε)-bounded if there exists a number m that lies between i and f, such that $(a_n)_{n=m}^{f}$ is (L, ε)-bounded.

The explanation: Consider any arbitrary sequence of rationals $(a_n)_{n=0}^{\infty}$ where the limit of a_n is L, shown in fig. 14.2. From the definition, there exists some N in which for all n that follows, $|a_n - L| \leq \varepsilon$. Expanding the absolute value yields

$$-\varepsilon \leq a_n - L \leq +\varepsilon \tag{14.68}$$

$$L - \varepsilon \leq a_n \leq L + \varepsilon. \tag{14.69}$$

I.e., for all $n \geq N$, a_n is between $L - \varepsilon$ and $L + \varepsilon$. Illustrated in fig. 14.2, it means that there exists some point a_N in which all the terms that follows it lies in the gray rectangle.

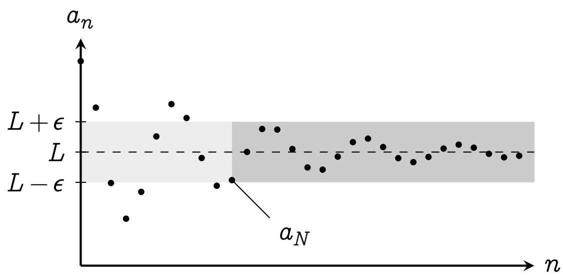


Figure 14.2 | CONVERGENCE OF SEQUENCES

14.11 Construction of the reals by Dedekind cuts

The construction of the reals by Dedekind cuts is arguably way simpler than the Cauchy sequences construction.

Part V

Sequences

CHAPTER 15

Limits of sequences

It's now time to study the sequence of reals in full. All the definitions about sequences in the chapter earlier that includes $\mathbb{Q}$ can now be replaced with $\mathbb{R}$.

15.1 Convergence of sequences

The definition of Cauchy sequence still persists:

Definition 15.1.1: *Cauchy sequence of the reals*

A sequence $(a_n)_{n=m}^{\infty}$ is Cauchy *iff*

$$\forall(\varepsilon > 0 \in \mathbb{R})\exists(N \in \mathbb{N})\forall(i, j \geq N \in \mathbb{N}) : |a_i - a_j| \leq \varepsilon \tag{15.1}$$

This is equivalent to the definition of Cauchy sequence used prior because the rationals are interspersed in the reals.

Since the reals were constructed from the limits of Cauchy sequences, all Cauchy sequences must converge onto a real number, and we define the notion of convergence as follows:

Definition 15.1.2: *Convergence of sequences*

A sequence $(a_n)_{n=m}^{\infty}$ converges to a rational L *iff* for all rational $\varepsilon > 0$, there exists a natural N such that for all natural $n \geq N$, $|a_n - L| \leq \varepsilon$. When this happens, L is called the limit of a_n, denoted $\lim_{n\to\infty}(a_n)$. Formally, this can be written as

$$\lim_{n\to\infty}(a_n) = L \iff \forall(\varepsilon > 0 \in \mathbb{R})\exists(N \in \mathbb{N})\forall(n \geq N \in \mathbb{N}) : |a_n - L| \leq \varepsilon \quad (15.2)$$

15.2 Boundedness and monotonicity

Sometimes, proving that a sequence is Cauchy, or that it converges to anything is hard. But maybe that's not what we want. We just want to know whether the sequence *converges* or not.

Consider, the sequence of logarithms:

$$\ln(1), \ln(2), \ln(3), \ln(4), \ln(5), \ldots \quad (15.3)$$

This sequence behaves like the logarithm. It looks like it will converge. The growth rate of the sequence is diminishingly slow. However, the sequence still manages to creep its way up and diverge to infinity. So, it'd be a waste of time trying to find the limit of this sequence. It'd be better if we can just tell that this sequence doesn't converge in the first place. Developing the tool to do so is the goal of this section.

Definition 15.2.1: *Boundedness of sequences*

A sequence $(a_n)_{n=m}^{\infty}$ is bounded by a real number M *iff*

$$\forall(n \geq m \in \mathbb{N}) : |a_n| < M, \quad (15.4)$$

The number M is the bound of the sequence. If it exists, the sequence is *bounded*

Similar to the notion of boundedness from $\mathbb{Q}$, finite sequences are always bounded (corollary 14.2.10), and every Cauchy sequences are bounded (prop. 14.2.11).

This bound can be made more specific by introducing upper and lower bounds:

Definition 15.2.2: *Upper bounds, lower bounds*

U is the upper bound of a sequence $(a_n)_{n=m}^{\infty}$ *iff*

$$\forall (n \geq m \in \mathbb{N}) : a_n < U \,. \tag{15.5}$$

If U exists, then the sequence is *bounded from above*. Similarly, L is the lower bound of the sequence *iff*

$$\forall (n \geq m \in \mathbb{N}) : L < a_n \,. \tag{15.6}$$

If L exists, then the sequence is *bounded from below*

Notice that a sequence can be bounded from above or below, but isn't overall bounded. E.g., the sequence

$$1, 2, 3, 4, 5, \ldots \tag{15.7}$$

is bounded from below by 0 but isn't bounded from above. It's also not bounded. However, a bounded sequence must be bounded from both below and above.

Corollary 15.2.3: *Boundedness and upper/lower bounds* If a sequence $(a_n)_{n=m}^{m'}$ has a bound M, then there must be a lower bound L and an upper bound U.

Proof. If $(a_n)_{n=m}^{m'}$ is bounded, then

$$\forall(n \in \mathbb{N}; m \leq n \leq m') : |a_n| < M, \quad \text{i.e.,} -M < a_n < M : \tag{15.8}$$

the sequence is bounded above by M and bounded below by $-M$ □

15.3 Divergence

15.4 Subsequences

15.5 Limits of functions

15.6 Continuity

CHAPTER 16

Approximation methods

Prior to this, I've been using a lot of "sloppy" approximations to simplify various calculations. There are two families of such sloppiness: *negligence* and *equivalence*.

Such example of negligence are: if $n \to \infty$ and r_1, r_2 are two finite real numbers, then

$$\frac{(n+r_1)(n+r_2)}{n^2} \approx \frac{(n)(n)}{n^2} = 1 \tag{16.1}$$

because when $n \to \infty$, r_1 and r_2 is negligible compared to n.

The *equivalence* type of approximation is reflected in the small angle approximation: when $\theta \to 0$,

$$\sin\theta \approx \tan\theta \approx \theta, \tag{16.2}$$

and

$$\cos\theta \approx 1 - \frac{\theta^2}{2} \tag{16.3}$$

These approximations are very useful in physical systems. But in mathematics, we ask a totally different set of questions: when is an approximation justified, when are approximations effective, what type of approximations are there, etc. These are the questions that will be answered in this chapter.

16.1 Approximation by negligence

To develop the theory, consider some examples of common approximations by negligence:

Example 16.1.1: *Large numbers* A moderately sized constant is negligible compared to a very large number. E.g., 37 is negligible when compared to 69 420 284 767. So, you could say that

$$69\,420\,284\,767 + 37 \approx 69\,420\,284\,767$$

Example 16.1.2: *Small numbers* A very small constant is negligible compared to a moderately sized constant. E.g., 0.000 000 013 is very small compared to 69, so you could say that

$$69 + 0.000\,000\,013 \approx 69 \tag{16.4}$$

Although these approximations are very useful in general science, it's not so acceptable in the land of mathematics. Mathematics is more interested in approximations that help us understand the behavior of functions. E.g., does the function plateaus into a single value, does it blow up, or does it decrease to zero.

Example 16.1.3: *Large functions* If a function $f(x)$ blows up to infinity when $x \to c$, then adding a constant wouldn't change the fact that the function is still blowing up:

$$\lim_{x\to c}\big(f(x)+N\big)=\lim_{x\to c}f(x)=\infty \tag{16.5}$$

for any finite constants N. However, note that

$$\lim_{x\to c}Nf(x)\neq\lim_{x\to c}f(x) \tag{16.6}$$

This can be further generalized. If a function $g(x)$ doesn't blow up to infinity at c, then

$$\lim_{x\to c}\big(f(x)+g(x)\big)=\lim_{x\to c}f(x), \tag{16.7}$$

and we can say that around c,

$$f(x)+g(x)\approx f(x) \tag{16.8}$$

or that $g(x)$ is negligible compared to $f(x)$.

It is very tempting to say that

$$\lim_{x\to c}\big(f(x)+g(x)\big)=\lim_{x\to c}f(x) \tag{16.9}$$

$$\lim_{x\to c}\big(f(x)+g(x)\big)-\lim_{x\to c}f(x)=0 \tag{16.10}$$

$$\lim_{x\to c}\big(f(x)+g(x)-f(x)\big)=0 \tag{16.11}$$

$$\lim_{x\to c}g(x)=0. \tag{16.12}$$

But this is wrong because the algebraic limit theorem works *iff* the limit exists as a real number. Here, $\lim_{x\to c}\big(f(x)+g(x)\big)=\infty$ is not a real number. Hence,

$$\lim_{x\to c}\big(f(x)+g(x)\big)-\lim_{x\to c}f(x)\neq\lim_{x\to c}\big(f(x)+g(x)-f(x)\big). \tag{16.13}$$

Example 16.1.4: *Small polynomials* If for $x \to c$, $g(x)$ is very small compared to $f(x)$, then

$$\lim_{x \to c} f(x) + g(x) = \lim_{x \to c} f(x) \tag{16.14}$$

16.2 Approximation by equivalence

CHAPTER 17

The derivative

17.1 Invitation: do you understand derivative?

For a full explanation and motivation on derivatives, I'd invite you to go back and read chapter 1. But just to remind you, the derivative of a function $f(x)$ is the rate of change, or the slope of the tangent line of f. It is defined as

$$f'(x) = \frac{\mathrm{d}f}{\mathrm{d}x} = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}, \tag{17.1}$$

which is just the ratio between the increase of the function f when x is incremented by a very small value h. It all makes sense, but why isn't the derivative defined this way instead?

$$\frac{\mathrm{d}f}{\mathrm{d}x} \stackrel{?}{=} \lim_{h\to 0} \frac{f(x+h) - f(x-h)}{2h}. \tag{17.2}$$

Geometrically, this is like comparing the point that succeeds x and the point that precedes x as shown on the left of fig. 17.1. When $h \to 0$, these two limits should intuitively converge to the same value. In fact, one can check that for the function in the example, $f(x) = \frac{x^2}{5}$, these two definitions give the same value. The standard derivative of $\frac{x^2}{5}$ gives $\frac{2}{5}x$. The alternate definition of the derivative can be calculated as follows:

$$\lim_{h\to 0} \frac{f(x+h)-f(x-h)}{2h} \tag{17.3}$$
$$= \lim_{h\to 0} \frac{\frac{1}{5}(x+h)^2 - \frac{1}{5}(x-h)^2}{2h} \tag{17.4}$$
$$= \frac{1}{10} \lim_{h\to 0} \frac{(x+h)^2-(x-h)^2}{h} \tag{17.5}$$
$$= \frac{1}{10} \lim_{h\to 0} \frac{x^2+2xh+h^2-x^2+2xh-h^2}{h} \tag{17.6}$$
$$= \frac{1}{10} \lim_{h\to 0} \frac{4xh}{h} \tag{17.7}$$
$$= \frac{1}{10}(4x) = \frac{2}{5}x \tag{17.8}$$

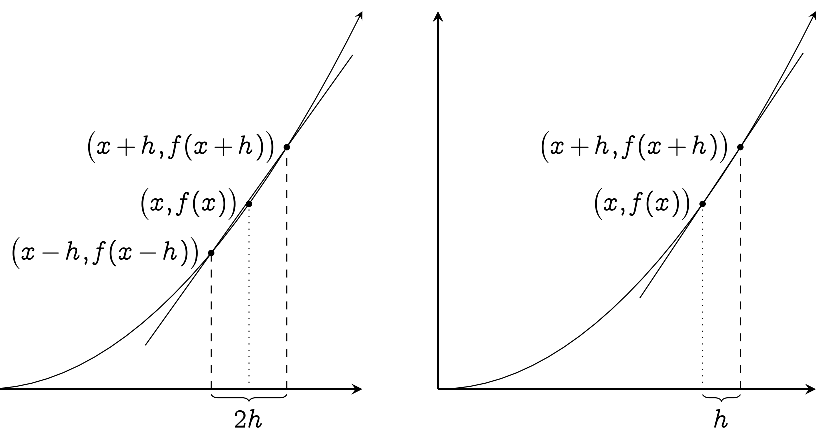


Figure 17.1 | GRAPH OF FUNCTION $\frac{x^2}{5}$ (LEFT) ALTERNATE DEFINITION OF THE DERIVATIVE (RIGHT) STANDARD DEFINITION OF THE DERIVATIVE

In fact, we can take this further and define the derivative as

$$\lim_{h\to 0} \frac{f(x+ah)-f(x-bh)}{(a+b)h}, \tag{17.9}$$

and for the most part, it will give the same result. So why do we not do this?

Is there some intrinsic properties that is only captured by the standard derivative but not captured at all in these other definitions?

The key lies in the notion of continuity and smoothness, which I shall break in two.

17.2 Mean value theorems

17.3 Graph sketching

In chapter 4, we've already seen how calculus can be used to sketch simple graphs of function and understand their geometry. However, I've left out the details of each proofs because real analysis still hasn't been developed. Now that we have those in our hands, let's go back and see why the methods of extrema and concavity actually works.

17.4 First derivatives: extrema

17.5 Second derivatives: concavity

Concavity is a geometrical feature of graphs, and is defined independently from derivatives.

First, let's consider the tangent line of a real function.

Definition 17.5.1: *Concavity*

A function f is *concave* in an interval I *iff*

$$\forall (x \in I) : \tag{17.10}$$

Part VI

Series: sums

CHAPTER 18

Finite series

18.1 Invitation: a treatise to integration

18.2 Convergence

CHAPTER 19

Theory of integration

On terminology: Antiderivatives and integrals are two different things that shouldn't be confused. Although, they will be connected with the fundamental theorem of calculus at the end of this chapter, which is often considered as the milestone of mathematical analysis.

Originally, derivatives and integrals were two completely different things. The ancient Greeks invented the integrals as a way to calculate areas bounded a non-rectangular perimeter. Then, Newton and Liebniz came along and invented derivatives as a measure of rate of change. There were no connections between derivatives and integrals until seventy years later when James Gregory discovered the fundamental theorem of calculus, which links derivatives and integrals together.

In chapter 2, I've glazed over the fundamental theorem of calculus like it is something that is *obviously true*. I've treated it like something that is just defined that way, when really it's not. So, in this chapter, I'd like to take you through the theory of integration. Defining integrals as a measure of area under the curve, unrelated to derivatives, then link it with the derivative later using the fundamental theorem of calculus. Then finally, we shall discuss why *antiderivatives* can be represented as integrals.

19.1 Riemann integrals

A partition $\mathcal{P}$ of a closed subset, a, b is an arranged list of number, $(x_0, x_1, \dots, x_n)$ such that

$$x_0 = a, x_n = b \quad \text{and} \quad x_0 < x_1 < \cdots < x_n. \tag{19.1}$$

19.2 Measures

Part VII

Linear algebra

CHAPTER 20

Geometric linear algebra

20.1 On points and vectors

Linear algebra is an *algebra* of linear things: a line, and an *algebra* is a rule that governs the manipulation and transformation of objects. Hence, linear algebra is a system that allows you to manipulate lines, along with other things that are constructed from lines: planes, cubes, polygons, etc. In some sense, it is a geometry toolkit which allows us to discuss geometry in a more natural way. In this chapter, I'll introduce a few geometrical objects and concepts, including

1. Points and lines,
2. Vectors, vector spaces, and basis vectors,
3. Span,
4. Vectors operations.

This will cover all the prerequisites of linear algebra. Then, we'll discuss what's so *linear* about a line in **??**. Linear algebra will take its form in **??**.

Prerequisites: *Analytical geometry, algebra: systems of linear equations, trigonometry*

20.2 Linear algebra and transformations

20.3 Rotating a graph

20.4 Coordinate dependent transformations

CHAPTER 21

Vector spaces

Now it's a good time to make the concepts of vectors and transformations more precise and abstract, starting with vectors itself.

What does it mean for something to *be* a vector? Normally, we'd say that a vector is something that has a size and a direction. But what are sizes and directions? The intuition of size and direction exists in the Euclidean space only.

21.1 Basic definition of vector spaces

Definition 21.1.1: *Definition of a vector space*

A vector space is a set over a field $\mathbb{F}$ equipped with two operations:

1. Addition: a binary operation from V to itself denoted $+$
2. Multiplication: a binary operation from V and $\mathbb{F}$ to V, denoted by $\cdot$ or by juxtaposition[a].

In order for a set V to be a vector space under $\mathbf{F}$, $+$, and

1. $\exists(\mathbf{0} \in V)\forall(\mathbf{v} \in V) : \mathbf{v} + \mathbf{0} = \mathbf{v}$
2. $\forall(\mathbf{u}, \mathbf{v} \in V) : \mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. $\forall(\mathbf{u}, \mathbf{v}, \mathbf{w} \in V) : \mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$
4. $\forall(\mathbf{u} \in V)\exists(\mathbf{v} \in V) : \mathbf{u} + \mathbf{v} = \mathbf{0}$
5. $\forall a \in \mathbb{F}$:

[a] Juxtaposition is a method of denoting multiplication by putting two variables next to each other. E.g., $a \cdot b$ is *juxtaposed* to ab

CHAPTER 22

Linear ordinary differential equations

Most of the intuitions of ODE's are already written in the first part of this book. Hence, I shall not go over those.

This chapter is on using the theory of linear algebra to solve **linear differential equations** which is defined as follows:

Definition 22.0.1: *Linear differential equations*

A linear differential equation is an equation of the form

$$\sum_{i=0}^{n} f_i(x)\frac{\mathrm{d}^i y}{\mathrm{d}x^i} = g(x) \tag{22.1}$$

where $f_i(x)$'s are i times differentiable w.r.t. x.

We've seen that derivatives are essentially linear operators; hence, we can factor them like one

$$\sum_{i=0}^{n} f_i(x)\frac{\mathrm{d}^i y}{\mathrm{d}x^i} = \sum_{i=0}^{n} f_i(x)D \tag{22.2}$$

Part VIII

Topology

CHAPTER 23

Metric spaces

23.1 Invitation: what is distance?

Mathematicians generalize a lot. And when they do, they strip down everything and boil down the mathematical structure to just a few axioms and definitions. This has happened with the real number system, which we've constructed from the Peano's axiom. If you're still reading here, you probably know that a lot of "elementary" mathematics is given to you for granted. And, I believe that you still remember every drop of sweat and tears that went into constructing the reals from the Peano's axiom.

The construction that we went through, even though sublime, is too specific to the Peano's axiom. It disregards every system that has a different behavior to the successor function. It disregards systems where distance between two numbers are different. Maybe a system without a distance? Maybe a system where the distance is one if two objects are the same, and zero otherwise. We'd have to p construct from scratch again. Therefore, we turn to abstraction.

The main focus of this chapter is on the metric space: a space where there is a notion of distance between two points that *makes sense*. It is a set X that's equipped with a **distance function** $d : X \times X \to \mathbb{R}_{\geq 0}$, aka **the metric**, which acts like a ruler that measures the separation between two points in X.

There is no such straightforward definition of the metric. However, there is a condition of *what* qualifies to be a metric. I could give you those condition now, but I encourage you to try to come up with it yourself first. As this is a practice of abstraction. What condition makes the distance function *acts* like one? Give it a try, and see how it compares with the traditional condition.

I've already spoiled that the metric maps two elements of X onto the $\mathbb{R}_{\geq 0}$. It'd be natural to ask what *should* happen when $d(x, y) = 0$? Well if the distance between two elements is zero, then the two objects should be the identical. Recall from elementary geometry that distance is relative. So, the distance between a and b should be identical to the distance from b to a.

The two condition above is still not enough to define the metric. E.g., $d(x, y) = \sin^2(x - y)$ where x and y are real numbers would still be a distance function according to the two properties above, but it doesn't make intuitive sense as a metric. As it doesn't make sense for distance to oscillate around. Therefore, we need another rule: the triangle inequality, which says that the distance taken by the *side-tracked* path should always be *more* than the direct path.

All in all, we crystallize those properties into the formal definition of metric space and distance function

Definition 23.1.1: *Metric spaces*

A **metric space** is a set-function pair (X, d) where $d : X \times X \to \mathbb{R}^+$ must have the following properties:

1. Nonnegativity: $d(x, y) \geq 0$
2. Injectivity of identity: $d(x, y) = 0 \implies x = y$,
3. Commutativity: $d(x, y) = d(y, x)$,
4. Triangle inequality: $d(x, z) \leq d(x, y) + d(y, z)$,

and is called the **metric**

The triangle inequality here also has its reversed counterpart:

Proposition 23.1.2: *Reversed triangle inequality* Let (X, d) be a metric space, then

$$\forall (x, y, z \in X) : |d(x, y) - d(y, z)| \leq d(x, z). \tag{23.1}$$

Proof. To prove that $|d(x, y) - d(y, z)| \leq d(x, z)$, we must prove that

$$d(x, y) - d(y, z) \leq d(x, z) \quad \text{and} \quad -d(x, z) \leq d(x, y) - d(y, z). \tag{23.2}$$

Using the triangle inequality,

$$d(x, y) \leq d(x, z) + d(z, y) \tag{23.3}$$

$$d(x, y) - d(y, z) \leq d(x, z) \tag{23.4}$$

Then, use the triangle inequality again to get

$$d(y, z) \leq d(y, x) + d(x, z) \tag{23.5}$$

$$-d(x, z) \leq d(x, y) - d(y, z). \tag{23.6}$$

Thus, we've proved the reversed triangle inequality □

Here are some examples of metric spaces to illustrate the capabilities of this abstraction.

Example 23.1.3: *Metric spaces* For all of these, the reader should verify that d qualifies as a metric unless said so.

The reals under absolute value: Let $X = \mathbb{R}$ and $d(x, y) = |x - y|$. This metric space is probably the easiest to understand, and shouldn't need more explanation. It is still a good exercise to verify that it $(\mathbb{R}, d)$ actually forms a metric space.

Proof.

1. The absolute value is never negative; thus, nonnegativity is satisfied.

2. $|x - y| = 0$ means $x - y = 0$, i.e., $x = y$; thus, the metric's identity is injective.

3. By the property of absolute value, $|x - y| = |y - x|$; thus, the metric is symmetric.

4. The triangle inequality of the metric follows from the triangle inequality of absolute value.

Since all the conditions are satisfied, $(\mathbb{R}, d)$ is a metric space. $\square$

The taxicab metric: Let $X = \mathbb{R}^n$, $\mathbf{x} = (x_1, \dots, x_n)$, $\mathbf{y} = (y_1, \dots, y_n)$ and $d(\mathbf{x}, \mathbf{y}) = \sum_{i=1}^{n} |x_n - y_n|$. This metric is also called the **Manhattan metric** because it represents the path that a car needs to take in a city that's obstructed by blocks of building. The distance measured by this of this space is shown as the black lines of fig. 23.1.

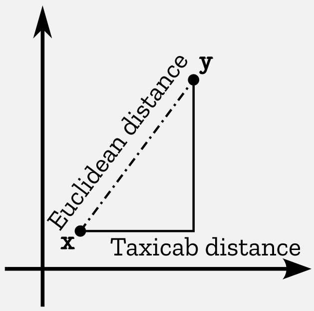


Figure 23.1 | EUCLIDEAN METRIC (IN DASHED) AND TAXICAB METRIC (IN BLACK) IN $\mathbb{R}^2$

Proof. Nonnegativity, injectivity of identity, and the symmetric property follow from the property of absolute values. What's left is for us to prove that the triangle inequality holds, i.e.,

$$\sum_{i=1}^{n} |x_i - z_i| \leq \sum_{i=1}^{n} |x_i - y_i| + \sum_{i=1}^{n} |y_i + z_i| \tag{23.7}$$

From the property of absolute values,

$$|x_i - z_i| \leq |x_i - y_i| + |y_i + z_i| \tag{23.8}$$

Summing up over all the indices will give us the triangle inequality of the taxicab metric. Therefore, the taxicab metric on $\mathbb{R}^n$ forms a metric space. □

The Euclidean metric: The Euclidean metric is a generalization of Pythagorean's theorem to n dimension. Let $X = \mathbb{R}^n$, $\mathbf{x} = (x_1, \dots x_n)$, $\mathbf{y} = (y_1, \dots, y_n)$, and let

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{i=1}^{n} (x_n - y_n)^2}. \tag{23.9}$$

This metric describes the shortest distance between two points in *flat* space, i.e., the Euclidean space. The distance measured is shown as dashed line in fig. 23.1. The proof is in section 23.2.

The discrete metric space: This metric space is arguably the most simple one yet. Simply, let X be a set of *anything*, and let

$$d(x, y) = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases} \tag{23.10}$$

This metric is used most in error correction where the only thing that matters is whether two codewords are the same or not.

Proof. Injectivity of identity and nonnegativity follows from the definition of d. If $x \neq y$, then $y \neq x$ as well, and the distance between them is one; therefore, d is symmetric. For the triangle inequality, if $x = z$, then

$$d(x, z) \leq d(x, y) + d(y, z) \tag{23.11}$$

for all $y \in X$. This is because the L.H.S. is 0, and the R.H.S. is non-negative. If $x \neq z$, then there are three cases.

1. $x \neq y$ and $y \neq z$, then the triangle inequality becomes $1 \geq 1+1$ which is true.
2. $x \neq y$ but $y = z$, then $1 \geq 1 + 0$ which is true.
3. $x = y$ but $y \neq z$, then $1 \geq 0 + 1$, which is also true.

Therefore, the discrete metric under any set forms a metric space.

□

Distance between two functions: This is the most common metric that's used to define correlation between function. If two functions are highly correlated, then the distance between them should be close to zero. Otherwise, it is far away from zero. Let X be the set of continuous functions $f : [\![a, b]\!] \to \mathbb{R}$, and let d be defined by

$$d(f, g) = \int_a^b |f(t) - g(t)| \, \mathrm{d}t \tag{23.12}$$

This integral sums up the absolute distance between two functions at each point. Obviously, this definition is nonnegative, injective, and symmetric. It is then left as an exercise to the reader to prove that the triangle inequality holds.

It would be quite a boilerplate to write "Let there be a metric space (X, d)" every time in a chapter. Therefore, let it be known that (X, d) refers to a metric space with the set X and metric d.

Now what comes next? The answer is: everything that you've seen in the reals. Open intervals, closed intervals, sequences, convergence, etc. We are able to generalize all of those into the notion of metric spaces. But before we do that, I'd like to interject with the proof that the Euclidean metric under the reals form a metric space by using the Cauchy-Schwarz' inequality.

23.2 Interlude: Cauchy-Schwarz' inequality

Since this is a calculus textbook, I wouldn't delve in too much about the interpretation of the Cauchy-Schwarz' inequality because it requires linear algebra. I would also not go into further generalizations of the inequality. For those who are curious, check out Hölder's inequality, Minkowski's inequality, and the Jensen's inequality. If you aren't comfortable with linear algebra, just read the theorem 23.2.1 and the proof that the Euclidean metric on $\mathbb{R}^n$ forms a metric space, and move on to the next section.

Let **a** and **b** be a vector in $\mathbb{R}^n$, the Cauchy-Schwarz' inequality relates the inner product of the two vectors to its length:

$$|\mathbf{a} \cdot \mathbf{b}|^2 = ab \tag{23.13}$$

When we're using linear algebra, this fact becomes trivial because $\mathbf{a} \cdot \mathbf{b}$ is the product of the two lengths, and is scaled by the angle between the two vectors. That scaling is given by $\cos(\theta)$ where θ is the difference in angles between the two vectors. Since the range of the cosine function is between -1 and 1,

$$ab\cos(\theta) \leq ab \tag{23.14}$$

When expanded out, we get the inequality in algebraic form:

Theorem 23.2.1: *Cauchy-Schwarz' inequality*

Let **a** and **b** be points in $\mathbb{R}^n$ with coordinates $a_1, \ldots, a_n)$ and $\mathbf{b} = (b_1, \ldots, b_n)$ respectively. Then,

$$\sum_{i=1}^{n} a_i b_i \leq \sqrt{\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right)} \tag{23.15}$$

Proof. Consider the following quadratic polynomial in x

$$\begin{aligned}\sum_{i=1}^{n}(a_i x + b_i)^2 &= \sum_{i=1}^{n}(a_i^2 x^2 + 2a_i b_i x + b_i^2) \\ &= x^2 \sum_{i=1}^{n} a_i^2 + \left(2\sum_{i=1}^{n} a_i b_i\right)x + \sum_{i=1}^{n} b_i^2\end{aligned} \tag{23.16}$$

This polynomial is always positive because it is a sum of squares of reals. Hence, it must have only one root or no root at all. I.e., the discriminant must be less than or equal to zero:

$$\left(2\sum_{i=1}^{n} a_i b_i\right)^2 - 4\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) \le 0 \tag{23.17}$$

$$4\left(\sum_{i=1}^{n} a_i b_i\right)^2 \le 4\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right) \tag{23.18}$$

$$\sum_{i=1}^{n} a_i b_i \le \sqrt{\left(\sum_{i=1}^{n} a_i^2\right)\left(\sum_{i=1}^{n} b_i^2\right)} \qquad \square$$

Corollary 23.2.2 $\mathbb{R}^n$ under the Euclidean metric forms a metric space.

Proof. Since $(a_i - b_i)^2 = (b_i - a_i)^2$ and $(a_i - b_i)^2 = 0$ *iff* $a_i = b_i$, injectivity of identity and the symmetric property are automatically satisfied.

To prove the triangle inequality, let $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$. Then,

$$\begin{aligned} d(\mathbf{a},\mathbf{c})^2 &= \sum_{i=1}^{n}(a_i - c_i)^2 && (23.19)\\ &= \sum_{i=1}^{n}\big((a_i - b_i) + (b_i - c_i)\big)^2 && (23.20)\\ &= \sum_{i=1}^{n}(a_i - b_i)^2 + \sum_{i=1}^{n}(b_i - c_i)^2 + 2\sum_{i=1}^{n}(a_i - b_i)(b_i - c_i) && (23.21)\\ &= d(\mathbf{a},\mathbf{b})^2 + d(\mathbf{b},\mathbf{c})^2 + \sum_{i=1}^{n}(a_i - b_i)(b_i - c_i) && (23.22) \end{aligned}$$

Using the Cauchy-Schwarz' inequality, we get that

$$\sum_{i=1}^{n}(a_i - b_i)(b_i - c_i) \le \sqrt{\left(\sum_{i=1}^{n}(a_i - b_i)^2\right)\left(\sum_{i=1}^{n}(b_i - c_i)^2\right)}; \tag{23.23}$$

therefore,

$$d(\mathbf{a},\mathbf{c})^2 \le d(\mathbf{a},\mathbf{b})^2 + d(\mathbf{b},\mathbf{c})^2 + 2d(\mathbf{a},\mathbf{b})d(\mathbf{b},\mathbf{c}) \tag{23.24}$$

$$d(\mathbf{a},\mathbf{c})^2 \leq (d(\mathbf{a},\mathbf{b}) + d(\mathbf{b},\mathbf{c}))^2 \tag{23.25}$$

$$d(\mathbf{a},\mathbf{c}) \leq d(\mathbf{a},\mathbf{b}) + d(\mathbf{b},\mathbf{c}). \qquad \square$$

23.3 Generalizing open and closed intervals

This section describes the generalization of open and closed intervals. I believe that if you're here, you already probably know what an interval on the reals is. But just to recapitulate, an interval on the reals is a set of all real numbers between a and b. A closed interval contains its endpoints, meanwhile an open interval doesn't. There are also intervals which aren't closed or open. Those are intervals which only contains one of its endpoints. Here, we extend the notion of open and closed intervals to open and closed sets. Loosely speaking, a subset of the reals is open if it can be written as an infinite union of open intervals, and is closed if it can be written as an intersection of closed intervals. In this section, we shall see how these concepts generalize onto the metric space, providing us with a foundation to build the notion of convergence and continuity later on.

Open and closed intervals are generalized to balls:

Definition 23.3.1: *Balls*

Let (X, d) be a metric space, then

1. An open ball of radius r centered at x is written as $B_X(x;r)$, and is the set $\{y \in X : d(x,y) < r\}$
2. A closed ball of radius r centered at x is written as $\overline{B}_X(x;r)$, and is the set $\{y \in X : d(x,y) \leq r\}$

Put simply,

> An open ball of radius r centered at x is *the set of all objects that is less than r units away from x.*
>
> A closed ball is an open ball which contains *all* its own boundaries.

To visualize this, we can draw a set diagram as shown in fig. 23.2. An open ball is shown as a circle with dashed perimeter. A closed ball is also a circle, but with blacked out perimeter.

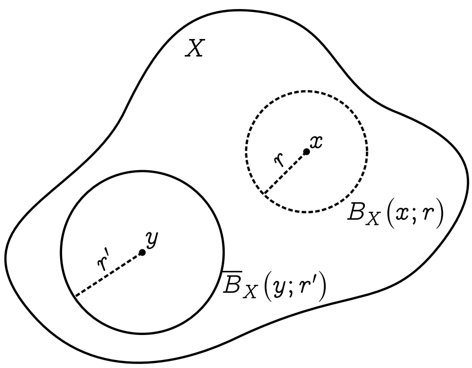


Figure 23.2 | OPEN AND CLOSED BALLS IN A METRIC SPACE

We've actually encountered this concept earlier, disguised in the definition of convergence. As a reminder, a sequence $(a_n)_{n=0}^{\infty}$ converges to L on the reals *iff*

$$\forall(\varepsilon > 0)\exists(N \in \mathbb{N})\forall(n > N \in \mathbb{N}) : |a_n - L| \leq \varepsilon \tag{23.26}$$

This can be generalized into the metric space. Let $(\mathbb{R}, d)$ be a metric space with $d(a_n, L) = |a_n - L|$, then

$$a_n \to L \iff \forall(\varepsilon > 0)\exists(N \in \mathbb{N})\forall(n > N \in \mathbb{N}) : d(a_n, L) \leq \varepsilon \tag{23.27}$$

or,

$$\forall(\varepsilon > 0)\exists(N \in \mathbb{N})\forall(n > N \in \mathbb{N}) : a_n \in \overline{B}_{\mathbb{R}}(L;\varepsilon) \tag{23.28}$$

The definition of open and closed balls is also analogous to the open and closed intervals on the reals:

$$[\![a,b]\!] = \overline{B}_{\mathbb{R}}\left(\frac{a+b}{2};\frac{a-b}{2}\right) \tag{23.29}$$

$$]\!]a,b[\![= B_{\mathbb{R}}\left(\frac{a+b}{2};\frac{a-b}{2}\right) \tag{23.30}$$

One immediate consequence is

Corollary 23.3.2 If $s < r$, then $\overline{B}_X(x;s) \subseteq B_X(x;r)$.

Proof. Let $a \in \overline{B}_X(x;s)$, then $d(a,x) \leq s$. Since $s < r$, $d(a,x) < r$. Therefore, $a \in B_X(x;r)$ aswell. □

We now define the notion of open and closed sets:

Definition 23.3.3: *Open and closed sets*

Let (X,d) be a metric space and let $G \subseteq X$.

1. G is open in X *iff* $\forall(x \in G)\exists(r > 0) : B_X(x;r) \subseteq G$
2. G is closed in X *iff* its complement, $X \backslash G$ is open.

An alternative definition for the closed set is:

Corollary 23.3.4 A set G is closed in X *iff* the following proposition is true:

$$x \notin G \implies \exists(r > 0) : B_X(x;r) \cap G = \emptyset \tag{23.31}$$

Proof. ($\Rightarrow$) Let G be a closed set. By definition, $X\backslash G$ is open, i.e.,

$$\forall(x \in (X\backslash G))\exists(r > 0) : B_X(x;r) \subseteq (X\backslash G) \tag{23.32}$$

Since $B_X(x;r) \subseteq (X\backslash G)$, then $B_X(x;r) \cap G = \emptyset$ as desired.

($\Leftarrow$) If $\forall(x \in (X\backslash G))\exists(r > 0) : B_X(x;r) \cap G = \emptyset$, then $B_X(x;r) \subseteq X\backslash G$, meaning that $X\backslash G$ is open. Thus, G is closed. □

Now that I've provided the definition of open and closed sets, it's time to discuss the interpretation of them. Frankly, there is a lot of nuances hidden in this simple definition.

Openness and interiors: As we shall see later, an open set contains only its interior, i.e., it doesn't have a boundary. Let's see how this intuition correspond to the definition of openness.

Consider a set G which has no boundary. I.e., the boundary of G lies in $X\backslash G$. Let $x \in G$, then there is always some distance between x and the boundary of G. This is because for the distance between x and the boundary to be 0, x *must* be on the boundary, which is not allowed. Therefore, there must be an element of G that's between x and the nearest boundary. I.e., exists an open ball around x such that the open ball is a subset of G.

Closed sets and boundaries: Intuitively, a closed set contains its own boundary. How does corollary 23.3.4 capture this behavior?

There are two ways to interpret this. Firstly, a set G is closed *iff* the complement is open by definition, i.e., the complement doesn't contain the boundary. If the complement doesn't contain the boundary, the boundary must be in the set G itself.

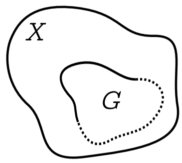


Figure 23.3 | A SET WHICH IS NEITHER OPEN NOR CLOSED

The second way is to use the contrapositive of corollary 23.3.4. A set is closed *iff*

$$\forall(r > 0) : B_X(x; r) \cap G \neq \emptyset \implies x \in G. \tag{23.33}$$

Let G be a closed subset of X, and let there be a point such that all open balls drawn around that point intersect the set. Loosely speaking, a set divides the space into three parts: its interior, boundaries, and its complement. If x is in the interior or the boundary of G, then all open balls must intersect the set. By the condition above, this means x must be in G, i.e., both the interior and boundary is contained in G. Because G contains its own boundary, it must be closed.

A set that's neither open nor closed: Notice that the definition didn't say that a closed set is a set that's not open, as there are some sets doesn't satisfy any of the condition. Such examples of these are half-open intervals on the reals: $[\![a, b[\![$, and sets that partially contains its boundary as shown in fig. 23.3.

Corollary 23.3.5 An empty set is always open, and a metric space is always open in itself.

Empty sets and the universe –clopen sets: There is no element in an empty set; thus, there is no open balls to be drawn. Therefore, the condition for openness is vacuously satisfied. An empty set is open. Thus, its complement, i.e., the universe, must be closed. However, if we look at the metric space as set, X is indeed open in itself. This is because $B_X(x; r)$ only contains the element of the space its in. Therefore, no open balls can escape X, i.e., X is open, and the empty set must be closed. How do we resolve this conflict?

Put simply, X and $\emptyset$ are *both* open and closed — they're **clopen sets**. This is different to a set that's neither open nor closed because a clopen set satisfies both condition of openness and closedness, but a set that's neither open nor closed doesn't satisfy any.

There are two rudimentary facts that I'd like to present. Firstly,

Corollary 23.3.6 An open ball is open, and a closed ball is closed.

Let $c \in X$. Firstly, we want to prove that $B_X(c; r_0)$ is open in X. In other words, we must show that for every $x \in B_X(c; r_0)$, there is an r such that $B_X(x; r) \subseteq B_X(c; r_0)$.

The intuition for this is that, wherever in an open ball you choose, there will always be some space between that point and the edge, even if it's an infinitesimally small one. This is because the only place where the distance would be zero is on the ball's edge itself, and as we've defined, that isn't included in the set.

In the upcoming proof, some geometric assistance will be used for clarity. However, it is important to remember that, at this point, these diagrams are just tools. While it is true that they are useful in supporting arguments and building intuition, pictures simply cannot tackle the problems we are facing here alone, especially with their exotic edge-cases. So, we will still have to rely on algebra to cross the finish line.

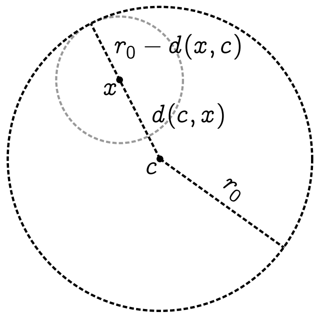


Figure 23.4 | OPEN BALLS ARE OPEN.

As shown in fig. 23.4, start off with an open ball centered at c with radius r_0. Then for any x, draw a line of length r_0 to the boundary of the ball. We see that x splits the line into two. To fit this ball inside the circle, we just have to set it's radius to the distance from x to the edge of the larger ball. This distance is exactly $r_0 - d(x,c)$. Equipped with the intuition, we now write the actual proof

Proof. We shall prove that $\forall\left(a \in B_X\left(x; r_0 - d(x,c)\right)\right) : a \in B_X\left(c; r_0\right)$.

If $a \in B_X\left(x; r_0 - d(x,c)\right)$, then

$$d(a,x) < r_0 - d(x,c) \tag{23.34}$$

$$d(a,x) + d(x,c) < r_0 \tag{23.35}$$

By the triangle inequality,

$$d(a,c) \leq d(a,x) + d(x,c) < r_0 \tag{23.36}$$

$$d(a,c) < r_0 \tag{23.37}$$

This qualifies for a to be in $B_X\left(c; r_0\right)$ by definition of open balls.

A similar proof can be given for closed balls. □

Secondly,

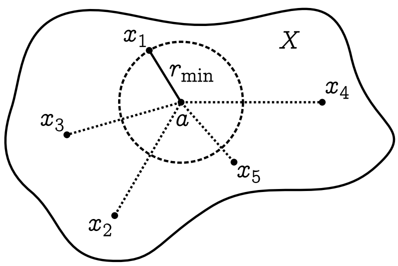


Figure 23.5 | AN EXAMPLE OF A FINITE SET $G = \{x_1, \dots, x_5\}$ IN A SPACE X.

Corollary 23.3.7 A set with a finite number of elements is closed.

Similarly, we use some geometric assistance. A set G with a finite number of elements can be represented as points in X as shown in fig. 23.5. To prove that G is closed is to prove that $X \backslash G$ is open, i.e.,

$$\forall (a \in G) \exists (r > 0) : B_X(a; r) \subseteq X \backslash G. \tag{23.38}$$

Geometrically, we want to draw an open ball that is in X but doesn't include elements of G. How do we do this? Well, just set

$$r = \min\{d(a, x_1), \dots, d(a, x_n)\} = r_{\min} \tag{23.39}$$

as shown in the diagram.

Proof. Let $a \in G$, $r_i = d(a, x_i)$, and $r = \min\{r_1, \dots, r_n\}$. We want to prove that $B_X(x; r) \subseteq X\ G$, i.e., $B_X(x; r) \cap G = \emptyset$.

$$\forall (1 \leq i \leq n) : d(x_i, a) \geq \min\{r_1, \dots, r_n\} \implies d(x_i, a) \geq r. \tag{23.40}$$

But since $B_X(x; r) = \{y \in X : d(y, a) < r\}$, none of x_i belongs in $B_X(x; r)$. Thus, $B_X(x; r)$ must be totally in $X \backslash G$, and G is closed. $\square$

Thirdly, the definition of open sets can also be restated as:

Corollary 23.3.8 Any arbitrary union of open balls form an open set.

As said earlier, the union of open intervals are open, and the union of closed intervals are closed. Well, it's actually not as simple as this: things do get weird when we deal with infinities.

Proposition 23.3.9: *Union of open and closed sets*

1. The union of any number of open sets is open. This includes infinite union of open sets as well.
2. The union of *finitely* many closed sets is closed.

Before we get to the proof, I'd like to give a counterexample to illustrate why we have to care about infinity.

Example 23.3.10: *Union of infinitely many closed sets can be open* Consider a sequence of sets on $\mathbb{R}$

$$F_n = \left[\!\left[-1 + \frac{1}{n}, 1 - \frac{1}{n} \right]\!\right]. \tag{23.41}$$

Then,

$$\bigcup_{n=1}^{\infty} F_n = \left]\!\right] -1, 1 \left[\!\left[, \tag{23.42}$$

which is an open set on $\mathbb{R}$. This is because even though the sequence F_n converges to the interval $]\!]0, 1[\![$, the endpoint is *never* reached. Therefore, even though we take the infinite union of F_n, there is no endpoints; thus, it is open.

This is the intuition. Now, for the proof:

Proof. Let $\bigcup_{n=1}^{\infty} F_n = \mathcal{F}$. To prove that $\mathcal{F} =]\!]-1,1[\![$, we need to prove that

$$\forall(x \in \mathcal{F}) : x \in]\!]-1,1[\![\quad \text{and} \quad \forall(x \in]\!]-1,1[\![) : x \in \mathcal{F} \tag{23.43}$$

($\Rightarrow$) When $n \geq 1$, we get

$$-1 + \frac{1}{n} > -1 \quad \text{and} \quad 1 - \frac{1}{n} < 1. \tag{23.44}$$

Therefore, each interval is contained in $]\!]-1,1[\![$, and their union must also be contained.

($\Leftarrow$) Let $x \in]\!]-1,1[\![$, we must show that

$$\exists(n \geq 1 \in \mathbb{N}) : x \in \left[\!\!\left[-1 + \frac{1}{n}, 1 - \frac{1}{n}\right]\!\!\right] \tag{23.45}$$

I.e., to find n as a function of x.

A number n satisfies the above condition *iff*

$$-1 + \frac{1}{n} \leq x \leq 1 - \frac{1}{n}. \tag{23.46}$$

Separate the equation into

$$-1 + \frac{1}{n} \leq x \quad \text{and} \quad \frac{1}{n} \leq 1 - x. \tag{23.47}$$

Choose

$$\frac{1}{n} \leq \min(x+1, 1-x) \tag{23.48}$$

and we're done. □

Proof of item 1 of prop. 23.3.9. Let $G_1, G_2, \ldots$ be a sequence of open sets. We want to prove that $\mathcal{G} = \bigcup_n G_n$ is open.

For $x \in \mathcal{G}$, $\exists i : x \in G_i$. Since G_i is open, $\exists(r > 0) : B_X(x; r) \subseteq G_i$.

And since $G_i \in \mathcal{G}$, we also get that $B_X(x;r) \in \mathcal{G}$. Hence, $\mathcal{G}$ is open. □

Proof of item 2 of prop. 23.3.9. Let $G_1, \ldots, G_n$ be a finite collection of closed sets, and let $\mathcal{G} = \bigcup_{i=1}^{n} G_n$. We want to prove that $\mathcal{G}$ is closed.

$$x \notin \mathcal{G} \implies \exists (r > 0) : B_X(x;r) \cap \mathcal{G} = \emptyset. \tag{23.49}$$

Let $x \notin \mathcal{G}$, i.e.,

$$x \notin G_1 \wedge \ldots \wedge x \notin G_n \tag{23.50}$$

Since all G_i's are closed,

$$\exists (r_i > 0) : B_X(x;r_i) \not\subseteq G_i. \tag{23.51}$$

Let $r = \min\{r_1, \ldots, r_n\}$. Since $r \le r_1, \ldots, r_n$, we get that $\forall 1 \le i \le n$: $B_X(x;r) \subseteq B_X(x;r_i)$. Hence, $B_X(x;r) \cap G_i = \emptyset$, and

$$B_X(x;r) \cap \mathcal{G} = B_X(x;r) \cap \bigcup_{i=1}^{n} G_i = \bigcup_{i=1}^{n} B_X(x;r) \cap G_i = \emptyset : \tag{23.52}$$

$\mathcal{G}$ is closed as desired. □

Applying de Morgan's law to prop. 23.3.9, we get

Proposition 23.3.11: *Intersection of open and closed sets*

1. The intersection of any number of closed sets is closed.
2. The intersection of *finitely* many open sets is open.

Similarly, there are also instances where infinite intersection of open sets is closed as shown here:

Example 23.3.12: *Infinite intersection of open sets can be closed* Consider a sequence of sets on $\mathbb{R}$

$$F_n = \left]\!\left] -\frac{1}{n}, \frac{1}{n} \right[\!\right[\tag{23.53}$$

Let $\mathcal{F} = \bigcap_{n=1}^{\infty} F_n$. It is left as an exercise to the reader to prove that $\mathcal{F} = \{0\}$. Since $\mathcal{F}$ is a finite set, it is closed by corollary 23.3.7.

23.4 Interiors, closures, and boundaries

I've been saying earlier that an open set contains only its interior and a closed set contains all its boundary. What does *interior* and *boundary* actually means? Here is where we define them.

Definition 23.4.1: *Interior, boundary, and closure*

Let (X, d) be a metric space, and let $A \subseteq X$. Then,

- The interior of A, denoted int A is defined as

$$\operatorname{int} A = \bigcup \{G : (G \text{ is open}) \wedge (G \subseteq A)\}.$$

- The closure of A, denoted cls A is defined as

$$\operatorname{cls} A = \bigcap \{G : (G \text{ is closed}) \wedge (A \subseteq G)\}.$$

- The boundary of A, denoted bnd A is defined as

$$\operatorname{bnd} A = \operatorname{cls} A \setminus \operatorname{int} A.$$

Similar to open and closed sets, there are also subtleties within this definition as we shall discuss here.

Interpreting the definition of interior: Notice that there are no indices in the definition of interior. The only thing that matters is that we're taking the union of *all* the possible open subsets of A. We've seen earlier that all open sets can be written as an infinite union of open balls. The definition of interior then simplifies to taking the union of infinitely open balls that are subset of A. In the diagram, this is like trying to cover A with infinitely many open balls as illustrated in fig. 23.6.

Obviously, all the interior of A gets covered. Good! But what about the boundary of A as shown in black. Is that also covered? The answer is no, since an open ball does not contain its own boundary; therefore, the union of open balls also doesn't contain its boundary. Even though A has a boundary, the boundary can't be reached by the open balls. Therefore, the definition of interior is essentially excluding the boundary from A, creating int A.

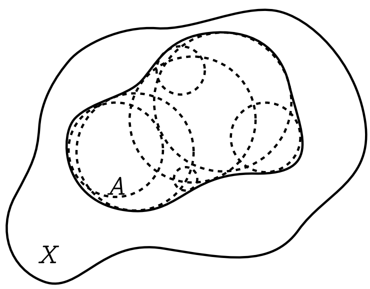


Figure 23.6 | A COVERING OF A BY OPEN BALLS.

Intuitively, this also means that int A is *always* an open subset. This is also proven by prop. 23.3.9. Turns out, the converse is also true. If a set is open, then it is equal to its interior. This will be proven later when we've discussed element chasing in interiors and closures.

Interpreting the closure and boundary: Similarly, there are no indices. We're taking the intersection of *all* the possible closed sets G that contains A. G is *closed* means G has a boundary. Each time you perform an intersection of closed set, it forms another closed set (prop. 23.3.9); therefore, the final set must be closed and contains a boundary. Hence, $\operatorname{cls} A$ is just A but with all the boundary elements.

Since $\operatorname{cls} A$ contains its boundary, and $\operatorname{int} A$ does not, we can subtract $\operatorname{int} A$ from $\operatorname{cls} A$ to obtain the boundary of A; hence, $\operatorname{bnd} A = \operatorname{cls} A \setminus \operatorname{int} A$.

Now, we get to the proof of our interpretations.

When we prove properties about sets, we prefer the method of elements chasing. I.e., to show that everything in the set has a certain property. The next two result gives us a way to chase elements of interiors and closures.

Lemma 23.4.2: *Interiors element chasing* Let (X, d) be a metric space with $A \subseteq X$.

$$x \in \operatorname{int} A \iff \exists (r > 0) : B_X(x; r) \subseteq A \tag{23.54}$$

Proof. $(\Rightarrow)$ Let $x \in \operatorname{int} A$, then by definition, there is an open set $G_i \subset A$ such that $x \in G_i$. Since G_i is open,

$$\exists (r > 0) : B_X(x; r) \subseteq G_i. \tag{23.55}$$

Since $G_i \subseteq \operatorname{int} A$, this ball must also be in $\operatorname{int} A$. Hence, it's possible to draw an open ball around any point $x \in \operatorname{int} A$ such that the open ball is in $\operatorname{int} A$: $\operatorname{int} A$ is open.

($\Leftarrow$) Let $\exists(r > 0) : B_X(x; r) \subseteq A$. By corollary 23.3.6, $B_X(x; r)$ is an open set. Thus, it is contained in $\operatorname{int} A = \bigcup\{G \subseteq A : G \text{ is open}\}$. And since $x \in B_X(x; r)$, $x \in \operatorname{int} A$ aswell. □

Lemma 23.4.3: *Closures element chasing* Let (X, d) be a metric space with $A \subseteq X$.

$$x \in \operatorname{cls} A \iff \forall(r > 0) : B_X(x; r) \cap A \neq \emptyset \tag{23.56}$$

Proof. ($\Rightarrow$) Let $x \in \operatorname{cls} A$. Since $\operatorname{cls} A = \bigcap\{G \supseteq A : G \text{ is closed}\}$, we get that $\forall i : x \in G_i$. □

The size of interiors and closures: Another interpretation that sheds light on the closure and interiors of set is that the interior of A is the largest open set that is a subset of A, and the closure of A is the smallest closed set that contains A.

Proposition 23.4.4 There is no open set containing $\operatorname{int} A$ that is also contained in A

Proof. Suppose that such subsets exists, i.e.,

$$\exists(B : B \text{ is open}) : \operatorname{int} A \subseteq B \subseteq A. \tag{23.57}$$

Recall the definition of interior:

$$\operatorname{int} A = \bigcap\{G : (G \text{ is open}) \wedge (G \subseteq A)\}. \tag{23.58}$$

If B is open and $B \subseteq A$, then B must belong in this union, i.e., $B \subseteq \operatorname{int} A$. Since $\operatorname{int} A \subseteq B$ and $B \subseteq \operatorname{int} A$, $B = \operatorname{int} A$: a contradiction. Hence, this set B must not exist. □

Proposition 23.4.5 There is no other closed subsets of $\operatorname{cls} A$ that also contains A

Proof. Suppose that such subsets exists, i.e.,

$$\exists(B : B \text{ is closed}) : A \subseteq B \subseteq \operatorname{cls} A \tag{23.59}$$

Recall the definition of closures:

$$\operatorname{cls} A = \bigcap \{G : (G \text{ is closed}) \wedge (A \subseteq G)\} \tag{23.60}$$

If B is closed, and $A \subseteq B$, then B must belong in this intersection. That means, $\operatorname{cls} A \subseteq B$ as well. Since $B \subseteq \operatorname{cls} A$ and $\operatorname{cls} A \subseteq B$, $B = \operatorname{cls} A$: a contradiction. Hence, this set B must not exist. □

23.5 Sequences, convergence, and completeness

The metric space also generalizes the notion of convergence:

Definition 23.5.1: *Convergence in metric space*

A sequence $(a_n)_{n=1}^{\infty}$ converges *iff*

$$\forall(\varepsilon > 0 \in \mathbb{R})\exists(N \in \mathbb{N})\forall(i, j > N \in \mathbb{N}) : |a_i - a_j| < \varepsilon \tag{23.61}$$

This is the usual definition of a Cauchy sequence. Like how in the rationals some sequence is Cauchy, i.e., it behaves like a convergent sequence, it doesn't always converge to a value that is in that space. This is like how the sequence

$$3, 3.1, 3.14, 3.141, 3.1415, \ldots \tag{23.62}$$

is *behaviorally convergent* in $\mathbb{Q}$ but doesn't converge anywhere in $\mathbb{Q}$. Rather, this sequence of number converges to π which isn't in $\mathbb{Q}$ but in $\mathbb{R}$. In some metric spaces, there might be some behaviorally convergent sequence that doesn't converge anywhere in the space. We call these space **incomplete metric spaces**. Such example is $\mathbb{Q}$, but there is also one other interesting example: the space of all polynomials under the inner product metric.

Corollary 23.5.2: *The space of all polynomial is an incomplete space*

Let $\mathcal{P}_{\mathbb{R}}$ be the space of all real polynomials confined to some domain $[\![a, b]\!]$, and define the metric to be

$$\sup_{x \in [\![a,b]\!]} |f(x) - g(x)|, \tag{23.63}$$

then there exists some sequence of polynomials which is Cauchy, but does not converge in $\mathcal{P}_{\mathbb{R}}$

23.6 Subspaces of metric spaces and composite metric spaces

23.7 Diameters of subsets and contractions

23.8 Continuity in a metric space

23.9 Heine-Borel property and completeness

Theorem 23.9.1: *Proper implies completeness*

Let (X, d) be a proper metric space, then it is also complete

CHAPTER 24

Topology

24.1 Prerequisites: preimages

Before getting into topology itself, let's familiarize ourselves with the notion of images and preimages.

Definition 24.1.1: *Images and preimages*

Let $f : X \to Y$ be a function. The image of $U \subseteq X$ of the function f is the set

$$f(U) = \{f(x) | x \in U\}. \tag{24.1}$$

The preimage of $V \subseteq Y$ of the function f is the set

$$f^{-1}(V) = \{x | x \in X \wedge f(x) \in V\}, \tag{24.2}$$

24.2 Prelude: continuity and openness

Yet again, topology is a byproduct of mathematicians trying to strip away and generalize things. Metric space generalizes *distance*. Vector spaces generalize *direction*. What more can we generalize? The answer is: *the idea of openness itself*. It might seem stupid at first, but with some insights, it quickly becomes one of the most useful ways to study *continuous functions*. In the view of topology, *continuity* can be built purely from *open sets* itself!

But what do we obtain from this generalization? Surprisingly, much of topology isn't about open set itself, but rather, the property of spaces and shapes that are preserved under *continuous deformations*. E.g., a circle can be continuously deformed into a square. The coordinates and points definitely changed, but the path remains connected: path connectedness is a topological property.

This entire chapter covers the following:

1. Topological spaces
2. Basis and prebasis - useful tool for constructing or destructing topological spaces
3. Continuity and homeomorphism - linking invariant properties of topological spaces
4. Hausdorff property - properties about sequences in a topological space

In the reals, continuity is defined like this:

> If $f : \mathbb{R} \to \mathbb{R}$ is continuous at a, then for all $\varepsilon > 0 \in \mathbb{R}$ there exists some $\delta > 0 \in \mathbb{R}$ in which for all $x \in \mathbb{R}$, $|f(a) - f(x)| < \varepsilon$ implies $|a - x| < \delta$

Metric spaces came along and rewritten our definition to:

> Let X and Y be a metric space. A function $f : X \to Y$ is continuous at a point $a \in X$ if for all open ball of size ε around $f(a)$ ($B_Y\left(f(a);\varepsilon\right)$), there exists an open ball $B_X\left(a;\delta\right)$ such that for all points in $x \in B_X\left(a;\delta\right)$, $f(x) \in B_Y\left(f(a);\varepsilon\right)$ as well.

And we say that a function is continuous if it is continuous at *every point*. This notion of continuity, although powerful, is still limiting as it only works for spaces that are "continuous." Every point in the space *must* form some continuous line, or planes, or volumes, etc. If the space itself is discrete, then it's impossible to tell whether a function is continuous or not. Saying "A continuous function from $\mathbb{R}$ to $\mathbb{Z}$," or "a continuous function from $\mathbb{N}$ to $\mathbb{N}$" doesn't really make sense because $\mathbb{N}$ and $\mathbb{Z}$ aren't continuous—they are discrete spaces.

Topology breaks the requirement of having a continuous space. It creates a new type of space called the **topological space** which encapsulates the metric space, and is capable of detaching the continuity from distance. And it all starts with a slight twist on the notion of continuity:

> Let X and Y be two topological spaces. A function $f : X \to Y$ is *continuous* (the entire function) if every open sets in Y has a preimage that's also open in X.

But what does it mean for a set to be open? Funnily enough, topology says that there are no standards on what sets should be open —the notion of openness is arbitrary, and depends entirely on the topology of interest. However, there are still rules that are imposed on open sets, which we shall discuss in the following section.

24.3 Topological spaces

Definition 24.3.1: *Topological spaces*

A **topology on a set** X is a collection $\mathcal{T}$ of subsets of X which has the following properties

T1) $\emptyset \in \mathcal{T}$

T2) $X \in \mathcal{T}$

T3) $\mathcal{T}$ is closed under any arbitrary unions

T4) $\mathcal{T}$ is closed under finite intersections

The pair $(X, \mathcal{T})$ forms a topology.

If you've read the chapter on metric spaces, the properties of sets in $\mathcal{T}$ should look familiar. In fact, the definition of open sets have gone full circle: the properties of open sets under unions and intersection came back to define what an open set is! Here, we've abstracted the properties of open sets to be:

> All open sets of a space X forms a collection that's closed under arbitrary unions and finite intersections.

An attempt of interpreting topological space: In the standard literature, a topological space is a structure that captures the *continuous coherence* between elements. If two elements a and b are *cohered* to each other in a continuous way, then a set containing a and b should be denoted *open*.

However, this interpretation doesn't work with rules of arbitrary unions and finite intersections. Say a and b cohere to each other, and c and d also coheres with each other. Would you say that a coheres with c, and b coheres with d? In the natural sense, this shouldn't happen because there is no *a priori* relationship between a and c, and b and d, etc. But if accept the notion of open sets as *the set of things that coheres to each other*, the rule of arbitrary union says that if a and b coheres, and c and d coheres, all of them must cohere to each other. So, maybe the notion of *elementwise coherence* isn't fitting for topological spaces. A stronger notion of coherence is needed to encompass the rule of arbitrary unions. It might be the case that topological spaces denote *universal coherence*; as if *coherence* is something that is inhibited by all elements in that space.

If we accept the idea of universal coherence, then topology does make a bit more sense, as you should check with the rule of finite intersections. If a and b coheres and b and c coheres, should b cohere with itself? It should! In some sense, this is saying that coherence is transitive. As far as this interpretation goes, it still has its downfall. What does it mean for the entire space X, and the empty set to be open? Does it have any geometrical meaning? Or was it devised as a mathematical trick to ease analysis?

At this point, I'd like to point out that the notion of topology isn't that well interpreted. It's not like the metric space, measure space, or vector spaces that have a clear interpretation. Metric space generalizes distance, measure spaces generalizes sizes, vector spaces generalizes direction. Usually, it's said that topology also generalizes open sets, and that's what I've used in the beginning of this chapter. But what exactly is an open set? No one seems to know. I've introduced my interpretation, and now it's up to you — the reader — to interpret this mathematical structure in your own way.

Now let's get back to mathematics: first, let's make sure that the notion of metric space can be *encoded* in topological spaces.

Corollary 24.3.2: *Metric spaces are topological spaces* If (X, d) is a metric space, and $\mathcal{T}$ is the collection of all open sets in X, then $(X, \mathcal{T})$ is a topological space

Proof. By corollary 23.3.5, both $\emptyset$ and X are open; hence, the first and second axiom of topological spaces is satisfied. By prop. 23.3.9, arbitrary union of open sets is open, and by prop. 23.3.11, finite intersections of open sets is open: the third and fourth axiom of topological space is satisfied. Therefore, a metric space is a topological space. □

Similar to how there are discrete and indiscrete metrics, there are also discrete topology and indiscrete topology.

Example 24.3.3: *Discrete and indiscrete topology* Let X be an arbitrary set. The **indiscrete topology** on X is given by $\mathcal{T} = \{\emptyset, X\}$, and the discrete topology is given by $\mathcal{T} = \mathcal{P}(X)$ where $\mathcal{P}(X)$ is the collection of every subset of X.

These topologies are usually not that interesting because they are *trivial* — it can be constructed for any sets. Most of the interesting stuffs happen in between where not all sets are open. I shall give a non-exhaustive example of these topologies here:

Example 24.3.4: *Examples of topology*

The standard topology on the reals: In the standard notion of the reals, sets that can be written as unions of open balls are automatically open. However, not all sets are open, e.g., closed intervals aren't open. Under the distance function $d(x, y) = |x - y|$, $\mathbb{R}$ and d forms a metric space. Hence, it is also a topological space.

The order topology: The standard topology on the reals can also be extended to other sets that's equipped with the notion of *order*. The naturals, integers, rationals are all example of sets with order. The subsets of those sets are also sets with order. E.g., for the set $\mathbb{Z}$, I can pull out every odd numbers:

$$\{\dots, -5, -3, -1, 1, 3, 5, \dots\}, \tag{24.3}$$

and this would form a set with order. Even finite subsets of such ordered sets are ordered. E.g.,

$$\{1, 4, 5, 9, 10, 58, 69\} \tag{24.4}$$

are ordered. There is a standard topology that one can impose on these sets: the **order topology**

Define an open ball to be all intervals of type $]\!]a, b[\![= \{x : a < x < b\}$. If the set has a lower bound L, then define any intervals of type $[\![a, b[\![= \{x : a \leq x < b\}$ to be open. Same if the set has an upper bound: define $]\!]a, b]\!] = \{x : a < x \leq U\}$. The order topology contains every sets that can be written as a union of these intervals.

Now let's prove that this is a topology.

Proof. W.I.P. □

Sierpiński's topology (Sierpiński's space): This is the topology that's used to demonstrate many topological properties because it's so simple. Let $X = \{0, 1\}$, and define $\mathcal{T} = \{\emptyset, X, \{1\}\}$. It's left to the reader to check that this forms a topology.

24.4 Basis and prebasis

In metric spaces, every open set can be constructed from an arbitrary union of open balls (corollary 23.3.8). In this sense, the open balls are more rudimentary than open sets, as it can be used to create *any* open sets in the metric space. A natural question to ask would be: is there any analogous *rudimentary objects* in a topological space, such that an arbitrary union of this rudimentary sets always create an open sets? That is the topic of this chapter: basis and prebasis.

Let's consider the set $X = \{a, b, c, d\}$, and a topology $\mathcal{T}$ on X such that

$$\mathcal{T} = \{\emptyset, X, \{a\}, \{c\}, \{a, b\}, \{a, c\}, \{c, d\}, \{a, b, c\}, \{a, c, d\}\} \tag{24.5}$$

Is there any collection $\mathcal{B}$ such that the collection of all arbitrary unions of sets in $\mathcal{B}$ create the set $\mathcal{T}$? Turns out, there is! You can easily check that

$$\mathcal{B} = \{\{a\}, \{c\}, \{a, b\}, \{c, d\}\} \tag{24.6}$$

forms the topology $\mathcal{T}$. Because other than the four sets that are already in $\mathcal{B}$,

$$\begin{aligned} \{a, c\} &= \{a\} \cup \{c\}, \\ \{a, b, c\} &= \{a, b\} \cup \{a, c\}, \\ \{a, c, d\} &= \{a, c\} \cup \{a, d\}, \\ X &= \bigcup_{B \in \mathcal{B}} B \\ \emptyset &= \text{Union of nothing} \end{aligned} \tag{24.7}$$

Hence, one could say that the topology of $\mathcal{T}$ is **generated** from $\mathcal{B}$.

There are many more sets like these that can generate a topology. E.g.,

$$\mathcal{B} = \{\{a\}, \{b\}, \{c\}, \{d\}\}, \quad \text{or} \quad \mathcal{B} = X \tag{24.8}$$

The first one generates the indiscrete topology of X, and the second one, the discrete topology of X.

These sets that generate a topology motivates us to define a **basis** for a topology, which will turn out to be very useful in proving the continuity of a function.

Definition 24.4.1: *Basis of a topology*

A **basis of a topology** $(X, \mathcal{T})$ is a collection of sets $\mathcal{B}$ such that every sets in $\mathcal{T}$ can be written as an arbitrary union of sets in $\mathcal{B}$. In other words,

$$\mathcal{T} = \left\{\bigcup B : B \subseteq \mathcal{B}\right\}. \tag{24.9}$$

We call elements in $\mathcal{B}$, a **basis element**.

The following proposition make use of this definition:

Lemma 24.4.2: *Continuity of basis and continuity of function* Let $f : X \to Y$ between topological spaces X and Y, and let $\mathcal{B}$ be a basis of Y. If for all $B \in \mathcal{B}$, $f^{-1}(B)$ is open, then f is continuous.

Proof. To show that f is continuous, we must show that for any open sets in $U \in Y$, $f^{-1}(U)$ is also open in X

Since U is open, it can be written as an arbitrary union of basis elements $B_1, B_2, B_3, \cdots \subseteq \mathcal{B}$:

$$U = B_1 \cup B_2 \cup B_3 \ldots \tag{24.10}$$

Using the properties of preimage,

$$f^{-1}(U) = f^{-1}(B_1 \cup B_2 \cup B_3 \cup \dots)$$
$$= f^{-1}(B_1) \cup f^{-1}(B_2) \cup f^{-1}(B_3) \dots \quad (24.11)$$

By the assumption that every preimage of a basis element is open, there union must be open; hence, $f^{-1}(U)$ must be open and f is continuous. □

24.5 Continuity and homeomorphism: linking topological structures

This section shall serve as a basic exposition, and also a purpose for studying topology.

In later chapters, you will see many useful topological properties; just to name a few:

1. Compactness: Closed and bounded subsets have an open cover - useful for doing induction
2. Connectedness: Intermediate value theorem and fixed point theorems
3. Separability: Every point in the set can be accessed by sequences

These properties are preserved under continuous function: suppose $(X, \mathcal{T}_X)$ is a topological space with these properties, and there exists a continuous function between $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$, then Y must also have those topological properties. Most of the time, showing that a continuous function exists is easier than showing that Y on its own has these topological properties. The model for X might could be the standard topology on $\mathbb{R}$, and other well-known topologies. Interestingly, the existence of continuous functions between two space can also tell us whether those two spaces are subspaces of each other or not.

In topology however, we are more interested in a special kind of continuous function: a **homeomorphism**, which is a function that's continuous, bijective, and has a continuous inverse. These functions act as "continuous deformations" between spaces, which preserves even more nice topological properties:

1. Hausdorff: The limit of a converging sequence is unique.
2. Countability: Spaces having countable basis, useful for proving properties about sequences as sequences are also countable

This section is just a basic exposition for so much more to come. It'd come back again when we have the tools to explore more.

Definition 24.5.1: *Continuity on a topology*

Let $(X, \mathcal{T}_X)$ and $(Y, \mathcal{T}_Y)$ be a topology, then a function $f : X \to Y$ is continuous *iff*

$$\forall (U \in \mathcal{T}_Y) : f^{-1}(U) \in \mathcal{T}_X \tag{24.12}$$

This is analogous to the notion of continuity in metric spaces. As we shall see, this definition allows us to say that certain functions, or even operations, are continuous.

Example 24.5.2: *Addition is continuous on the standard topology of* $\mathbb{R}$

Note that topological continuity does not focus on *pointwise continuity*. Rather, it focuses on the function as a whole on a topological space.

As you shall see later, this is more useful in practice.

24.6 Convergence, and "housed off": Hausdorff

We first start with the definition of convergence. In the metric space setting, a sequence $(x_n)_{n=0}^{\infty}$ converges to L *iff*

$$\forall(\varepsilon \in \mathbb{R}_{>0})\exists(N \in \mathbb{N})\forall(n \in \mathbb{N}_{>N}) : |x_n - L| < \varepsilon \tag{24.13}$$

In topological spaces, we don't have the glory of the metric that allows us to denote distance. To translate metric convergence to topological convergence, we must abstract the distance into *sets*. The definition prior is equivalent to:

$$\forall(\varepsilon \in \mathbb{R})\exists(N \in \mathbb{N})\forall(n \in \mathbb{N}_{>N}) : x_n \in B_{\mathbb{R}}(L; \varepsilon). \tag{24.14}$$

There's only one thing left: the epsilon.

In a metric space, a set A is open if every point in A has an open ball situated that point which is contained in A. Hence, if A is an open set that contains L, then we can always find an open ball of size ε at L. Since ε is arbitrary, this must work for all open ball; hence, *all open sets as well*. Thus, we've arrived at our definition of topological convergence:

Definition 24.6.1: *Topological convergence*

Let $(X, \mathcal{T})$ be a topological space with a sequence $(x_n)_{n=0}^{\infty}$ in X. We say that $x_n \to L \in \mathbb{X}$ *iff*

$$\forall(U \in \mathcal{T})\exists(N \in \mathbb{N})\forall(n \in \mathbb{N}_{>N}) : x_n \in U. \tag{24.15}$$

Here we arrive at a problem: there are some topological spaces in which under this definition, the limits aren't unique. E.g.,

24.7 Subspaces of topological spaces and composite topological spaces

24.8 Interiors, closures, and boundaries

The definition that I've used in the metric space gracefully generalizes here to topological spaces as well:

Definition 24.8.1: *Interiors, closures, and boundaries*

Let $(X, \mathcal{T})$ be a topological space, and let $A \subseteq X$. Then,

- The interior of A, $\operatorname{int} A = \bigcup\{G : G \in \mathcal{T} \wedge G \subseteq A\}$
- The closure of A, $\operatorname{cls} A = \bigcap\{G : X \backslash G \in \mathcal{T} \wedge A \subseteq G\}$
- The boundary of A, $\operatorname{bnd} A = \operatorname{cls} A \backslash \operatorname{int} A$

These definitions also have the same interpretation as those in metric spaces. The interior of A is the largest open set which is contained in A (prop. 23.4.4), and the closure of A is the smallest closed set which is contained in A (prop. 23.4.5).[1] However, these come with a different rule for element chasing, as there are no more standard definition of open balls. We introduce two more notions:

Definition 24.8.2: *Interior point and adherent point*

Let $(X, \mathcal{T})$ be a topology and let $A \subseteq X$. A point x is an **interior point** of A *iff* there exists some open set $U \in \mathcal{T}$ such that $x \in U \subseteq A$, i.e.,

$$\exists(U \in \mathcal{T}; U \subseteq A) : x \in U \subseteq A. \tag{24.16}$$

[1]The proof that I've used in metric spaces is independent of the metric, but is intrinsic to the nature of sets. Hence, the same proof can also be used in topological spaces.

The point x is an **adherent point** of A *iff* for all open sets $U \subset \mathcal{T}$ that contains x, $U \cap A \neq \emptyset$, i.e.,

$$\forall (U \in \mathcal{T}; x \in U) : U \cap A \neq \emptyset. \tag{24.17}$$

Note that x doesn't have to be in A.

Sometimes, working with basis is easier. Hence, I shall also give an alternative definition of these notions using basis.

Corollary 24.8.3: *Alternative definition of interior and adherent points*

Let $(X, \mathcal{T})$ be a topology, let $A \subseteq X$, and let $x \in A$.

1. x is an interior point of A *iff* there exists some basis element B that contains x and also is contained in A, i.e., $x \in B \subseteq A$
2. x is an adherent point of A *iff* for all basis B that contains x, $B \cap A \neq \emptyset$

Proof of item 1. ($\Rightarrow$) If x is an interior point, it's contained in some open set $U \subseteq A$. By the property of basis, every point in an open set has a basis element B such that $x \in B \subseteq U$.

($\Leftarrow$) If x is contained in some basis element that's also contained in A, x is contained in an open set because a basis element is always open. Hence, x must be the interior point of A $\square$

Proof of item 2. ($\Rightarrow$) If x is an adherent point, then all sets $U \subset \mathcal{T}$ that contains x, $U \cap A \neq \emptyset$. Since all basis are open, they must also have this property; hence, for all basis B that contains x, $x \in B \subseteq U$.

($\Leftarrow$) If for all basis element B contained in A that also contains x, $B \cap A \neq \emptyset$, then intersection between any open set U that contains in x and A is then non-empty because U can be written as an arbitrary union of those bases. $\square$

Lemma 24.8.4: *Interior and closures element chasing*

1. An element x is an interior point of A *iff* it belongs in $\operatorname{int} A$.
2. An element x is adherent to A *iff* it belongs in $\operatorname{cls} A$.

24.9 Countability, and denseness

24.10 Between continuous spaces and discrete spaces: connectivity

Part IX

Differential equations: analysis

CHAPTER 25

Differential forms in Euclidean space. First order differential equations

The equation of particular interest in this chapter is the first order equations of the form:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = f(x, y). \tag{25.1}$$

In most texts

25.1 Separable equations

The R.H.S. of this equation might be tangled with y, which makes it hard to solve. For some differential equations, the function in the R.H.S. can be written as a product of two functions: one dependent only on x and one depedendent only on y:

$$f(x, y) = X(x)Y(y) \tag{25.2}$$

If this is the case, then the equation is **separable**, and solving it is easy.

Definition 25.1.1: *Separable differential equation*

A first order differential equation is separable if it can be written in the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} = X(x)Y(y) \tag{25.3}$$

where X and Y are functions that are dependent only on x and y respectively.

Book III

Functional analysis

CHAPTER 26

Normed space

Like how the metric is a generalization of distance, the normed space is a generalization of *size*.

Definition 26.0.1: *Normed space*

A pair $(E, \|\|)$ where $\|\cdot\| : E \to \mathbb{R}$ is a normed space if $\forall(v, w \in E \wedge a, b \in \mathbb{K})$,

1. $\|v\| \geq 0$ (Positive definiteness)
2. $a\|v\| = a\|v\|$
3. $\|v + w\| \leq \|v\| + \|w\|$

26.1 Normed space from metric space

Index

Bibliography

1. M. Alkousa, *MIT Integration Bee, Solutions of Qualifying Tests from 2010 to 2023, 249 Pages, 15 Chapters.* (Independently Published, May 22, 2023).
2. J. Foster, *u-Substitution* (2024; https://people.math.sc.edu/josephcf/Teaching/142/Files/Worksheets/U-Substitution.pdf).
3. D. Goldrei, *Classic set theory, For Guided Independent Study* (CRC Press, July 1, 1996).
4. E. Landau, *Foundations of analysis* (American Mathematical Soc., Jan. 1, 2001).
5. C. H. C. Little, K. L. Teo, B. Van Brunt, *The number systems of analysis* (World Scientific, Jan. 1, 2003).
6. Math Visualized, *Galois theory explained simply*, Nov. 10, 2020, (https://www.youtube.com/watch?v=Ct2fyigNgPY).
7. T. Tao, *Analysis I, Fourth Edition* (Springer Singapore, sg, ed. 4, Aug. 29, 2022), vol. 1, (https://link.springer.com/book/10.1007/978-981-19-7261-4).
8. J. Turnbull *et al.*, *Oxford Advanced Learner's Dictionary, 8th Edition: Paperback* (OUP Oxford, 2010), ISBN: 9780194799003, (https://books.google.co.th/books?id=t90oQgAACAAJ).

9. user121799, *Graphing the Thomae Function*, Website address for user not found, eprint: https://tex.stackexchange.com/a/461611(2026; https://tex.stackexchange.com/a/461611).

10. user21820 (https://math.stackexchange.com/users/21820/user21820\bibnamedelimb), *How do we know what natural numbers are?*, eprint: https://math.stackexchange.com/q/2251401(2025; https://math.stackexchange.com/q/2251401).

11. Wikipedia contributors, *Summation: Powers and logarithm of arithmetic progressions* (https://en.wikipedia.org/wiki/Summation#Powers_and_logarithm_of_arithmetic_progressions).